

Linear Algebra and Beyond

Joseph Breen

Last updated: January 12, 2023

Department of Mathematics
University of California, Los Angeles

Abstract

This is a set of course notes on (abstract, proof-based) linear algebra. It consists of two parts. Part [I](#) covers the standard theory of linear algebra as is typically taught in a Math 115A course at UCLA. Part [II](#) covers a selection of advanced topics in linear algebra not typically seen in an undergraduate course, but written at a level which is accessible to a 115A graduate. Part [I](#) was originally birthed as a set of rough lecture notes from a 115A course I taught in the spring 2021 term; this part is in the process of being polished and bolstered over the course of the winter 2022 quarter. Part [II](#) is more ambitious and will likely be written over the course of some number of years, partially in collaboration with undergraduate students as part of individual study projects.

Contents

1	Introduction	4
1.1	Organization and information about these notes	7
1.1.1	An important convention regarding purple text	8
1.2	Some history	9
1.3	Acknowledgements	11
I	The Standard Theory	12
2	Vector Spaces	13
2.1	Fields	14
2.1.1	Sets	14
2.1.2	Fields	15
2.1.3	Properties of fields	17
2.2	Vector spaces	23
2.3	Subspaces	28
2.4	Direct sums of subspaces	33
2.5	Span and linear independence	37
2.6	Bases and dimension	43
2.6.1	An aside on proof writing and Principle 2.6.5	51
*2.7	Quotient vector spaces	55
*2.8	Polynomial interpolation	60
2.8.1	The approach using the standard basis	61
2.8.2	The approach using Lagrange polynomials	62
3	Linear Transformations	68
3.1	Linear transformations	68
3.1.1	Kernels and images	70
3.1.2	Rank-nullity theorem	73
3.1.3	Injectivity and surjectivity	74
3.2	Coordinates and matrix representations	78
3.3	Composition of linear maps	84
3.4	Invertibility	88
3.4.1	Invertible matrices and coordinate representations	93
3.5	Change of coordinates	97

*3.6	Dual spaces	103
3.6.1	The transpose of a linear map	107
4	Eigentheory	112
4.1	Invariant subspaces	112
4.2	Diagonalization, eigenvalues and eigenvectors	115
4.3	Diagonalizability as a direct sum of eigenspaces	122
4.4	Computational aspects of eigenvalues via the determinant	124
5	Inner Product Spaces	126
5.1	Inner products	127
5.2	The Cauchy-Schwarz inequality	134
5.3	Orthonormal bases and orthogonal complements	136
5.4	Adjoint	139
5.4.1	The adjoint and the transpose	140
5.5	Self-adjoint operators and the spectral theorem	142
*5.6	Orthogonalization methods	143
5.6.1	Householder transformations	143
5.6.2	QR Decomposition	145
*5.7	An aside on quantum physics	146
6	Determinants	147
6.1	Multilinear maps	147
6.2	Definition of the determinant (proof of uniqueness)	147
6.3	Construction of the determinant (proof of existence)	147
6.4	Computing the determinant	147
6.5	Applications to eigenvalues	147
II	A Taste of Advanced Topics	148
7	Multilinear Algebra and Tensors	149
8	Symplectic Linear Algebra	150
8.1	Motivation via Hamiltonian mechanics	151
8.2	Motivation via complex inner products	154
8.3	Symplectic vector spaces	158
8.3.1	The standard symplectic vector space	159
8.3.2	Properties of symplectic vector spaces	163
8.3.3	Symplectic forms induce volume measurements	165
8.3.4	The cotangent symplectic vector space	167
8.4	Symplectic linear transformations	169
8.4.1	Symplectic matrices	174
8.5	Special subspaces	179
8.6	Compatible complex structures	186
8.6.1	Complex structures	187

8.6.2	Compatible complex structures	189
8.7	Where to from here? Survey articles on symplectic geometry	190
8.7.1	Symplectic manifolds	190
8.7.2	Contact manifolds	190
9	Fredholm Index Theory	191
10	Lie Algebras	192
III	Appendix	193
A	Miscellaneous Review Topics	194
A.1	Induction	194
A.1.1	The size of power sets	195
A.1.2	Some numerical inequalities	197
A.2	Equivalence relations	198
A.2.1	The definition of an equivalence relation	199
A.2.2	An example involving finite fields	200
A.2.3	An example involving subspaces of vector spaces	201
A.3	Row reduction	202
	Bibliography	203

Chapter 1

Introduction

An introduction to Part I

Math 115A is a second course in linear algebra that develops the subject from a rigorous, abstract, and proof-based perspective. If you are reading this, you likely have already taken a course in linear algebra and are wondering why you are taking a second one. This is a valid concern, and in this introduction I hope to suggest that an explanation exists. Answering the question *why is this useful* about any topic in mathematics is difficult to do on an intrinsic and satisfying level; all too often, the importance of definitions, theorems, or even entire fields of math only becomes clear after years of further study. It is hard to appreciate parenting when you are a kid, and this just as true in mathematics. This phenomenon is also not contained to the early years of math education. Professional mathematicians are constantly realizing the importance of definitions they learned years prior; they frequently regret not paying more attention in some specialized course they sat through in grad school, and they constantly wish they had a stronger background in some aspect of math that unexpectedly appeared in their own work.¹ With all these being said, I will do my best to pitch the importance of redeveloping linear algebra — a subject you already know — at an abstract and proof-based level. Know that you may not truly appreciate anything I say until years down the line.

Broadly speaking, there are two main goals of a class like 115A.

- (1) Develop linear algebra from scratch in an abstract setting.
- (2) Improve logical thinking and mathematical communications skills.

Goal (2) can be disparagingly be interpreted as *learn to write the p-word*,² but I don't like describing math this way for reasons that may or may not become clear over the course of these notes. In any case, I'll discuss each of these goals separately, and you should keep them in the back of your mind as you wade deeper into the subject of linear algebra.

(1) Develop linear algebra from scratch in an abstract setting.

In a typical first-exposure linear algebra class, the subject is usually presented as the study of matrices, or at the least it tends to come off in this way. In reality, linear algebra is the study of *vector spaces and their transformations*.

¹To be transparent, this claim is based on personal experience. I am boldly (but confidently) extrapolating to all professional mathematicians!

²Proofs.

I haven't told you what a *vector space* is yet, so currently this sentence should mean very little to you. To continue saying temporarily meaningless things, a *vector space* is simply a universe in which one can do linear algebra. We'll talk about this carefully soon enough, but for now I'll tell you about a vector space that you're already familiar with: $\mathbb{R}^n$, the set of all n -tuples of real numbers. You might be tempted to call this *Euclidean space*, but in my opinion that name encodes much more information than just its vector space structure. In any case, $\mathbb{R}^n$ is baby's first vector space, and in a first-exposure linear algebra class it is usually the *only* vector space that you encounter. In 115A we will develop the theory of linear algebra in other many other vector spaces. Actually, we will develop the theory of linear algebra in *all possible vector spaces that ever have and could ever exist* by working at an extremely abstract level, which turns out to be a useful thing to do. Here are some examples to convince you that this is a worthwhile pursuit. I'll emphasize that I haven't told you what a vector space is, so you should interpret all of these examples as mysterious movie trailers for a variety of mathematical films that star linear algebra.

Example 1.0.1. *Infinite dimensional vector spaces* are important and come up in math all the time. The one vector space you have seen before, $\mathbb{R}^n$, is definitely not infinite dimensional. For example, consider the partial differential equation called the *Laplace equation*:

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0.$$

Don't worry if you don't know anything about partial differential equations — you can just trust me that they are important. The above equation, and many other differential equations, can be presented as a transformation of an infinite dimensional vector space. In particular, the elements of the vector space are the functions $f(x, y)$.

Example 1.0.2. In a similar vein, you may have heard of the *Fourier transform*. Here is the Fourier transform of a function $f(x)$:

$$\mathcal{F}(f)(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} dx.$$

Again, don't worry if this means nothing to you; just trust me that the Fourier transform is important. Looking at the above formula — with an integral, an exponential, and imaginary numbers — it may seem like the Fourier transform is as far from “linear algebra” as possible. In reality, the Fourier transform is just another transformation of an infinite dimensional vector space of functions!

Example 1.0.3. Infinite dimensional vector spaces arise naturally in physics as well. For example, in quantum mechanics, the set of possible states of a quantum mechanical system forms an infinite dimensional vector space. An *observable* in quantum mechanics is just a transformation of that infinite dimensional vector space. Don't ask me too many questions about this, because I don't know anything about quantum physics. But I can read articles and textbooks on quantum physics, *purely because I am very comfortable with abstract linear algebra*.

Example 1.0.4. Finite dimensional vector spaces other than $\mathbb{R}^n$ are also important. There are a number of simple examples I could give, but I'll describe something a little more exotic and

near to my heart. In geometry and topology — the mathematical study of the nature of shape itself — mathematicians are usually interested in detecting when two complicated shapes are either the same or different. One fancy way of doing this is with something called *homology*. You can think of homology as a complicated mathematical machine that eats in a shape and spits out a bunch of data. Oftentimes, that data is a list of vector spaces. In other words, if S_1 and S_2 are two complicated mathematical shapes, and H is a homology machine, you can feed S_1 and S_2 to H to get:

$$\begin{aligned} H(S_1) &= \{V_1, \dots, V_n\} \\ H(S_2) &= \{W_1, \dots, W_n\}. \end{aligned}$$

Here, $V_1, \dots, V_n$ and $W_1, \dots, W_n$ are all vector spaces. If the homology machine spits out different lists for the two shapes, then those shapes must be different! This might sound ridiculous (because it is ridiculous) but if your shapes live in high dimensions that cannot be visualized, it is usually easier to distinguish them by comparing the vector spaces output by a homology machine, rather than trying to distinguish them in some geometric way.

Example 1.0.5. There is an entire field of math called *representation theory* that is built upon the idea of taking some complicated mathematical object and turning into a linear algebra problem. In particular, you *represent* your object as some collection of transformations of a vector space.

My point is that vector spaces of all sizes and shapes are extremely common in math, physics, statistics, engineering, and life in general, so it is important to develop a theory of linear algebra that applies to all of these, rather than just $\mathbb{R}^n$. We will approach the subject by starting from square one. A healthy perspective to take is to forget almost all math you've ever done and treat 115A like a foundational axiomatic course to develop a particular field of math. This is the first goal of 115A.

The last remark about goal (1) that I'll make is the following. You might be thinking: *Wow, linear algebra in vector spaces other than $\mathbb{R}^n$ must be wild and different from what I'm used to! I can't wait to learn all of the new interesting theory that Joe is hyping up!* If you are thinking this, then I'm going to burst your bubble and spoil the punchline of 115A: *Abstract linear algebra in general vector spaces is basically the same as linear algebra in $\mathbb{R}^n$. Nothing new or interesting happens.* We will talk about linear independence, linear transformations, kernels and images, eigenvectors and diagonalization, all topics that you are familiar with in the context of $\mathbb{R}^n$, and everything will work the same way in 115A.

(2) Improve logical thinking and technical communication skills.

At some level, this goal is a flowery way of referring to *proof-writing*, but I don't like boiling it down to something as simple as that. Upper division math (and real math in general) is different than lower division math because of the focus on *discovering and communicating truth*, rather than computation. As such, you should treat every solution you write in 115A (and any other math class, ever) as a mini technical essay. Long gone are the days where you do scratch work to figure out the answer to some problem and then just submit that. High level math is all about polished, logical, and clear communication of truth.

This is difficult to do well and it takes a lot of time and practice! Learning to communicate sophisticated mathematics in a professional and logical way is very much like learning a language. You will not be very good at the beginning, and you cannot become fluent by just reading or watching other people do it well. You must actively practice your mathematical communication skills and get feedback from your instructors and mentors.

An introduction to Part II

The world probably does not need another text on linear algebra, and I can't say that I offer anything unique in the first part of these notes. If anything, I offer some helpful exposition and interesting exercises. The second part of the notes is hopefully a new offering into the world of undergraduate linear algebra, with a biased selection of advanced topics written at an accessible level for a 115A-level student. Some of these topics likely have accessible introductions somewhere, but others — like symplectic linear algebra — do not. As much as I preach about the importance of linear algebra, the best way to internalize this is to get a taste of its appearance and fundamental presence in other areas of math. Part II is meant to be exactly this.

1.1 Organization and information about these notes

Part I. Part I consists of six chapters that develop abstract linear algebra in a more-or-less standard way. There are exercises at the end of each section, some of which are standard (and shamelessly lifted from standard references like [Ax14] and [FIS14]), while others are of my own invention. Some are fairly involved and develop interesting techniques and topics themselves. There are also optional sections in each chapter, indicated by a star next to the section name in the table of contents. Currently, the content of Part I is given by the following chapters.

- ◇ Chapter 2 develops the basic theory of vector spaces over abstract fields. This includes the study of fields, vector spaces, subspaces, linear independence, bases, and dimension. A notable point of emphasis is on the direct sum of subspaces. Optional sections at the end discuss quotient vector spaces and Lagrange interpolation.
- ◇ Chapter 3 develops the theory of linear transformations between vector spaces. This includes injectivity, surjectivity, Rank-Nullity, the notion of invertibility, and a study of coordinates. There is an optional section that discusses dual spaces.
- ◇ Chapter 4 develops the theory of eigenvectors and eigenvalues, notably without the use of determinants.
- ◇ Chapter 5 covers abstract inner product spaces. This includes basic concepts, Cauchy-Schwarz and other inequalities, the theory of orthonormal bases, and some brief theory on adjoints. It will likely be fleshed out more and more over the years.
- ◇ Chapter 6 has yet to be written, but will eventually develop the theory of determinants in a rigorous way in the language of alternating multilinear maps. It will then conclude with some applications to eigenvalue computation.

Part II. Part II will eventually consist of a selection of chapters that introduce advanced topics in math in an accessible way from the viewpoint of a freshly-minted 115A graduate. For the most part, each chapter can be read independently, but will require a good chunk of Part I as a prerequisite. The selection of topics is highly biased toward what I find interesting and what I know well. There are also a number of topics that seem natural to include (like the theory of Hilbert spaces and Banach spaces) Also, the chapters on the front end of Part II will feel closer in spirit to the thoroughly-developed theory of Part I, where the chapters toward the back end will be more casual and expository.

Currently, there is only one chapter (mostly) written.

- ◊ Chapter 8 develops the theory of symplectic linear algebra. Linear symplectic theory is commonly introduced in texts on symplectic geometry, but I am not aware of an introduction to the subject at an accessible undergraduate level. I hope that this chapter can not only serve this purpose, but also be a helpful reference and resource for graduate students that are first exposed to symplectic geometry. I certainly learned a few things in the process of writing this chapter. At the end is a casually written section that surveys some ideas in symplectic geometry, using the linear algebraic theory as a starting point.

Appendix. Part I is mostly self-contained, and also does not use much induction. Thus, I have pushed some review topics from a first linear algebra course (like row reduction, which is also never used in Part I) along with induction to an appendix.

1.1.1 An important convention regarding purple text

One way to become a better writer is to read great authors from the past, study their use of language, and take inspiration from them to build your own style. This advice applies to learning proof-writing and professional mathematical communication. At the risk of pretentiously suggesting that I am a “great mathematical author”, I suggest that you read these notes with the following goals in mind, in exact correspondence with the broad goals I described above:

- (1) Read the notes to learn definitions, concepts, ideas, and examples.
- (2) Pay attention to how I communicate, how I write proofs, and what I write when I solve example problems. This includes my use of language and the format of my writing.

But there is an important caveat, because there is a difference between what I considered good, professional mathematical communication, and casual expository prose. **Any instance of what I consider quality professional mathematical writing will be highlighted in purple**, which is my favorite color. Much of my writing in between formal proofs and solved examples will be pretty casual, more so than in a typical math textbook. This is intentional, because I believe the cold and professional tone of mathematical writing can be difficult to learn from at an early stage. I still want to present you with examples of quality mathematical writing for you to take inspiration from, and thus you should pay special attention to anything written in purple. Just to point out an example of what to pay attention to — in my normal prose, I will use “I” when referring to myself; however, you will notice that I *never* do this in any purple text. Anyway, just keep in mind that non-purple text is me talking with you, and purple text is me demonstrating quality professional mathematical writing.

1.2 Some history

An interesting feature of mathematics (and probably any academic field, but I can only personally speak about math) that is both good and bad is that the way it is taught usually does not reflect how it was first discovered. This is good, in the sense that mathematical discovery is often messy, convoluted, and inefficient. Clarity and elegance in a mathematical theory arises out of years of hindsight and reevaluation. If you were to teach a bread-making class for amateur bakers, you would not begin by having the class mimic ancient recipes and antique techniques from 10,000 years ago — you would teach modern bread-making techniques and follow modern recipes that have had the benefit of thousands of years of scientific and culinary progress. Math is the same way. One of our first definitions in this set of notes will be that of an abstract vector space, but this notion only arose after hundreds of years of messy mathematical discovery in systems of linear equations, the study of matrices, and even physics.

With that being said, while learning math from a modern perspective is important for the sake of the theory, it can obfuscate the underlying history and context. It is hard to appreciate the power and beauty of the notion of a vector space without understanding the hundreds of years of discovery that preceded its inception. Speaking from personal experience, I have found it difficult to learn and appreciate a mathematical theory without knowing the *story* of why it exists. Thus, I hope to give some interesting historical context for our brand of modern abstract linear algebra in this (optional) section.

Egyptian, Babylonian, and Chinese methods (2000 BC — 200 BC)

The history of linear algebra, unsurprisingly, begins with the desire to solve linear equations. Systems like

$$\begin{cases} 2x + y = 8 \\ 3x - y = 2 \end{cases}$$

are natural in both practice (solving problems relating to commerce or measurement) and in theory, and ancient Egyptian, Babylonian, and Chinese mathematicians were studying solutions to systems like this as far as 4000 years ago. Of course, none of these cultures had access to the algebraic notation or manipulations that we have today, so they would have expressed the problems and solutions in a much different format. For example, an Egyptian document called the *Rhind Papyrus*, dating back to 1650 BC and discovered in the mid 1800's [RS90], stated problems mostly verbally and with hieroglyphics from right to left. Linear equations were solved using the method of *false position*, which is essentially a “guess and check (and adjust accordingly)” method. This is also indicative of how the Babylonians solved simple systems of equations [Kle07].

Across the world, Chinese mathematicians were busy composing a book now known in English as *The Nine Chapters on the Mathematical Art*, completed in 200 BC, which is as important to the history of mathematics in the east as Euclid's *Elements* is in the west [SCLL99]. In the eighth chapter, there is a theory of “rectangular arrays”, which is a prototype for solving systems of linear equations using only the coefficients — a precursor to Gaussian elimination, nearly 2000 years before the technique was developed in modern form in the west. *The Nine Chapters on the Mathematical Art* contains systems of equations as complicated as three equa-

tions with five unknowns.

Early linear algebra in the west (1600 — 1850)

It should be emphasized that the coherent theory of linear algebra as we know it now did not start as a coherent theory. That is, it was *not* the case that a group of mathematicians sat around and thought up the basic ideas and concepts of linear algebra and then figured out all of the different ways they could apply it. No classical mathematical theory began this way. Instead, the theory developed in the opposite direction. People from all walks of mathematical life developed techniques and theories to solve (at the time) unrelated problems, and only after hundreds of years did these converge into a coherent theory. In particular, there was a flurry of activity in Europe in the 17th and 18th centuries that birthed many linear algebraic ideas that are very familiar to us now.

Systems of equations

In the mid 1700's, Swiss mathematicians like Gabriel Cramer were interested in geometric questions related to curves in the plane [Kat95]. For example, one could consider a general degree 2 curve in the (x, y) -plane given by the equation

$$A + Bx + Cy + Dx^2 + Exy + Fy^2 = 0$$

for various choices of numbers A, B, C, D, E, F . Different choices of these numbers will yield different equations and thus different curves drawn in the plane. For example, choosing $A = -1, D = 1, F = 1$, and setting the rest of the constants to be 0 gives the equation

$$-1 + x^2 + y^2 = 0$$

which cuts out the unit circle. Or, setting $C = 1, D = -1$, and the rest to be 0 gives

$$y - x^2 = 0$$

which is a parabola. Cramer was interesting in the following question: How many prescribed points are required to *uniquely* describe a degree 2 curve? For example, suppose we pick two of our favorites points, $P = (-1, 0)$ and $Q = (1, 0)$. Do these two points *uniquely* determine a degree 2 curve passing through them? The answer is no. For example, the distinct curves

$$-1 + y - x^2 = 0 \quad \text{and} \quad -1 + x^2 + y^2 = 0$$

both pass through P and Q . In 1750, Cramer published a paper that more-or-less described a general solution to this problem for any degree, not just 2. For the case of a degree 2 curve, he started out by assuming $F = 1$, which is reasonable due to certain symmetries of the equation, so that he was considering curves of the form

$$A + Bx + Cy + Dx^2 + Exy + y^2 = 0$$

He then deduced that if you generically prescribe 5 points

$$(x_1, y_1), \dots, (x_5, y_5)$$

in the plane, there will be a unique such curve degree 2 curve passing through them. He did this by plugging in each (x_j, y_j) into the above equation to generate a system of 5 linear equations with 5 unknowns, A, B, C, D, E :

$$\begin{cases} A + Bx_1 + Cy_1 + Dx_1^2 + Ex_1y_1 + y_1^2 = 0 \\ \vdots \\ A + Bx_5 + Cy_5 + Dx_5^2 + Ex_5y_5 + y_5^2 = 0 \end{cases}.$$

Cramer's contribution was the development of a technique — now known as *Cramer's rule* — to determine when and how such systems could be solved.

For various other reasons, the systematic study of systems of linear equations was also undertaken and further developed by mathematicians like Euler and Gauss in the late 1700's and early 1800's [Kle07]. Euler was the first to study systems that did not necessarily admit unique solutions, and Gauss developed what we now know as *Gaussian elimination*, the successor to the methods of ancient Chinese mathematicians, in an 1811 paper that studied the orbits of asteroids.

Determinants

FINISH

Matrices

FINISH

1.3 Acknowledgements

The overall organization of Part I of this text is partially indebted to the official textbook for Math 115A at UCLA, which is *Linear Algebra* by Friedberg, Insel, and Spence [FIS14]. I also owe an acknowledgement to *Linear Algebra Done Right* by Axler [Ax14], which was a source of inspiration for many of the exercises here and is one of the standard references for linear algebra at this level. Likewise, I'll give a shout out to John Alongi and his unpublished set of linear algebra course notes. I learned linear algebra at this level from John, and many of the mathematical habits that he instilled in me are undoubtedly present here (the good ones — all of the bad habits are my own).

Thank you to Shayan Saadat for providing inspiration for some exercises.

Part I

The Standard Theory

Chapter 2

Vector Spaces

I want to begin by describing the mindset you should be in when approaching linear algebra at this level. If you are reading this, you have already studied a certain brand of linear algebra, along with many other topics in math — calculus, statistics, geometry, differential equations, etc. **I want you to formally forget everything you’ve ever learned about math.** Forget that you’ve learned calculus; forget that you know what a matrix is. Forget even basic algebra, like multiplication and division. In fact, forget even the concept of a number! Pretend you know nothing about mathematics. We are going to wipe the slate nearly clean and build a theory of linear algebra from the ground up. This includes developing simple concepts like that of *numbers* and *addition* and *algebra*.

Of course, I don’t actually want you to forget everything you’ve ever learned, and I will make reference to topics in calculus other areas of math throughout the notes. Don’t take this too literally. This is purely formal, and to get you in the right mindset. I want you to secretly remember all the math you know, but on the face of it you should approach the subject like we are building an abstract theory from first principles.¹

With this in mind, recall that in the introduction I described linear algebra as *the study of vector spaces and their transformations*. Our first order of business is to define the notion of a *vector space* — what I previously referred to as simply an abstract universe in which you can do linear algebra — and study its properties. However, the notion of a vector space encapsulates a great deal of algebraic data in various levels. Because we have wiped the slate clean, we have some preliminary mathematical concepts to build before we can get to vector spaces. In particular, we need some notion of a “number system” that resembles the number systems that you are (secretly) familiar with: the real numbers, the rational numbers, the complex numbers, and so on. The abstract redevelopment of these concepts for our wiped-clean mind (or our alien friend) will lead us to the notion of a *field*. Fields are the basic building block of vector spaces, and they will be the star of the first section of this chapter.

¹Another description of this mindset that I sometimes give is as follows. Pretend that Earth was visited by some bizarre alien being from a faraway galaxy in which there is no concept of math, at all. No numbers, no algebra, nothing. Also suppose that this alien being is highly intelligent and can speak English. The alien being says to you: “I’ve heard about this wonderful thing called *linear algebra*. Please develop this theory for me as rigorously as possible.” To calibrate your exposition, you ask the alien what it knows about math. Does it know how to take a derivative? “Never heard of it.” Does it know how to solve quadratic equations? “What does this word *quadratic* mean?” Does it know what $2 + 2$ equals? “What is this symbol ‘2?’” You soon realize that, although the alien is intelligent and capable of logical reasoning, it has *no* knowledge of any concept in math. So, you really need to start from the very beginning.

2.1 Fields

Before we even discuss fields, I want to introduce you to some notation and basic concepts that will be central to this set of notes, linear algebra, and math as a whole. Even though I suggested that you tactfully forget everything you've ever learned about math, I will assume that you are somewhat familiar with basic set theoretic notions to some extent. For this reason, the first subsection below will be on the brief side and generally will not resemble the rest of the text. For a more thorough development of these notions, I suggest consulting Chapter 1 of [Kap01].

2.1.1 Sets

One of the most fundamental objects of interest in all of math is that of a *set*. A set is just a collection — possibly infinite — of things. For example,

$$S_1 = \{1, 2, -400\}$$

is a set consisting of the numbers 1, 2, and -400 . As another example,

$$S_2 = \{\text{blue}, :, \pi\}$$

is a set consisting of the color blue, a smiley face, and the number π . Believe it or not, this is perfectly valid mathematical set.

Set notation.

For larger sets that are hard to write down explicitly, we will sometimes use the following notation:

$$S_3 = \{n \mid n \text{ is a positive, even number}\}.$$

This notation is read as follows: S_3 is the set consisting of elements of the form n , such that n is any positive, even number. The vertical bar is read as “such that”, and the stuff after the vertical bar is a set of conditions that all elements of the set must satisfy. Unraveling the set S_3 above, this means that

$$S_3 = \{n \mid n \text{ is a positive, even number}\} = \{2, 4, 6, 8, \dots\}.$$

We use the symbol $\in$ to indicate if something is an element of a set. For example, recall the set $S_1 = \{1, 2, -400\}$ from above. We could write

$$2 \in S_1$$

because 2 is an element of S_1 . We could also write

$$3 \notin S_1$$

because 3 is *not* an element of S_1 . The objects of sets can even be as complicated as sets themselves: for example, we could write

$$\{1, 2\} \in \{\text{green}, \pi, \{1, 2\}, -34\}.$$

Unions and intersections.

We can define operations on sets. For example, if A and B are sets, then we define

$$\begin{aligned} A \cup B &:= \{x \mid x \in A \text{ or } x \in B\}, \\ A \cap B &:= \{x \mid x \in A \text{ and } x \in B\}. \end{aligned}$$

The first is the *union* of the sets A and B , and the second is the *intersection*. For example, using S_1 and S_3 from above,

$$\begin{aligned} S_1 \cup S_3 &= \{-400, 1, 2, 4, 6, 8, \dots\} \\ S_1 \cap S_3 &= \{2, -400\}. \end{aligned}$$

The *empty set*, denoted $\emptyset$, is the set consisting of no elements. That is, $\emptyset := \{\}$. Thus, we could write

$$S_1 \cap S_2 = \emptyset.$$

When two sets have empty intersection, we say that they are *disjoint*.

We can also discuss *subsets*. In particular, if A and B are two sets, then we say $A \subset B$ (or $A \subseteq B$, both notations confusingly mean the same thing) if every element of A is also an element of B . For example,

$$\{4, 6, e\} \subset \{4, 6, e, 10, 24\}.$$

One thing that you will have to do often in linear algebra, and in life, is show that two sets are the same. The following is a fundamental principle in math.

Principle 2.1.1. To show that $A = B$, you should separately show that $A \subset B$ and $B \subset A$.

Here are some important sets that you are (secretly!) already familiar with.

$$\begin{aligned} \mathbb{N} &:= \{x \mid x \text{ is a natural number}\} = \{0, 1, 2, 3, \dots\} \\ \mathbb{Z} &:= \{x \mid x \text{ is an integer}\} = \{\dots, -2, -1, 0, 1, 2, \dots\} \\ \mathbb{R} &:= \{x \mid x \text{ is a real number}\} \\ \mathbb{Q} &:= \{x \mid x \text{ is a rational number}\} = \left\{ \frac{p}{q} \mid p, q \in \mathbb{Z}, q \neq 0 \right\} \\ \mathbb{C} &:= \{x \mid x \text{ is a complex number}\} = \{a + bi \mid a, b \in \mathbb{R}\}. \end{aligned}$$

2.1.2 Fields

Some sets are just simple collections of elements with no extra structure. Other sets naturally admit an extra amount of structure and interaction (i.e., *algebra*.) For example, in the set of integers, $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, we are (secretly!) familiar with the algebraic operations of addition (+) and multiplication ($\cdot$). That is, given two elements $n, m \in \mathbb{Z}$, we can construct a third element $n + m \in \mathbb{Z}$ by adding them together, and likewise a fourth element $n \cdot m \in \mathbb{Z}$. Furthermore, these algebraic operations obey a handful of rules (commutativity, distribution,

etc.) that you learned when you were little.

In contrast, consider the set $S_2 = \{\text{blue square}, :, \pi\}$ from before. We don't have any familiar algebraic structure on this set, so for now it will be just be an unstructured collection of random elements.

Recall, again, that you have formally forgotten any notion of algebraic structure. Currently, all we have at our disposal is the notion of a set. We wish to do algebra, and so it is up to us to *define* the correct notions of algebra. In other words, we get to impose our own algebraic rules on an arbitrary set. An important example of this construction is the notion of a *field*

In 115A there is a particular type of set called a *field* that will be of utmost importance. It is a set with two operations that satisfy a bunch of rules. I'll give you the formal definition, and then we'll look at some examples.

Definition 2.1.2. A **field** is a set $\mathbb{F}$ with two operations, addition (+) and multiplication ($\cdot$), that take a pair of elements $x, y \in \mathbb{F}$ and produce new elements $x + y, x \cdot y \in \mathbb{F}$. Furthermore, these operations satisfy the following properties:

- (1) For all $x, y \in \mathbb{F}$,

$$x + y = y + x \quad \text{and} \quad x \cdot y = y \cdot x.$$

- (2) For all $x, y, z \in \mathbb{F}$,

$$(x + y) + z = x + (y + z) \quad \text{and} \quad (x \cdot y) \cdot z = x \cdot (y \cdot z)$$

- (3) For all $x, y, z \in \mathbb{F}$,

$$x \cdot (y + z) = x \cdot y + x \cdot z.$$

- (4) There are elements $0, 1 \in \mathbb{F}$ such that, for all $x \in \mathbb{F}$,

$$0 + x = x \quad \text{and} \quad 1 \cdot x = x.$$

The element 0 is called an **additive identity** and the element 1 is called a **multiplicative identity**.

- (5) For each $x \in \mathbb{F}$, there is an element $x' \in \mathbb{F}$, called an **additive inverse**, such that $x + x' = 0$. Similarly, for every $y \neq 0 \in \mathbb{F}$, there is an element $y' \in \mathbb{F}$, called a **multiplicative inverse**, such that $y \cdot y' = 1$.

Remark 2.1.3. Property (1) is called *commutativity of addition and multiplication*, property (2) is called *associativity of addition and multiplication*, and property (3) is called *distributivity of multiplication over addition*. I'll emphasize that these properties do not comprise a theorem, or a claim about something you already know. We are defining the concept of algebra by arbitrarily imposing these conditions on our given set $\mathbb{F}$.

Example 2.1.4. The main example of a field that you are (secretly!) familiar with is $\mathbb{R}$, the set of real numbers, with the usual operations of addition and multiplication. All of the above properties should look familiar to you, precisely because they are modeled after the behavior of $\mathbb{R}$. Throughout abstract linear algebra we will work with abstract fields $\mathbb{F}$, but when you read the symbol " $\mathbb{F}$ " you can (usually) pretend this is something like " $\mathbb{R}$ " instead.

Example 2.1.5. Other familiar examples of fields are $\mathbb{Q}$ and $\mathbb{C}$; the exercises will define multiplication and addition on $\mathbb{C}$, in case you have not seen this before.

Example 2.1.6. The set of integers, $\mathbb{Z}$, with the usual operations of addition and multiplication, is *not* a field. Almost all of the field properties are satisfied, except for the multiplicative inverse property. In particular, it is *not* the case that for any $y \neq 0 \in \mathbb{Z}$, there is a $y' \in \mathbb{Z}$ such that $y \cdot y' = 1$. For example, the element $2 \in \mathbb{Z}$ does not have a multiplicative inverse. We (secretly) know that such a number would have to be $\frac{1}{2}$, but that number doesn't exist in $\mathbb{Z}$. Similarly, $\mathbb{N}$ is not a field. Not only does $\mathbb{N}$ not in general have multiplicative inverses, but it also doesn't have additive inverses!

Example 2.1.7. Here is an example of a field that you may not have seen before. Let $\mathbb{F}_2 := \{0, 1\}$ be the set consisting of 2 elements, 0 and 1. Define addition in the following way:

$$0 + 0 := 0$$

$$0 + 1 := 1$$

$$1 + 1 := 0$$

and define multiplication as follows:

$$0 \cdot 0 := 0$$

$$0 \cdot 1 := 0$$

$$1 \cdot 1 := 1.$$

I claim that $\mathbb{F}_2$ is a field! I won't verify all of the properties here (that will be an exercise), but I will point out the most interesting aspect: each element has an additive inverse (the additive inverse of 0 is 0, and the additive inverse of 1 is 1), and each nonzero element has a multiplicative inverse (the multiplicative inverse of 1 is 1).

2.1.3 Properties of fields

Next, we will prove some basic properties of fields. Not only will this build up the theory of fields, but it will be our first exposure to writing proofs and communicating mathematics in a professional and rigorous way.

There are many algebraic operations in fields that will (secretly!) be familiar to you, as they will mimic the algebraic rules you learned when you were a baby. But remember — we have formally forgotten everything we have ever known about math, so we must first prove all of these properties using the definition. For example, you (secretly!) know that in an algebraic equation, if you have the same term that appears on both sides then you can “cancel it out.” I didn't say anything about this property in the definition of a field! It turns out that you *can* do this in an abstract field, but we have to prove that we can do so first.

Proposition 2.1.8 (Cancellation laws). *Let $\mathbb{F}$ be a field. Let $x, y, z \in \mathbb{F}$.*

(i) *If $x + y = x + z$, then $y = z$.*

(ii) *If $x \cdot y = x \cdot z$ and $x \neq 0$, then $y = z$.*

Proof. First, we prove (i). Suppose that $x+y = x+z$. By existence of additive inverses (property (5) in the definition of a field), there exists an element x' such that $x+x' = 0$. Adding x' to both sides of the assumed equality gives

$$x' + (x + y) = x' + (x + z).$$

By associativity of addition in a field, this is equivalent to

$$(x' + x) + y = (x' + x) + z.$$

Using the fact that $x' + x = 0$, we have

$$0 + y = 0 + z \quad \text{and thus} \quad y = z.$$

Next, we prove (ii). Suppose that $x \cdot y = x \cdot z$ and $x \neq 0$. By the existence of multiplicative inverses, there is an element x' such that $x' \cdot x = 1$. Multiplying both sides of the assumed equality by x' gives

$$x' \cdot (x \cdot y) = x' \cdot (x \cdot z).$$

By associativity of multiplication, it follows that

$$(x' \cdot x) \cdot y = (x' \cdot x) \cdot z$$

and thus $1 \cdot y = 1 \cdot z$. This implies $y = z$, as desired. $\square$

As a corollary of this result, we can prove another fact that you have likely already taken for granted. Note that in the definition of a field, I never said anything about there only being *one* element called “0”, or only *one* element called “1”. In the number systems that we are used to, there is of course only one of each. But a priori, the definition of a field allows for the possibility that there are multiple 0 elements and multiple 1 elements. Fortunately, we can definitively prove, using the above properties, that these elements are unique.

Corollary 2.1.9. *The elements 0 and 1 in a field are unique. That is, there is only one element satisfying the additive identity property, and only one element satisfying the multiplicative identity property.*

Proof. Suppose that $0' \in \mathbb{F}$ is another additive identity, so that $0' + x = x$ for all $x \in \mathbb{F}$. Then since $0 + x = x$, we have

$$0' + x = 0 + x$$

for all $x \in \mathbb{F}$. By the previous proposition, it follows that $0' = 0$.

Similarly, suppose that there is an element $1' \in \mathbb{F}$ such that $1' \cdot x = x$ for all $x \in \mathbb{F}$. Then since $1 \cdot x = x$, we have

$$1' \cdot x = 1 \cdot x$$

for all $x \in \mathbb{F}$. In particular, we may choose $x = 1$. By the previous proposition, it follows that $1 = 1'$. $\square$

This proof is the first instance of the following general mathematical principle. You will encounter this multiple times over the course of these notes, and you will also be asked to

execute this principle yourself.

Principle 2.1.10. To prove that an object $\star$ is unique, assume that you have another object $\star'$ that satisfies the same properties. Then show that $\star = \star'$.

Using this principle, we can prove a similar statement on the uniqueness of multiplicative and additive inverses.

Corollary 2.1.11. *For each $x \in \mathbb{F}$, the element x' satisfying $x + x' = 0$ is unique. If $x \neq 0$, the element x' satisfying $x' \cdot x = 1$ is unique.*

Proof. Exercise. □

These corollaries allow us to talk about *the* additive identity, *the* multiplicative identity, *the* additive inverse of an element, etc. Furthermore, we can make the following notational definition.

Definition 2.1.12. Let $\mathbb{F}$ be a field, and let $x \in \mathbb{F}$. The unique additive inverse of x is denoted $-x$, and the unique multiplicative inverse (if $x \neq 0$) is denoted x^{-1} or $\frac{1}{x}$.

We continue by proving more familiar properties of real numbers that are true in the general setting of fields.

Proposition 2.1.13. *Let $\mathbb{F}$ be a field, and let $x, y \in \mathbb{F}$. Then*

- (i) $0 \cdot x = 0$,
- (ii) $(-x) \cdot y = x \cdot (-y) = -(x \cdot y)$,
- (iii) $(-x) \cdot (-y) = x \cdot y$,
- (iv) *and 0 has no multiplicative inverse.*

Proof. Exercise. □

We can also now define the notions of subtraction and division in a field, more things that you've taken for granted!

Definition 2.1.14. Let $\mathbb{F}$ be a field. For $x, y \in \mathbb{F}$, define

$$x - y := x + (-y).$$

Similarly, if $y \neq 0$, define

$$\frac{x}{y} := x \cdot \frac{1}{y}.$$

I'll repeat one last time that the point of this section is to abstractly re-develop the concept of a number system from the ground up. We formally forget everything we knew about the real numbers, algebraic manipulation, or even basic concepts like addition and multiplication. Thus, we begin with a raw collection of elements (a set) and we impose our own axioms of algebra on this set, leading to the completely abstract notion of a field. This abstract notion of a field generalizes familiar number systems like $\mathbb{R}$, $\mathbb{Q}$, and $\mathbb{C}$, and it also leads to some surprising new structures like the field with two elements $\mathbb{F}_2 = \{0, 1\}$.

Exercises

1. For each of the following statements, say whether it is *true* or *false*. If a statement is true, prove it. If a statement is false, provide an explicit counter example.

(a) For all subsets A, B, C of some larger set X ,

$$A \cap (B \cup C) = (A \cap B) \cup C.$$

(b) For all subsets A, B, C of some larger set X ,

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

(c) Let $\{A_1, A_2, A_3, \dots\}$ be an infinite collection of sets (so that each A_n is a set). If $A_n \cap A_m \neq \emptyset$ for all $n, m \geq 1$, then

$$\bigcap_{n=1}^{\infty} A_n \neq \emptyset.$$

2. Let $\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$ be the field of complex numbers with addition defined as

$$(a_1 + b_1 i) + (a_2 + b_2 i) := (a_1 + a_2) + (b_1 + b_2) i$$

and multiplication defined as

$$(a_1 + b_1 i)(a_2 + b_2 i) := (a_1 a_2 - b_1 b_2) + (a_1 b_2 + a_2 b_1) i$$

Suppose that $a + bi \neq 0$. Prove that the multiplicative inverse of $a + bi$ is

$$\frac{a}{a^2 + b^2} - \frac{b}{a^2 + b^2} i.$$

3. For each of the following sets, determine whether it is a field. If it is, prove it. If it isn't, explain why.

(a) The set $\mathbb{R}^2 = \{(x, y) \mid x, y \in \mathbb{R}\}$ with addition defined as

$$(x_1, y_1) + (x_2, y_2) := (x_1 + x_2, y_1 + y_2)$$

and multiplication defined as

$$(x_1, y_1) \cdot (x_2, y_2) := (x_1 \cdot x_2, y_1 \cdot y_2).$$

(b) The set of $n \times n$ matrices with real entries

$$M_n(\mathbb{R}) := \{A \mid A \text{ is an } n \times n \text{ real matrix}\}$$

with addition defined as usual matrix addition, and multiplication defined as matrix multiplication.

(c) The set of $n \times n$ invertible matrices with real entries

$$GL_n(\mathbb{R}) := \{ A \mid A \text{ is an } n \times n \text{ invertible real matrix} \}$$

with addition defined as usual matrix addition, and multiplication defined as matrix multiplication.

4. Verify that $\mathbb{F}_2 = \{0, 1\}$ as defined in this section is a field.
5. Verify the rest of the field axioms for the complex numbers $\mathbb{C}$.
6. Let $\mathbb{F}$ be a field and let $x, y \in \mathbb{F}$. Prove that if $x \cdot y = 0$, then either $x = 0$ or $y = 0$.
7. Let $D_n(\mathbb{F})$ be the set of diagonal $n \times n$ matrices with entries in a field $\mathbb{F}$. Define addition in the usual way, and define multiplication as matrix multiplication. Is $D_n(\mathbb{F})$ a field? How is this question different than 3(b) and 3(c) from above?
8. Let $\mathbb{F}$ be a field, and let $x \in \mathbb{F}$. Prove that the element x' satisfying $x + x' = 0$ is unique. If $x \neq 0$, prove that the element x' satisfying $x \cdot x' = 1$ is unique.
9. Let $\mathbb{F}$ be a field. Prove the following statements.
 - (a) For all $x \in \mathbb{F}$, $0 \cdot x = 0$.
 - (b) For all $x, y \in \mathbb{F}$, $(-x) \cdot y = x \cdot (-y) = -(x \cdot y)$.
 - (c) For all $x, y \in \mathbb{F}$, $(-x) \cdot (-y) = x \cdot y$.
 - (d) The additive identity 0 has no multiplicative inverse.
10. Let $\mathbb{F}$ be a field with finitely many elements. Prove that there is a smallest nonzero number $p \in \mathbb{N}$ with $p > 1$ such that

$$\underbrace{1 + 1 + \cdots + 1}_{p \text{ times}} = 0.$$

11. Let $\mathbb{F}$ be a field with finitely many elements, and let p be the smallest number such that

$$\underbrace{1 + 1 + \cdots + 1}_{p \text{ times}} = 0.$$

(See the previous problem.) This number is called the *characteristic* of a finite field. Prove that the characteristic of a finite field must be prime.

12. Let p be a prime number and let $\mathbb{F}_p := \{0, 1, 2, \dots, p-1\}$. Define addition and multiplication in the usual way *modulo* p , that is,

$$x + y := r$$

where $r \in \mathbb{F}_p$ is the unique number such that $x + y = p + r$ (in this last equality, addition is the usual addition). Similarly,

$$x \cdot y := r$$

where $r \in \mathbb{F}_p$ is the unique number such that $x \cdot y = np + r$ for some n . For example, if $p = 5$, then $2 + 3 = 0$, $4 + 4 = 3$, and $3 \cdot 3 = 4$.

- (a) Prove that $\mathbb{F}_p$ is a field.
- (b) Prove that if k is not prime, then the set $\mathbb{F}_k$ defined as above (with k in place of p) is not a field.

13. Can you find a field with 4 elements?

14. Can you find a field with 6 elements?

15. Consider the following set of *Laurent polynomials*:

$$S = \left\{ \sum_{n=-M}^N a_n x^n \mid a_n \in \mathbb{R}, N, M \in \mathbb{N} \right\}$$

Define addition and multiplication in the usual way. Is S a field? (Hint: think about the element $1 + x \in S$.)

16. It is easy to construct a field with a prime number of elements by simply doing modular arithmetic (see the above exercises). If k is not prime, however, modular arithmetic will not produce a field of size k (again, see the above exercises). However, that does not mean that all finite fields have a prime number of elements! It is possible to construct fields with a non-prime number of elements, though in general this can be pretty complicated. In this exercise, I will cryptically walk you through the construction of a field with 9 elements.

- (a) We'll begin by considering $\mathbb{F}_3 = \{0, 1, 2\}$, the field with 3 elements, where arithmetic is modular. If you haven't done this already, prove that $\mathbb{F}_3$ is a field.
- (b) Next, we will consider polynomials with coefficients coming from $\mathbb{F}_3$. For example, $p(x) = 2x^3 + x + 1$ is a valid polynomial with coefficients in $\mathbb{F}_3$, while $q(x) = \pi x^2 - 4.3x + 7$ is not.

Consider the polynomial $p(x) = x^2 + 1$ as a function on $\mathbb{F}_3$. Prove that $p(x)$ does not have any roots, and conclude that $p(x)$ cannot be factored over $\mathbb{F}_3$. In other words, show that $p(0) \neq 0$, $p(1) \neq 0$, and $p(2) \neq 0$.

- (c) Next, we will define a set $\mathbb{F}_9$ as follows.

$$\mathbb{F}_9 = \{ a + bx \mid a, b \in \mathbb{F}_3 \}.$$

Prove that $\mathbb{F}_9$ has 9 elements. List them all out.

- (d) Prove that the equation $x^2 + 1 = 0$ is equivalent to the equation $x^2 = 2$, when viewed over $\mathbb{F}_3$.
- (e) Next, we define addition and multiplication on $\mathbb{F}_9$. In words, addition and multiplication is defined in the usual way for polynomials, except anytime you get an x^2 you replace it with 2. To be explicit, define

$$(a + bx) + (c + dx) := (a + c) + (b + d)x$$

and

$$(a + bx) \cdot (c + dx) := (ac + 2bd) + (ad + bc)x.$$

- (f) Compute the product $(1 + 2x)(2 + x)$.
- (g) Prove that $\mathbb{F}_9$ with addition and multiplication defined above is a field. (The main thing to do is verify that every nonzero element has a multiplicative inverse.)
- (h) This is an intentionally vague and hopefully provocative question. Do you notice any similarities between this exercise and the construction of the complex numbers?

2.2 Vector spaces

Equipped with the notion of a field, we can now define the notion of a vector space. These are the central objects of interest in linear algebra. Just as a field is an abstraction of $\mathbb{R}$, a vector space will be an abstraction of our understanding of $\mathbb{R}^n$. Also like fields, a vector space will just be a raw set equipped with a handful of algebraic axioms, though in a more intricate fashion than in the definition of a field. In particular, to every vector space there is an associated field $\mathbb{F}$ that interacts with the vector space in a special way.

Definition 2.2.1. A **vector space over a field** $\mathbb{F}$, also referred to as an **$\mathbb{F}$ -vector space**, or simply a **vector space** if the field $\mathbb{F}$ is clear from context, is a set V with two operations, addition $(+)$ and scalar multiplication $(\cdot)$, the first of which takes a pair of elements $v, w \in V$ and produces a new element $v + w \in V$, and the second of which takes an element $\lambda \in \mathbb{F}$ and an element $v \in V$ and produces a new element $\lambda \cdot v \in V$. Moreover, these operations satisfy the following properties:

- (1) For all $v, w \in V$, $v + w = w + v$.
- (2) For all $u, v, w \in V$, $(u + v) + w = u + (v + w)$.
- (3) There exists an element $0 \in V$ such that $v + 0 = v$ for all $v \in V$.
- (4) For each $v \in V$, there is an element $v' \in V$ such that $v + v' = 0$.
- (5) For all $v \in V$, $1 \cdot v = v$.
- (6) For all $\lambda, \mu \in \mathbb{F}$ and $v \in V$,
$$(\lambda \cdot \mu) \cdot v = \lambda \cdot (\mu \cdot v).$$
- (7) For all $\lambda \in \mathbb{F}$ and $v, w \in V$,
$$\lambda \cdot (v + w) = \lambda \cdot v + \lambda \cdot w.$$
- (8) For all $\lambda, \mu \in \mathbb{F}$ and $v \in V$,
$$(\lambda + \mu) \cdot v = \lambda \cdot v + \mu \cdot v.$$

Remark 2.2.2. An important but subtle point is that the operations that define a vector space are *distinct* from those that define a field. For example, in (6) in the definition above, there are

two completely different types of multiplication happening on each side of the equation. For example, in the expression

$$(\lambda \cdot \mu) \cdot v$$

the multiplication in the parantheses is the *multiplication operation in the field $\mathbb{F}$* . The second $\cdot$ represents *scalar multiplication in the vector space V* , which is a completely different operation. In contrast, in the expression

$$\lambda \cdot (\mu \cdot v)$$

both multiplication operations are scalar multiplication in the vector space V . As another example, in (8) above there are two completely different addition operations occurring. On the left hand side, the addition operation in

$$(\lambda + \mu) \cdot v$$

is the addition in the field $\mathbb{F}$ (and the multiplication is the scalar multiplication in the vector space). On the right hand side, the addition operation in

$$\lambda \cdot v + \mu \cdot v$$

is the addition operation in the vector space, not in $\mathbb{F}$.

Definition 2.2.3. A **vector** is an element of a vector space.

I will continue to make annoying philosophical remarks about definitions like this, because I believe they are crucial for keeping all of this content organized correctly in your mind. You have formally forgotten all math you’ve ever learned, so the word “vector” means nothing to you. The word “vector” doesn’t refer to an arrow, because you don’t what arrows are. It also doesn’t refer to a tuple or column of numbers like $(1, 5, 6)$, because at the moment you don’t know what these things are. Now that we have defined the notion of a vector space, we *define* the word “vector” to simply mean “an element of an abstract vector space.”

Example 2.2.4. The set

$$\mathbb{R}^n := \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_j \in \mathbb{R} \right\}$$

is a vector space over $\mathbb{R}$ with addition defined as

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} := \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

and scalar multiplication defined as

$$\lambda \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} := \begin{pmatrix} \lambda \cdot x_1 \\ \vdots \\ \lambda \cdot x_n \end{pmatrix}.$$

All of the above vector space axioms are the usual familiar algebraic rules in $\mathbb{R}^n$. Note that we will also refer to element of $\mathbb{R}^n$ as $(x_1, \dots, x_n)$ when convenient. The difference between $(x_1, \dots, x_n)$ and the column version is purely notational, and they are defined to mean the same thing.

Example 2.2.5. More generally, we can consider

$$\mathbb{F}^n := \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \mid x_j \in \mathbb{F} \right\}$$

where $\mathbb{F}$ is any field. Then $\mathbb{F}^n$ is a vector space over $\mathbb{F}$ with addition defined as

$$\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} := \begin{pmatrix} x_1 + y_1 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

and scalar multiplication defined as

$$\lambda \cdot \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} := \begin{pmatrix} \lambda \cdot x_1 \\ \vdots \\ \lambda \cdot x_n \end{pmatrix}.$$

Example 2.2.6. To give an explicit version of the above example, consider the field $\mathbb{F}_2 = \{0, 1\}$ and the vector space $\mathbb{F}_2^3$. We can actually entirely list out all elements of this vector space.

$$\mathbb{F}_2^3 = \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

Then, for example,

$$\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1+0 \\ 1+1 \\ 1+0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}.$$

The abstract “0” element in $\mathbb{F}_2^3$ is the vector $(0, 0, 0)$.

Example 2.2.7. For this example, you can use the fact that you (secretly) remember what a matrix is. Let

$$M_{m \times n}(\mathbb{F}) := \{ A \mid A \text{ is an } m \times n \text{ matrix with entries in } \mathbb{F} \}.$$

Define addition in the usual way: if $A, B \in M_{m \times n}(\mathbb{F})$, then

$$(A + B)_{ij} := A_{ij} + B_{ij}.$$

Here, A_{ij} is the (i, j) -th entry of the matrix A . Likewise, define scalar multiplication for $\lambda \in \mathbb{F}$ and $A \in M_{m \times n}(\mathbb{F})$ as

$$(\lambda A)_{ij} := \lambda A_{ij}.$$

Then $M_{m \times n}(\mathbb{F})$ is a vector space over $\mathbb{F}$. Again, I won't verify all of the properties here, but these are the usual algebraic operations on matrices that you already know. The zero matrix $0 \in M_{m \times n}(\mathbb{F})$ satisfies (3) in the definition of a vector space.

Example 2.2.8. We can endow $\mathbb{C}$ with the structure of a vector space in a few ways. First, With the usual operations of complex addition and multiplication, $\mathbb{C}$ is a vector space over $\mathbb{C}$. (In this case, both field multiplication *and* scalar multiplication in the vector space are given by the usual multiplication of complex numbers).

Example 2.2.9. We can also endow $\mathbb{C}$ with the structure of a *real* vector space; that is, a vector space over $\mathbb{R}$. This time, scalar multiplication for $\lambda \in \mathbb{R}$ and $a + bi \in \mathbb{C}$ is defined as

$$\lambda \cdot (a + bi) = \lambda a + \lambda bi.$$

This is a *different* vector space structure than in the previous example, though we will have to wait until later in this chapter to understand why these are different vector space structures (besides being over different fields).

Example 2.2.10. In general, if $\mathbb{F}$ is a field, then $\mathbb{F}$ is a vector space over $\mathbb{F}$.

Example 2.2.11. Let S be a set, and let $\mathbb{F}$ be a field. Define

$$F(S, \mathbb{F}) := \{f : S \rightarrow \mathbb{F}\}.$$

The notation $f : S \rightarrow \mathbb{F}$ means a function f whose domain is S and whose codomain is $\mathbb{F}$. That is, f is an object that eats elements of the set S and returns values in the set $\mathbb{F}$. To define addition of elements $f, g \in F(S, \mathbb{F})$, we need to define a new function $f + g \in F(S, \mathbb{F})$. We do this by describing what the new function $f + g$ does to any element $s \in S$:

$$(f + g)(s) := f(s) + g(s).$$

Similarly, define scalar multiplication of $\lambda \in \mathbb{F}$ and $f \in F(S, \mathbb{F})$ as

$$(\lambda f)(s) := \lambda f(s).$$

Then $F(S, \mathbb{F})$ is a vector space over $\mathbb{F}$. In the exercises, you will verify this for a very similar vector space.

Just like with fields, there are multiple algebraic operations and techniques that hold in vector spaces, like cancellation laws. Also as with fields, we have to prove that these laws are true — I said nothing about such things in the definition of a vector space.

Proposition 2.2.12 (Cancellation law). *Let v be a vector space, and let $u, v, w \in V$. Suppose that $u + v = u + w$. Then $v = w$.*

Proof. Exercise. □

Corollary 2.2.13. *In a vector space, the element 0 is unique. Likewise, for each $v \in V$, the element $v' \in V$ satisfying $v + v' = 0$ is unique.*

Definition 2.2.14. Let $v \in V$. Define $-v \in V$ to be the unique element satisfying $v + (-v) = 0$.

As in a field, we can then define *subtraction* in a vector space using the previous definition: $v - w := v + (-w)$. Note that there is no such thing as division of vectors in a vector space! This is because there is no notion of *vector multiplication* (only scalar multiplication).

Proposition 2.2.15. Let V be a vector space over a field $\mathbb{F}$.

(i) For each $v \in V$, $0 \cdot v = 0$.

(ii) For each $v \in V$ and $\lambda \in \mathbb{F}$,

$$(-\lambda)v = \lambda(-v) = -(\lambda v).$$

(iii) For each $\lambda \in \mathbb{F}$, $\lambda \cdot 0 = 0$.

Proof.

(i) Exercise.

(ii) Recall that $-(\lambda v)$ is the unique element in V such that $\lambda v + -(\lambda v) = 0$. Note that

$$\lambda v + (-\lambda)v = (\lambda + -\lambda)v$$

by property (8) in the definition of a vector space. But $\lambda + -\lambda = 0$ in $\mathbb{F}$, so

$$\lambda v + (-\lambda)v = 0v = 0.$$

The last equality follows from (i). By uniqueness of $-(\lambda v)$, it follows that $(-\lambda)v = -(\lambda v)$.

Setting $\lambda = 1$ gives $(-1)v = -v$. Thus,

$$\lambda(-v) = \lambda \cdot ((-1) \cdot v) = (\lambda \cdot -1) \cdot v = (-\lambda)v$$

where in the second equality we have used property (6) in the definition of a vector space. This completes the proof that

$$(-\lambda)v = \lambda(-v) = -(\lambda v).$$

(iii) Exercise, similar to (i). Note that the 0 in this statement is the vector $0 \in V$, not the scalar $0 \in \mathbb{F}$!

□

Exercises

1. Let $\mathbb{F}$ be a field and let $V = \{ (x, y) \mid x, y \in \mathbb{F} \}$. Define addition in the usual way as

$$(x_1, y_1) + (x_2, y_2) := (x_1 + x_2, y_1 + y_2),$$

and define scalar multiplication as

$$\lambda(x, y) := (\lambda x, y).$$

Is V a vector space over $\mathbb{F}$?

2. Prove that the 0 vector in a vector space is unique.
3. Let $V := \{ f : \mathbb{R} \rightarrow \mathbb{R} \mid f(-x) = f(x) \}$. Define addition as

$$(f + g)(x) := f(x) + g(x)$$

and define scalar multiplication as

$$(\lambda f)(x) := \lambda f(x).$$

Prove² that V is a vector space over $\mathbb{R}$.

4. For each of the following statements, say whether it is *true* or *false*. If a statement is true, prove it. If it is false, explain why and/or provide an explicit counter example.
 - (a) The set of integers $\mathbb{Z}$ is a vector space over $\mathbb{R}$.
 - (b) The set of integers $\mathbb{Z}$ is a vector space over $\mathbb{Z}$.
 - (c) The set $\mathbb{F}_2^3$ can be made a vector space over $\mathbb{R}$.
 - (d) Every nontrivial³ vector space has infinitely many elements.

5. Let $\mathbb{F}$ be a field and let

$$P(\mathbb{F}) := \{ a_0 + a_1x + \cdots + a_nx^n \mid n \in \mathbb{N}, a_j \in \mathbb{F} \}$$

be the set of polynomials with coefficients in $\mathbb{F}$. Define addition of polynomials and scalar multiplication in the usual way. Verify that $P(\mathbb{F})$ is a vector space over $\mathbb{F}$. Be careful to observe that the polynomials don't have a fixed degree!

6. Let $\mathbb{F}$ be a field. Let $\mathbb{E} \subset \mathbb{F}$ be a subset of $\mathbb{F}$ such that $\mathbb{E}$ is also a field with the same operations as in $\mathbb{F}$. Prove that $\mathbb{F}$ is a vector space over $\mathbb{E}$. (This implies that $\mathbb{C}$ is a vector space over $\mathbb{R}$, $\mathbb{R}$ is a vector space over $\mathbb{Q}$, etc.)

2.3 Subspaces

Whenever you define a mathematical object (like a *field*, or a *vector space*, or a *manifold*, or a *category*, whatever those last two things are), there are two natural ways to investigate it:

- ◇ Study *subsets* of your object that “respect the structure” of the object.

²Yes, you do have to verify all of the vector space axioms. Yes, I know this is annoying and tedious — but it's a rite of passage you need to go through, so be careful and thorough.

³Here, *nontrivial* means any vector space that isn't just $\{0\}$.

◇ Study the *functions* into and out of your object that “respect the structure” of your object.

I’m being intentionally vague about what I mean by “respect the structure” so that I can phrase this as a very general principle in mathematics. In any case, in this section we will start investigating the first of these principles with respect to vector spaces; the next chapter will be devoted to the second principle.

To be less vague, in this section we want to identify certain subsets of vector spaces that play nicely with the structure of the vector space. In particular, we want to consider subsets of vector spaces that *are themselves naturally vector spaces*. It turns out that this will be an extremely fruitful concept, as we can decompose a vector space into simpler pieces in a variety of ways.

Definition 2.3.1. Let V be a vector space over a field $\mathbb{F}$. A **subspace** of V is a subset $W \subset V$ such that W is a vector space over $\mathbb{F}$ with the operations inherited from V .

Example 2.3.2. Let V be a vector space. Then $\{0\} \subset V$ and $V \subset V$ are both subspaces of V .

Note that, for *any* subset of a vector space, the axioms (1), (2), (5), (6), (7), and (8) in the definition of a vector space automatically hold, because the subset inherits the operations of V . Thus, to determine if a given subset W of V is a subspace, one needs to only verify that addition and scalar multiplication are well-defined when restricted to the subset, along with following axioms:

(3) There exists an element $0 \in W$ such that $w + 0 = w$ for all $w \in W$.

(4) For each $w \in W$, there is an element $w' \in W$ such that $w + w' = 0$.

In fact, we can identify subspaces in the following efficient manner.

Proposition 2.3.3. Let V be a vector space over $\mathbb{F}$. A subset $W \subset V$ is a subspace if and only if the following three properties hold for the operations defined in V :

(i) $0 \in W$,

(ii) if $v, w \in W$, then $v + w \in W$, and

(iii) if $\lambda \in \mathbb{F}$ and $v \in W$, then $\lambda v \in W$.

Proof. First, assume that $W \subset V$ is a subspace. Then since W is a vector space with the operations inherited from V , (ii) and (iii) automatically hold. Furthermore, since W is a vector space, there exists an element $0' \in W$ such that $0' + w = w$ for all $w \in W$. But $0 + v = v$ for all $v \in V$, so by the cancellation law it follows that $0 = 0'$. Thus, $0 \in W$ and so (i) holds.

Next, assume that (i), (ii), and (iii) hold. We wish to show that W is a subspace of V . Because W inherits the operations of V , it follows that (1), (2), (3), (5), (6), (7), and (8) automatically hold in W in the definition of a vector space. Thus, we only need to verify (4), the existence of additive inverses. Fix $w \in W$. By (iii), $(-1)w = -w \in W$, and thus w has an additive inverse in W . Therefore, W is a subspace. $\square$

Using these, we can easily identify more examples (and non-examples) of subspaces.

Example 2.3.4. Consider the vector space $M_n(\mathbb{F})$ of $n \times n$ matrices with entries in a field $\mathbb{F}$. Let $W \subset M_n(\mathbb{F})$ be the set of all *upper-triangular* matrices, i.e.,

$$W := \{ A \in M_n(\mathbb{F}) \mid A_{ij} = 0 \text{ if } i > j \}.$$

Note that the 0-matrix is upper triangular. Also, if A and B are upper triangular, then $A + B$ and λA are upper triangular. Thus, W is a subspace of $M_n(\mathbb{F})$.

Example 2.3.5. Consider the vector space $M_n(\mathbb{R})$ of $n \times n$ matrices with entries in $\mathbb{R}$. Let $S \subset M_n(\mathbb{R})$ be the set of matrices with nonnegative entries:

$$S := \{ A \in M_n(\mathbb{R}) \mid A_{ij} \geq 0 \}.$$

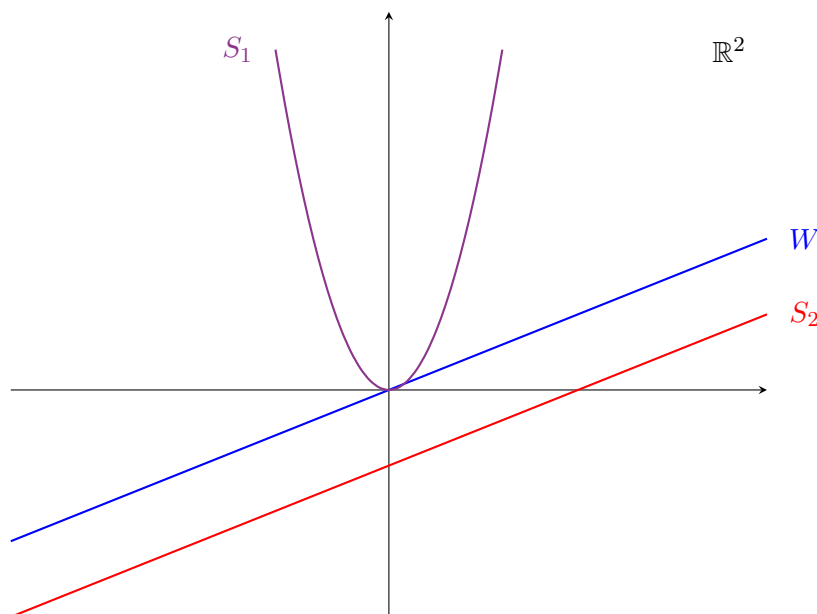
The set S is not a subspace because it isn't closed under scalar multiplication. For example, if $A \in M_n(\mathbb{R})$ is any nonzero matrix, then $(-1)A \notin S$.

Example 2.3.6. The intuitive notion of a subspace should be familiar in the setting of $\mathbb{R}^n$. Namely, any hyperplane passing through the origin is a subspace. A hyperplane *not* passing through the origin is *not* a subspace, in part because it doesn't contain 0. A set like

$$S = \{ (x, y) \in \mathbb{R}^2 \mid y = x^2 \}$$

is not a subspace of $\mathbb{R}^2$, because it isn't closed under addition or scalar multiplication (though it does contain $(0, 0)$). *To be explicit, consider the element $(1, 1)$. Note that $1 = 1^2$, and so $(1, 1) \in S$. However, $2(1, 1) = (2, 2)$, and $2 \neq 2^2$, so $2(1, 1) \notin S$. This implies that S is not closed under scalar multiplication and thus is not a subspace.*

The figure below depicts a subspace of $\mathbb{R}^2$, W , and two sets S_1 and S_2 that are *not* subspaces.



I should remark that this figure and the discussion above about subspaces in $\mathbb{R}^n$ being hyperplanes through the origin obviously does not *literally* apply in the setting of an abstract vector

space. For example, subspaces in a weird finite vector space like $\mathbb{F}_2^n$ could be described as certain lattices (as a fun challenge exercise, see if you can make sense of what I mean here), while most subspaces of infinite-dimensional (whatever that means) vector spaces like $F(\mathbb{R}, \mathbb{R})$ are essentially impossible to draw at all. However, the great thing about linear algebra is that the intuition you have in $\mathbb{R}^2$ and $\mathbb{R}^3$ usually applies to vector spaces in general. You have to be careful, and things can be deceiving, but usually when I am working with abstract vector spaces I am picturing $\mathbb{R}^2$ and $\mathbb{R}^3$ in my head. This is an important enough point to me that I will state it as a vague principle. Over the course of these notes, you will see this principle rear its head over and over again.

Principle 2.3.7. In abstract mathematics, you should always have a picture in mind, even if it is not entirely representative of the abstract concepts. In particular, in linear algebra, your intuition about $\mathbb{R}^2$ and $\mathbb{R}^3$ often generalizes correctly to statements about abstract vector spaces.

Let's continue by investigating how to produce new subspaces out of old subspaces. There are two main operations on sets that we discussed earlier in this chapter, which are the operations of intersecting and taking unions. The former operation is a legitimate way to produce new subspaces.

Proposition 2.3.8. *Let V be a vector space, and let $W_1, W_2 \subset V$ be subspaces. Then $W_1 \cap W_2$ is a subspace.*

Proof. To prove that $W_1 \cap W_2$ is a subspace, we verify the three conditions in Proposition 2.3.3.

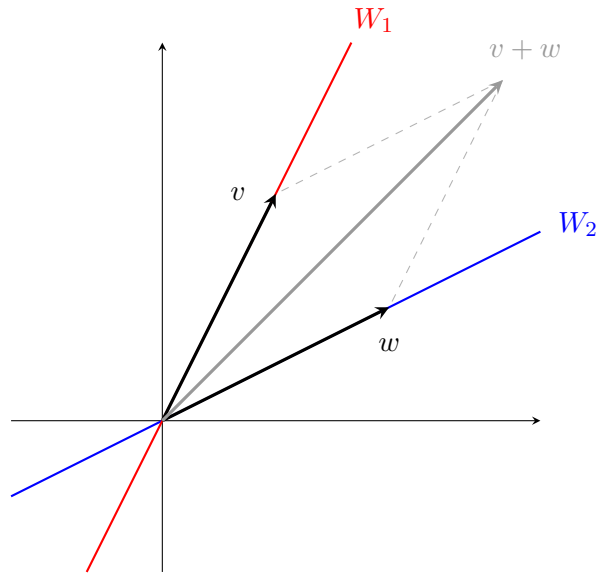
- (i) First, since W_1 and W_2 are subspaces, $0 \in W_1$ and $0 \in W_2$. Thus, $0 \in W_1 \cap W_2$.
- (ii) Next, let $v, w \in W_1 \cap W_2$. Since $v, w \in W_1$, and since W_1 is a subspace, we have $v + w \in W_1$. Likewise, since $v, w \in W_2$ and W_2 is a subspace, we have $v + w \in W_2$. Thus, $v + w \in W_1 \cap W_2$.
- (iii) Finally, let $v \in W_1 \cap W_2$ and $\lambda \in \mathbb{F}$. Since W_1 is a subspace and $v \in W_1$, we have $\lambda v \in W_1$. Likewise, since W_2 is a subspace and $v \in W_2$, we have $\lambda v \in W_2$.

By Proposition 2.3.3, $W_1 \cap W_2$ is a subspace. □

In general, the union of two subspaces is not a subspace!

Example 2.3.9. Consider $\mathbb{R}^2$, and let W_1 and W_2 be the x and y axes, respectively. Then $W_1 \cup W_2$ is not closed under addition. For example, consider $(1, 0) + (0, 1) = (1, 1)$. We have $(1, 0) \in W_1$ and $(0, 1) \in W_2$, so in particular $(1, 0), (0, 1) \in W_1 \cup W_2$. However, $(1, 1) \notin W_1 \cup W_2$ because $(1, 1) \notin W_1$ and $(1, 1) \notin W_2$. Thus $W_1 \cup W_2$ is not a subspace.

The figure below gives a visual depiction of the argument in the above example to show why $W_1 \cup W_2$ is not, in general, a subspace. Note that $W_1 \cup W_2$ below is simply the two lines together, and nothing else.



Exercises

1. Let S be a (nonempty) set, and let $\mathbb{F}$ be a field. Consider the vector space $F(S, \mathbb{F}) = \{f : S \rightarrow \mathbb{F}\}$. Fix $s_0 \in S$ and define

$$W = \{f \in F(S, \mathbb{F}) \mid f(s_0) = 0\}.$$

Prove that W is a subspace of $F(S, \mathbb{F})$.

2. Let $\mathbb{F}$ be a field, and consider the vector space $\mathbb{F}^n = \{(x_1, \dots, x_n) \mid x_j \in \mathbb{F}\}$. Prove that

$$W := \{(x_1, \dots, x_n) \in \mathbb{F}^n \mid x_1 + x_2 + \dots + x_n = 0\}$$

is a subspace of $\mathbb{F}^n$.

3. Let V be a vector space, and let $W_1, W_2 \subseteq V$ be subspaces. Prove that $W_1 \cup W_2$ is a subspace of V if and only if $W_1 \subseteq W_2$ or $W_2 \subseteq W_1$.
4. Prove that a subset W of a vector space V is a subspace if and only if $0 \in W$, and $\lambda v + w \in W$ whenever $\lambda \in \mathbb{F}$ and $v, w \in W$.
5. Let $C(\mathbb{R}) \subset F(\mathbb{R}, \mathbb{R})$ be the subset of continuous functions of $F(\mathbb{R}, \mathbb{R})$, the vector space of functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Prove that $C(\mathbb{R})$ is a subspace.
6. Let $F(\mathbb{R}, \mathbb{R})$ be the (real) vector space of functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Is the set

$$W = \{f \in F(\mathbb{R}, \mathbb{R}) \mid f(x) \leq 1 \text{ for all } x \in \mathbb{R}\}$$

a subspace of $F(\mathbb{R}, \mathbb{R})$? If it is prove, it. If it isn't, prove it.

7. Let $\mathbb{F}$ be a field with infinitely many elements. Let V be a nonzero vector space. Prove that V has infinitely many elements.

8. Prove that if $\mathbb{F}$ has infinitely many elements, then the only subspace of a vector space V with finitely many elements is $\{0\}$.
9. Can you find a vector space V with infinitely many elements and a nonzero subspace W with finitely many elements?

2.4 Direct sums of subspaces

In the previous section we saw that the union of two subspaces is not, in general, another subspace. However, a general theme in math is the desire to build new objects out of old. In our case, we would like a way to combine two subspaces to produce a new one. This will lead us to our first definition, which is that of the *sum* of two subspaces. This will naturally lead into the main topic of this section, which is the slightly more refined notion of a *direct sum*. Direct sums describe a vector space as a decomposition of subspaces.

First, we'll consider the normal sum operation.

Definition 2.4.1. Let $S_1, S_2 \subset V$ be subsets of a vector space V . The **sum** of S_1 and S_2 is

$$S_1 + S_2 := \{ s_1 + s_2 \mid s_1 \in S_1, s_2 \in S_2 \}.$$

Remark 2.4.2. Note that this definition is phrased in terms of *subsets*, not necessarily subspaces! We will generally be interested in the sum of subspaces, but the definition holds in more generality.

Definition 2.4.3. A vector space V is the **direct sum** of two subspaces $W_1, W_2 \subset V$, written $V = W_1 \oplus W_2$, if $W_1 \cap W_2 = \{0\}$ and $W_1 + W_2 = V$.

Example 2.4.4. Consider the vector space $\mathbb{R}^2$ and the following subspaces:

$$\begin{aligned} W_1 &= \{ (x, 0) \mid x \in \mathbb{R} \}, \\ W_2 &= \{ (0, y) \mid y \in \mathbb{R} \}. \end{aligned}$$

In words, these are the x and y axes, respectively. **Fix any element $(x, y) \in \mathbb{R}^2$. Then**

$$(x, y) = (x, 0) + (0, y).$$

Since $(x, 0) \in W_1$ and $(0, y) \in W_2$, this means that $\mathbb{R}^2 = W_1 + W_2$. Moreover, note that $W_1 \cap W_2 = \{(0, 0)\}$. Thus, $\mathbb{R}^2 = W_1 \oplus W_2$.

More generally, if W_1 and W_2 are any two non-parallel lines passing through the origin in $\mathbb{R}^2$, then $\mathbb{R}^2 = W_1 \oplus W_2$. Even more generally and informally, you can think about direct sums as being “completely non parallel” subspaces whose “dimensions” (which we haven't formally defined yet!) add up to the “dimension” of the vector space.

Example 2.4.5. Let $W_1 = \{ (x, 0) \mid x \in \mathbb{R} \}$ be the subspace of $\mathbb{R}^2$ from above. This time, let $W'_2 = \mathbb{R}^2$. Fix $(x, y) \in \mathbb{R}^2$. It is still the case that, for example,

$$(x, y) = (x, 0) + (0, y).$$

Since $(x, 0) \in W_1$ and $(0, y) \in W'_2$, this means that $\mathbb{R}^2 = W_1 + W'_2$. However, it is *not* the case that $\mathbb{R}^2 = W_1 \oplus W'_2$, because $W_1 \cap W'_2 = W_1 \neq \{(0, 0)\}$.

An important point here is that the decomposition $(x, y) = (x, 0) + (0, y)$ from above is not unique in this setting. For example, we could have also written $(x, y) = (0, 0) + (x, y)$. Then $(0, 0) \in W_1$ and $(x, y) \in W'_2$. We will return to this point shortly.

Example 2.4.6. Consider the vector space $\mathbb{R}^3$. Let

$$\begin{aligned} W_1 &= \{ (x, 0, 0) \mid x \in \mathbb{R} \} \\ W_2 &= \{ (0, y, z) \mid y, z \in \mathbb{R} \} \\ W_3 &= \{ (x, 0, z) \mid x, z \in \mathbb{R} \}. \end{aligned}$$

In words, W_1 is the x -axis, W_2 is the (y, z) -plane, and W_3 is the (x, z) -plane. These are all subspaces of $\mathbb{R}^3$. We will consider the interactions of each of these subspaces with respect to the notion of sums and direct sums.

- ◇ First, consider W_1 and W_2 . We claim that $\mathbb{R}^3 = W_1 \oplus W_2$. Indeed, note that any $(x, y, z) \in \mathbb{R}^3$ can be written

$$(x, y, z) = (x, 0, 0) + (0, y, z)$$

where $(x, 0, 0) \in W_1$ and $(0, y, z) \in W_2$. Thus, $\mathbb{R}^3 = W_1 + W_2$. Since $W_1 \cap W_2 = \{(0, 0, 0)\}$, it then follows that $\mathbb{R}^3 = W_1 \oplus W_2$.

- ◇ Next, consider W_1 and W_3 . We claim that $W_1 + W_3 = W_3$, but it is *not* the case that W_2 is the direct sum of W_1 and W_2 . This last statement is immediate from the fact that $W_1 \cap W_3 = W_1 \neq \{(0, 0, 0)\}$, so it suffices to show that $W_1 + W_3 = W_3$. First, note that $W_1 \subset W_3$. Thus, if $v \in W_1 \subset W_3$ and $w \in W_3$, then $v + w \in W_3$ because W_3 is a subspace. This shows that $W_1 + W_3 \subset W_3$. The reverse inclusion is clear, because if $w \in W_3$ we can always write $w = 0 + w$ where $0 \in W_1$. Thus, $W_3 = W_1 + W_3$.
- ◇ Finally, consider W_2 and W_3 . We claim that $\mathbb{R}^3 = W_2 + W_3$, but that it is *not* the case that $\mathbb{R}^3$ is the direct sum of W_2 and W_3 . Like before, this last claim is immediate from the observation that $W_2 \cap W_3 = \{ (0, 0, z) \mid z \in \mathbb{R} \} \neq \{(0, 0, 0)\}$. In words, the intersection of the (y, z) -plane and the (x, z) -plane is the z -axis. Thus, it remains to show that $\mathbb{R}^3 = W_2 + W_3$. Let $(x, y, z) \in \mathbb{R}^3$. Note that

$$(x, y, z) = (x, 0, 0) + (0, y, z)$$

and that $(x, 0, 0) \in W_2$ and $(0, y, z) \in W_3$. Thus, $(x, y, z) \in W_2 + W_3$ and so $\mathbb{R}^3 = W_2 + W_3$. Observe that, just like in the previous example, this decomposition is not unique. We could also consider, for instance, $(x, y, z) = (x, 0, \frac{z}{2}) + (0, y, \frac{z}{2})$.

Example 2.4.7. Consider the vector space $F(\mathbb{R}, \mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R}\}$ of functions from $\mathbb{R}$ back to itself. Define the following subspaces of even and odd functions:

$$\begin{aligned} W_1 &= \{ f \in F(\mathbb{R}, \mathbb{R}) \mid f(-x) = x \}, \\ W_2 &= \{ f \in F(\mathbb{R}, \mathbb{R}) \mid f(-x) = -x \}. \end{aligned}$$

We claim that $F(\mathbb{R}, \mathbb{R}) = W_1 + W_2$. To show this, let $f \in F(\mathbb{R}, \mathbb{R})$. Define $g, h \in F(\mathbb{R}, \mathbb{R})$ as follows:

$$g(x) := \frac{f(x) + f(-x)}{2} \quad \text{and} \quad h(x) := \frac{f(x) - f(-x)}{2}.$$

Note that $f = g + h$. Indeed,

$$\begin{aligned} g(x) + h(x) &= \frac{f(x) + f(-x)}{2} + \frac{f(x) - f(-x)}{2} \\ &= \frac{f(x)}{2} + \frac{f(x)}{2} \\ &= f(x). \end{aligned}$$

Next, note that

$$g(-x) = \frac{f(-x) + f(-(-x))}{2} = \frac{f(-x) + f(x)}{2} = g(x)$$

so that $g \in W_1$. Similarly, note that

$$h(-x) = \frac{f(-x) - f(-(-x))}{2} = -\frac{f(x) - f(-x)}{2} = -h(x)$$

so that $h \in W_2$. Since $f = g + h$, this shows that $F(\mathbb{R}, \mathbb{R}) = W_1 + W_2$.

Next, we claim that, in fact, $F(\mathbb{R}, \mathbb{R}) = W_1 \oplus W_2$. To prove this, it remains to show that $W_1 \cap W_2 = \{0\}$, where $0 : \mathbb{R} \rightarrow \mathbb{R}$ is the 0-function. To do this, suppose that $f \in W_1 \cap W_2$. Then $f(x) = f(-x)$ and $f(x) = -f(-x)$ for all $x \in \mathbb{R}$. Thus, $f(-x) = -f(-x)$ for all $x \in \mathbb{R}$. The only number satisfying this property is 0, and so $f(x) = f(-x) = 0$ for all $x \in \mathbb{R}$. Thus, $W_1 \cap W_2 = \{0\}$, and therefore $F(\mathbb{R}, \mathbb{R}) = W_1 \oplus W_2$.

The following proposition gives another perspective on the meaning of a direct sum.

Proposition 2.4.8. *Let V be a vector space and let $W_1, W_2 \subset V$ be subspaces. Then $V = W_1 \oplus W_2$ if and only if each element of v can be uniquely written as $w_1 + w_2$ for some $w_1 \in W_1, w_2 \in W_2$.*

Proof. First, suppose that $V = W_1 \oplus W_2$. Fix $v \in V$. Since $V = W_1 \oplus W_2$, in particular $V = W_1 + W_2$, and therefore there exists $w_1 \in W_1$ and $w_2 \in W_2$ such that $v = w_1 + w_2$. We need to show that this decomposition is unique. To do this, suppose that there is another decomposition $v = w'_1 + w'_2$ such that $w'_1 \in W_1$ and $w'_2 \in W_2$. Then $w_1 + w_2 = w'_1 + w'_2$, which implies $w_1 - w'_1 = w'_2 - w_2$. Because W_1 and W_2 are subspaces (and hence closed under addition), the left side of the equality is an element in W_1 , and the right side is an element in W_2 . So the equality represents an element of $W_1 \cap W_2$. But since $V = W_1 \oplus W_2$, we know that $W_1 \cap W_2 = \{0\}$. Thus, $w_1 - w'_1 = 0 = w'_2 - w_2$, and so $w_1 = w'_1$ and $w_2 = w'_2$. Thus, the decomposition $v = w_1 + w_2$ is unique.

Next, suppose that $W_1, W_2 \subset V$ are subspaces such that every $v \in V$ has a unique decomposition $v = w_1 + w_2$ for some $w_1 \in W_1$ and $w_2 \in W_2$. We need to show that $V = W_1 \oplus W_2$. Since every element of v admits a decomposition $v = w_1 + w_2$, it follows immediately that $V = W_1 + W_2$. It remains to show that $W_1 \cap W_2 = \{0\}$. To do this, suppose that $v \in W_1 \cap W_2$. Note that

$$v = v + 0.$$

Since $v \in W_1$ and $0 \in W_2$, this is a decomposition of v as $w_1 + w_2$ with $w_1 = v$ and $w_2 = 0$. By

assumption, this decomposition is unique. On the other hand, we can view

$$v = 0 + v$$

as a decomposition with $w_1 = 0 \in W_1$ and $w_2 = v \in W_2$. By uniqueness, it follows that $v = 0$. Thus, $W_1 \cap W_2 = \{0\}$ and so $V = W_1 \oplus W_2$. $\square$

Remark 2.4.9. Note that, in the first direction of the proof above, we used Principle 2.1.10 to prove that the decomposition was unique.

Exercises

- Let $W_1, W_2 \subset V$ be subspaces of a vector space V .
 - Prove that $W_1 + W_2$ is a subspace of V that contains both W_1 and W_2 .
 - Prove that any subspace of V containing both W_1 and W_2 must also contain $W_1 + W_2$.
- Give an example of *proper, nontrivial*⁴ subspaces $W_1, W_2 \subset \mathbb{R}^4$ such that $\mathbb{R}^4 = W_1 + W_2$, but that $\mathbb{R}^4$ is *not* the direct sum of W_1 and W_2 .

- Consider the vector space of polynomials of degree at most n with coefficients in a field $\mathbb{F}$:

$$P_n(\mathbb{F}) := \{ a_0 + a_1x + \cdots + a_nx^n \mid a_j \in \mathbb{F} \}.$$

Define the following subspaces of $P_n(\mathbb{F})$:

$$W_1 := \{ a_0 + a_1x + \cdots + a_nx^n \mid a_j = 0 \text{ if } j \text{ is even} \}$$

$$W_2 := \{ a_0 + a_1x + \cdots + a_nx^n \mid a_j = 0 \text{ if } j \text{ is odd} \}.$$

Here, 0 counts as an even number. Prove that $P_n(\mathbb{F}) = W_1 \oplus W_2$.

- Give an example of subspaces $W_1, W_2, W_3 \subset V$ of a vector space V such that

$$W_1 + W_3 = W_2 + W_3$$

but that $W_1 \neq W_2$.

- Give examples of subspaces $W_1, W_2 \subset \mathbb{R}^3$ such that $W_1 \neq \mathbb{R}^3$, $W_2 \neq \mathbb{R}^3$, $W_1 + W_2 = \mathbb{R}^3$, but $\mathbb{R}^3$ is *not* the direct sum of W_1 and W_2 .
- Recall that an $n \times n$ matrix $A \in M_n(\mathbb{R})$ is *symmetric* if $A^T = A$, where A^T is the transpose of A . A matrix is *skew-symmetric* if $A^T = -A$. Define the following subspaces of $M_n(\mathbb{R})$:

$$W_1 := \{ A \in M_n(\mathbb{R}) \mid A^T = A \},$$

$$W_2 := \{ A \in M_n(\mathbb{R}) \mid A^T = -A \}.$$

Prove that $M_n(\mathbb{R}) = W_1 \oplus W_2$.

⁴This just means you are not allowed to pick $\mathbb{R}^4$ or $\{0\}$ as subspaces.

7. Recall that if $A \in M_n(\mathbb{R})$, then A_{ij} is the (i, j) th entry of A . Let

$$W_1 = \{ A \in M_n(\mathbb{R}) \mid A_{ij} = 0 \text{ if } i > j \},$$

$$W_2 = \{ A \in M_n(\mathbb{R}) \mid A_{ij} = 0 \text{ if } i < j \}$$

be the subspaces of upper-triangular matrices and lower-triangular matrices, respectively.

- (a) Prove that $M_n(\mathbb{R}) = W_1 + W_2$.
- (b) Prove that $M_n(\mathbb{R})$ is *not* the direct sum of W_1 and W_2 .

2.5 Span and linear independence

In the previous section, I expressed the desire to build new subspaces out of old. More generally, it is desirable to have concrete ways of building subspaces by “generating” them. We don’t want to have to write down all elements of a subspace explicitly, every time we want to describe or construct a subspace.

For example, suppose we have some abstract vector space V over $\mathbb{F}$ and a collection of three vectors $\{v_1, v_2, v_3\} \subset V$. It is very unlikely that $\{v_1, v_2, v_3\}$ will be a subspace — for example, we know that every subspace must contain the 0 vector. What if neither of our three vectors is the 0 vector? More vaguely, subspaces tend to have many elements (in fact, unless $\mathbb{F}$ is a finite field, a nontrivial subspace will always have infinitely many elements) because subspaces are closed under addition and scalar multiplication. The question then becomes the following: How can we build a subspace using our initial collection $\{v_1, v_2, v_3\}$?

Well, since subspaces are closed under scalar multiplication, we know that, for example av_1 would need to be in our hypothetical subspace for all $a \in \mathbb{F}$. Likewise with all scalar multiples of v_2 and v_3 . Subspaces are also closed under addition, in our hypothetical subspace we should further be able to *add* any scalar multiple of any of the v_j ’s together to get a new element of the subspace. In other words, the property of being a subspace forces all of the elements of the form

$$a_1v_1 + a_2v_2 + a_3v_3$$

for $a_1, a_2, a_3 \in \mathbb{F}$ to be in our subspace. This motivates the following definition.

Definition 2.5.1. Let V be a vector space over a field $\mathbb{F}$. A **linear combination** of vectors $v_1, \dots, v_k \in V$ is an element of the form

$$a_1v_1 + \dots + a_kv_k \in V$$

where $a_1, \dots, a_n \in \mathbb{F}$.

In words, our previous discussion then suggests that if we have a collection of vectors $\{v_1, \dots, v_k\}$ and we want to build a subspace using those vectors, our hypothetical subspaces should contain all *linear combinations* of $\{v_1, \dots, v_k\}$.

Example 2.5.2. Consider the real vector space $\mathbb{R}^2$. The element $(1, 6) \in \mathbb{R}^2$ is a linear combination of $(1, 2)$ and $(-1, 0)$:

$$3 \begin{pmatrix} 1 \\ 2 \end{pmatrix} + 2 \begin{pmatrix} -1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 6 \end{pmatrix}.$$

Example 2.5.3. Consider the vector space $F(\mathbb{R}, \mathbb{R})$ of functions $f : \mathbb{R} \rightarrow \mathbb{R}$. The element $2x + 3 \sin x$ is a linear combination of $x \in F(\mathbb{R}, \mathbb{R})$ and $\sin x \in F(\mathbb{R}, \mathbb{R})$.

The following definition gives a name to the operation of *taking all possible linear combinations of a collection of vectors*.

Definition 2.5.4. Let V be a vector space and let $S \subset V$ be a subset. The **span** of S , denoted $\text{Span}(S)$, is the set of linear combinations of all elements of S :

$$\text{Span}(S) := \{ a_1 v_1 + \cdots + a_k v_k \mid a_j \in \mathbb{F}, v_j \in S \}.$$

If $S = \emptyset$, then we define $\text{Span}(\emptyset) := \{0\}$.

Remark 2.5.5. There is a subtlety to this definition that is likely to fool you on first reading. In the above definition, I am *not* suggesting that

$$S = \{v_1, \dots, v_k\}.$$

I repeat, this is *not* the case. For instance, the set S could have infinitely many elements. The above definition says that to build the span of S , you consider all possible (finite) linear combinations of finite sub-collections from S . For some clarification on this remark, see the next example.

Example 2.5.6. Consider the vector space $F(\mathbb{R}, \mathbb{R})$, the set of all functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Let $S \subset F(\mathbb{R}, \mathbb{R})$ be the subset

$$S = \{1, x, x^2, \dots\}.$$

We want to consider $\text{Span}(S)$, the span of this set. Note that

$$2 \cdot 1 + 3 \cdot x + 7 \cdot x^{36} \quad \text{and} \quad 3 \cdot x^3 + 25 \cdot x^5 + 3 \cdot x^7$$

are both elements of $\text{Span}(S)$. The underlying vectors in the collection S that are used to form the corresponding linear combinations are completely different for each. The first one is a linear combination of $1, x$, and x^{36} , while the second is a linear combination of x^3, x^5 , and x^7 . Finally, consider the expression

$$1 + x + x^2 + \cdots$$

or any similar expression consisting of infinitely many elements added together. This doesn't make sense! In a vector space, addition is a finite *algebraic* operation, and so you are only allowed to add together finitely many elements. It would be incorrect to call the above expression a linear combination of elements a linear combination of elements of S . In fact, it is incorrect to even write or consider an expression like this at all!

Example 2.5.7. Consider the vector space $\mathbb{R}^3$ and the set $S = \{(1, 0, 0), (0, 1, 0)\}$. The span of S

is geometrically described as the (x, y) -plane. Formally,

$$\begin{aligned}\text{Span}(S) &= \left\{ a_1 \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + a_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \mid a_1, a_2 \in \mathbb{R} \right\} \\ &= \left\{ \begin{pmatrix} a_1 \\ a_2 \\ 0 \end{pmatrix} \mid a_1, a_2 \in \mathbb{R} \right\}.\end{aligned}$$

Next, we confirm that taking the span of a set of vectors is a way to build a subspace of a vector space.

Proposition 2.5.8. *Let V be a subspace, and let $S \subset V$ be a subset. Then $\text{Span}(S)$ is a subspace of V . Moreover, it is the smallest subspace containing S .*

Proof. The case when $S = \emptyset$ is trivial, so it suffices to assume that S is nonempty. First, we show that $\text{Span}(S)$ is a subspace. Let $s \in S$ be any element. Then $0 = 0s \in \text{Span}(S)$. Next, let $v, w \in \text{Span}(S)$. By definition of the span, there exists $s_1, \dots, s_k \in S$ and $a_1, \dots, a_k \in \mathbb{F}$ such that

$$v = a_1 s_1 + \dots + a_k s_k.$$

Likewise, there exists $s'_1, \dots, s'_n \in S$ and $a'_1, \dots, a'_n \in \mathbb{F}$ such that

$$w = a'_1 s'_1 + \dots + a'_n s'_n.$$

Then

$$v + w = a_1 s_1 + \dots + a_k s_k + a'_1 s'_1 + \dots + a'_n s'_n$$

and so $v + w$ is a linear combination of elements in S . Thus, $v + w \in \text{Span}(S)$. Finally, let $v \in \text{Span}(S)$ and $\lambda \in \mathbb{F}$. As before, there exists $s_1, \dots, s_k \in S$ and $a_1, \dots, a_k \in \mathbb{F}$ such that

$$v = a_1 s_1 + \dots + a_k s_k.$$

Then

$$\lambda v = (\lambda a_1) s_1 + \dots + (\lambda a_k) s_k.$$

Thus, λv is a linear combination of elements of S , and so $\lambda v \in \text{Span}(S)$. By Proposition 2.3.3, it follows that $\text{Span}(S)$ is a subspace.

Next, we prove that $\text{Span}(S)$ is the smallest subspace containing S . Suppose that W is any subspace such that $S \subset W$. We will show that $\text{Span}(S) \subset W$. Let $v \in \text{Span}(S)$. There exists $s_1, \dots, s_k \in S$ and $a_1, \dots, a_k \in \mathbb{F}$ such that

$$v = a_1 s_1 + \dots + a_k s_k.$$

Because each $s_j \in S \subset W$ and W is a subspace, $v = a_1 s_1 + \dots + a_k s_k \in W$. Thus, $v \in \text{Span}(S)$ implies $v \in W$, and so $\text{Span}(S) \subset W$. $\square$

The last part of this proof (where we show that $\text{Span}(S)$ is the smallest subspace containing

S) deserves a bit more exposition, because I have seen this kind of argument confuse students in the past. I will start by phrasing this as a very general mathematical principle.

Principle 2.5.9. To show that an object $\star$ is the smallest object of some kind, let $\star'$ be another such object and show that $\star \subset \star'$.

Next, we discuss a concept that should be familiar from linear algebra in $\mathbb{R}^n$: that of linear independence. The motivation is the following. Recall that, for example, if $S \subset \mathbb{R}^3$ is the set given by $S = \{(1, 0, 0), (0, 1, 0)\}$, then

$$\text{Span}(S) = \left\{ \begin{pmatrix} a_1 \\ a_2 \\ 0 \end{pmatrix} \mid a_1, a_2 \in \mathbb{R} \right\}.$$

On the other hand, if we add the vector $(1, 1, 0)$ to S to get $S' = \{(1, 0, 0), (0, 1, 0), (1, 1, 0)\}$, the span does not increase. That is,

$$\text{Span}(S') = \left\{ \begin{pmatrix} a_1 \\ a_2 \\ 0 \end{pmatrix} \mid a_1, a_2 \in \mathbb{R} \right\}.$$

Roughly this is because the third vector is *already* a linear combination of the first two, and so it doesn't add "new information." To rephrase this, there is a *nontrivial* linear combination of the three vectors that produces the zero vector:

$$\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} - \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Definition 2.5.10. Let S be a subset of a vector space V . We say that S is (or the elements S are) *linearly dependent* if there are vectors $v_1, \dots, v_n \in S$ and scalars $a_1, \dots, a_n \in \mathbb{F}$, not all 0, such that

$$a_1 v_1 + \dots + a_n v_n = 0.$$

A set S is **linearly independent** if it is not linearly dependent.

Remark 2.5.11. In a similar spirit to Remark 2.5.5, note that we have made no assumption about S being finite in this definition! The definition holds for sets of infinite vectors.

Example 2.5.12. Consider the vector space $P(\mathbb{R})$ of polynomials with coefficients in $\mathbb{R}$. Consider the set

$$S = \{1 + x, x - 2x^2, 1 + 2x^2\}.$$

Then S is linearly dependent, because

$$1(1 + x) - 1(x - 2x^2) - 1(1 + 2x^2) = 0.$$

On the other hand, the set $S' = \{1, x, x^2\}$ is linearly independent. Indeed, suppose that there was a linear combination of these vectors that produced the zero vector. That is, suppose that there are constants a_1, a_2, a_3 such that

$$a_1 + a_2x + a_3x^2 = 0.$$

By an elementary fact about polynomials, the only way this is possible is if all the coefficients are 0.

Example 2.5.13. Suppose that $0 \in S$. Then S is linearly dependent. Indeed, note that

$$1 \cdot 0 = 0.$$

This is a nontrivial linear combination of elements in S that produces the zero vector, thus, S is linearly dependent.

The first example leads to an important equivalent way to think about linear independence, which is essentially just a reformulation of the above definition. See also the principle below about how to prove linear independence.

Proposition 2.5.14. A set $S \subset V$ is linearly independent if and only if, for any $v_1, \dots, v_n \in S$,

$$a_1v_1 + \dots + a_nv_n = 0$$

implies $a_1 = \dots = a_n = 0$.

Example 2.5.15. Consider the vector space $F(\mathbb{R}, \mathbb{R})$. The vectors $\sin x$ and $\cos x$ are linearly independent. Indeed, suppose that

$$a \sin x + b \cos x = 0$$

for some constants a, b . Note that this is an equality of *functions*, not of numbers! This means that $a \sin x + b \cos x = 0$ for all $x \in \mathbb{R}$. Plugging in $x = 0$ gives $b = 0$, and plugging in $x = \frac{\pi}{2}$ gives $a = 0$. Therefore, $\sin x$ and $\cos x$ are linearly independent.

The definition of linear independence is a little bizarre, because it is defined in a negative way. That is, it is defined as the property of *not* being linearly dependent. However, the above examples suggest the following direct way to think about (and prove) that a set is linearly independent. This is the content of the following principle, which is one of the most important principles in linear algebra. You will do well to learn this principle well!

Principle 2.5.16 (Proving linear independence). To prove that a set $\{v_1, \dots, v_k\}$ is linearly independent,

- (1) start out by supposing you have a linear combination of the form $a_1v_1 + \dots + a_kv_k = 0$, and then
- (2) try to prove that $a_1 = \dots = a_k = 0$.

This principle will rear its head often throughout these notes. For now, I will present one purple-quality example of this in action.

Example 2.5.17. Let V be a vector space, and suppose that $\{v_1, v_2, v_3\} \subset V$ is a linearly independent set of vectors. We claim that

$$\{v_1 + v_2 + v_3, v_2 + v_3, v_3\}$$

is another linearly independent set.

To see this, suppose that $a_1, a_2, a_3 \in \mathbb{F}$ are scalars such that

$$a_1(v_1 + v_2 + v_3) + a_2(v_2 + v_3) + a_3v_3 = 0.$$

Then

$$a_1v_1 + (a_1 + a_2)v_2 + (a_1 + a_2 + a_3)v_3 = 0.$$

Since $\{v_1, v_2, v_3\}$ is linearly independent, it must be the case that

$$\begin{cases} a_1 = 0 \\ a_1 + a_2 = 0 \\ a_1 + a_2 + a_3 = 0 \end{cases}.$$

Plugging $a_1 = 0$ into the second equation gives $a_2 = 0$. Plugging $a_1 = 0$ and $a_2 = 0$ into the third equation gives $a_3 = 0$. Thus, $a_1 = a_2 = a_3 = 0$. Since

$$a_1(v_1 + v_2 + v_3) + a_2(v_2 + v_3) + a_3v_3 = 0$$

implies $a_1 = a_2 = a_3 = 0$, it follows that

$$\{v_1 + v_2 + v_3, v_2 + v_3, v_3\}$$

is a linearly independent set, as desired.

Exercises

1. Let $S_1, S_2 \subset V$ be two subsets of a vector space V . Prove that $\text{Span}(S_1 \cap S_2) \subset \text{Span}(S_1) \cap \text{Span}(S_2)$.
2. Let $S_1, S_2 \subset V$ be two subsets of a vector space V . Prove that $\text{Span}(S_1 \cup S_2) = \text{Span}(S_1) + \text{Span}(S_2)$.
3. Prove that subset W of a vector space V is a subspace if and only if $\text{Span}(W) = W$.
4. For the following statement, say whether it is *true* or *false*. If it is true, prove it. If it is false, provide an explicit counter example.

Statement: If $S_1, S_2 \subset V$ are subsets of a vector space V , then $\text{Span}(S_1 \cup S_2) = \text{Span}(S_1) \cup \text{Span}(S_2)$.

5. Two vectors $v, w \in V$ in a vector space are **parallel** if there exists some $\lambda \in \mathbb{F}$ such that either $v = \lambda w$ or $w = \lambda v$.
- (a) Let V be a vector space over $\mathbb{F}$. Prove that $v, w \in V$ are linearly independent if and only if v and w are not parallel.
- (b) Let V be a vector space over $\mathbb{R}$. Prove that $v, w \in V$ are linearly independent if and only if $v + w$ and $v - w$ are linearly independent.
- [Optional challenge question: is this still true over a general field $\mathbb{F}$?]*
6. Let V be a vector space over $\mathbb{F}$. Let S_1 and S_2 be two subsets of V such that $S_1 \subset S_2$.
- (a) Prove that if S_1 is linearly dependent, then S_2 is also linearly dependent.
- (b) Prove that if S_2 is linearly independent, then S_1 is also linearly independent.
7. Prove that if a set $S \subset V$ in a vector space is linearly dependent then either $S = \{0\}$ or there exist distinct vectors $v, u_1, \dots, u_k \in S$ such that v is a linear combination of $u_1, \dots, u_k$.
8. Consider the vector space $M_{3 \times 2}(\mathbb{F})$ of 3×2 matrices with entries in a field $\mathbb{F}$. Prove that

$$\left\{ \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} \right\}$$

is linearly dependent.

9. Consider the vector space $F(\mathbb{R}, \mathbb{R})$ of functions $f : \mathbb{R} \rightarrow \mathbb{R}$.
- (a) Fix $a \neq b \in \mathbb{R}$. Prove that the set $\{e^{ax}, e^{bx}\} \subset F(\mathbb{R}, \mathbb{R})$ is linearly independent.
- (b) Prove that $\{|x|, \sin x\} \subset F(\mathbb{R}, \mathbb{R})$ is linearly independent.
- (c) Prove that $\{1, \sin^2 x, \cos^2 x\} \subset F(\mathbb{R}, \mathbb{R})$ is linearly dependent.

2.6 Bases and dimension

The idea of linear independence was motivated by the desire to identify a “minimal spanning set.” In words, given a vector space or subspace, we wanted to find a set S with as few elements as possible that spanned the space. The notion of a “minimal spanning set” is important enough to warrant a special name.

Definition 2.6.1. Let V be a vector space. A **basis** of V is a linearly independent set $\mathcal{B}$ such that $\text{Span}(\mathcal{B}) = V$.

Example 2.6.2. Consider the vector space $\mathbb{F}^n$, where $\mathbb{F}$ is any field. Let

$$\mathcal{B} = \{e_1, \dots, e_n\}$$

where $e_j = (0, \dots, 0, \underbrace{1}_{j\text{th entry}}, 0, \dots, 0)$. We claim that $\mathcal{B}$ is a basis of $\mathbb{F}^n$, called the **standard basis**. Note that this is the exact same standard basis of $\mathbb{R}^n$ that you are familiar with from a prior mathematical lifetime in the case of $\mathbb{F} = \mathbb{R}$.

To prove that $\mathcal{B}$ is a basis, we must prove that it is both linearly independent and that it spans $\mathbb{F}^n$. We first prove linear independence. Suppose that $a_1, \dots, a_n \in \mathbb{F}$ are constants such that $a_1 e_1 + \dots + a_n e_n = 0$. This means that

$$a_1 \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + a_n \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \quad \text{and thus} \quad \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix}.$$

It follows that $a_1 = \dots = a_n = 0$, and so $\mathcal{B}$ is linearly independent.

Next, we prove that $\text{Span } \mathcal{B} = \mathbb{F}^n$. Let $v = (a_1, \dots, a_n) \in \mathbb{F}^n$. Note that

$$v = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} = a_1 \begin{pmatrix} 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + a_n \begin{pmatrix} 0 \\ \vdots \\ 1 \end{pmatrix}$$

and so $v \in \text{Span } \mathcal{B}$. It follows that $\text{Span } \mathcal{B} = \mathbb{F}^n$, as desired. Thus, $\mathcal{B}$ is a basis.

Example 2.6.3. A vector space may have many different bases. For example, consider $\mathbb{R}^2$. By the previous example, one basis is the standard basis, which in this case is

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}.$$

We can also show that

$$\mathcal{B}' = \left\{ \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \begin{pmatrix} -1 \\ 1 \end{pmatrix} \right\}$$

is a different basis for $\mathbb{R}^2$. To see this, we first show that $\mathcal{B}'$ is linearly independent. Suppose that $a_1, a_2 \in \mathbb{R}$ are scalars such that

$$a_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + a_2 \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Then

$$\begin{pmatrix} a_1 - a_2 \\ a_1 + a_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{and hence} \quad \begin{cases} a_1 - a_2 = 0 \\ a_1 + a_2 = 0 \end{cases}.$$

The first equation implies $a_1 = a_2$, and then plugging this into the second equation we have $2a_1 = 0$. Thus, $a_1 = 0$ and $a_2 = 0$, and so $\mathcal{B}'$ is linearly independent.

Next, we show that $\text{Span}(\mathcal{B}') = \mathbb{R}^2$. Fix $(a, b) \in \mathbb{R}^2$. Note that

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{a+b}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{-a+b}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

This implies that $(a, b) \in \text{Span}(\mathcal{B}')$, and since $(a, b) \in \mathbb{R}^2$ was arbitrary it follows that $\text{Span}(\mathcal{B}') = \mathbb{R}^2$. Thus, $\mathcal{B}'$ is a basis for $\mathbb{R}^2$.

Remark 2.6.4. I have a remark about the previous example — more specifically, my proof that $\mathcal{B}'$ is a basis — that I will discuss at length in Subsection 2.6.1. For now, I will simply state the principle and move on as to not interrupt the flow of the content of this section, but I highly encourage you to read Subsection 2.6.1 either now or later as it relates to my solution above, because the following principle is an important one that is not emphasized enough (or at all) in lower division math courses.

Principle 2.6.5. The way you write up a proof or solution may often look very different from how you actually figured out the proof or solved the problem. In other words, there are (at least) two distinct phases to doing math: (1) a “rough draft” phase, where you tinker around and discover a solution, and (2) a “final draft” phase, where you take your jumbled discoveries from phase (1) and present them in a streamlined coherent form. Crucially, this final draft presentation will often not reflect your process in (1).

Example 2.6.6. Consider the vector space $M_2(\mathbb{F})$ of 2×2 matrices with entries in a field $\mathbb{F}$. Let

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}.$$

One can show using an argument very similar to the one above (the standard basis of $\mathbb{F}^n$) that $\mathcal{B}$ is a basis for $M_2(\mathbb{F})$.

More generally, consider the vector space $M_{m \times n}(\mathbb{F})$ of $m \times n$ matrices with $\mathbb{F}$ -entries. For each $1 \leq i \leq m$ and $1 \leq j \leq n$, define the matrix E_{ij} by

$$(E_{ij})_{k\ell} := \begin{cases} 1 & \text{if } (k, \ell) = (i, j) \\ 0 & \text{otherwise} \end{cases}.$$

In words, the matrix E_{ij} has a 0 everywhere except for in the (i, j) slot, where there is a 1. Then one can show that the set

$$\mathcal{B} = \{ E_{ij} \mid 1 \leq i \leq m, 1 \leq j \leq n \}$$

is a basis of $M_{m \times n}(\mathbb{F})$.

Example 2.6.7. A basis can include infinitely many elements. This gets subtle, for reasons that will be discussed later, but consider for now the vector space $P(\mathbb{R})$ of polynomials with real coefficients of any degree. In the exercises, you will show that

$$\{1, x, x^2, \dots\}$$

is a basis of $P(\mathbb{R})$. It will be helpful to reference the discussion in the previous section. Also, note that this is *not* saying that every element of $P(\mathbb{R})$ can be written

$$a_0 + a_1x + a_2x^2 + \dots.$$

The point is similar to the one I made in the previous section — the expression I just wrote is an infinite sum, and that doesn't make sense in $P(\mathbb{R})$. The elements of $P(\mathbb{R})$ are *finite* degree polynomials.

Remark 2.6.8. I should remark that, in general, the order of the elements of a basis *does* matter. This is subtle, because when we refer to sets, the order does *not* matter. That is, the set $\{1, 2, 3\}$ is the same as the set $\{3, 2, 1\}$. However, the bases

$$\mathcal{B} = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\} \quad \text{and} \quad \mathcal{B} = \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$$

should be considered distinct. The reasons for this will become clear in the next chapter, when we discuss coordinates. Note that some texts emphasize this distinction more clearly with the notion of an *ordered basis*, which means exactly that you think of a basis as a set with a prescribed order. I have opted to not bother with this notion, as it usually isn't a big deal.

Next, we will prove some abstract properties of bases. The next proposition gives a nice way to conceptualize the point of bases.

Proposition 2.6.9. *Let V be a vector space and let $\mathcal{B} = \{u_1, \dots, u_n\}$ be a subset of V . Then $\mathcal{B}$ is a basis of V if and only if every element $v \in V$ admits a unique decomposition of the form*

$$v = a_1 u_1 + \dots + a_n u_n$$

for some $a_1, \dots, a_n \in \mathbb{F}$.

Proof. We will prove one direction, and leave the other as an exercise. Suppose that $\mathcal{B}$ is a basis. Let $v \in V$. Because $\text{Span}(\mathcal{B}) = V$, there exist scalars $a_1, \dots, a_n \in \mathbb{F}$ such that

$$v = a_1 u_1 + \dots + a_n u_n.$$

Suppose that $v = a'_1 u_1 + \dots + a'_n u_n$ is another such decomposition. Then since

$$a_1 u_1 + \dots + a_n u_n = a'_1 u_1 + \dots + a'_n u_n$$

we have

$$(a_1 - a'_1)u_1 + \dots + (a_n - a'_n)u_n = 0.$$

Since $\{u_1, \dots, u_n\}$ is linearly independent, it follows that $a_1 - a'_1 = 0, \dots, a_n - a'_n = 0$. Thus, $a'_1 = a_1, \dots, a'_n = a_n$, and so the above decomposition is unique.

The other direction is left as an exercise. □

One important feature of bases is that, in a given vector space, every (finite) basis has the same number of elements. For example, above I gave you two examples of bases of $\mathbb{R}^2$, both with two vectors. You should play around with some other examples and convince yourself that it is impossible to have a basis of $\mathbb{R}^2$ with only one element, and also impossible to have a basis of $\mathbb{R}^2$ with three (or more) elements. Perhaps surprisingly, it takes a decent amount of work to prove that all bases must have the same number of elements. To do this, we will first prove a difficult and technical theorem referred to as the *replacement theorem*. If this is your first

pass through this material, you can skip the statement and proof of the replacement theorem. In particular, the proof uses induction, which we have not yet discussed in the main content of these notes; refer to [A.1](#) in the Appendix. The main reason why the replacement theorem is stated and proved is for the purpose of Corollary [2.6.11](#).

Theorem 2.6.10 (Replacement theorem). *Let V be a vector space, and suppose that $S \subset V$ is a finite subset with n elements such that $\text{Span}(S) = V$. Let $L \subset V$ be a set of linearly independent vectors with m elements. Then*

- (i) $m \leq n$, and
- (ii) there is a subset $S' \subset S$ of $n - m$ vectors such that $\text{Span}(L \cup S') = V$.

Proof. ⁵ Fix $n \geq 0$. We proceed by induction on m .

As a base case, we consider $m = 0$. A subset $L \subset V$ with $m = 0$ elements is necessarily the empty set. Note that $0 \leq n$ so that (i) holds, and further note that $\text{Span}(L \cup S) = \text{Span}(S) = V$ so that (ii) holds with $S' = S$.

Now we perform the inductive step. Suppose the theorem is true for some $m \geq 0$. Let L be a set of linearly independent vectors with $m + 1$ elements. We wish to show that $m + 1 \leq n$ and that there is a subset $S' \subset S$ with $n - (m + 1)$ elements such that $\text{Span}(L \cup S') = V$.

Denote the elements of L by $L = \{v_1, \dots, v_{m+1}\}$. By a previous exercise, since L is linearly independent it follows that $\{v_1, \dots, v_m\} \subset L$ is linearly independent. By the inductive hypothesis, we have $m \leq n$. Also by the inductive hypothesis, there exists a subset $S' := \{u_1, \dots, u_{n-m}\} \subset S$ such that

$$\text{Span}(v_1, \dots, v_m, u_1, \dots, u_{n-m}) = V.$$

Note that, currently, it may be the case that $n - m = 0$.

Since $v_{m+1} \in V$, it then follows that

$$v_{m+1} = a_1 v_1 + \dots + a_m v_m + b_1 u_1 + \dots + b_{n-m} u_{n-m}$$

for some scalars $a_j, b_j \in \mathbb{F}$. Suppose that $n - m = 0$. Then v_{m+1} is a linear combination of $v_1, \dots, v_m$, which contradicts that fact that L is linearly independent. Thus, $n - m > 0$, and so it follows that $m + 1 \leq n$. This proves (i) in the statement of the theorem.

Finally, we need to identify a subset $S' \subset S$ with $n - (m + 1)$ elements such that $L \cup S'$ spans V . To do this, note that at least one of the b_j coefficients must be nonzero by the above discussion. Assume without loss of generality (possibly after reordering elements) that $b_1 \neq 0$. Let $S' = \{u_2, \dots, u_{n-m}\}$. Note that S' has $n - (m + 1)$ elements. We claim that

$$\text{Span}(L \cup S') = V.$$

Note that

$$v_1, \dots, v_m, u_2, \dots, u_{n-m} \in \text{Span}(L \cup S').$$

⁵This proof uses induction and is optional on a first reading.

Also note that $u_1 \in \text{Span}(L \cup S')$, since

$$u_1 = \frac{1}{b_1}v_{m+1} - \frac{a_1}{b_1}v_1 - \cdots - \frac{a_m}{b_1}v_m - \frac{b_2}{b_1}u_2 - \cdots - \frac{b_{n-m}}{b_1}u_{n-m}.$$

Therefore,

$$\{v_1, \dots, v_m, u_1, \dots, u_{n-m}\} \subset \text{Span}(L \cup S').$$

But the span of the leftmost set is V (by the inductive hypothesis) and so it follows that $V \subset \text{Span}(L \cup S')$, which implies $V \subset \text{Span}(L \cup S')$, as desired. This concludes the induction and proves the theorem. $\square$

Corollary 2.6.11. *Let V be a vector space with a finite basis. Then every basis of V contains the same number of elements.*

Proof. Let $\mathcal{B}$ be a basis of V with n elements. Let $\mathcal{B}'$ be another basis. We wish to show that $\mathcal{B}'$ also has n elements. Our first order of business is actually to show that $\mathcal{B}'$ has a finite number of elements (we cannot assume this!). Suppose that $\mathcal{B}'$ has more than n elements (possibly infinitely many). Let S be a subset of $\mathcal{B}'$ with $n + 1$ elements. Since $\mathcal{B}'$ is linearly independent, S is linearly independent, and so by the replacement theorem it follows that $n + 1 \leq n$. But this is a contradiction. Thus, $\mathcal{B}'$ cannot have more than n elements.

Now let m denote the (finite) number of elements of $\mathcal{B}'$. By the above argument, $m \leq n$. But we can now reverse the roles of $\mathcal{B}$ and $\mathcal{B}'$: since $\text{Span}(\mathcal{B}') = V$ and since $\mathcal{B}$ is linearly independent, the replacement theorem implies that $n \leq m$. Since $m \leq n$ and $n \leq m$, we have $n = m$. $\square$

Using this corollary, we can now formally define the notion of *dimension*.

Definition 2.6.12. A vector space V is **finite dimensional** if it has a basis with finitely many elements. The number of vectors in any basis for a finite dimensional vector space V is the **dimension** of V , denoted $\dim V$. If a vector space does not have a finite basis, it is **infinite dimensional**.

Example 2.6.13. The vector space $\{0\}$ has dimension 0, because of the silly reason that we define $\text{Span}(\emptyset) = \{0\}$.

Example 2.6.14. The dimension of $\mathbb{F}^n$ is n , because the standard basis from above has n elements.

Example 2.6.15. The vector space $P(\mathbb{R})$ is infinite dimensional, because the set $\{1, x, x^2, \dots\}$ is a basis with infinitely many elements (see the exercises).

Example 2.6.16. Recall that $\mathbb{C}$ can be given a vector space structure in two different ways: as a real vector space (over $\mathbb{R}$) and as a complex vector space (over $\mathbb{C}$). Lets first consider the vector space structure over $\mathbb{C}$. Then according the above example ($\mathbb{F}^n$), this vector space has dimensions 1 because a basis is $\{1\}$. This is just $\mathbb{F}^n$ when $\mathbb{F} = \mathbb{C}$ and $n = 1$.

Example 2.6.17. Next, view $\mathbb{C}$ as a vector space over $\mathbb{R}$. We claim that $\{1, i\}$ is a basis for this vector space, and so $\mathbb{C}$ has dimension 2 as a real vector space. Indeed, note that any complex number $a + bi \in \mathbb{C}$ is a linear combination of 1 and i , since

$$a + bi = (a)1 + (b)i.$$

Now we show that $\{1, i\}$ is linearly independent (over $\mathbb{R}$!). Suppose that $a_1, a_2 \in \mathbb{R}$ are real numbers such that

$$a_1 \cdot 1 + a_2 \cdot i = 0.$$

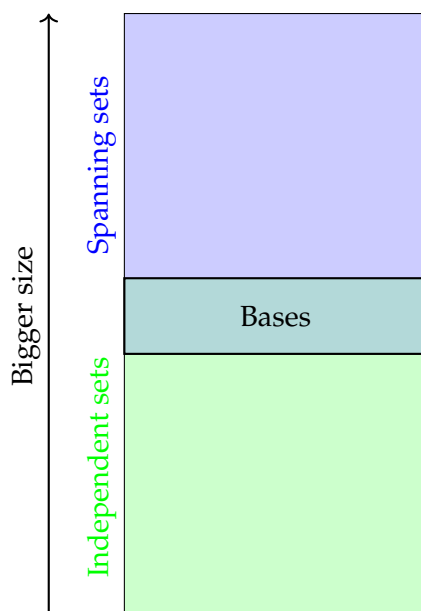
Then $a_1 + a_2i = 0$, and so it must be the case that $a_1 = a_2 = 0$. Thus, $\{1, i\}$ is linearly independent. Since $\{1, i\}$ spans $\mathbb{C}$ and is linearly independent, it is a basis.

In general, one can show that $\mathbb{C}^n$ is an n -dimensional complex vector space — again, this is just the example of $\mathbb{F}^n$ above with $\mathbb{F} = \mathbb{C}$ — while $\mathbb{C}^n$ is a $2n$ -dimensional real vector space.

Proposition 2.6.18. *Let V be a vector space with dimension n .*

- (i) *If $S \subset V$ is a subset such that $\text{Span}(S) = V$, then $|S| \geq n$. If $|S| = n$, then S is a basis. If $|S| > n$, then S can be reduced to a basis (that is, there is a basis $\mathcal{B}$ such that $\mathcal{B} \subset S$).*
- (ii) *If $L \subset V$ is a linearly independent subset of V , then $|L| \leq n$. If $|L| = n$, then S is a basis. If $|L| < n$, then L can be extended to a basis (that is, there is a basis $\mathcal{B}$ such that $L \subset \mathcal{B}$).*

Remark 2.6.19. In words, this proposition describes a basis as a sweet spot between the two extremes of linearly independent sets and spanning sets. The former is always smaller than the latter, and bases lie in their overlap.



Proof. Let $\mathcal{B}_0$ be a basis of V (with n elements).

- (i) Suppose for the sake of contradiction that $|S| < n$. Then in the replacement theorem we may take $L = \mathcal{B}_0$ to conclude that $n \leq |S| < n$, a contradiction. Thus, $|S| \geq n$.

Next, suppose that $|S| = n$. Suppose for the sake of contradiction that S is not a basis. Since S spans V , it must be the case that S is linearly dependent. Then there exists an element $s \in S$ which is a linear combination of the remaining elements in S . Removing this element gives a subset $S' \subset S$ with $n - 1$ elements such that $\text{Span}(S') = V$. But this contradicts the replacement theorem, since $\mathcal{B}_0$ has n linearly independent vectors.

Finally, suppose that $|S| > n$. Then S is not a basis. Since it spans V , it must be linearly dependent. Thus, there is at least one element of S which is a linear combination of the other elements. Removing this element gives a spanning set of size $|S| - 1$. We can continue this process until we obtain a spanning set of size n , which by the above, must be a basis.

- (ii) Let L be a linearly independent subset of V . First, we claim that $|L|$ is finite. Suppose not for the sake of contradiction. Then we could select a linearly independent subset of size $n + 1$, which contradicts (i) of the replacement theorem. Now that $|L|$ is finite, the same result then immediately implies that $|L| \leq n$.

Next, suppose that $|L| = n$. By the replacement theorem, there is a subset S' of $\mathcal{B}_0$ of size $n - n = 0$ such that $L \cup S'$ spans V . Then necessarily $S' = \emptyset$, and so $L \cup S' = L$. Thus, L is a linearly independent spanning set and so L is a basis.

Finally, suppose that $|L| < n$. By the replacement theorem, there is a subset $S' \subset \mathcal{B}_0$ with $n - |L|$ elements such that $L \cup S'$ spans V . Since $L \cup S'$ has n elements, by (i) it follows that $\mathcal{B} := L \cup S'$ is a basis.

□

Proposition 2.6.20. *Let V be a finite dimensional vector space, and let $W \subset V$ be a subspace. then W is finite dimensional, and $\dim W \leq \dim V$.*

Proof. Let S be a spanning set for W (for example, one could take $S = W$). By the above proposition, this set can be reduced to a basis of W ; that is, there is a basis $\mathcal{B}_W$ of W such that $\mathcal{B}_W \subset S$. Since $\mathcal{B}_W$ is a basis for W , it is linearly independent. In particular, it is linearly independent in V . By the above proposition, it then follows that $|\mathcal{B}_W| \leq n$. Therefore, W is finite dimensional, and

$$\dim W = |\mathcal{B}_W| \leq n = \dim V.$$

□

Example 2.6.21. Consider the vector space $P(\mathbb{R})$ of polynomials (of any degree) with coefficients in $\mathbb{R}$. Define the following vectors:

$$\begin{aligned} p_1(x) &= x + 2x^3 \\ p_2(x) &= 1 + x + x^2 \\ p_3(x) &= 3 + 4x + 3x^2 + 2x^3 \\ p_4(x) &= 0. \end{aligned}$$

Let $W := \text{Span}(p_1(x), p_2(x), p_3(x), p_4(x))$. We claim that $\dim W = 2$. In particular, we claim that $\{p_1(x), p_2(x)\}$ is a basis for W . To prove this, we will show that $\{p_1(x), p_2(x)\}$ is linearly independent and that $W = \text{Span}(p_1(x), p_2(x))$.

First, we will show that $\{p_1(x), p_2(x)\}$ is linearly independent. Suppose that $a_1, a_2 \in \mathbb{R}$ such that

$$a_1 p_1(x) + a_2 p_2(x) = 0$$

Then

$$a_1 (x + 2x^3) + a_2 (1 + x + x^2) = 0$$

and so

$$a_2 + (a_1 + a_2)x + a_2x^2 + 2a_1x^3 = 0.$$

For a polynomial to be the zero polynomial, all of its coefficients must be 0. Thus,

$$\begin{cases} a_2 = 0 \\ a_1 + a_2 = 0 \\ a_2 = 0 \\ 2a_1 = 0 \end{cases}.$$

In particular, the first and fourth equations imply that $a_1 = a_2 = 0$. Therefore, $\{p_1(x), p_2(x)\}$ is linearly independent.

Next, we show that $\text{Span}(p_1(x), p_2(x)) = W$. To see this, first note that

$$\begin{aligned} p_3(x) &= 3 + 4x + 3x^2 + 2x^3 \\ &= (x + 2x^3) + 3(1 + x + x^2) \\ &= p_1(x) + 3p_2(x). \end{aligned}$$

Next, let $v(x) \in W$. Then

$$v(x) = a_1 p_1(x) + a_2 p_2(x) + a_3 p_3(x) + a_4 p_4(x)$$

for some $a_1, a_2, a_3, a_4 \in \mathbb{R}$. By the above observation,

$$\begin{aligned} v(x) &= a_1 p_1(x) + a_2 p_2(x) + a_3 (p_1(x) + 3p_2(x)) + a_4 \cdot 0 \\ &= (a_1 + a_3) p_1(x) + (a_2 + 3a_3) p_2(x). \end{aligned}$$

Thus, $v \in \text{Span}(p_1(x), p_2(x))$. This shows that $W \subset \text{Span}(p_1(x), p_2(x))$. Since $\text{Span}(p_1(x), p_2(x))$ is clearly a subset of W , it follows that $W = \text{Span}(p_1(x), p_2(x))$. Since $\{p_1(x), p_2(x)\}$ is a linearly independent spanning set of W , it is a basis. Thus, $\dim W = 2$.

2.6.1 An aside on proof writing and Principle 2.6.5

Oftentimes, the clearest way to present a proof is not necessarily to reformulate how you discovered the solution. For example, sometimes there is a lot of messy work that goes in to solving a computation, that in hindsight, is easier to present without showing all of those details. Some of the examples in this section demonstrate this principle. For instance, in my solution to Example 2.6.3, I have the following observation to prove that the vectors $(1, 1)$ and

$(-1, 1)$ span $\mathbb{R}^2$: “Fix $(a, b) \in \mathbb{R}^2$. Note that

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{a+b}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{-a+b}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

This implies that $(a, b) \in \text{Span}(\mathcal{B}')$, and since $(a, b) \in \mathbb{R}^2$ was arbitrary it follows that $\text{Span}(\mathcal{B}') = \mathbb{R}^2$.” From the readers perspective, I have very clearly and definitively demonstrated that (a, b) is in the span of the two vectors, but I mysteriously pulled the coefficients $\frac{a+b}{2}$ and $\frac{-a+b}{2}$ out of nowhere. They undeniably work, but it is not clear from my solution where they came from. This was intentional on my part, and the truth is that I did some computations on the side to figure out ahead of time. This is a personal choice and comes with its own costs and benefits, but this is exactly the principle that I’m referring to. The way that I actually solved this part of the example does not reflect the absolute clearest way of presenting it.

TO be completely clear, let me show you the computations I did on the side that do not appear in my solution above. Given a vector $(a, b) \in \mathbb{R}^2$, I wanted to find constants C and D such that

$$\begin{pmatrix} a \\ b \end{pmatrix} = C \begin{pmatrix} 1 \\ 1 \end{pmatrix} + D \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

I didn’t know what those coefficients were, so I simplified and set up the corresponding system of equations:

$$\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} C - D \\ C + D \end{pmatrix} \Rightarrow \begin{cases} C - D = a \\ C + D = b \end{cases}.$$

Adding the two equations give $2C = a + b$, and hence $C = \frac{a+b}{2}$, and subtracting the equations gives $2D = -a + b$, and so $D = \frac{-a+b}{2}$. Thus, this tells me that

$$\begin{pmatrix} a \\ b \end{pmatrix} = \frac{a+b}{2} \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \frac{-a+b}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

When I sat down to write up the my solution, I made the conscious decision to omit the calculations above. *For the purpose of convincing the reader that the two vectors span*, that calculation is unnecessary. It doesn’t matter where those coefficients came from; maybe they came to me in a dream. They evidently work, and for the purpose of convincing a reader that the two vectors span $\mathbb{R}^2$ that is all that is necessary.

With that being said, there is certainly something lost with a solution like this. Though my original solution is completely convincing in showing that the vectors span, it doesn’t indicate how you might make a similar argument for other sets of vectors. This would be more apparent had I included the computations I described above (at the cost of cluttering the argument and introducing technically unnecessary data in the form of the constants C and D). Is one kind of solution better? It entirely depends on the context and the personal style of the author.

Perhaps my main point with this subsection is the following. Unlike in lower division mathematics, there is a very intentional process of crafting a proof and deciding what to communicate and how to communicate it, and importantly, this process is mostly distinct from the actual process of solving the problem. A proof is not just “show your work”. A proof is a final

draft reformulation of whatever content is needed to convince the reader of the claim. As a result of this, a professional proof may look very different than the actual process of doing the mathematics to solve the problem!

Exercises

1. Consider the vector space

$$P(\mathbb{F}) = \{ a_0 + a_1x + \cdots + a_nx^n \mid n \in \mathbb{N}, a_j \in \mathbb{F} \}$$

of polynomials (of any degree) with coefficients in a field $\mathbb{F}$. Prove that

$$\mathcal{B} = \{1, x, x^2, x^3, \dots\}$$

is a basis of $P(\mathbb{F})$.

2. Let V be a vector space and let $W_1, W_2 \subset V$ be subspaces. Prove that

$$\dim(W_1 \cap W_2) = \dim(W_1)$$

if and only if $W_1 \subset W_2$.

3. Consider the vector space $\mathbb{R}^n$. Let

$$W := \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n \mid x_1 + \cdots + x_n = 0 \right\}.$$

Recall that W is a subspace. Compute, with proof, the dimension of W .

[Hint: First, identify a collection of vectors that you think should be a basis for W . Then prove that the collection spans W , and then prove that it is linearly independent.]

4. Consider the vector space $F(\mathbb{R}, \mathbb{R})$ of functions $\mathbb{R} \rightarrow \mathbb{R}$. Compute, with proof, the dimension of the subspace $\text{Span}(\{1, \sin^2 x, \cos^2 x\})$.
5. Let V be a vector space, and let $\mathcal{B} = \{u_1, \dots, u_n\} \subset V$. Prove that if every element $v \in V$ admits a unique decomposition of the form

$$v = a_1u_1 + \cdots + a_nu_n$$

then $\mathcal{B}$ is a basis of V .

6. Prove that the vector space $M_{m \times n}(\mathbb{F})$ has dimension mn .
7. Consider the vector space $\mathbb{R}^4$. Let

$$v_1 = \begin{pmatrix} 1 \\ 3 \\ -1 \\ 4 \end{pmatrix}, v_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \\ 2 \end{pmatrix}, v_3 = \begin{pmatrix} 1 \\ -1 \\ 1 \\ 0 \end{pmatrix},$$

Let $W := \text{Span}(v_1, v_2, v_3)$. Compute, with proof, $\dim W$.⁶

8. Consider the set of complex numbers $\mathbb{C}$. Recall that there are two natural ways to endow $\mathbb{C}$ with a vector space structure: over the field $\mathbb{C}$, and over the field $\mathbb{R}$.
 - (a) Prove that a basis for the vector space $\mathbb{C}$ over the field $\mathbb{C}$ is $\{1\}$. Conclude that the dimension of this vector space is 1.
 - (b) Prove that a basis for the vector space $\mathbb{C}$ over the field $\mathbb{R}$ is $\{1, i\}$. Conclude that the dimension of this vector space is 2.
9. For each of the following statements, determine whether it is **true** or **false**. If it is true, prove it. If it is false, give an explicit counter example or an explicit reason why.
 - (a) Let V be a vector space and let $W_1, W_2 \subset V$ be subspaces. If $\mathcal{B}_1$ is a basis for W_1 and $\mathcal{B}_2$ is a basis for W_2 , then $\mathcal{B}_1 \cup \mathcal{B}_2$ is a basis for $W_1 + W_2$.
 - (b) Let V be a vector space. If $S_1, S_2 \subset V$ are (nonempty) subsets of V such that $S_2 \subset \text{Span}(S_1)$, then $S_1 \cup S_2$ is linearly dependent.
10. Let V be a vector space. Prove that if $\{u, v, w\}$ is a basis of V , then $\{u + v + w, v + w, w\}$ is also a basis of V .
11. Let V be a finite dimensional vector space. Suppose that W_1 and W_2 are subspaces such that $V = W_1 \oplus W_2$.
 - (a) Suppose that $\mathcal{B}_1$ is a basis for W_1 and $\mathcal{B}_2$ is a basis for W_2 . Prove that $\mathcal{B}_1 \cup \mathcal{B}_2$ is a basis for V .
 - (b) Use the previous part to conclude that $\dim(W_1 \oplus W_2) = \dim W_1 + \dim W_2$.
12. Consider the vector space $P_n(\mathbb{R})$ of polynomials of degree at most n with coefficients in $\mathbb{R}$. Fix $a \in \mathbb{R}$. Let

$$W = \{p \in P_n(\mathbb{R}) \mid p(a) = 0\}.$$
 - (a) Prove that $\dim P_n(\mathbb{R}) = n + 1$.
 - (b) Prove that W is a subspace of $P_n(\mathbb{R})$.
 - (c) Prove that $\dim W = n$.
13. Prove that the vector space $F(\mathbb{R}, \mathbb{R})$ of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ is infinite dimensional.
14. Let V be a vector space, and let $W_1, W_2 \subset V$ be finite dimensional subspaces. Prove that

$$\dim(W_1 + W_2) = \dim(W_1) + \dim(W_2) - \dim(W_1 \cap W_2).$$

[Hint: Start with a basis of the subspace $W_1 \cap W_2$. You can extend this to a basis of W_1 , and you can also extend this to a basis of W_2 .]
15. Let V be a vector space, and let $W_1, W_2 \subset V$ be finite dimensional subspaces such that $V = W_1 + W_2$. Prove that $V = W_1 \oplus W_2$ if and only if $\dim(W_1) + \dim(W_2) = \dim V$.

⁶Remember that we have not defined or used row reduction at all in 115A!

16. Let V be a finite dimensional vector space. Let W_1 be any subspace. Prove that there is a subspace W_2 such that $V = W_1 \oplus W_2$.

2.7 Quotient vector spaces

In this optional section, we will discuss the notion of a *quotient vector space*. Informally, a quotient vector space arises by way of “dividing” a vector space by a subspace. To formulate this precisely, we need the language of equivalence relations and equivalence classes, so if this is unfamiliar to you I suggest reading Appendix A.2.

An important example of an equivalence relation — also discussed in Appendix A.2, so I will refer you to there for the details — is the following. Let V be a vector space and let $W \subset V$ be a subspace. Define an equivalence relation $\sim$ on V by

$$v \sim w \quad \text{if and only if } v - w \in W.$$

The equivalence classes (the various collections of equivalent vectors) are then “parallel shifts” of the subspace W . That is, one equivalence class is $[0]$, which is just the subspace W . If you pick a vector $v \notin W$, then the equivalence class $[v]$ is given by $\{v\} + W$, a shift of the entire subspace W by the vector v . Using this equivalence relation, we can construct a new vector space.

Definition 2.7.1. Let V be a vector space and $W \subset V$ a subspace. Define the **quotient of V by W** , also read as $V \bmod W$, written V/W , to be the set of equivalence classes of V under $\sim$ as defined above. That is,

$$V/W := \{ [v] \mid v \in V, v \sim w \text{ if and only if } v - w \in W \}.$$

Although I have already claimed that V/W is a vector space, this is not at all clear from the above definition. We still have a lot of work to do to actually endow this with a vector space structure. How does one add equivalence classes? How is scalar multiplication defined? What is the 0 vector? The vector space structure of the quotient is given by the following theorem.

Theorem 2.7.2. Let V be a vector space and $W \subset V$ a subspace. Then V/W is a vector space under the following operations of addition and scalar multiplication. For $v, w \in V$ and $\lambda \in \mathbb{F}$, define

$$[v] + [w] := [v + w]$$

and

$$\lambda \cdot [v] := [\lambda v].$$

Proof. There is a lot to prove in this theorem; more than you might expect if this is the first time you’ve seen an algebraic structure defined using equivalence classes. We will eventually have to verify all of the vector space axioms (commutativity of addition, existence of a zero vector, etc.) but before even that we have to verify that the operations defined above are actually well-defined. By this I mean the following. Suppose we selected two equivalence classes $[v], [w] \in$

V/W and wanted to add them together to get a new vector in the quotient, $[v] + [w] \in V/W$. But because addition of equivalence classes is defined using the representatives v and w , there is a possible issue: what if we selected different representatives for the same equivalence classes, $v', w' \in V$ with $[v'] = [v]$ and $[w'] = [w]$ and formed the addition $[v'] + [w']$. In theory, this *should* be the same object as $[v] + [w]$. Is it? We have to prove that it is! Likewise with scalar multiplication.

Proving that addition is well-defined.

First, we prove that the addition operation in the theorem is well-defined on equivalence classes. Let $v, v', w, w' \in V$ such that $[v'] = [v]$ and $[w'] = [w]$. We need to prove that

$$[v] + [w] = [v'] + [w'].$$

By definition of addition in V/W , to do this we must prove that

$$[v + w] = [v' + w'].$$

By definition of the equivalence relation, this is the case precisely when

$$(v + w) - (v' + w') \in W.$$

Since $[v'] = [v]$, we know that $v - v' \in W$. Likewise, since $[w'] = [w]$, we know that $w - w' \in W$. Thus, it is in fact the case that

$$(v + w) - (v' + w') = (v - v') + (w - w') \in W.$$

Therefore, $[v] + [w] = [v'] + [w']$ and so addition of vectors in V/W is well-defined.

Proving that scalar multiplication is well-defined.

To prove that scalar multiplication is well-defined, we must show that for $v, v' \in V$ with $[v'] = [v]$ and $\lambda \in \mathbb{F}$, we have

$$\lambda[v] = \lambda[v'].$$

This is similar to the previous step and is left as an exercise.

Verifying the vector space axioms.

Having proved that addition and scalar multiplication are well-defined in V/W , we now have to verify all of the vector space axioms. I will only do a few of them here, and I will leave the rest as exercises.

First, we prove that addition in V/W is commutative. For $[v], [w] \in V/W$, note that

$$\begin{aligned} [v] + [w] &= [v + w] \\ &= [w + v] \\ &= [w] + [v]. \end{aligned}$$

The other algebraic axioms (distributivity, associativity of scalar multiplication, etc.) are similar and are left as exercises.

Next, we verify the existence of the zero vector in V/W . We claim that $[0]$, the equivalence class containing $0 \in V$, is the zero vector. To see this, note that

$$\begin{aligned}[0] + [v] &= [0 + v] \\ &= [v].\end{aligned}$$

Similarly, we can verify the existence of additive inverses in V/W . For $[v] \in V/W$, note that

$$[v] + [-v] = [v + -v] = [0].$$

This completes the proof that V/W is a vector space. □

Remark 2.7.3. Note that, by the above proof, the zero vector in V/W is $[0] = W$. That is, the entire subspace W represents the zero vector in the quotient.

Example 2.7.4. Let $V = \mathbb{R}^3$, and let $W = \text{Span}(e_1, e_2)$ be the subspace which is the (x, y) -plane. Note that $\dim V = 3$ and $\dim W = 2$. The zero vector in V/W is the (x, y) -plane, i.e., W itself, and every other vector in V/W is given by a vertical shift of the (x, y) -plane. In particular, we claim that V/W is a 1-dimensional vector space (given by the 1-dimensional family of planes stacked on top of each other). To prove this, we can find a basis for V/W with one element.

Consider the set

$$\left\{ \left[\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \right] \right\}.$$

This is a set consisting of a single element, which is the equivalence class of vectors equivalent to $(0, 0, 1)$. Because this set has a single nonzero vector (note that it is nonzero because $(0, 0, 1) \notin W$) it is certainly linearly independent. We just have to show that the span of this singleton set is all of V/W . To do this, let $[v] \in V/W$ be an arbitrary equivalence class. We can write

$$v = \begin{pmatrix} a \\ b \\ c \end{pmatrix}$$

for some $a, b, c \in \mathbb{R}$. Note that $(a, b, 0) \in W$. Thus, since

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} - \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix} = \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} \in W$$

it follows that $[v] = [(0, 0, c)] = c[(0, 0, 1)]$. Thus, $[v]$ is in the span of $\{[(0, 0, 1)]\}$ and so $\{[(0, 0, 1)]\}$ is a basis for V/W .

We can generalize the observations from the previous example with the following proposition, which computes the dimension of a quotient V/W in terms of the dimensions of V and W .

Proposition 2.7.5. *Let V be a finite dimensional vector space, and let $W \subset V$ be a subspace. Then*

V/W is finite dimensional, and

$$\dim V/W = \dim V - \dim W.$$

Proof. Let $k = \dim W$, and let $\{w_1, \dots, w_k\}$ be a basis for W . We can extend this basis to a basis of V :

$$\{w_1, \dots, w_k, v_1, \dots, v_\ell\}.$$

Note that, with this notation, we have $\dim V = k + \ell$, so that $\dim V - \dim W = \ell$. Thus, to prove the proposition it suffices to find a basis for V/W with exactly ℓ elements.

We claim that $\{[v_1], \dots, [v_\ell]\}$ is such a basis. First, we verify that this set is linearly independent. Suppose that

$$a_1[v_1] + \dots + a_\ell[v_\ell] = [0].$$

Then

$$[a_1v_1 + \dots + a_\ell v_\ell] = [0].$$

By definition of the equivalence relation, this means that

$$a_1v_1 + \dots + a_\ell v_\ell \in W.$$

Since $\{w_1, \dots, w_k\}$ is a basis for W , we can then write

$$a_1v_1 + \dots + a_\ell v_\ell = b_1w_1 + \dots + b_kw_k$$

for some $b_1, \dots, b_k \in \mathbb{F}$. This implies

$$a_1v_1 + \dots + a_\ell v_\ell + (-b_1)w_1 + \dots + (-b_k)w_k = 0.$$

Since $\{w_1, \dots, w_k, v_1, \dots, v_\ell\}$ is a basis for V , it is linearly independent. This implies that $a_1 = \dots = a_\ell = 0$, and so $\{[v_1], \dots, [v_\ell]\}$ is linearly independent, as desired.

Next, we prove that $\{[v_1], \dots, [v_\ell]\}$ spans V/W . Let $[v] \in V/W$. Since $v \in V$, we can write

$$a_1v_1 + \dots + a_\ell v_\ell + b_1w_1 + \dots + b_kw_k$$

for some scalars $a_j, b_j \in \mathbb{F}$. Note that $b_1w_1 + \dots + b_kw_k \in W$, and so

$$\begin{aligned} [v] &= [a_1v_1 + \dots + a_\ell v_\ell + b_1w_1 + \dots + b_kw_k] \\ &= [a_1v_1 + \dots + a_\ell v_\ell] \\ &= a_1[v_1] + \dots + a_\ell[v_\ell]. \end{aligned}$$

Thus, $[v] \in \text{Span}([v_1], \dots, [v_\ell])$. This completes the proof that $\{[v_1], \dots, [v_\ell]\}$ is a basis. Thus, V/W is finite dimensional and $\dim V/W = \ell = (k + \ell) - k = \dim V - \dim W$. $\square$

Example 2.7.6. It is still possible to take quotients of infinite dimensional subspaces. In fact, it is even possible for V/W to be finite dimensional when V and W are infinite dimensional! Here is an example. Let $V = P(\mathbb{F})$, the vector space of all polynomials with coefficients in $\mathbb{F}$,

and let

$$W = \{ a_0 + a_1x + \cdots + a_nx^n \mid a_0 = a_1 = 0, n \in \mathbb{N} \}.$$

In words, W is the subspace of all polynomials such that the first two coefficients (the constant term and the degree 1 coefficient) are 0. Note that V is an infinite dimensional vector space, and W is an infinite dimensional subspace of V . Indeed, $\{x^2, x^3, x^4, \dots\}$ is an infinite set of linearly independent vectors in W . We claim that $\dim V/W = 2$. To prove this, we will find a basis for V/W with two elements. Indeed, consider the set

$$\{[1], [x]\}.$$

This set consists of two vectors in V/W , the equivalence classes of 1 and x , respectively. First, we will show that this set is linearly independent. Suppose that $a[1] + b[x] = [0]$. Then $[a + bx] = [0]$, which means $a + bx \in W$. But by definition of W , this forces $a = b = 0$, and so $\{[1], [x]\}$ is linearly independent, as desired. Next, so show that $V/W = \text{Span}(\{[1], [x]\})$, let $[v] \in V/W$. Then $v = a_0 + a_1x + \cdots + a_nx^n$ for some $n \in \mathbb{N}$ and some $a_j \in \mathbb{F}$. Note that

$$v - (a_0 + a_1x) = a_2x^2 + \cdots + a_nx^n \in W$$

and so

$$[v] = [a_0 + a_1x] = a_0[1] + a_1[x].$$

Thus, $V/W = \text{Span}(\{[1], [x]\})$ and so $\{[1], [x]\}$ is a basis for V/W .

The examples we have seen above suggest the following intuition about quotient vector spaces, which is essentially just the intuition about equivalence classes. The quotient vector space V/W is obtained by “squashing” W down to the zero vector. That is, you squash out of all W and call the entire subspace “0”. This has a ripple effect throughout the whole vector space, and whatever you’re left with is the quotient vector space V/W . Moreover, by squashing out the subspace W , you are removing $\dim W$ directions from V and you are left with a new vector space of dimension $\dim V - \dim W$.

Exercises

1. Prove that scalar multiplication in V/W is well-defined.
2. Verify the rest of the vector space axioms for V/W .
3. Let $V = \mathbb{R}^2$, and let $W = \text{Span}(e_1 + e_2)$ be the subspace given by the line $y = x$. Describe the elements of V/W . What is $\dim V/W$? Can you give a basis for V/W ?
4. Prove that $V/\{0\} = V$. Likewise, prove that $V/V = \{0\}$.
5. Let $W \subset P(\mathbb{R})$ be defined as follows:

$$W = \{ p(x) \in P(\mathbb{R}) \mid p'''(x) = 0 \}.$$

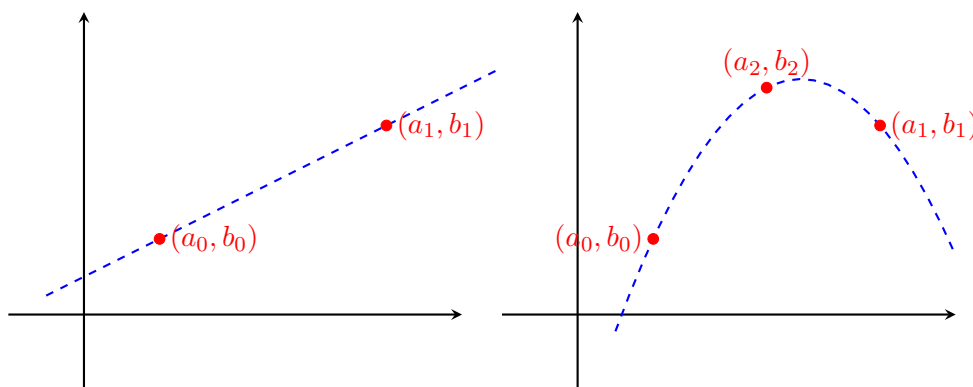
Prove that V/W is finite dimensional by computing a basis.

6. Can you find a vector space V and a nonzero subspace W such that V/W is infinite dimensional?

2.8 Polynomial interpolation

This optional section is devoted to the idea of polynomial interpolation from a linear algebraic perspective. You can interpret this section as an “application” section, but I don’t necessarily see it that way. The actual linear algebra involved is relatively simple, only using the notion of linear combinations, systems of equations, and bases. Rather than being viewed as an application, this discussion of polynomial interpolation should be taken as an example of how mathematical theories and levels of abstraction can offer clarity and insight into situations that could otherwise be solved in down to earth terms. Do you need an understanding of abstract vector spaces to understand polynomial interpolation? No. Does it help to understand the big picture, quickly absorb the ideas, and retain the information without getting lost in formulas and details? Yes!

In any case, here is the set up. This harkens back to the historical origins of linear algebra; see Section 1.2. To warm up, suppose we were to draw two points in the plane with distinct x values, say, (a_0, b_0) and (a_1, b_1) with $a_0 \neq a_1$. There is a unique line, given as a function of x , passing through these two points. That is, there is a unique function of the form $f(x) = \lambda_0 x + \lambda_1$ with $\lambda_0, \lambda_1 \in \mathbb{R}$, such that $f(a_0) = b_0$ and $f(a_1) = b_1$. It is not hard to find a formula for this function f using elementary geometry or point-slope format.



Likewise, if we prescribe three points (a_0, b_0) , (a_1, b_1) , and (a_2, b_2) in the plane, all with distinct x values (so that $a_1 \neq a_2$, $a_1 \neq a_3$, and $a_2 \neq a_3$), then there is a unique quadratic polynomial $f(x) = \lambda_0 + \lambda_1 x + \lambda_2 x^2$ that passes through the three points. In general, it turns out that if you prescribe $n + 1$ points in the plane

$$(a_0, b_0), \dots, (a_n, b_n) \in \mathbb{R}^2$$

with pairwise distinct x coordinates (so that $a_i \neq a_j$ if $i \neq j$), then there is a unique polynomial of degree n passing through all of the points. This section is devoted to solving this problem from various perspectives in linear algebra.

2.8.1 The approach using the standard basis

First, instead of considering points in the plane $\mathbb{R}^2$, we may as well generalize to $\mathbb{F}^2$, where $\mathbb{F}$ can be any field. The most natural way to solve this problem is as follows. Given $n + 1$ points $(a_0, b_0), \dots, (a_n, b_n) \in \mathbb{F}^2$ with $a_i \neq a_j$ for $i \neq j$, we seek a polynomial

$$f(x) = \lambda_0 + \lambda_1 x + \dots + \lambda_n x^n$$

such that $f(a_j) = b_j$ for all $0 \leq j \leq n$. We can set up a system of equations as follows:

$$\begin{cases} \lambda_0 + a_0 \lambda_1 + \dots + a_0^n \lambda_n = b_0 \\ \lambda_0 + a_1 \lambda_1 + \dots + a_1^n \lambda_n = b_1 \\ \vdots \\ \lambda_0 + a_n \lambda_1 + \dots + a_n^n \lambda_n = b_n \end{cases}.$$

There are a lot of letters flying around, but remember that the constants a_j and b_j are given to us. They are not the variables. The variables we wish to solve for are the λ_j 's which represent the unknown coefficients of our desired polynomial. Thus, this is a system of $n + 1$ linear equations in $n + 1$ variables $(\lambda_0, \dots, \lambda_n)$. In particular, we could write this as a matrix-vector equation as follows:

$$\begin{pmatrix} 1 & a_0 & \dots & a_0^n \\ 1 & a_1 & \dots & a_1^n \\ \vdots & \vdots & \ddots & \vdots \\ 1 & a_n & \dots & a_n^n \end{pmatrix} \begin{pmatrix} \lambda_0 \\ \lambda_1 \\ \vdots \\ \lambda_n \end{pmatrix} = \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{pmatrix}.$$

The $(n + 1) \times (n + 1)$ matrix built out of the a_j 's is called the *Vandermonde* matrix.

Example 2.8.1. Let $(1, 1), (2, 3), (3, -1) \in \mathbb{R}^2$ be three points in the plane. We seek the unique quadratic polynomial that passes through these three points. Let $f(x) = \lambda_0 + \lambda_1 x + \lambda_2 x^2$. The desired system of equations is

$$\begin{cases} \lambda_0 + \lambda_1(1) + \lambda_2(1)^2 = 1 \\ \lambda_0 + \lambda_1(2) + \lambda_2(2)^2 = 3 \\ \lambda_0 + \lambda_1(3) + \lambda_2(3)^2 = -1 \end{cases} \iff \begin{cases} \lambda_0 + \lambda_1 + \lambda_2 = 1 \\ \lambda_0 + 2\lambda_1 + 4\lambda_2 = 3 \\ \lambda_0 + 3\lambda_1 + 9\lambda_2 = -1 \end{cases}.$$

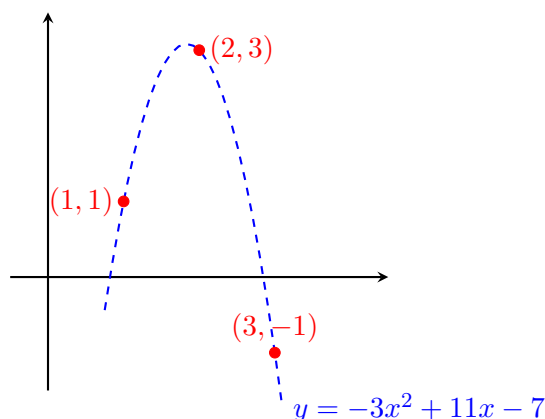
Subtracting the first equation from the second to get a new second equation, and subtracting the first equation from the third equation to get a new third equation, we have

$$\begin{cases} \lambda_0 + \lambda_1 + \lambda_2 = 1 \\ \lambda_1 + 3\lambda_2 = 2 \\ \lambda_1 + 5\lambda_2 = -4 \end{cases}.$$

Subtracting the second and third equation above then gives $2\lambda_2 = -6$, and hence $\lambda_2 = -3$. With this, the second equation becomes $\lambda_1 = 2 - 3(-3) = 11$, and the the first equation gives

$\lambda_0 = 1 - 11 - (-3) = -7$. Therefore, the polynomial we seek is

$$f(x) = -3x^2 + 11x - 7.$$



Remark 2.8.2. You could, of course, solve the above system of equations by row-reducing the Vandermonde matrix. Since I haven't used techniques like that in this text, I opted for a more direct method of calculation.

To phrase this approach in language more akin to that of this textbook, here's what we have done. Consider the vector space $P_n(\mathbb{F})$ of polynomials of degree n . This is a vector space of dimension $n+1$, as we have seen. We are looking for a specific polynomial $f \in P_n(\mathbb{F})$ satisfying $f(a_j) = b_j$ for all $0 \leq j \leq n$. If we pick a basis for $P_n(\mathbb{F})$, say, the standard basis

$$\mathcal{B} = \{1, x, x^2, \dots, x^n\}$$

then we know that f must be a linear combination of the above elements. That is, there must be some scalars $\lambda_0, \dots, \lambda_n \in \mathbb{F}$ such that

$$f(x) = \lambda_0 + \lambda_1 x + \dots + \lambda_n x^n.$$

This was our starting point above that led to the Vandermonde system of equations. This approach has the benefit of being easy to set up, but at the cost of being a bit more computationally intensive to actually solve (you must solve the resulting system of equations). The next approach — using something called Lagrange polynomials — takes a bit more work to set up, but then finding the solution is much simpler.

Remark 2.8.3. At this point in the text, we have not discussed determinants or invertibility of matrices, but there is much to say about this regarding the Vandermonde matrix. In particular, it turns out that if $a_i \neq a_j$ for $i \neq j$ — the precise condition we have been assuming — then the associated Vandermonde matrix is invertible (refer to Chapter 6 for a proof of this fact), and thus, there is in fact a unique solution to the above system of equations.

2.8.2 The approach using Lagrange polynomials

Lagrange interpolation is based off of the idea of using a different basis than the standard basis $\mathcal{B} = \{1, x, x^2, \dots, x^n\}$ as in the above subsection. Instead, if we choose a basis which is correctly

tuned to the prescribed points $(a_0, b_0), \dots, (a_n, b_n)$, determining the coefficients in the resulting linear combination is actually trivial. To see how this might work, let's motivate the abstract theory with a concrete example. For consistency, let's consider one of the same examples that we did with the first approach.

Consider three points in the plane $(1, 1)$, $(2, 3)$, and $(3, -1)$. As before, we want to find a quadratic polynomial $f(x)$ whose graph passes through all three of these points. That is, we want to find a quadratic polynomial such that $f(1) = 1$, $f(2) = 3$, and $f(3) = -1$. Our approach will be as follows. Suppose that we can find three quadratic polynomials p_1, p_2, p_3 satisfying the following:

$$\begin{aligned} p_1(1) &= 1 & p_1(2) &= 0 & p_1(3) &= 0 \\ p_2(1) &= 0 & p_2(2) &= 1 & p_2(3) &= 0 \\ p_3(1) &= 0 & p_3(2) &= 0 & p_3(3) &= 1 \end{aligned}$$

In words, we want p_1 to be a quadratic polynomial such that p_1 evaluates to 1 on the x -coordinate of the first point, and evaluates to 0 on the x -coordinates of all the other points. Likewise with the other polynomials. Let's assume for the moment that we've found such polynomials. Then if we set

$$f(x) = 1 \cdot p_1(x) + 3 \cdot p_2(x) + -1 \cdot p_3(x)$$

then $f(x)$ will be the quadratic polynomial we are looking for! Indeed, we can quickly check

$$\begin{aligned} f(1) &= p_1(1) + 3p_2(1) - p_3(1) = 1 + 0 + 0 = 1 \\ f(2) &= p_1(2) + 3p_2(2) - p_3(2) = 0 + 3 + 0 = 3 \\ f(3) &= p_1(3) + 3p_2(3) - p_3(3) = 0 + 0 + -1 = -1. \end{aligned}$$

Thus, all we need to do is construct the polynomials p_1, p_2 , and p_3 above. Let's consider how to build p_1 . We want p_1 to be a quadratic polynomial which vanishes at $x = 2$ and $x = 3$, and equals 1 when $x = 1$. We can easily build a quadratic polynomial that vanishes at $x = 2, 3$ by considering the expression

$$(x - 2)(x - 3).$$

The problem with this is that it does not evaluate to 1 when $x = 1$. It evaluates to $(1 - 2)(1 - 3) = 2$. To then define p_1 , we can simply divide the above expression by 2! Indeed, if we set

$$p_1(x) := \frac{1}{2}(x - 2)(x - 3) = \frac{1}{2}x^2 - \frac{5}{2}x + 3$$

then $p_1(1) = 1$, $p_1(2) = 0$, and $p_1(3) = 0$. We can build p_2 and p_3 with similar logic:

$$\begin{aligned} p_2(x) &:= \frac{1}{(2 - 1)(2 - 3)}(x - 1)(x - 3) = -x^2 + 4x - 3 \\ p_3(x) &:= \frac{1}{(3 - 1)(3 - 2)}(x - 1)(x - 2) = \frac{1}{2}x^2 - \frac{3}{2}x + 1. \end{aligned}$$

Then, according to our discussion above, the polynomial we seek is

$$\begin{aligned} f(x) &= 1 \cdot p_1(x) + 3 \cdot p_2(x) + -1 \cdot p_3(x) \\ &= \left(\frac{1}{2}x^2 - \frac{5}{2}x + 3 \right) + 3(-x^2 + 4x - 3) - \left(\frac{1}{2}x^2 - \frac{3}{2}x + 1 \right) \\ &= -3x^2 + 11x - 7. \end{aligned}$$

This is indeed the same polynomial as we got before, using totally different logic! The point here is that instead of expressing f as a linear combination of the standard basis $\{1, x, x^2\}$ of $P_2(\mathbb{R})$, we chose a different collection of polynomials $\{p_1, p_2, p_3\}$, dependent on the prescribed points, which makes the resulting linear combination very easy to identify. To formalize this discussion in the language of bases, we first give the following general definition.

Definition 2.8.4. Let $a_0, \dots, a_n \in \mathbb{F}$ be scalars such that $a_i \neq a_j$ for $i \neq j$. The **Lagrange polynomials associated to** $a_0, \dots, a_n$ are the polynomials $p_j(x) \in P_n(\mathbb{F})$ defined as follows:

$$p_j(x) := \left(\frac{1}{(a_j - a_1)(a_j - a_2) \cdots (a_j - a_n)} \right) \cdot (x - a_1) \cdots (x - a_{j-1})(x - a_{j+1}) \cdots (x - a_n).$$

In words, the Lagrange polynomial $p_j(x)$ is given by the product of factors of the form $(x - a_i)$, except for the factor corresponding to a_j , together with a scalar out front which is chosen so that $p_j(a_j) = 1$. Indeed, we have the following lemma.

Lemma 2.8.5. Let $a_0, \dots, a_n \in \mathbb{F}$ be scalars such that $a_i \neq a_j$ for $i \neq j$, and let $p_0(x), \dots, p_n(x)$ be the associated Lagrange polynomials. Then

- (i) for $i \neq j$, we have $p_j(a_i) = 0$, and
- (ii) $p_j(a_j) = 1$.

Proof. Suppose that $i \neq j$. Note that

$$p_j(a_i) = \left(\frac{1}{(a_j - a_1)(a_j - a_2) \cdots (a_j - a_n)} \right) \cdot (a_i - a_1) \cdots (a_i - a_{j-1})(a_i - a_{j+1}) \cdots (a_i - a_n).$$

Since $i \neq j$, one of the expressions $(a_i - a_k)$ above will in fact be $(a_i - a_i) = 0$. Thus, $p_j(a_i) = 0$, which proves (i). Similarly, note that

$$\begin{aligned} p_j(a_j) &= \left(\frac{1}{(a_j - a_1)(a_j - a_2) \cdots (a_j - a_n)} \right) \cdot (a_j - a_1) \cdots (a_j - a_{j-1})(a_j - a_{j+1}) \cdots (a_j - a_n) \\ &= 1 \end{aligned}$$

which proves (ii). □

As we discussed above, the point of the Lagrange polynomials is to identify a basis of $P_n(\mathbb{F})$ which is particularly suited to the problem of polynomial interpolation. Indeed, our main theorem is the following.

Theorem 2.8.6. Let $a_0, \dots, a_n \in \mathbb{F}$ be scalars such that $a_i \neq a_j$ for $i \neq j$, and let $p_0(x), \dots, p_n(x)$ be the associated Lagrange polynomials. Then

$$\mathcal{B}_L := \{p_0, \dots, p_n\}$$

is a basis of $P_n(\mathbb{F})$.

Proof. Because $\dim P_n(\mathbb{F}) = n + 1$, it suffices to prove that $\{p_0, \dots, p_n\}$ is linearly independent. Suppose that

$$\lambda_0 p_0 + \dots + \lambda_n p_n = 0$$

for some $\lambda_0, \dots, \lambda_n \in \mathbb{F}$. Then

$$\lambda_0 p_0(x) + \dots + \lambda_n p_n(x) = 0$$

for all $x \in \mathbb{F}$. In particular, when $x = a_j$, Lemma 2.8.5 gives

$$\lambda_0 p_0(a_j) + \dots + \lambda_{j-1} p_{j-1}(a_j) + \lambda_j p_j(a_j) + \lambda_{j+1} p_{j+1}(a_j) + \dots + \lambda_n p_n(a_j) = 0$$

which is equivalent to

$$0 + \dots + 0 + \lambda_j \cdot 1 + 0 + \dots + 0 = 0$$

and thus $\lambda_j = 0$. This proves that $\mathcal{B}_L$ is linearly independent, and thus a basis. $\square$

With this theorem in hand, we can present what is known as the *Lagrange interpolation formula*, which is the formalization of the above procedure we discussed up above.

Corollary 2.8.7 (Lagrange interpolation formula). Let $(a_0, b_0), \dots, (a_n, b_n) \in \mathbb{F}^2$ be $n + 1$ points in the plane with $a_i \neq a_j$ for $i \neq j$. Then there is a unique polynomial $f(x)$ of degree n passing through all points (a_j, b_j) and it is given by

$$f(x) := b_0 p_0(x) + b_1 p_1(x) + \dots + b_n p_n(x)$$

where $p_0(x), \dots, p_n(x)$ are the Lagrange polynomials associated to $a_0, \dots, a_n$.

Proof. Let

$$f(x) := b_0 p_0(x) + b_1 p_1(x) + \dots + b_n p_n(x)$$

as in the statement of the corollary. Note that

$$\begin{aligned} f(a_j) &= b_0 p_0(a_j) + \dots + b_{j-1} p_{j-1}(a_j) + b_j p_j(a_j) + b_{j+1} p_{j+1}(a_j) + \dots + b_n p_n(a_j) \\ &= 0 + \dots + 0 + b_j \cdot 1 + 0 + \dots + 0 \\ &= b_j. \end{aligned}$$

Thus, the polynomial f does pass through each point (a_j, b_j) . Next, we show that it is unique. Suppose g is another such polynomial. Because $\mathcal{B}_L$ is a basis of $P_n(\mathbb{F})$, there exist scalars $\lambda_0, \dots, \lambda_n \in \mathbb{F}$ such that

$$g(x) = \lambda_0 p_0(x) + \dots + \lambda_n p_n(x).$$

Note that

$$\begin{aligned} g(a_j) &= \lambda_0 p_0(a_j) + \cdots + \lambda_{j-1} p_{j-1}(a_j) + \lambda_j p_j(a_j) + \lambda_{j+1} p_{j+1}(a_j) + \cdots + \lambda_n p_n(a_j) \\ &= 0 + \cdots + 0 + \lambda_j \cdot 1 + 0 + \cdots + 0 \\ &= \lambda_j. \end{aligned}$$

By assumption, $g(a_j) = b_j$, so $\lambda_j = b_j$. Thus,

$$\begin{aligned} g(x) &= b_0 p_0(x) + \cdots + b_n p_n(x) \\ &= f(x) \end{aligned}$$

and so $f(x)$ is unique, as desired. □

Exercises

1. Let $(-1, 0), (2, 0), (4, 3) \in \mathbb{R}^2$ be three prescribed points in the plane.
 - (a) Find the unique quadratic polynomial passing through these points using the standard basis (a.k.a. Vandermonde) approach.
 - (b) Alternatively, find the unique quadratic polynomial passing through these points using Lagrange interpolation.
2. Let $(0, 0), (1, 1), (2, -1), (3, 3) \in \mathbb{R}^2$ be four prescribed points in the plane.
 - (a) Find the unique cubic polynomial passing through these points using the standard basis (a.k.a. Vandermonde) approach.
 - (b) Alternatively, find the unique cubic polynomial passing through these points using Lagrange interpolation.
3. This exercise will describe a third approach to polynomial interpolation using *Newton polynomials*. As before, let $(a_0, b_0), \dots, (a_n, b_n) \in \mathbb{F}^2$ be $n+1$ prescribed points with $a_i \neq a_j$ for $i \neq j$. For $0 \leq j \leq n$, define

$$q_0(x) = 1$$

and

$$q_j(x) = (x - a_0)(x - a_1) \cdots (x - a_{j-1}).$$

These are the **Newton polynomials** associated to $(a_0, b_0), \dots, (a_n, b_n)$.

- (a) Prove that $\{q_0, \dots, q_n\}$ is a basis for $P_n(\mathbb{F})$.

Next, we determine the corresponding coefficients that give rise to the polynomial f passing through the prescribed points. This involves a new concept called the **divided difference coefficients**. We will define a coefficient $[b_0, \dots, b_j] \in \mathbb{F}$ recursively as follows. Set $[b_0] := b_0$. Then for $j > 0$, define

$$[b_0, \dots, b_j] := \frac{[b_0, \dots, b_j] - [b_0, \dots, b_{j-1}]}{a_j - a_0}.$$

(b) Prove that $[b_0, b_1] = \frac{b_1 - b_0}{a_1 - a_0}$.

(c) Prove that

$$[b_0, b_1, b_2] = \frac{b_2 - b_1}{(a_2 - a_1)(a_2 - a_0)} + \frac{b_1 - b_0}{(a_1 - a_0)(a_2 - a_0)}.$$

With this notation, we can now state the *Newton interpolation formula*. It turns out that the unique polynomial $f(x)$ of degree n passing through $(a_0, b_0), \dots, (a_n, b_n)$ is

$$f(x) := [b_0] q_0(x) + [b_0, b_1] q_1(x) + \dots + [b_0, \dots, b_n] q_n(x).$$

(d) Consider the prescribed points $(1, 1), (2, 3), (3, -1) \in \mathbb{R}^2$ as in the section above. Compute the unique quadratic polynomial passing through the three points using Newton interpolation, and verify that it agrees with the Vandermonde and Lagrange approaches.

Next, we will walk through the proof of the Newton interpolation formula.

(e) Use induction to prove that $[b_0, \dots, b_k] = [b_k, \dots, b_0]$ for all $k \geq 1$.

(f) Use induction to prove that

$$[b_0, \dots, b_k] (a_k - a_0) \cdots (a_k - a_{k-1}) = b_k - P(a_k)$$

where $P(x)$ is the unique degree $k-1$ polynomial passing through $(a_0, b_0), \dots, (a_{k-1}, b_{k-1})$. As a base case, prove that

$$[b_0, b_1] (a_1 - a_0) = b_1 - P(a_1)$$

where $P(x) = y_0$ is the constant polynomial at y_0 .

(g) Conclude Newton's interpolation formula.

Chapter 3

Linear Transformations

Up until this point, we have followed a very typical structure for developing a mathematical theory.

- ◊ We have axiomatically defined a class of mathematical objects (vector spaces).
- ◊ We have developed a base of concrete examples ($\mathbb{R}^n$, vector spaces of functions, etc.)
- ◊ We have investigated basic properties and consequences of the definition (the concept of linear independence, the notion of dimension, etc.)
- ◊ We have studied sub-objects of our objects (subspaces of vector spaces).

The naturally trajectory from here is to ask the following types of questions: *How do we compare two of our mathematical objects? How can we tell when or measure how two of our objects are similar or different?* These questions lead to the study of functions that map from one object to another object.

In the context of linear algebra, so far we have only developed the theory of a single vector space. We want to expand our horizons and communicate between two different vector spaces, perhaps to understand how similar or how different they are. Can we identify when two vector spaces are fundamentally the “same”? What does “same” even mean? If two vector spaces are “different”, whatever that means, can we develop a theory that measures how “different” they are? To answer these questions rigorously, we need to study functions $f : V \rightarrow W$ whose domain is some vector space V and whose codomain is another vector space W . But in order for this to be a useful endeavor, we don’t want to consider any functions between vector spaces. We want our functions to *respect the vector space structure*. This gives rise to the definition of a *linear transformation*, which is the central definition of this chapter. Arguably, the notion of a linear transformation is the most fundamental definition in all of linear algebra.

3.1 Linear transformations

To *respect the vector space structure* means to be compatible with the operations of addition and scalar multiplication, the fundamental operations that define the algebraic structure of a vector space. This compatibility is described formally by the following definition.

Definition 3.1.1. Let V, W be vector spaces over the same field $\mathbb{F}$. A function $T : V \rightarrow W$ is called a **linear transformation**, or a **linear map**, or a **linear function**, or simply **linear**, if the following properties hold.

(1) For all $v, v' \in V$,

$$T(v + v') = T(v) + T(v').$$

(2) For all $v \in V$ and $\lambda \in \mathbb{F}$,

$$T(\lambda v) = \lambda T(v).$$

Remark 3.1.2. Note that the addition and multiplication on the left side of each equality above is occurring in V , while the addition and multiplication on the right side of each equality is occurring in W . To emphasize, the vectors v and v' live in V , and so $v + v'$ is the addition operation in V , whereas $T(v)$ and $T(v')$ live in W , and so $T(v) + T(v')$ is the addition operation in W .

Example 3.1.3. Consider the map $T : \mathbb{R} \rightarrow \mathbb{R}$ given by $T(x) = 3x$. Then T is linear. Indeed, note that

$$\begin{aligned} T(x + y) &= 3(x + y) \\ &= 3x + 3y \\ &= T(x) + T(y) \end{aligned}$$

and

$$\begin{aligned} T(\lambda x) &= 3(\lambda x) \\ &= \lambda \cdot 3x \\ &= \lambda T(x) \end{aligned}$$

for all $x, y \in \mathbb{R}$ and $\lambda \in \mathbb{R}$.

On the other hand, consider $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 3x + 1$. The function f is *not* linear. Indeed, note that

$$\begin{aligned} f(2 \cdot 1) &= 3(2 \cdot 1) + 1 = 7 \\ 2 \cdot f(1) &= 2(3 \cdot 1 + 1) = 8. \end{aligned}$$

Since $f(2 \cdot 1) \neq 2 \cdot f(1)$, it follows that f is not linear.

This last non-example is important: the concept of linearity in linear algebra is more restrictive than your colloquial understanding of linearity from a calculus perspective. In calculus, the function $f(x) = 3x + 1$ would likely be thought of as “linear”, but it is *not* linear in the technical sense.¹ We will consider more examples soon, but one important and identifying feature of a linear transformation is that it always takes the zero vector to the zero vector. This is another way to immediately recognize that f above is not linear.

Proposition 3.1.4. Suppose that $T : V \rightarrow W$ is linear. Then $T(0) = 0$.

¹We would call such a function *affine*.

Proof. Note that $T(0) = T(0 \cdot 0) = 0 \cdot T(0) = 0$. Here, linearity of T is used in the second equality. $\square$

Remark 3.1.5. Again I will point out the subtlety of the notation. In the above proof, there are two different objects called “0”. One of them is the scalar 0, and the other is the 0 vector.

Example 3.1.6. Let $A \in M_{m \times n}(\mathbb{F})$. Let $T : \mathbb{F}^n \rightarrow \mathbb{F}^m$ be given by $T(x) = Ax$, where Ax is the usual matrix-vector multiplication of an $m \times n$ matrix by an $n \times 1$ vector. Then T is a linear transformation.

Example 3.1.7. Here is a more interesting example. Let $T : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be the map given by $T(p)(x) = p'(x)$ for all $p \in P(\mathbb{R})$. In words, T is the map that eats a polynomial and takes the derivative. Then T is linear. Indeed, let $p, q \in P(\mathbb{R})$ and $\lambda \in \mathbb{R}$. To prove that T is linear, we need to prove that $T(p + q) = T(p) + T(q)$ and $T(\lambda p) = \lambda T(p)$. Using familiar properties of the derivative from calculus, we have, for any $x \in \mathbb{R}$,

$$\begin{aligned} T(p + q)(x) &= (p + q)'(x) \\ &= p'(x) + q'(x) \\ &= T(p)(x) + T(q)(x) \end{aligned}$$

and

$$\begin{aligned} T(\lambda p)(x) &= (\lambda p)'(x) \\ &= \lambda p'(x) \\ &= \lambda T(p)(x). \end{aligned}$$

Thus, $T(p + q) = T(p) + T(q)$ and $T(\lambda p) = \lambda T(p)$. This proves that T is linear.

Example 3.1.8. Let $0 : V \rightarrow W$ be the zero transformation, i.e., the map such that $0(v) = 0$ for all $v \in V$. Then 0 is linear. Indeed,

$$0(v + w) = 0 = 0 + 0 = 0(v) + 0(w)$$

and

$$0(\lambda v) = 0 = \lambda \cdot 0 = \lambda \cdot 0(v).$$

Example 3.1.9. Let $I : V \rightarrow V$ be the identity map, i.e., the map defined by $I(v) = v$ for all $v \in V$. Then I is linear. Indeed,

$$I(v + w) = v + w = I(v) + I(w)$$

and

$$I(\lambda v) = \lambda v = \lambda I(v).$$

3.1.1 Kernels and images

Given a linear transformation $T : V \rightarrow W$, we want to understand its behavior in systematic ways. To this end, two important features of a linear transformation are its *kernel*, also called the *null space*, and the *image*, also called the *range*.

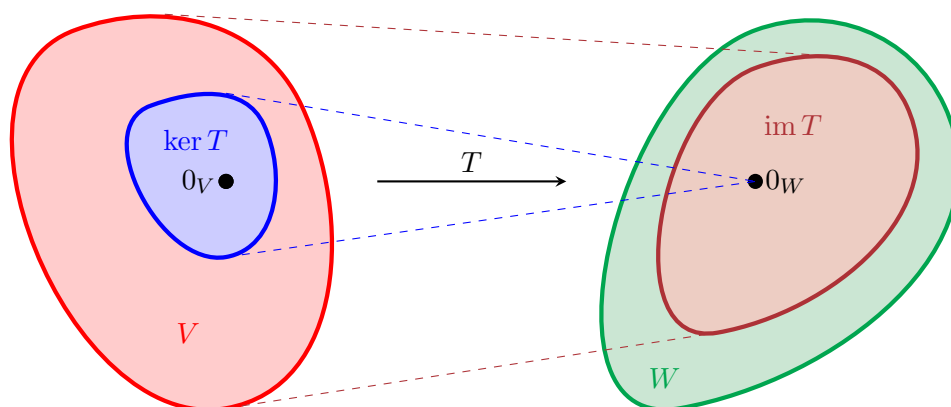
Definition 3.1.10. Let $T : V \rightarrow W$ be a linear transformation. The **kernel** of T is the set $\ker T \subset V$ defined by

$$\ker T := \{ v \in V \mid T(v) = 0 \}.$$

The **image** of T is the set $\operatorname{im} T \subset W$ defined by

$$\operatorname{im} T := \{ w \in W \mid w = T(v) \text{ for some } v \in V \}.$$

In words, the kernel is the set of stuff in the domain that gets mapped to the 0 element of W , and the image is the set of all outputs of T in W . See the figure below for a heuristic depiction of these sets. Note that other references use the notation $N(T)$ in place of $\ker T$, and $R(T)$ in place of $\operatorname{im} T$.



Example 3.1.11. Let $P : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the linear map given by $P(x, y, z) = (x, y, 0)$. In words, P is projection onto the (x, y) -plane. To compute the kernel of P , we need to identify the set of vectors $(x, y, z) \in \mathbb{R}^3$ such that $P(x, y, z) = (0, 0, 0)$. Using the formula for P , we get

$$\begin{aligned} \ker P &= \{ (x, y, z) \mid P(x, y, z) = (0, 0, 0) \} \\ &= \{ (x, y, z) \mid (x, y, 0) = (0, 0, 0) \} \\ &= \{ (x, y, z) \mid x = y = 0 \} \\ &= \operatorname{Span}\{(0, 0, 1)\}. \end{aligned}$$

Thus, in words, $\ker P$ is the z -axis. This makes sense geometrically: if you do a perpendicular² projection of three dimensional space onto the (x, y) -plane, anything on the z -axis will get squashed to the origin. To compute the image of P , we proceed as follows:

$$\operatorname{im} P = \{ P(x, y, z) \mid x, y, z \in \mathbb{R} \} = \{ (x, y, 0) \mid x, y, z \in \mathbb{R} \} = \operatorname{Span}\{(1, 0, 0), (0, 1, 0)\}.$$

In words, $\operatorname{im} P$ is the (x, y) -plane. Again, this makes sense geometrically: the set of all things that a perpendicular projection onto the (x, y) -plane can produce is exactly the (x, y) -plane!

Example 3.1.12. Let V be a vector space. Let $0 : V \rightarrow V$ be the 0-map, i.e., the linear transformation that sends everything to 0 defined by $0(v) := 0$ for all $v \in V$. Then $\ker(0) = V$, and

²Technically, we don't what this means yet!

$\text{im}(0) = \{0\}$. On the other end of the spectrum is the identity map $I : V \rightarrow V$, defined by $I(v) = v$ for all $v \in V$. We have $\ker I = \{0\}$ and $\text{im } I = V$.

Proposition 3.1.13. *Let $T : V \rightarrow W$ be linear. Then $\ker T$ is a subspace of V , and $\text{im } T$ is a subspace of W .*

Proof. First, we show that $\ker T$ is a subspace of V . Note that $T(0_V) = 0_W$ by the above proposition, so $0_V \in \ker T$. Next, suppose that $v, v' \in \ker T$. Then

$$T(v + v') = T(v) + T(v') = 0 + 0 = 0$$

so that $v + v' \in \ker T$. Similarly, if $v \in \ker T$ and $\lambda \in \mathbb{F}$, then

$$T(\lambda v) = \lambda T(v) = \lambda \cdot 0 = 0$$

so that $\lambda v \in \ker T$. Thus, $\ker T$ is a subspace of V .

Next, we show that $\text{im } T$ is a subspace of W . Since $0_W = T(0_V)$, $0_W \in \text{im } T$. Next, suppose that $w, w' \in \text{im } T$. Then there are $v, v' \in V$ such that $w = T(v)$ and $w' = T(v')$. So

$$w + w' = T(v) + T(v') = T(v + v').$$

Thus, $w + w'$ is the image of an element in V , it follows that $w + w' \in \text{im } T$. Similarly, if $w \in \text{im } T$ and $\lambda \in \mathbb{F}$, then writing $w = T(v)$ for some $v \in V$ gives

$$\lambda w = \lambda T(v) = T(\lambda v)$$

and so $\lambda w \in \text{im } T$. Thus, $\text{im } T$ is a subspace of W . □

Now that we know that $\ker T$ and $\text{im } T$ are subspaces, we can make the following definition. I won't use this language at all throughout this text, but I bring it up because it is standard and explains the name of the upcoming theorem.

Definition 3.1.14. Let $T : V \rightarrow W$ be linear, and suppose that $\ker T$ and $\text{im } T$ are finite dimensional. The **nullity** of T is

$$\text{null}(T) := \dim \ker T$$

and the **rank** of T is

$$\text{rank}(T) := \dim \text{im } T.$$

Example 3.1.15. Let $P : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be the projection onto the (x, y) -plane as in Example 3.1.11, given by $P(x, y, z) = (x, y, 0)$. Then

$$\text{null}(P) = \dim \ker P = \dim \text{Span}\{(0, 0, 1)\} = 1$$

and

$$\text{rank}(P) = \dim \text{im } P = \dim \text{Span}\{(1, 0, 0), (0, 1, 0)\} = 2.$$

3.1.2 Rank-nullity theorem

The following theorem is one of the most important results in linear algebra. In fact, it is sometimes referred to as the *fundamental theorem of linear algebra*. In words, it describes a natural balance between the relative size of the kernel and image of any transformation. In particular, it says that if $T : V \rightarrow W$ has a large kernel (so that it squashes a lot of stuff), then the image (the rest of the stuff T spits out) can't be very big. Likewise, if the kernel of T is small (so that not much stuff got squashed), then the image of T will consequently be larger. The rank-nullity goes further and gives a quantitative measurement of this phenomenon.

Theorem 3.1.16 (Rank-nullity theorem). *Let $T : V \rightarrow W$ be linear. Suppose that V is finite dimensional. Then*

$$\dim \ker T + \dim \operatorname{im} T = \dim V.$$

Remark 3.1.17. Note that there is no assumption of finite dimensionality of W ! We only need to assume V is finite dimensional.

Proof. Let $n = \dim V$ and let $k = \dim \ker T$. Let $\{u_1, \dots, u_k\}$ be a basis for $\ker T$. Because $\{u_1, \dots, u_k\}$ is a basis of $\ker T$, it is a linearly independent set in V . We can extend this set to a basis $\{u_1, \dots, u_k, \tilde{u}_{k+1}, \dots, \tilde{u}_n\}$ of V . We claim that $\{T(\tilde{u}_{k+1}), \dots, T(\tilde{u}_n)\}$ is a basis for $\operatorname{im} T$.

First, we prove linear independence of $\{T(\tilde{u}_{k+1}), \dots, T(\tilde{u}_n)\}$. Suppose that

$$a_{k+1}T(\tilde{u}_{k+1}) + \dots + a_nT(\tilde{u}_n) = 0.$$

Then by linearity of T , we have

$$T(a_{k+1}\tilde{u}_{k+1} + \dots + a_n\tilde{u}_n) = 0.$$

Thus, $a_{k+1}\tilde{u}_{k+1} + \dots + a_n\tilde{u}_n \in \ker T$. Since $\{u_1, \dots, u_k\}$ is a basis for $\ker T$, there exists $a_1, \dots, a_k \in \mathbb{F}$ such that

$$a_{k+1}\tilde{u}_{k+1} + \dots + a_n\tilde{u}_n = a_1u_1 + \dots + a_ku_k.$$

This implies

$$a_1u_1 + \dots + a_ku_k + (-a_{k+1})\tilde{u}_{k+1} + \dots + (-a_n)\tilde{u}_n = 0.$$

Since $\{u_1, \dots, u_k, \tilde{u}_{k+1}, \dots, \tilde{u}_n\}$ is linearly independent, $a_1 = \dots = a_k = -a_{k+1} = \dots = -a_n = 0$. In particular, $a_{k+1} = \dots = a_n = 0$. Thus, $\{T(\tilde{u}_{k+1}), \dots, T(\tilde{u}_n)\}$ is linearly independent.

Next, we show that $\{T(\tilde{u}_{k+1}), \dots, T(\tilde{u}_n)\}$ spans $\operatorname{im} T$. Let $w \in \operatorname{im} T$. Then there exists some $v \in V$ such that $w = T(v)$. Since $\{u_1, \dots, u_k, \tilde{u}_{k+1}, \dots, \tilde{u}_n\}$ is a basis for V , there exist $b_1, \dots, b_n \in \mathbb{F}$ such that

$$v = b_1u_1 + \dots + b_ku_k + b_{k+1}\tilde{u}_{k+1} + \dots + b_n\tilde{u}_n.$$

Then

$$\begin{aligned}
w &= T(v) \\
&= T(b_1u_1 + \cdots + b_ku_k + b_{k+1}\tilde{u}_{k+1} + \cdots + b_n\tilde{u}_n) \\
&= b_1T(u_1) + \cdots + b_kT(u_k) + b_{k+1}T(\tilde{u}_{k+1}) + \cdots + b_nT(\tilde{u}_n) \\
&= b_{k+1}T(\tilde{u}_{k+1}) + \cdots + b_nT(\tilde{u}_n).
\end{aligned}$$

The last equality follows from the fact that $u_j \in \ker T$ for $1 \leq j \leq k$. Thus,

$$w \in \text{Span}(\{T(\tilde{u}_{k+1}), \dots, T(\tilde{u}_n)\}).$$

This shows that $\text{im } T = \text{Span}(\{T(\tilde{u}_{k+1}), \dots, T(\tilde{u}_n)\})$, and so $\{T(\tilde{u}_{k+1}), \dots, T(\tilde{u}_n)\}$ is a basis for $\text{im } T$.

Since $\{T(\tilde{u}_{k+1}), \dots, T(\tilde{u}_n)\}$ has $n - k$ vectors, it follows that $\dim \text{im } T = n - k$. Thus,

$$\dim \ker T + \dim \text{im } T = k + (n - k) = n = \dim V$$

as desired. □

Example 3.1.18. Consider again the projection map $P : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ from Example 3.1.11 given by $P(x, y, z) = (x, y, 0)$. We have previously computed that $\dim \ker P = 1$ and $\dim \text{im } P = 2$, so it is indeed the case that $\dim \ker P + \dim \text{im } P = \dim \mathbb{R}^3 = 3$.

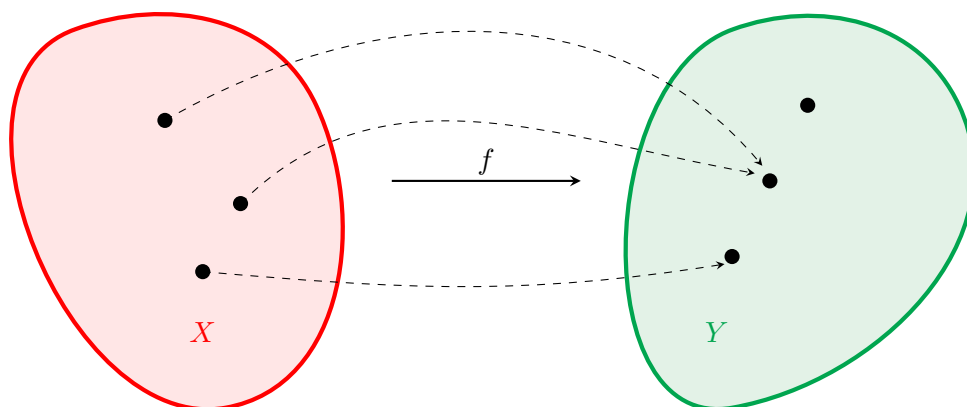
Example 3.1.19. Let $R : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be the transformation given by $R(x, y) := (-y, x)$. Geometrically, this is counterclockwise rotation by 90 degrees. First we compute the kernel of R . Suppose that $R(x, y) = (0, 0)$. Then $(-y, x) = (0, 0)$, and hence $x = y = 0$. Thus, if $(x, y) \in \ker R$ then $(x, y) = (0, 0)$, hence $\ker R = \{(0, 0)\}$. It follows that $\dim \ker R = 0$. Although we could compute the image of R directly — and geometrically, it should be clear that $\text{im } R = \mathbb{R}^2$ — we could apply Rank-Nullity to conclude that

$$\dim \text{im } R = \dim \mathbb{R}^2 - \dim \ker R = 2 - 0 = 2.$$

Thus, since $\text{im } R \subset \mathbb{R}^2$ is a 2-dimensional subspace of $\mathbb{R}^2$, it must be the case that $\text{im } R = \mathbb{R}^2$.

3.1.3 Injectivity and surjectivity

In order to develop more consequences of the rank-nullity theorem, we recall the notions of *injectivity* (being one-to-one) and *surjectivity* (being onto). The following definition is very general, and holds for any function between sets. Before I state the precise definition, it is helpful to give an informal description based on the following picture.



The above figure depicts a function $f : X \rightarrow Y$, where X and Y are both sets with three elements, indicated by the black dots. The dashed arrows show where f sends each of the dots in X . This function f is neither injective nor surjective.

- ◇ The function f is not injective, because two of the dots on the left get sent to the same dot on the right. In words, a function f is injective if this never happens; that is, a function is injective if, whenever you have two distinct points in the domain, their images under the function are distinct.
- ◇ The function f is not surjective, because the topmost dot in Y , the codomain, is not produced by f . In words, a function is surjective if it spits up everything in the codomain.

With this general figure in mind, here are the precise definitions of injectivity and surjectivity. It is a good exercise to reconcile them with the picture above.

Definition 3.1.20. Let X, Y be sets, and let $f : X \rightarrow Y$.

- (i) The function $f : X \rightarrow Y$ is **injective** if $f(x) = f(x')$ implies $x = x'$.
- (ii) The function $f : X \rightarrow Y$ is **surjective** if, for all $y \in Y$, there is an $x \in X$ such that $f(x) = y$.

Remark 3.1.21. We can unravel the definitions of injective and surjective to get the following equivalent notions:

- (i) A function $f : X \rightarrow Y$ is *not* injective if and only if there exist $x \neq x' \in X$ such that $f(x) = f(x')$.
- (ii) A function $f : X \rightarrow Y$ is surjective if and only if $\text{im } f = Y$.

If we know that a function is linear between two vector spaces, we can say even more about the relation of injectivity to the linear structure. The following proposition is frequently used in practice to determine injectivity of a linear transformation.

Proposition 3.1.22. A linear map $T : V \rightarrow W$ is injective if and only if $\ker T = \{0\}$.

Proof. First, suppose that T is injective. Let $v \in \ker T$. Then note that $T(v) = 0 = T(0)$. By injectivity, it follows that $v = 0$. Thus, $\ker T = \{0\}$.

Next, suppose that $\ker T = \{0\}$. Suppose that $v, v' \in V$ are elements such that $T(v) = T(v')$. Then, by linearity of T , we have $T(v - v') = 0$. Thus, $v - v' \in \ker T$. This implies $v - v' = 0$, and so $v = v'$. Thus, T is injective. $\square$

If we know that the domain and codomain are finite dimensional and of the same dimension, we can give an even stronger relation between injectivity and surjectivity.

Proposition 3.1.23. *Suppose that V, W are finite dimensional vector spaces such that $\dim V = \dim W$ and that $T : V \rightarrow W$ is linear. Then T is injective if and only if T is surjective.*

Proof. First, suppose that T is injective. Then $\dim \ker T = 0$. By the rank-nullity theorem, it follows that $0 + \dim \operatorname{im} T = \dim V$. Since $\dim W = \dim V$, it follows that $\dim \operatorname{im} T = \dim W$. Since $\operatorname{im} T$ is a subspace of W , it follows that $\operatorname{im} T = W$, and so T is surjective.

Next, suppose that T is surjective. Then $\operatorname{im} T = W$, so that $\dim \operatorname{im} T = \dim W = \dim V$. Then the rank-nullity theorem implies

$$\dim \ker T = \dim V - \dim \operatorname{im} T = 0.$$

Thus, $\ker T = \{0\}$, and so T is injective. $\square$

ADD EXAMPLES AND DISCUSSION OF INTUITION

Exercises

- For each of the following functions, determine if it is linear or not. If it is linear, prove it. If it is not linear, prove it.

(a) The function $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x, y, xy)$.

(b) The function $S : P(\mathbb{F}) \rightarrow P(\mathbb{F})$ defined by $S(p)(x) = x^2 p(x)$.

- Let V, W be vector spaces, and let $T : V \rightarrow W$ be a surjective linear transformation. Suppose that $\{v_1, \dots, v_n\} \subset V$ is a set of vectors such that $\operatorname{Span}(\{v_1, \dots, v_n\}) = V$. Prove that

$$\operatorname{Span}(\{T(v_1), \dots, T(v_n)\}) = W.$$

- Let V, W be vector spaces. Let $T : V \rightarrow W$ be a linear map.

(a) Suppose that $\{v_1, \dots, v_n\} \subset V$ is a collection of vectors such that $\{T(v_1), \dots, T(v_n)\} \subset W$ is linearly independent. Prove that $\{v_1, \dots, v_n\}$ is linearly independent.

(b) Assume that T is injective. Prove that if $\{v_1, \dots, v_n\}$ is linearly independent, then

$$\{T(v_1), \dots, T(v_n)\}$$

is linearly independent.

Is this statement still true if you don't assume that T is injective? Can you come up with a counter example?

4. In this problem, you'll need to remember a few basic facts from calculus. Let V be the (real) vector space of continuously differentiable³ functions $f : \mathbb{R} \rightarrow \mathbb{R}$, and let W be the (real) vector space of continuous functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Let $T : V \rightarrow W$ be the linear⁴ map defined by $T(f)(x) := f'(x)$.

(a) Prove that T is not injective.

(b) Prove that T is surjective. [Hint: Think about the fundamental theorem of calculus!]

5. Let V and W be finite dimensional vector spaces, and suppose that $T : V \rightarrow W$ is linear. Prove that if $\dim V < \dim W$, then T is not surjective.

6. Let $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be given by $T(x, y, z) = (x - y, 2z)$.

(a) Prove that T is linear.

(b) Compute $\ker T$.

(c) Compute $\operatorname{im} T$.

7. Suppose that $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is a linear transformation such that $T(1, 0) = (3, 4)$ and $T(1, 1) = (2, 2)$. Compute $T(2, 3)$.

8. Give an example of a linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ such that $\ker T = \operatorname{im} T$.

9. Let V, W be vector spaces, and let W_0 be a subspace of W . Prove that

$$T^{-1}(W_0) := \{v \in V \mid T(v) \in W_0\}$$

is a subspace of V .

10. Let V and W be finite dimensional vector spaces, and suppose that $T : V \rightarrow W$ is linear. Prove that if $\dim V > \dim W$, then T is not injective.

11. Let V be a finite dimensional vector space, and let $T : V \rightarrow V$ be linear.

(a) Suppose that $V = \ker T + \operatorname{im} T$. Prove that $V = \ker T \oplus \operatorname{im} T$.

(b) Suppose that $\ker T \cap \operatorname{im} T = \{0\}$. Prove that $V = \ker T \oplus \operatorname{im} T$.

12. Let V be a vector space, and let W_1, W_2 be subspaces such that $V = W_1 \oplus W_2$. Define a linear map $P_1 : V \rightarrow V$ as follows. For each $v \in V$, let $v = w_1 + w_2$ be the unique decomposition of v with $w_1 \in W_1$ and $w_2 \in W_2$. Then define $P(v) := w_1$. This map is called the *projection* onto W_1 .

(a) Prove that $P_1 : V \rightarrow V$ is linear.

(b) Compute $\ker P_1$ and $\operatorname{im} P_1$.

(c) Think about why this is called a “projection,” and reconcile this with your previous understanding of a “projection.”

³A function is *continuously differentiable* if $f'(x)$ exists and is continuous. These words are here just to ensure that what I'm about to define makes sense, so don't let this scare you!

⁴You don't have to prove T is linear, you can assume this.

13. True or false: there exists a linear transformation $T : V \rightarrow V$ which is injective, but not surjective.
14. Let $F(\mathbb{R}, \mathbb{R})$ be the real vector space of functions $f : \mathbb{R} \rightarrow \mathbb{R}$ (note that this includes discontinuous and piecewise functions!). Define a map $T : F(\mathbb{R}, \mathbb{R}) \rightarrow \mathbb{R}$ by

$$T(f) := f(1) + f(-1).$$

- (a) (4 points) Prove that T is linear.
- (b) (4 points) Prove that T is not injective.
- (c) (4 points) Prove that T is surjective.
15. Let $T : V \rightarrow V$ be a linear map. Prove that $T^2 = 0$ if and only if $\text{im } T \subset \ker T$.
16. True or false: If $v_1, v_2 \in V$ are linearly independent and $T : V \rightarrow W$ is a surjective linear map, then $\{T(v_1), T(v_2)\}$ is linearly independent.
17. True or false: Suppose that $T : V \rightarrow W$ is a linear map such that $\ker T$ is finite dimensional. Then V is finite dimensional.
18. True or false: If $T, S : V \rightarrow V$ are operators such that $\ker T = \ker S$ and $\text{im } T = \text{im } S$, then $T = S$.

3.2 Coordinates and matrix representations

We have seen a number of vector spaces in class so far. For example, there are the familiar vector spaces like

$$\mathbb{R}^3 = \left\{ \begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix} \mid a_j \in \mathbb{R} \right\}$$

and there are some vector spaces that, perhaps initially, were less familiar, like

$$P_2(\mathbb{R}) = \{ a_0 + a_1x + a_2x^2 \mid a_j \in \mathbb{R} \}.$$

If you've thought to yourself *these two vector spaces feel like the same thing, up to renaming*, then you're correct. The next few topics we discuss (coordinates, isomorphisms, and so on) will make this notion formal.

We begin with the notion of *coordinates* in an abstract vector space. From lower-division linear algebra, we are comfortable with numerical vectors like

$$\begin{pmatrix} 1 \\ 2 \\ -\pi \end{pmatrix}.$$

Moreover, they are useful for computations. In an abstract vector space V , the elements of V may not at all look like the numerical vectors above. For example, if V is a vector space of

colors, then the only true way we can reference an element of V is by referring to the abstract color it represents. While is this fine for many purposes, it would be nice to *produce numerical data that represents the abstract data of a vector space*. This is the slogan of this section, and leads to the notion of coordinates. The way that we do this is with a basis of V .

Definition 3.2.1. Let $\mathcal{B} = \{u_1, \dots, u_n\}$ be an (ordered) basis for a vector space V . Let $v \in V$, and let $v = a_1u_1 + \dots + a_nu_n$ be the unique expression of v as a linear combination of elements of $\mathcal{B}$. The **coordinate vector of v with respect to $\mathcal{B}$** is

$$[v]_{\mathcal{B}} = \begin{pmatrix} a_1 \\ \vdots \\ a_n \end{pmatrix} \in \mathbb{F}^n.$$

Remark 3.2.2. Note that the order of the elements in the basis matters! When we talk about coordinates, we need to keep track of this order, which is why the above definition refers to the basis as an “ordered basis.”

Remark 3.2.3. Note that this definition is only made for *finite dimensional* vector spaces. Using coordinates in infinite dimensional vector spaces can be delicate.

In words, the above definition says the following: once you choose a basis $\mathcal{B}$, you can pick your favorite abstract vector $v \in V$ and expand it as a linear combination of the elements of V . If you collect all of the coefficients and put them into a numerical vector, you get the coordinate vector of v with respect to $\mathcal{B}$. Let’s look at some concrete examples.

Example 3.2.4. Let $p(x) = 1 - 2x^2 \in P_2(\mathbb{R})$. Let $\mathcal{B} = \{1, x, x^2\}$. Then since $p(x) = 1 \cdot 1 + 0 \cdot x + (-2) \cdot x^2$, we have

$$[p(x)]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 0 \\ -2 \end{pmatrix}.$$

Note that the resulting coordinate vector depends on the choice of basis. For instance,

$$\mathcal{C} = \{1, 1 + x, 1 + x^2\}$$

is another basis for $P_2(\mathbb{R})$. With $p(x) = 1 - 2x^2$ as above, note that

$$p(x) = (3)(1) + (0)(1 + x) + (-2)(1 + x^2).$$

Thus,

$$[p(x)]_{\mathcal{C}} = \begin{pmatrix} 3 \\ 0 \\ -2 \end{pmatrix}.$$

To emphasize, we have produced two different numerical coordinate vectors that represent the same abstract vector $p(x)$. The difference comes from the choice of basis.

Example 3.2.5. For a second simple but possibly bizarre example, consider the vector space $\mathbb{R}$.

Let $2 \in \mathbb{R}$. Let $\mathcal{B} = \{8\}$. Then since $2 = \frac{1}{4} \cdot 8$, it follows that

$$[2]_{\mathcal{B}} = \frac{1}{4}.$$

We can go even further and find numerical representations (i.e., matrices) of linear transformations. Before doing this, we will prove a result that, in words, says that the action of a linear transformation on a vector space is *entirely determined by what it does to a basis*.

Proposition 3.2.6. *Let $T : V \rightarrow W$ be linear and suppose that V is finite dimensional with a basis $\{u_1, \dots, u_n\}$. Suppose that $S : V \rightarrow W$ is another linear transformation such that $S(u_j) = T(u_j)$ for all $1 \leq j \leq n$. Then $S = T$.*

Proof. Let $v \in V$. Because $\{u_1, \dots, u_n\}$ is a basis for V , we can write $v = a_1u_1 + \dots + a_nu_n$ for some $a_1, \dots, a_n \in \mathbb{F}$. Then

$$\begin{aligned} S(v) &= S(a_1u_1 + \dots + a_nu_n) \\ &= a_1S(u_1) + \dots + a_nS(u_n) \\ &= a_1T(u_1) + \dots + a_nT(u_n) \\ &= T(a_1u_1 + \dots + a_nu_n) \\ &= T(v). \end{aligned}$$

Since $S(v) = T(v)$ for all $v \in V$, it follows that $S = T$. □

Next, let V, W be finite dimensional vector spaces. Let $\mathcal{B} = \{u_1, \dots, u_n\}$ be a basis for V , and let $\mathcal{C} = \{w_1, \dots, w_m\}$ be a basis for W . For each $1 \leq j \leq n$, we can consider the vector $T(u_j) \in W$. Since $\mathcal{C}$ is a basis for W , we have a unique decomposition

$$T(u_j) = a_{1j}w_1 + \dots + a_{mj}w_m$$

for some scalars $a_{1j}, \dots, a_{mj}$. Here we have kept track of the j index in this list of coefficients, because the scalars will depend on which vector u_j we feed to T . This tells us that the coordinate vector of $T(u_j)$ with respect to $\mathcal{C}$ is

$$[T(u_j)]_{\mathcal{C}} = \begin{pmatrix} a_{1j} \\ \vdots \\ a_{mj} \end{pmatrix}.$$

If we collect all of these vectors into one matrix, we get a numerical representation of the abstract linear transformation T .

Definition 3.2.7. The **coordinate matrix of T with respect to $\mathcal{B}, \mathcal{C}$** , denoted $[T]_{\mathcal{B}}^{\mathcal{C}}$, is the $m \times n$ matrix defined by

$$([T]_{\mathcal{B}}^{\mathcal{C}})_{ij} := a_{ij}.$$

In other words, to describe the matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ by its columns, we have

$$[T]_{\mathcal{B}}^{\mathcal{C}} = \left(\begin{array}{c|c|c} | & & | \\ [T(u_1)]_{\mathcal{C}} & \cdots & [T(u_n)]_{\mathcal{C}} \\ | & & | \end{array} \right) = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix}.$$

In words, the j th column of $[T]_{\mathcal{B}}^{\mathcal{C}}$ is $[T(u_j)]_{\mathcal{C}}$.

Example 3.2.8. Let $T : P_3(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be the derivative operator, i.e., the map defined by $T(p)(x) = p'(x)$, and let $\mathcal{B} = \{1, x, x^2, x^3\}$ and $\mathcal{C} = \{1, x, x^2\}$ be (ordered) bases of $P_3(\mathbb{R})$ and $P_2(\mathbb{R})$, respectively. We will compute $[T]_{\mathcal{B}}^{\mathcal{C}}$.

First, note that $T(1) = 0 = 0 \cdot 1 + 0 \cdot x + 0 \cdot x^2$. So $[T(1)]_{\mathcal{C}} = (0, 0, 0)$. Similarly,

$$\begin{aligned} T(x) &= 1 = 1 \cdot 1 + 0 \cdot x + 0 \cdot x^2 \\ T(x^2) &= 2x = 0 \cdot 1 + 2 \cdot x + 0 \cdot x^2 \\ T(x^3) &= 3x^2 = 0 \cdot 1 + 0 \cdot x + 3 \cdot x^2 \end{aligned}$$

so that $[T(x)]_{\mathcal{C}} = (1, 0, 0)$, $[T(x^2)]_{\mathcal{C}} = (0, 2, 0)$, and $[T(x^3)]_{\mathcal{C}} = (0, 0, 3)$. Thus,

$$[T]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Example 3.2.9. Consider the identity map $I : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, that is, the linear transformation defined by $I(x, y) := (x, y)$ for all $(x, y) \in \mathbb{R}^2$. For those of you reading that thought *ah, so the identity matrix*, I urge you to banish that thought from your mind. The map $I : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ as defined above is an *abstract linear transformation*, a priori not associated to any matrix. While it is possible to associate the identity matrix to this transformation, there are multiple other matrices that can be associated as well. For example, let $\mathcal{B} = \{(1, 0), (0, 1)\}$ be the standard basis, and let $\mathcal{C} = \{(2, 1), (1, 2)\}$. Then

$$\begin{aligned} [I]_{\mathcal{B}}^{\mathcal{C}} &:= \begin{pmatrix} | & | \\ [I(1, 0)]_{\mathcal{C}} & [I(0, 1)]_{\mathcal{C}} \\ | & | \end{pmatrix} \\ &= \begin{pmatrix} | & | \\ [(1, 0)]_{\mathcal{C}} & [(0, 1)]_{\mathcal{C}} \\ | & | \end{pmatrix}. \end{aligned}$$

Next, note that $(1, 0) = \frac{2}{3}(2, 1) - \frac{1}{3}(1, 2)$ and $(0, 1) = -\frac{1}{3}(2, 1) + \frac{2}{3}(1, 2)$. Thus,

$$[I]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ -\frac{1}{3} & \frac{2}{3} \end{pmatrix}.$$

To emphasize, this matrix above — definitively not the identity matrix — is a numerical representation of the identity transformation. In linear algebra at this level, it is important to

not immediately and intrinsically associate a matrix with an abstract linear transformation $\mathbb{R}^n \rightarrow \mathbb{R}^m$. This association comes from a choice of coordinates, and this will be explored more in the rest of this section.

Before we continue exploring coordinate representation of linear transformations, we have the following important fact.

Proposition 3.2.10. *Let V, W be vector spaces, and let $T, S : V \rightarrow W$ be linear. Let $\lambda \in F$. Then $T + S$ and λT are both linear as well.*

Proof. We prove the addition statement, and leave scalar multiplication as an exercise. Let $v, v' \in V$. Then

$$\begin{aligned} (T + S)(v + v') &= T(v + v') + S(v + v') \\ &= T(v) + T(v') + S(v) + S(v') \\ &= [T(v) + S(v)] + [T(v') + S(v')] \\ &= (T + S)(v) + (T + S)(v'). \end{aligned}$$

Thus, $T + S$ is additive. The scalar multiplication property is similar, and the argument for λT is also similar. $\square$

Corollary 3.2.11. *Let $\mathcal{L}(V, W) := \{ T : V \rightarrow W \mid T \text{ is linear} \}$. Then $\mathcal{L}(V, W)$ is a vector space over $\mathbb{F}$.*

Proof. Exercise. $\square$

I'll emphasize in words the point of the above corollary. We have seen in the past that spaces of functions can form vector spaces, so that a “vector” in such a vector space is a function. Linear transformations are functions, and so we can consider the space of all linear transformations between two vector spaces, denoted $\mathcal{L}(V, W)$. The above results show that, indeed, we can view $\mathcal{L}(V, W)$ as a vector space. The process of taking a coordinate representation of a transformation respects the linear structure of $\mathcal{L}(V, W)$, in the following sense.

Proposition 3.2.12. *Let V, W be finite dimensional vector spaces with (ordered) bases $\mathcal{B}, \mathcal{C}$, respectively. Let $T, S \in \mathcal{L}(V, W)$. Then $[T + S]_{\mathcal{B}}^{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} + [S]_{\mathcal{B}}^{\mathcal{C}}$ and $[\lambda T]_{\mathcal{B}}^{\mathcal{C}} = \lambda [T]_{\mathcal{B}}^{\mathcal{C}}$ for all $\lambda \in \mathbb{F}$.*

Proof. We show that addition statement, and leave the multiplication statement as an exercise. Let $\mathcal{B} = \{u_1, \dots, u_n\}$ and $\mathcal{C} = \{w_1, \dots, w_m\}$. Then

$$T(u_j) = a_{1j}w_1 + \dots + a_{mj}w_m \quad \text{and} \quad S(u_j) = b_{1j}w_1 + \dots + b_{mj}w_m$$

for some scalars $a_{ij}, b_{ij} \in \mathbb{F}$. Then

$$(T + S)(u_j) = (a_{1j} + b_{1j})w_1 + \dots + (a_{mj} + b_{mj})w_m.$$

Thus,

$$\begin{aligned} ([T + S]_{\mathcal{B}}^{\mathcal{C}})_{ij} &= a_{ij} + b_{ij} \\ &= ([T]_{\mathcal{B}}^{\mathcal{C}})_{ij} + ([S]_{\mathcal{B}}^{\mathcal{C}})_{ij}. \end{aligned}$$

This proves that $[T + S]_{\mathcal{B}}^{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} + [S]_{\mathcal{B}}^{\mathcal{C}}$. The other statement is similar, and is left as an exercise. $\square$

This proposition can be phrased in a more abstract way as follows:

Corollary 3.2.13. *Let V, W be finite dimensional vector spaces. Fix ordered bases $\mathcal{B}$ and $\mathcal{C}$ of V and W , respectively. Define $\phi : \mathcal{L}(V, W) \rightarrow M_{m \times n}(\mathbb{F})$ by $\phi(T) = [T]_{\mathcal{B}}^{\mathcal{C}}$. Then ϕ is a linear transformation.*

Soon we will show that ϕ is in fact something called an *isomorphism*, which means that $\mathcal{L}(V, W)$ and $M_{m \times n}(\mathbb{F})$ are essentially the “same” as vector spaces!

Exercises

1. Let V, W be finite dimensional vector spaces such that $\dim V = \dim W$. Let $T : V \rightarrow W$ be linear. Prove that there are (ordered) bases $\mathcal{B}$ and $\mathcal{C}$ for V and W respectively, such that $[T]_{\mathcal{B}}^{\mathcal{C}}$ is a diagonal matrix.
2. Let $\mathcal{B}_1 = \{1, x, x^2\}$ and $\mathcal{B}_2 = \{1, 1 + x, 1 + x + x^2\}$ be two (ordered) bases of $P_2(\mathbb{R})$ (you may assume that they are bases, you don't have to prove this.) Let

$$\mathcal{C} = \left\{ \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \right\}$$

be the standard (ordered) basis of $M_2(\mathbb{R})$.

- (a) Define $T : P_2(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ by $T(p)(x) := p'(x)$. Compute $[T]_{\mathcal{B}_1}^{\mathcal{B}_2}$. (You may assume T is linear.)
- (b) Define $S : P_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ by

$$S(p) = \begin{pmatrix} p'(0) & 2p(1) \\ 0 & p''(2) \end{pmatrix}$$

Compute $[S]_{\mathcal{B}_1}^{\mathcal{C}}$. (You may assume S is linear.)

3. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be defined by $T(x, y) = (2x + y, x - y, 4y)$. Let $\mathcal{B}_2$ and $\mathcal{B}_3$ denote the standard (ordered) bases of $\mathbb{R}^2$ and $\mathbb{R}^3$, respectively. Compute $[T]_{\mathcal{B}_2}^{\mathcal{B}_3}$.
4. True or false: If $T, S \in \mathcal{L}(V)$ are operators such that $T \neq S$, then T and S are linearly independent.
5. True or false: If $T \in \mathcal{L}(V)$ is a linear operator such that, for bases $\mathcal{B}_1, \mathcal{B}_2$, $[T]_{\mathcal{B}_1}^{\mathcal{B}_2} = I_n$ is the $n \times n$ identity matrix, then $T = I_V$ is the identity transformation on V .
6. Let V, W be vector spaces. Let $U \subset V$ be a subset (not necessarily a subspace). Define the following subset of $\mathcal{L}(V, W)$:

$$U^0 := \{ T \in \mathcal{L}(V, W) \mid T(u) = 0 \text{ for all } u \in U \}.$$

- (a) Prove that U^0 is a subspace of $\mathcal{L}(V, W)$.

- (b) Prove that if $U_1 \subset U_2$, then $U_1^0 \subset U_2^0$.
- (c) Prove that if U_1, U_2 are subspaces of V , then $(U_1 + U_2)^0 = U_1^0 \cap U_2^0$.

3.3 Composition of linear maps

We saw in the previous section that two linear maps from the same domain can be added together to produce another linear map. Although multiplication doesn't make sense in a vector space, there is a notion of "multiplication" at the level of linear maps. More precisely, one can *compose* two linear maps when appropriate, by doing one after another. This is the subject of the present section.

Definition 3.3.1. Let $T : V \rightarrow W$ and $S : W \rightarrow U$ be functions. Then the **composition of S and T** is the function $S \circ T : V \rightarrow U$ defined by $(S \circ T)(v) = S(T(v))$ for all $v \in V$.

At this level of math, it is helpful to visualize composition by the following diagram:

$$V \xrightarrow{T} W \xrightarrow{S} U.$$

Notice also that function composition notation can be confusing: $S \circ T$ means "do T first, then do S ."

Proposition 3.3.2. Let $T : V \rightarrow W$ and $S : W \rightarrow U$ be linear. Then $S \circ T : V \rightarrow U$ is linear.

Proof. We verify additivity. The scalar multiplication property is similar, and is left as an exercise. Let $v, v' \in V$. Then

$$\begin{aligned} (S \circ T)(v + v') &= S(T(v + v')) \\ &= S(T(v) + T(v')) \\ &= S(T(v)) + S(T(v')) \\ &= (S \circ T)(v) + (S \circ T)(v'). \end{aligned}$$

Thus, $S \circ T$ is additive. □

Proposition 3.3.3. Let T, S_1, S_2 be linear transformations between (varying) domains such that each of the compositions below makes sense. Then

- (i) $T \circ (S_1 + S_2) = T \circ S_1 + T \circ S_2$ and $(S_1 + S_2) \circ T = S_1 \circ T + S_2 \circ T$.
- (ii) $(T \circ S_1) \circ S_2 = T \circ (S_1 \circ S_2)$.
- (iii) $\lambda(S_1 \circ S_2) = (\lambda S_1) \circ S_2 = S_1 \circ (\lambda S_2)$.

Proof. Exercise. □

Remark 3.3.4. Note that composition is not commutative. If $S \circ T$ is defined, the other composition $T \circ S$ may not even be well-defined, let alone equal to $S \circ T$! For example, if we were given the set up

$$\mathbb{R}^3 \xrightarrow{T} \mathbb{R}^2 \xrightarrow{S} \mathbb{R}$$

then the composition $S \circ T$ makes sense, and is a function $S \circ T : \mathbb{R}^3 \rightarrow \mathbb{R}$. However, $T \circ S$ does not make sense. If you feed something in $\mathbb{R}^2$ to S , the output lives in $\mathbb{R}$, and this cannot be fed to T .

An important question to ask is how coordinate representations interact with composition. In particular, if we have two linear maps T and S between finite dimensional vector spaces, can we determine a relationship between the coordinate matrices of T , S , and $S \circ T$? Indeed, we have the following result, which is the underlying motivation for the familiar and initially-odd definition of matrix multiplication.

Theorem 3.3.5. *Let V, W, U be finite dimensional vector spaces with (ordered) bases $\mathcal{B}, \mathcal{C}, \mathcal{D}$, respectively. Let $T \in \mathcal{L}(V, W)$ and $S \in \mathcal{L}(W, U)$. Then*

$$[S \circ T]_{\mathcal{B}}^{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}} [T]_{\mathcal{B}}^{\mathcal{C}}.$$

Remark 3.3.6. In words, this theorem says that composition of linear maps corresponds to multiplication of the corresponding coordinate matrices.

Proof. Recall that the (i, j) -th entry of a matrix product AB is given

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}.$$

Enumerate the elements of the bases as follows:

$$\begin{aligned}\mathcal{B} &= \{v_1, \dots, v_n\} \\ \mathcal{C} &= \{w_1, \dots, w_m\} \\ \mathcal{D} &= \{u_1, \dots, u_\ell\}.\end{aligned}$$

Let $a_{ij} := ([S]_{\mathcal{C}}^{\mathcal{D}})_{ij}$ and let $b_{ij} := ([T]_{\mathcal{B}}^{\mathcal{C}})_{ij}$. Note that

$$\begin{aligned}(S \circ T)(v_j) &= S(T(v_j)) \\ &= S\left(\sum_{k=1}^m b_{kj} w_k\right) \\ &= \sum_{k=1}^m b_{kj} S(w_k) \\ &= \sum_{k=1}^m b_{kj} \sum_{i=1}^{\ell} a_{ik} u_i \\ &= \sum_{k=1}^m \sum_{i=1}^{\ell} b_{kj} a_{ik} u_i.\end{aligned}$$

By commutativity and associativity of addition, we can reverse the order of the summations

to get

$$(S \circ T)(v_j) = \sum_{i=1}^{\ell} \left(\sum_{k=1}^m a_{ik} b_{kj} \right) u_j.$$

Thus, the (i, j) -th entry of $[S \circ T]_{\mathcal{B}}^{\mathcal{D}}$ is $\sum_{k=1}^m a_{ik} b_{kj}$. By definition of matrix multiplication, this is precisely $([S]_{\mathcal{C}}^{\mathcal{D}} [T]_{\mathcal{B}}^{\mathcal{C}})_{ij}$. Thus,

$$[S \circ T]_{\mathcal{B}}^{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}} [T]_{\mathcal{B}}^{\mathcal{C}}.$$

□

Example 3.3.7. Let $T : P_3(\mathbb{R}) \rightarrow P_2(\mathbb{R})$ be the map given by $T(p)(x) = p'(x)$. Let $S : P_2(\mathbb{R}) \rightarrow P_3(\mathbb{R})$ be the map given by $S(p)(x) = \int_0^x p(t) dt$. Let $\mathcal{B} = \{1, x, x^2, x^3\}$ and $\mathcal{C} = \{1, x, x^2\}$. Recall from Example 3.2.8 that

$$[T]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

Similarly, we can compute $[S]_{\mathcal{C}}^{\mathcal{B}}$. Note that

$$\begin{aligned} S(1) &= \int_0^x 1 dt = x \\ S(x) &= \int_0^x t dt = \frac{1}{2}x^2 \\ S(x^2) &= \int_0^x t^2 dt = \frac{1}{3}x^3. \end{aligned}$$

Thus,

$$[S]_{\mathcal{C}}^{\mathcal{B}} = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix}.$$

Recall that the fundamental theorem of calculus says that

$$\frac{d}{dx} \int_0^x p(t) dt = p(x).$$

Phrased differently, this says that $(T \circ S)(p)(x) = p(x)$, and so $T \circ S = I_{P_2(\mathbb{R})}$. Indeed, we can verify that

$$\begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Since $[T \circ S]_{\mathcal{C}}^{\mathcal{C}} = [I_{P_2(\mathbb{R})}]_{\mathcal{C}}^{\mathcal{C}} = I_3$, this verifies that $[T \circ S]_{\mathcal{C}}^{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}} [S]_{\mathcal{C}}^{\mathcal{B}}$. Note that, in contrast, the composition $S \circ T$ — differentiating first, and then integrating — corresponds via coordinate

matrices to the product

$$\begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Here, we are detecting the fact that $T \circ S$ is *not* the identity map on $P_3(\mathbb{R})$. Indeed, when you differentiate first you kill off all constant information, which is impossible to recover using definite integration. Specifically, this is why the top left corner of the product above is 0. Anyway, this example gives a purely linear algebraic discussion of the fundamental theorem of calculus!

Finally, we state a result which captures the utility of passing to a coordinate matrix. In words, it says the following: If you want to compute the coordinates of $T(v)$, then multiply the coordinate matrix of T by the coordinate vector of v .

Proposition 3.3.8. *Let V, W be finite dimensional vector spaces with (ordered) bases $\mathcal{B}, \mathcal{D}$, respectively. Let $T : V \rightarrow W$ be linear, and let $v \in V$. Then*

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[v]_{\mathcal{B}}.$$

Proof. Enumerate the elements of the bases as follows:

$$\begin{aligned} \mathcal{B} &= \{v_1, \dots, v_n\}, \\ \mathcal{C} &= \{w_1, \dots, w_m\}. \end{aligned}$$

Write $v = a_1v_1 + \dots + a_nv_n$. Let $b_{ij} := ([T]_{\mathcal{B}}^{\mathcal{C}})_{ij}$. Then

$$\begin{aligned} T(v) &= T\left(\sum_{j=1}^n a_j v_j\right) \\ &= \sum_{j=1}^n a_j T(v_j) \\ &= \sum_{j=1}^n a_j \sum_{i=1}^m b_{ij} w_i \\ &= \sum_{i=1}^m \left(\sum_{j=1}^n b_{ij} a_j\right) w_i. \end{aligned}$$

Thus, the i th entry of the coordinate vector $[T(v)]_{\mathcal{C}}$ is $\sum_{j=1}^n b_{ij} a_j$. But this is precisely the i th entry of the matrix vector product $[T]_{\mathcal{B}}^{\mathcal{C}}[v]_{\mathcal{B}}$. Thus,

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[v]_{\mathcal{B}}.$$

□

Exercises

1. Let $A : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be the linear map defined by $A(p(x)) = xp(x)$. Define a map $\phi_A : \mathcal{L}(P(\mathbb{R})) \rightarrow \mathcal{L}(P(\mathbb{R}))$ by $\phi_A(T) := T \circ A$.

- (a) Prove that ϕ_A is linear.
- (b) Prove that ϕ_A is not injective.

[Hint: think about polynomials in terms of their coefficients.]

2. This problem is mostly an exercise in processing and working with multiple layers of abstraction. This is a difficult and important skill that arises all over the place in math! If you get stumped, focus on working with the relevant definitions as directly as possible.
 - (a) Let V, W be vector spaces. Let $T, S \in \mathcal{L}(V, W)$ such that $T \neq 0$, $S \neq 0$, and $\text{im } T \cap \text{im } S = \{0\}$. Prove that $\{T, S\}$ is a linearly independent subset of $\mathcal{L}(V, W)$.
 - (b) Let U, V, W be vector spaces. For any $A \in \mathcal{L}(W, U)$, define a function $F_A : \mathcal{L}(V, W) \rightarrow \mathcal{L}(V, U)$ as follows: for all $T \in \mathcal{L}(V, W)$,

$$F_A(T) := A \circ T.$$

- (i) Prove that $F_A : \mathcal{L}(V, W) \rightarrow \mathcal{L}(V, U)$ is linear.
 - (ii) Suppose that A is injective. Prove that F_A is injective.
3. Let F_A be the same linear map defined in the previous exercise. Define a map

$$F : \mathcal{L}(W, U) \rightarrow \mathcal{L}(\mathcal{L}(V, W), \mathcal{L}(V, U))$$

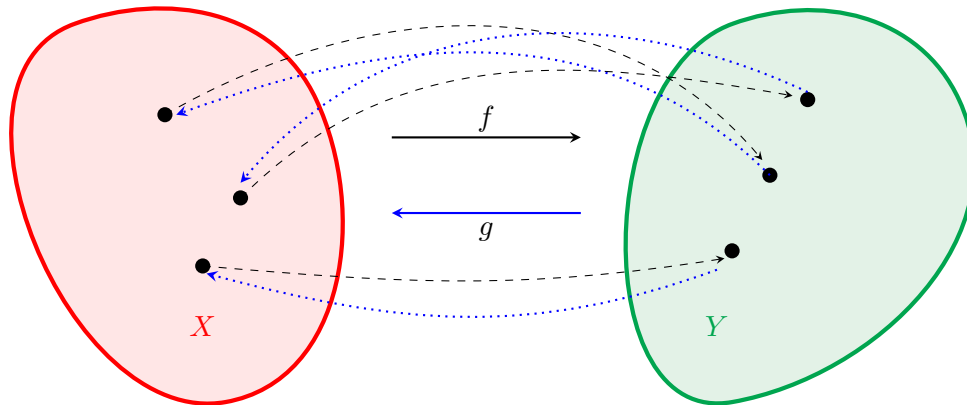
by $F(A) := F_A$. Prove that F is linear. Think about whether F could be surjective or injective.

3.4 Invertibility

The notion of *invertibility* is central in linear algebra, and more generally in all of math. There are many ways to motivate invertibility, but perhaps the most natural is to consider the setting of solving equations. Suppose we have a function $f : X \rightarrow Y$ and a fixed element $y_0 \in Y$, and we are interested in solving the equation $f(x) = y_0$. That is, finding all elements $x \in X$ such that $f(x) = y_0$. We have the following heuristics:

- ◇ *Existence* of solutions corresponds to *surjectivity* of f . That is, if f is surjective, then there will always be at least one $x \in X$ such that $f(x) = y_0$.
- ◇ *Uniqueness* of solutions corresponds to *injectivity* of f . That is, if a solution to $f(x) = y_0$ exists (and it may not!) then it must be unique.
- ◇ *Existence and uniqueness together* corresponds to *invertibility* of f . That is, if f is invertible, not only can you always find a solution to $f(x) = y_0$, but there will always be only one.

Next, before defining invertibility formally, I will show you a picture of what an invertible function looks like.



In words, a function is invertible if it can be “undone”, i.e., if you can completely reverse the effect of applying f . In the picture above, the function f , depicted by the left-to-right dashed black arrows, can be “undone” by defining a function $g : Y \rightarrow X$, given by the right-to-left dotted blue arrows, which just sends everything back the way it came. Crucially, the existence of such a function g requires f to be both surjective and injective. With that being said, existence of such a function g is precisely the formal definition of invertibility.

Definition 3.4.1. Let $f : X \rightarrow Y$. We say that f is **invertible** if there is another function $g : Y \rightarrow X$ such that $g \circ f = I_X$ and $f \circ g = I_Y$.

Example 3.4.2. First, we consider an example out of the realm of linear algebra. Let $f : [0, \infty) \rightarrow [0, \infty)$ be given by $f(x) = x^2$. Let $g : [0, \infty) \rightarrow [0, \infty)$ be defined by $g(x) = \sqrt{x}$. Then

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= (\sqrt{x})^2 \\ &= x. \end{aligned}$$

Since $(f \circ g)(x) = x$ for all $x \in [0, \infty)$, $f \circ g = I_{[0, \infty)}$. Similarly, one can check that $(g \circ f)(x) = x$ for all $x \in [0, \infty)$. Thus, f is invertible.

Although invertibility is a completely general notion, from now on we will specialize to considering invertibility of linear functions between vector spaces. Thus, given a linear map $T : V \rightarrow W$, we are interested in the existence of a function (which, a priori, may or may not be linear!) $S : W \rightarrow V$ such that $S \circ T = I_V$ and $T \circ S = I_W$.

Example 3.4.3. Let V be a vector space over $\mathbb{F}$, and let $\lambda \neq 0 \in \mathbb{F}$. Define a linear map $T : V \rightarrow V$ by $T(v) := \lambda v$. We claim that T is invertible. Indeed, define $S : V \rightarrow V$ by $S(v) := \lambda^{-1}v$. Then

$$(T \circ S)(v) = T(S(v)) = T(\lambda^{-1}v) = \lambda\lambda^{-1}v = v$$

and

$$(S \circ T)(v) = S(T(v)) = S(\lambda v) = \lambda^{-1}\lambda v = v.$$

Thus, $T \circ S = I_V$ and $S \circ T = I_W$, and so T is in fact invertible.

Proposition 3.4.4. *Let $T : V \rightarrow W$ be linear. Suppose that T is invertible. Then the map $S : W \rightarrow V$ satisfying $S \circ T = I_V$ and $T \circ S = I_W$ is unique and linear.*

Proof. First, we prove uniqueness. Suppose that $S' : W \rightarrow V$ also satisfies $S' \circ T = I_V$ and $T \circ S' = I_W$. Then note that

$$\begin{aligned} S' &= S' \circ I_W \\ &= S' \circ (T \circ S) \\ &= (S' \circ T) \circ S \\ &= I_V \circ S \\ &= S. \end{aligned}$$

Thus, $S' = S$ and so S is unique.

Next, we prove that S is linear. As usual, we show that S is additive and leave the scalar multiplication property as an exercise. Let $w, w' \in W$. Then note that

$$\begin{aligned} S(w + w') &= S(I_W(w) + I_W(w')) \\ &= S((T \circ S)(w) + (T \circ S)(w')) \\ &= S(T(S(w)) + T(S(w'))) \\ &= S(T(S(w) + S(w'))) \\ &= (S \circ T)(S(w) + S(w')) \\ &= I_V(S(w) + S(w')) \\ &= S(w) + S(w'). \end{aligned}$$

Thus, S is additive, as desired. □

Now that we have proven that the function $S : W \rightarrow V$ that undoes an invertible map $T : V \rightarrow W$ is unique, we are allowed to give it a special notation and name.

Definition 3.4.5. Let $T : V \rightarrow W$ be an invertible linear map. The **inverse** of T , denoted T^{-1} , is the unique linear map $T^{-1} : W \rightarrow V$ such that $T^{-1} \circ T = I_V$ and $T \circ T^{-1} = I_W$.

Next, we formalize the heuristic stated at the beginning of this section, namely, that invertibility is equivalent to being both injective and surjective.

Proposition 3.4.6. *Let $T : V \rightarrow W$. Then T is invertible if and only if T is both injective and surjective.*

Proof. First, suppose that T is invertible. Suppose that $T(v) = 0$ for some $v \in V$. Applying T^{-1} to both sides gives $T^{-1}(T(v)) = T^{-1}(0)$. Since $T^{-1} \circ T = I_V$, the left side of this equality is simply $T^{-1}(T(v)) = I_V(v) = v$. Since T^{-1} is linear, the right side is $T^{-1}(0) = 0$. It follows that $v = 0$. Thus, $\ker T = \{0\}$ and so T is injective. Next, let $w \in W$. Then note that $T(T^{-1}(w)) = w$, and so T is surjective. Therefore, T is both injective and surjective.

The other direction is left as an exercise. □

Proposition 3.4.7. Suppose that V, W are finite dimensional vector spaces. If $T : V \rightarrow W$ is an invertible linear transformation, then $\dim V = \dim W$.

Proof. Since T is invertible, it is injective. Thus, $\ker T = \{0\}$ and so $\dim \ker T = 0$. Since T is also surjective, $\text{im } T = W$ and so $\dim \text{im } T = \dim W$. By the rank nullity theorem, it then follows that

$$0 + \dim W = \dim V$$

and so $\dim V = \dim W$. □

Example 3.4.8. Let $P(\mathbb{R})$ denote the vector space of polynomials of any degree with coefficients in $\mathbb{R}$. Consider the derivative map $\frac{d}{dx} : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ defined by $\frac{d}{dx}(p(x)) := p'(x)$. Note that $\frac{d}{dx}(1) = 0$, and so $\ker \frac{d}{dx} \neq \{0\}$. Thus, $\frac{d}{dx}$ is not injective and hence is not invertible.

Example 3.4.9. Let $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $T(x, y) := (-y, x)$. Geometrically, T is counter clockwise rotation by 90 degrees. We claim that T is injective. Indeed, suppose that $T(x, y) = (0, 0)$. Then $(-y, x) = (0, 0)$ and so $x = y = 0$. This implies that $\ker T = \{(0, 0)\}$ and hence T is injective. Since the domain and codomain of T are both 2-dimensional, it follows that T is also surjective and thus invertible. Moreover, one can verify that $T^{-1} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is given by $T^{-1}(x, y) = (y, -x)$. Indeed, note that

$$(T \circ T^{-1})(x, y) = T(y, -x) = (x, y)$$

and

$$(T^{-1} \circ T)(x, y) = T^{-1}(-y, x) = (x, y).$$

Next, we formalize the idea that we have referenced above where we can identify two vector spaces as “essentially the same, up to a change in clothes”.

Definition 3.4.10. Let V, W be vector spaces. We say that V and W are **isomorphic** if there is an invertible linear map $T : V \rightarrow W$. In this case, we say that T is an **isomorphism** from V to W . We also sometimes write $V \cong W$.

Remark 3.4.11. This language can be a little confusing the first time you see it, but for our purposes you can take “isomorphism” and “invertible linear map” to be synonyms. We use this word simply to distinguish functions $f : V \rightarrow W$ between vector spaces that are invertible, but not linear.

Example 3.4.12. Consider the vector spaces

$$\mathbb{R}^3 = \left\{ \begin{pmatrix} a \\ b \\ c \end{pmatrix} \mid a, b, c \in \mathbb{R} \right\}$$

and

$$P_2(\mathbb{R}) = \{ a + bx + cx^2 \mid a, b, c \in \mathbb{R} \}.$$

We claim that $\mathbb{R}^3 \cong P_2(\mathbb{R})$, that is, $\mathbb{R}^3$ and $P_2(\mathbb{R})$ are isomorphic as vector spaces. Indeed, to prove this it suffices to provide an invertible linear transformation $T : \mathbb{R}^3 \rightarrow P_2(\mathbb{R})$. Define such a map as follows:

$$T \begin{pmatrix} a \\ b \\ c \end{pmatrix} := a + bx + cx^2.$$

Note that T is a linear map. Recall that $\dim \mathbb{R}^3 = 3 = \dim P_2(\mathbb{R})$, and so to prove that T is invertible it suffices to prove that T is injective. Suppose that $T(a, b, c) = 0$. Then since 0 is the 0-polynomial, it must be the case that $a = b = c = 0$. Thus, $\ker T = \{(0, 0, 0)\}$ and so T is injective. It follows that $T : \mathbb{R}^3 \rightarrow P_2(\mathbb{R})$ is an isomorphism, and so $\mathbb{R}^3 \cong P_2(\mathbb{R})$.

Note that $\mathbb{R}^3$ and $P_2(\mathbb{R})$ are isomorphic in many different ways. For example, another isomorphism is given by $S : \mathbb{R}^3 \rightarrow P_2(\mathbb{R})$ defined by

$$S \begin{pmatrix} a \\ b \\ c \end{pmatrix} := c + ax + (a + b + c)x^2.$$

Next is one of the main punchlines of abstract linear algebra, namely, that finite dimensional vector spaces are characterized up to isomorphism by their dimension.

Theorem 3.4.13. *Let V, W be finite dimensional vector spaces. Then V and W are isomorphic if and only if $\dim V = \dim W$.*

Proof. First, suppose that $V \cong W$. By definition, there exists an invertible linear map $T : V \rightarrow W$. By Proposition 3.4.7, $\dim V = \dim W$.

Next, suppose that $\dim V = \dim W$. Let

$$\begin{aligned} \mathcal{B} &:= \{v_1, \dots, v_n\} \\ \mathcal{C} &:= \{w_1, \dots, w_n\} \end{aligned}$$

be bases for V and W , respectively. Define $T : V \rightarrow W$ as follows. For $a_1v_1 + \dots + a_nv_n \in V$, define

$$T(a_1v_1 + \dots + a_nv_n) = a_1w_1 + \dots + a_nw_n.$$

In other words, we are defining T by $T(v_j) := w_j$ and “extending by linearity.” In particular, recall by Proposition 3.2.6 that the behavior of a linear map is completely determined by its behavior on a basis.

First, note that T is clearly linear. Next, we claim that T is injective. Suppose that $T(a_1v_1 + \dots + a_nv_n) = 0$. Then $a_1w_1 + \dots + a_nw_n = 0$. Since $\mathcal{C}$ is a basis, it is linearly independent, and so $a_1 = \dots = a_n = 0$. Thus, $a_1v_1 + \dots + a_nv_n = 0$. This shows that $\ker T = \{0\}$, and so T is injective. Since $\dim V = \dim W$, T is also surjective by Rank-Nullity. Thus, T is invertible, and therefore is an isomorphism from V to W . $\square$

Corollary 3.4.14. *Let V be an n -dimensional vector space over $\mathbb{F}$. Then $V \cong \mathbb{F}^n$.*

3.4.1 Invertible matrices and coordinate representations

Closely related to the notion of invertibility for abstract linear transformations is the notion of invertibility for matrices.

Definition 3.4.15. A matrix $A \in M_{m \times n}(\mathbb{F})$ is **invertible** if there is another matrix $B \in M_{n \times m}(\mathbb{F})$ such that $AB = I_m$ and $BA = I_n$.

The following proposition summarizes properties of invertible matrices in analogy to the theory developed above. The proof is left as an exercise.

Proposition 3.4.16. Let $A \in M_{m \times n}(\mathbb{F})$ be an invertible matrix. Then $m = n$ and the matrix $B \in M_{n \times m}(\mathbb{F}) = M_{n \times n}$ satisfying $AB = I_n$ and $BA = I_n$ is unique. We then use the notation $A^{-1} := B$ and call A^{-1} the **inverse** of A .

Example 3.4.17. Let $A \in M_2(\mathbb{R})$ be given by

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix}.$$

Note that

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

and

$$\begin{pmatrix} 1 & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Thus, A is invertible and $A^{-1} = \begin{pmatrix} 1 & -\frac{2}{3} \\ 0 & \frac{1}{3} \end{pmatrix}$.

Remark 3.4.18. You are likely familiar with other techniques for investigating invertibility of matrices (determinants, formulas, row reduction). As usual, this will not be the focus of this chapter. In Chapter 6 we will formally develop the theory of determinants and return to the notion of invertibility for matrices.

The relation between invertibility of abstract linear transformations and matrices is, unsurprisingly, given by the transformation-matrix correspondence coming from coordinate representations. Formally, we have the following proposition.

Proposition 3.4.19. Let V, W be finite dimensional vector spaces with (ordered) bases $\mathcal{B}, \mathcal{C}$, respectively. A linear map $T : V \rightarrow W$ is invertible if and only if the coordinate matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$ is invertible. Moreover,

$$[T^{-1}]_{\mathcal{C}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{C}})^{-1}.$$

Proof. First, suppose that T is invertible. Then, by definition, $T^{-1} \circ T = I_V$ and $T \circ T^{-1} = I_W$. Taking coordinate matrices gives

$$[T^{-1} \circ T]_{\mathcal{B}}^{\mathcal{B}} = [I_V]_{\mathcal{B}}^{\mathcal{B}} \quad \text{and} \quad [T \circ T^{-1}]_{\mathcal{C}}^{\mathcal{C}} = [I_W]_{\mathcal{C}}^{\mathcal{C}}.$$

By Theorem 3.3.5, and the fact that $[I_V]_{\mathcal{B}}^{\mathcal{B}} = [I_W]_{\mathcal{C}}^{\mathcal{C}} = I_n$, where $n = \dim V = \dim W$, we then have

$$[T^{-1}]_{\mathcal{C}}^{\mathcal{B}} [T]_{\mathcal{B}}^{\mathcal{C}} = I_n \quad \text{and} \quad [T]_{\mathcal{B}}^{\mathcal{C}} [T^{-1}]_{\mathcal{C}}^{\mathcal{B}} = I_n.$$

Thus, $[T]_{\mathcal{B}}^{\mathcal{C}}$ is invertible and $([T]_{\mathcal{B}}^{\mathcal{C}})^{-1} = [T^{-1}]_{\mathcal{B}}^{\mathcal{C}}$.

Next, suppose that $[T]_{\mathcal{B}}^{\mathcal{C}}$ is invertible. Let $b_{ij} := ([T]_{\mathcal{B}}^{\mathcal{C}})^{-1}_{ij}$. Let $\mathcal{B} = \{v_1, \dots, v_n\}$ and $\mathcal{C} = \{w_1, \dots, w_n\}$. Define $S : W \rightarrow V$ on $\mathcal{C}$ as

$$S(w_j) := b_{1j}v_1 + \dots + b_{nj}v_n$$

and extend the definition of S to W by linearity. Then, by definition, $[S]_{\mathcal{C}}^{\mathcal{B}} = ([T]_{\mathcal{B}}^{\mathcal{C}})^{-1}$. Then

$$[S \circ T]_{\mathcal{B}}^{\mathcal{B}} = [S]_{\mathcal{C}}^{\mathcal{B}}[T]_{\mathcal{B}}^{\mathcal{C}} = ([T]_{\mathcal{B}}^{\mathcal{C}})^{-1}[T]_{\mathcal{B}}^{\mathcal{C}} = I_n.$$

This means that $(S \circ T)(v_j) = v_j$ for all $1 \leq j \leq n$. By linearity, it follows that $S \circ T = I_V$. By a similar argument, one can show that $T \circ S = I_W$, and so T is invertible. $\square$

Now that we have the language of isomorphisms, we can rephrase the nature of the correspondence between transformations and matrices in a rigorous way. This is a perfect example of the power of abstraction and the reason why it is worth developing linear algebra abstractly.

Proposition 3.4.20. *Let V be an n -dimensional vector space with basis $\mathcal{B}$, and W an m -dimensional vector space with basis $\mathcal{C}$, both over $\mathbb{F}$. Then the linear map $\phi : \mathcal{L}(V, W) \rightarrow M_{m \times n}(\mathbb{F})$ from Corollary 3.2.13 given by $\phi(T) = [T]_{\mathcal{B}}^{\mathcal{C}}$ is an isomorphism.*

Proof. As we have already proven that ϕ is linear (Corollary 3.2.13) it suffices to prove that ϕ is invertible. To do this, we will show that ϕ is injective and surjective. First, let $\mathcal{B} = \{v_1, \dots, v_n\}$ and $\mathcal{C} = \{w_1, \dots, w_m\}$.

First we prove injectivity of ϕ . Suppose that $\phi(T) = 0_{m \times n}$. Let $v \in V$. Then

$$[T(v)]_{\mathcal{C}} = [T]_{\mathcal{B}}^{\mathcal{C}}[v]_{\mathcal{B}} = 0_{m \times n}[v]_{\mathcal{B}} = \begin{pmatrix} 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{F}^m.$$

This implies that $T(v) = 0w_1 + \dots + 0w_m = 0 \in W$. Thus, $T = 0 \in \mathcal{L}(V, W)$, and so $\ker \phi = \{0\}$. Thus, ϕ is injective.

Next, we show that ϕ is surjective. Let $A \in M_{m \times n}(\mathbb{F})$. Define $T : V \rightarrow W$ on $\mathcal{B}$ by

$$T(v_j) := a_{1j}w_1 + \dots + a_{mj}w_m$$

and extend the definition of T to V by linearity. By definition, $\phi(T) = A$. Thus, ϕ is surjective. This proves that ϕ is an isomorphism, and so $\mathcal{L}(V, W) \cong M_{m \times n}(\mathbb{F})$. $\square$

Since $\dim M_{m \times n}(\mathbb{F}) = mn$, we have the following immediate corollary.

Corollary 3.4.21. *Let V and W be $\mathbb{F}$ -vector spaces of dimensions n and m , respectively. Then*

$$\mathcal{L}(V, W) \cong M_{m \times n}(\mathbb{F}) \cong \mathbb{F}^{mn}$$

and in particular $\dim \mathcal{L}(V, W) = mn$.

Remark 3.4.22. All of this stuff about $\mathcal{L}(V, W)$ has been phrased in very abstract language. To emphasize the point in down-to-earth language, these results show that there is perfect

correspondence between abstract linear maps $V \rightarrow W$ and matrices. This correspondence is not unique, though, and depends on the choice of bases in the domain and codomain.

Exercises

1. Let $P(\mathbb{R})$ be the vector space of all polynomials of any degree. Define $T : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ by $T(p(x)) := \frac{d}{dx}(xp(x))$. Prove that T is invertible.
2. Let V, W be finite dimensional vector spaces, and let $T : V \rightarrow W$ be linear. Let $\mathcal{B}$ be a basis for V . Prove that T is an isomorphism if and only if the set

$$T(\mathcal{B}) := \{ T(u) \mid u \in \mathcal{B} \}$$

is a basis for W .

3. For each of the following linear maps, determine whether it is an isomorphism. Justify your answers with proof.

(a) $T : \mathbb{R}^3 \rightarrow P_2(\mathbb{R})$ defined by $T(a_1, a_2, a_3) = (a_1 - a_2) + a_2x + a_1x^2$.

(b) $T : P_3(\mathbb{F}) \rightarrow M_2(\mathbb{F})$ defined by

$$T(a_0 + a_1x + a_2x^2 + a_3x^3) = \begin{pmatrix} a_0 & a_1 \\ a_2 & a_3 \end{pmatrix}.$$

(c) $T : M_n(\mathbb{F}) \rightarrow M_n(\mathbb{F})$ defined by $T(A) = A^T$. Here, A^T is the *transpose* of A , i.e., the matrix defined by $(A^T)_{ij} := A_{ji}$.

4. Let V, W be vector spaces and let $T : V \rightarrow W$ be linear. Prove that T is surjective if and only if there exists a linear map $S : W \rightarrow V$ such that $T \circ S = I_W$.
5. Let V, W be vector spaces and let $T : V \rightarrow W$ be linear. Prove that T is injective if and only if there exists a linear map $S : W \rightarrow V$ such that $S \circ T = I_V$.
6. Let V, W, U be vector spaces, and let $T : V \rightarrow W$ and $S : W \rightarrow U$ be invertible. Prove that $S \circ T : V \rightarrow U$ is invertible.
7. Suppose that $T : V \rightarrow W$ and $S : W \rightarrow V$ are maps such that $T \circ S : W \rightarrow W$ is invertible. Is it true that T and S are both invertible?⁵
8. Suppose that $T : V \rightarrow V$ and $S : V \rightarrow V$ are maps such that $T \circ S : V \rightarrow V$ is invertible. Is it true that T and S are both invertible?⁶
9. Let V be finite dimensional. Suppose that $T : V \rightarrow V$ and $S : V \rightarrow V$ are maps such that $T \circ S : V \rightarrow V$ is invertible. Is it true that T and S are both invertible?⁷

⁵Hint: The answer is no. Can you find an example?

⁶Hint: The answer is still no. Can you find an example?

⁷Hint: Finally, the answer is yes! Try to prove it.

10. Let V be a vector space of dimension n , and let $v_0 \in V$ be a fixed nonzero vector. Let

$$W := \{v_0\}^0 = \{T \in \mathcal{L}(V, V) \mid T(v_0) = 0\} \subset \mathcal{L}(V, V)$$

be the subspace of $\mathcal{L}(V, V)$ consisting of transformations that send v_0 to 0. In this exercise, we will compute $\dim W$.

- (a) Let $\phi : \mathcal{L}(V, V) \rightarrow V$ be given by $\phi(T) := T(v_0)$. Prove that ϕ is linear and surjective.
- (b) Use the Rank-Nullity theorem to prove that $\dim W = n^2 - n$.
- (c) Consider a set of independent vectors $\{v_1, \dots, v_m\} \subset V$. Define

$$W_m := \{v_1, \dots, v_m\}^0 = \{T \in \mathcal{L}(V, V) \mid T(v_j) = 0 \text{ for all } 1 \leq j \leq m\} \subset \mathcal{L}(V, V)$$

as a generalization of W above. Adapt the above argument to prove that $\dim W_m = n^2 - mn$. You will have to change the codomain of ϕ !

11. Let $F(\mathbb{R}, \mathbb{R})$ be the real vector space of functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Given $g \in F(\mathbb{R}, \mathbb{R})$, define a map $T_g : F(\mathbb{R}, \mathbb{R}) \rightarrow F(\mathbb{R}, \mathbb{R})$ by $T_g(f)(x) := g(x)f(x)$.

- (a) Prove that T_g is linear for any $g \in F(\mathbb{R}, \mathbb{R})$.
- (b) Let $g \in F(\mathbb{R}, \mathbb{R})$ be defined as

$$g(x) = \begin{cases} 1 & x \geq 0 \\ 0 & x < 0 \end{cases}.$$

Prove that T_g is not injective.

- (c) Let $g(x) = x^2$. Prove that T_g is not surjective.
- (d) Let $g(x) = e^x$. Prove that T_g is an isomorphism.

12. Let V, W be vector spaces over the same field $\mathbb{F}$. The **product of V and W** is the set

$$V \times W := \{(v, w) \mid v \in V, w \in W\}.$$

- (a) With addition on $V \times W$ defined as $(v, w) + (v', w') := (v + v', w + w')$ and scalar multiplication defined as $\lambda(v, w) := (\lambda v, \lambda w)$, $V \times W$ is a vector space over $\mathbb{F}$.
- (b) Suppose that V, W are finite dimensional and that $\dim V = n$ and $\dim W = m$. Prove that

$$V \times W \cong \mathbb{F}^{n+m}.$$

13. This exercise requires knowledge of quotient vector spaces; see 2.6.1. In particular, this exercise will introduce you to an important theorem called the *first isomorphism theorem*. Let $T : V \rightarrow W$ be linear, and suppose that T is surjective. The first isomorphism theorem says that

$$V / \ker T \cong W.$$

We will prove this as follows. First, recall that elements of $V/\ker T$ are equivalence classes denoted by $[v]$, where v is a choice of representative for that equivalence class. Define a map $\phi : V/\ker T \rightarrow W$ by $\phi([v]) := T(v)$.

- (a) Show that ϕ is well-defined. That is, show that if $v, v' \in V$ such that $[v] = [v']$, then $\phi([v]) = \phi([v'])$.
- (b) Prove that T is linear.
- (c) Prove that T is injective.
- (d) Prove that T is surjective, and conclude the statement of the first isomorphism theorem. Note that we have not assumed finite dimensionality!

One can actually conclude the Rank-Nullity theorem as a consequence of the first isomorphism theorem.

- (e) Assume now that V and W are finite dimensional, and that $T : V \rightarrow W$ is a linear map (non necessarily surjective). By considering the restricted linear map $T : V \rightarrow \text{im } T$, use the first isomorphism theorem together with facts about the dimension of a quotient vector space to conclude that

$$\dim V - \dim \ker T = \dim \text{im } T.$$

3.5 Change of coordinates

Now that we have the theory of compositions and invertibility under our belt, we will return to the idea of coordinate representations of abstract linear transformations. In particular, this section will discuss the notion of *change of coordinates*. For the time being, we will focus on linear maps from a single vector space back to itself: $T : V \rightarrow V$. In this case we will call T a **linear operator** on V , we will write

$$\mathcal{L}(V) := \mathcal{L}(V, V)$$

to denote the space of all linear operators on V , and if $\mathcal{B}$ is a basis for a finite dimensional space V then we will write

$$[T]_{\mathcal{B}} := [T]_{\mathcal{B}}^{\mathcal{B}}$$

for the coordinate matrix with respect to $\mathcal{B}$ in both the domain and codomain.

To motivate change of coordinates, I will start off with a simple example. Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (2x + y, x + 2y)$. Currently, this is an abstract linear transformation. One of the things we like to do with abstract linear transformations is to obtain a numerical representative (a matrix) of that transformation by picking a basis; i.e., computing the coordinate matrix of T . This is nice, because matrices are easy to do computations with. But, as has been observed, *picking different bases will lead to different coordinate matrices*. To be concrete, let's compute a coordinate matrix for T using two different bases.

- ◇ The easiest basis to use is the standard basis $\mathcal{B} = \{(1, 0), (0, 1)\}$. To compute $[T]_{\mathcal{B}}$, note that

$$T(1, 0) = (2, 1) = 2(1, 0) + 1(0, 1) \quad \text{and so} \quad [T(1, 0)]_{\mathcal{B}} = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

Similarly,

$$T(0, 1) = (1, 2) = 1(1, 0) + 2(0, 1) \quad \text{and so} \quad [T(0, 1)]_{\mathcal{B}} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Therefore,

$$[T]_{\mathcal{B}} = \begin{pmatrix} | & | \\ [T(1, 0)]_{\mathcal{B}} & [T(0, 1)]_{\mathcal{B}} \\ | & | \end{pmatrix} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}.$$

- ◇ Next, let's use the basis $\mathcal{B}' = \{(1, 1), (1, -1)\}$. To compute $[T]_{\mathcal{B}'}$, note that

$$T(1, 1) = (3, 3) = 3(1, 1) + 0(1, -1) \quad \text{and so} \quad [T(1, 1)]_{\mathcal{B}'} = \begin{pmatrix} 3 \\ 0 \end{pmatrix}.$$

Similarly,

$$T(1, -1) = (1, -1) = 0(1, 1) + 1(1, -1) \quad \text{and so} \quad [T(1, -1)]_{\mathcal{B}'} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Therefore,

$$[T]_{\mathcal{B}'} = \begin{pmatrix} | & | \\ [T(1, 1)]_{\mathcal{B}'} & [T(1, -1)]_{\mathcal{B}'} \\ | & | \end{pmatrix} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

So with our two choices of bases, $\mathcal{B}$ and $\mathcal{B}'$, we have produced two different coordinate matrices for the same transformation:

$$[T]_{\mathcal{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad \text{and} \quad [T]_{\mathcal{B}'} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

I will keep emphasizing the point that even though these are different matrices, they represent the same abstract linear transformation. Picking different coordinate systems and switching between them is often a useful endeavor in linear algebra. For example, the standard coordinate system given by $\mathcal{B}$ above is nice because the basis $\mathcal{B}$ is simple, but the resulting matrix $[T]_{\mathcal{B}}$ is comparatively complicated. The coordinate system induced by $\mathcal{B}'$, while using a more complicated basis, results in a diagonal matrix $[T]_{\mathcal{B}'}$ which is much simpler than $[T]_{\mathcal{B}}$. In general, we want to understand how to translate between two different coordinate systems. In other words, in our example above, we want to identify an explicit relationship between the matrices

$$[T]_{\mathcal{B}} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad \text{and} \quad [T]_{\mathcal{B}'} = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

We can do so by remembering how coordinate matrices interact with composition of transfor-

mations. In particular, recall from Theorem 3.3.5 that in the general situation $V \xrightarrow{T} W \xrightarrow{S} U$ where V , W , and U are finite dimensional vector spaces with bases $\mathcal{B}$, $\mathcal{C}$, and $\mathcal{D}$ respectively, we have the formula

$$[S \circ T]_{\mathcal{B}}^{\mathcal{D}} = [S]_{\mathcal{C}}^{\mathcal{D}} [T]_{\mathcal{B}}^{\mathcal{C}}.$$

Returning to our motivating example of $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ where $T(x, y) = (2x + y, x + 2y)$, I will cleverly apply this formula to relate the two different coordinate matrices $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{B}'}$. In particular, note that

$$T = I_{\mathbb{R}^2} \circ T \circ I_{\mathbb{R}^2}$$

where $I_{\mathbb{R}^2} : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is the identity map. In words, the above formula just says that if you

(1) do nothing, then

(2) do T , then

(3) do nothing again,

that this is the same as just doing T . Next, I will visualize the composition $I_{\mathbb{R}^2} \circ T \circ I_{\mathbb{R}^2}$ using the following diagram:

$$(\mathbb{R}^2, \mathcal{B}) \xrightarrow{I_{\mathbb{R}^2}} (\mathbb{R}^2, \mathcal{B}') \xrightarrow{T} (\mathbb{R}^2, \mathcal{B}') \xrightarrow{I_{\mathbb{R}^2}} (\mathbb{R}^2, \mathcal{B}).$$

Here, the notation $(\mathbb{R}^2, \mathcal{B})$ just means I am thinking about the vector space $\mathbb{R}^2$ being equipped with the basis $\mathcal{B}$, and $(\mathbb{R}^2, \mathcal{B}')$ means I am thinking about the vector space $\mathbb{R}^2$ being instead equipped with the basis $\mathcal{B}'$. Applying Theorem 3.3.5 to this composition, we have

$$\begin{aligned} [T]_{\mathcal{B}} &= [T]_{\mathcal{B}}^{\mathcal{B}} \\ &= [I_{\mathbb{R}^2} \circ T \circ I_{\mathbb{R}^2}]_{\mathcal{B}}^{\mathcal{B}} \\ &= [I_{\mathbb{R}^2}]_{\mathcal{B}}^{\mathcal{B}'} [T]_{\mathcal{B}'}^{\mathcal{B}'} [I_{\mathbb{R}^2}]_{\mathcal{B}}^{\mathcal{B}'} . \end{aligned}$$

This gives us a concrete relationship between the coordinate matrices $[T]_{\mathcal{B}}$ and $[T]_{\mathcal{B}'}$, and this motivates the main definition of this section.

Definition 3.5.1. Let V be a finite dimensional vector space. Let $\mathcal{B}, \mathcal{B}'$ be two different bases of V . The matrix $Q := [I_V]_{\mathcal{B}}^{\mathcal{B}'}$ is the **change of coordinate matrix from $\mathcal{B}$ to $\mathcal{B}'$** .

Example 3.5.2. Consider the linear transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (2x + y, x + 2y)$ as above, and let $\mathcal{B} = \{(1, 0), (0, 1)\}$ and $\mathcal{B}' = \{(1, 1), (1, -1)\}$. We will compute the change of coordinate matrix Q from $\mathcal{B}$ to $\mathcal{B}'$. By the above definition, this is the change of coordinate matrix $Q = [I]_{\mathcal{B}}^{\mathcal{B}'}$, where $I = I_{\mathbb{R}^2}$ is the identity map. Note that

$$I(1, 0) = (1, 0) = \frac{1}{2}(1, 1) + \frac{1}{2}(1, -1)$$

and so

$$[I(1, 0)]_{\mathcal{B}'} = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix}.$$

Similarly, note that

$$I(0, 1) = (0, 1) = \frac{1}{2}(1, 1) + (-\frac{1}{2})(1, -1)$$

and so

$$[I(0, 1)]_{\mathcal{B}'} = \begin{pmatrix} \frac{1}{2} \\ -\frac{1}{2} \end{pmatrix}.$$

Therefore, the change of coordinate matrix from $\mathcal{B}$ to $\mathcal{B}'$ is

$$\begin{aligned} Q &= [I]_{\mathcal{B}}^{\mathcal{B}'} \\ &= \begin{pmatrix} | & | \\ [I(1, 0)]_{\mathcal{B}'} & [I(0, 1)]_{\mathcal{B}'} \\ | & | \end{pmatrix} \\ &= \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}. \end{aligned}$$

The following theorem summarizes and generalizes many of the observations we've made so far about changing coordinates.

Theorem 3.5.3. *Let V be a finite dimensional vector space, and let $\mathcal{B}, \mathcal{B}'$ be two different bases of V . Let $Q = [I_V]_{\mathcal{B}}^{\mathcal{B}'}$ be the change of coordinate matrix from $\mathcal{B}$ to $\mathcal{B}'$.*

(i) *The matrix Q is invertible.*

(ii) *For any $v \in V$,*

$$[v]_{\mathcal{B}'} = Q[v]_{\mathcal{B}}.$$

(iii) *For any $T \in \mathcal{L}(V)$,*

$$[T]_{\mathcal{B}} = Q^{-1}[T]_{\mathcal{B}'}Q.$$

Remark 3.5.4. In words, part (ii) of this theorem justifies the “change of coordinate matrix” name: it says that if you have the $\mathcal{B}$ -coordinates of an abstract vector v , given by the coordinate vector $[v]_{\mathcal{B}}$, and you want the $\mathcal{B}'$ -coordinates of the same vector, you multiply $[v]_{\mathcal{B}}$ by Q .

Proof. This proof is a compact summary of many of the computations we did in the above motivating example.

(i) Since $I_V : V \rightarrow V$ is clearly invertible, by a proposition from the previous lecture it follows that Q is invertible.

(ii) Note that

$$[v]_{\mathcal{B}'} = [I_V(v)]_{\mathcal{B}'} = [I_V]_{\mathcal{B}}^{\mathcal{B}'}[v]_{\mathcal{B}} = Q[v]_{\mathcal{B}}.$$

(iii) Note that

$$[T]_{\mathcal{B}} = [I_V \circ T \circ I_V]_{\mathcal{B}} = [I_V]_{\mathcal{B}}^{\mathcal{B}'}[T]_{\mathcal{B}'}[I_V]_{\mathcal{B}}^{\mathcal{B}'}$$

Since $Q^{-1} = ([I_V]_{\mathcal{B}}^{\mathcal{B}'})^{-1} = [I_V^{-1}]_{\mathcal{B}'}^{\mathcal{B}} = [I_V]_{\mathcal{B}'}^{\mathcal{B}}$, it follows that

$$[T]_{\mathcal{B}} = Q^{-1}[T]_{\mathcal{B}'}Q.$$

□

Example 3.5.5. Suppose that V is a vector space over $\mathbb{R}$ with $\dim V = 3$. Let $\mathcal{B} = \{v_1, v_2, v_3\}$ and $\mathcal{B}' = \{v'_1, v'_2, v'_3\}$ be two bases of V . Suppose the change of coordinates matrix from $\mathcal{B}$ to $\mathcal{B}'$ is

$$Q = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 0 & -1 \\ 0 & 2 & 0 \end{pmatrix}.$$

Suppose also that $v \in V$ is the abstract vector $v = 2v_1 - v_2 + 3v_3$, expressed as a linear combination of $\mathcal{B}$. If we want to write v instead as a linear combination of $\mathcal{B}'$, we can use the change of coordinate matrix. The fact that $v = 2v_1 - v_2 + 3v_3$ tells us that

$$[v]_{\mathcal{B}} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}.$$

By Theorem 3.5.3, we have

$$[v]_{\mathcal{B}'} = Q[v]_{\mathcal{B}} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 0 & -1 \\ 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix} = \begin{pmatrix} 9 \\ 5 \\ -2 \end{pmatrix}.$$

Thus, this tells us that, abstractly, $v = 9v'_1 + 5v'_2 - 2v'_3$.

Here is a fancy way to visualize all of this nonsense that we've talked about with coordinate representations and coordinate changes. Suppose we have a linear operator $T \in \mathcal{L}(V)$ on a finite dimensional vector space. Diagrammatically, this is represented by the following diagram:

$$V \xrightarrow{T} V$$

Next, choose a basis $\mathcal{B}$ for V , and let $f_{\mathcal{B}} : V \rightarrow \mathbb{F}^n$ be the map that sends a vector $v \in V$ to its coordinate vector $[v]_{\mathcal{B}}$. That is, $f_{\mathcal{B}}(v) = [v]_{\mathcal{B}}$. This choice of coordinates allows us to pass to a new "equivalent" row of the diagram. Corresponding to the map T is the map $\mathbb{F}^n \rightarrow \mathbb{F}^n$ given by multiplication by $[T]_{\mathcal{B}}$.

$$\begin{array}{ccc} V & \xrightarrow{T} & V \\ \downarrow f_{\mathcal{B}} & & \downarrow f_{\mathcal{B}} \\ \mathbb{F}^n & \xrightarrow{[T]_{\mathcal{B}}} & \mathbb{F}^n \end{array}$$

On the other hand, we could pick a *different* basis $\mathcal{B}'$ and get a different, yet also equivalent row!

$$\begin{array}{ccc} \mathbb{F}^n & \xrightarrow{[T]_{\mathcal{B}'}} & \mathbb{F}^n \\ \uparrow f_{\mathcal{B}'} & & \uparrow f_{\mathcal{B}'} \\ V & \xrightarrow{T} & V \\ \downarrow f_{\mathcal{B}} & & \downarrow f_{\mathcal{B}} \\ \mathbb{F}^n & \xrightarrow{[T]_{\mathcal{B}}} & \mathbb{F}^n \end{array}$$

The change of coordinate matrix is the direct translation between the bottom and top row.

$$\begin{array}{ccc}
 \mathbb{F}^n & \xrightarrow{[T]_{\mathcal{B}'}} & \mathbb{F}^n \\
 \uparrow f_{\mathcal{B}'} & & \uparrow f_{\mathcal{B}'} \\
 Q \left(\begin{array}{ccc} & & \\ & T & \\ & & \end{array} \right) Q & & \\
 \downarrow f_{\mathcal{B}} & & \downarrow f_{\mathcal{B}} \\
 \mathbb{F}^n & \xrightarrow{[T]_{\mathcal{B}}} & \mathbb{F}^n
 \end{array}$$

Another way to capture many of the ideas discussed in this section is to introduce the notion of *similarity*. Recall that in the motivating example at the beginning of this section, we obtained two different matrix representatives for a single linear transformation:

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

These matrices both represent the same transformation, after a change in coordinates. This is precisely what it means to say that the two matrices are *similar*, as given by the following definition.

Definition 3.5.6. Two operators $T \in \mathcal{L}(V)$ and $S \in \mathcal{L}(W)$ are **similar** if there is an isomorphism $\phi : V \rightarrow W$ such that $T = \phi^{-1} \circ S \circ \phi$. Likewise, two matrices $A, B \in M_n(\mathbb{F})$ are **similar** if there is an invertible matrix $Q \in M_n(\mathbb{F})$ such that $A = Q^{-1}BQ$.

To emphasize, you should think about similar operators or similar matrices as being objects that represent the “same” transformation, up to a change in coordinates or change in perspective in a different vector space.

Example 3.5.7. Let $P(\mathbb{R})$ be the vector space of polynomials in any degree with coefficients in $\mathbb{R}$. Let $\mathbb{R}^\infty$ be the vector space of sequences with finitely many nonzero entries. That is,

$$\mathbb{R}^\infty := \{ (a_0, a_1, a_2, \dots) \mid a_j \neq 0 \text{ for finitely many } j \}.$$

First, we claim that $P(\mathbb{R}) \cong \mathbb{R}^\infty$. Let $\phi : P(\mathbb{R}) \rightarrow \mathbb{R}^\infty$ be given by

$$\phi(a_0 + a_1x + \dots + a_nx^n) = (a_0, a_1, \dots, a_n, 0, \dots).$$

It is straightforward to check that ϕ is an isomorphism.

Next, we define two operators, one on $P(\mathbb{R})$ and one on $\mathbb{R}^\infty$. Let $T : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be given by $T(p(x)) := xp(x)$. Let $S : \mathbb{R}^\infty \rightarrow \mathbb{R}^\infty$ be the “right-shift” operator given by

$$S(a_0, a_1, a_2, \dots) := (0, a_0, a_1, \dots).$$

We claim that T and S are similar, so that they fundamentally represent the “same” transfor-

mation. Indeed, note that

$$\begin{aligned}
 (\phi^{-1} \circ S \circ \phi)(a_0 + a_1x + \cdots + a_nx^n) &= \phi^{-1}(S(\phi(a_0 + a_1x + \cdots + a_nx^n))) \\
 &= \phi^{-1}(S(a_0, a_1, \dots, a_n, 0, 0, \dots)) \\
 &= \phi^{-1}(0, a_0, \dots, a_{n-1}, a_n, 0, \dots) \\
 &= a_0x + a_1x^2 + \cdots + a_nx^{n+1} \\
 &= x(a_0 + a_1x + \cdots + a_nx^n) \\
 &= T(a_0 + a_1x + \cdots + a_nx^n).
 \end{aligned}$$

Thus, $T = \phi^{-1} \circ S \circ \phi$, and so T and S are similar. The diagram representing this situation is the following:

$$\begin{array}{ccc}
 \mathbb{R}^\infty & \xrightarrow{S} & \mathbb{R}^\infty \\
 \phi \uparrow & & \uparrow \phi \\
 P(\mathbb{R}) & \xrightarrow{T} & P(\mathbb{R})
 \end{array}$$

Exercises

- Let $\mathcal{B}_1 = \{2x^2 - x, 3x^2 + 1, x^2\}$ and $\mathcal{B}_2 = \{1, x, x^2\}$ be two bases of $P_2(\mathbb{R})$ (you don't have to prove that they are bases).
 - Compute the change of coordinate matrix from $\mathcal{B}_1$ -coordinates to $\mathcal{B}_2$ -coordinates.
 - Let $p(x) \in P_2(\mathbb{R})$ such that $[p(x)]_{\mathcal{B}_1} = (1, 2, 3)$. What is $[p(x)]_{\mathcal{B}_2}$? What is $p(x)$?
- Let V be an n -dimensional vector space over $\mathbb{F}$. Let $\mathcal{B}_1 = \{v_1, \dots, v_n\}$ be a basis of V . Let Q be an $n \times n$ invertible matrix with entries in $\mathbb{F}$. Define a set $\mathcal{C} = \{w_1, \dots, w_n\}$ by

$$w_j := \sum_{i=1}^n Q_{ij}v_i.$$

Prove that $\mathcal{C}$ is a basis for V , and compute the change of coordinate matrix from $\mathcal{B}$ to $\mathcal{C}$.

- True or false: Let V be an n -dimensional real vector space with (ordered) bases $\mathcal{B}_1$ and $\mathcal{B}_2$, and let Q be the change of coordinate matrix from $\mathcal{B}_1$ to $\mathcal{B}_2$. Then the transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^n$ defined by

$$T \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} = Q \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

is invertible.

3.6 Dual spaces

There is a meta-principle in mathematics that if you are interested in an object X — this could be a vector space, a geometric object, or anything at all — it is insightful to study the

functions out of that object rather than the object itself. For example, if X is a geometric blob, like a sphere, or a donut, or a Klein bottle, you can identify various properties of that blob by considering the nature functions $\psi : X \rightarrow \mathbb{R}$. In the context of linear algebra, given a vector space V we have been studying linear functions $\psi : V \rightarrow W$ from V to some other vector space W . If our goal is to understand V , then there are many choices of codomain vector spaces W , and in this section we will make the simplest choice and let $W = \mathbb{F}$, the ambient field. The collection of all linear maps $\psi : V \rightarrow \mathbb{F}$ is called the *dual space* of V , and it is the focus of this section.

Definition 3.6.1. Let V be a vector space over a field $\mathbb{F}$. The **dual space** of V , denoted V^* , is the vector space

$$V^* := \mathcal{L}(V, \mathbb{F}) = \{ \psi : V \rightarrow \mathbb{F} \mid \psi \text{ is linear} \}.$$

An element $\psi \in V^*$ is often called a **linear functional** on V .

Example 3.6.2. Consider the case $V = \mathbb{R}^3$. Fix a row vector

$$\begin{pmatrix} 4 & 2 & -3 \end{pmatrix}$$

and define $\psi : \mathbb{R}^3 \rightarrow \mathbb{R}$ by

$$\psi \begin{pmatrix} x \\ y \\ z \end{pmatrix} := \begin{pmatrix} 4 & 2 & -3 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = 4x + 2y - 3z.$$

Then $\psi \in (\mathbb{R}^3)^*$. We have not discussed the dot product (or inner products, which generalize the dot product), but as a reader you are likely familiar with the dot product. In words, this example suggests that the operation of taking the dot product with a fixed vector — equivalently, multiplication on the left by a fixed row vector — is one natural way to produce linear functionals on $\mathbb{R}^n$. We will see below that *every* linear functional on $\mathbb{R}^n$ arises this way! Thus, if you view $\mathbb{R}^n$ as the space of column n -vectors, then $(\mathbb{R}^n)^*$ can be identified with the space of row n -vectors. In particular, this suggests that $(\mathbb{R}^n)^* \cong \mathbb{R}^n$.

Example 3.6.3. Here is an infinite dimensional riff on the above example, which suggests some very interesting differences in the nature of dual spaces in infinite dimensions. Let

$$V := \{ (a_0, a_1, a_2, \dots) \mid a_j \in \mathbb{F}, \text{ and } a_j \neq 0 \text{ for only finitely many } j \}.$$

In words, V is the set of all sequences which are eventually 0 forever. For instance, the sequence

$$(1, 1, 1, 1, 1, \dots)$$

would *not* be an element of V , because it is *not* the case the a_j is nonzero for only finitely many j . On the other hand,

$$(1, 1, 1, 0, \dots) \in V$$

is an element of V because $a_j \neq 0$ for only three indices: $j = 0, 1, 2$. However, we can define a linear functional — an element of the dual space V^* — as follows. Begin by choosing literally

any sequence. For example, consider the sequence

$$(1, 1, 1, 1, 1, \dots).$$

I'll emphasize again that this sequence is not an element of V . However, we can define a linear functional $\psi : V \rightarrow \mathbb{R}$ by taking the “dot product” of any sequence in V with the above sequence. To be more formal, for any sequence $(b_0, b_1, b_2, \dots)$, not necessarily in V , define $\psi : V \rightarrow \mathbb{R}$ by

$$\psi(a_0, a_1, a_2, \dots) := \sum_{n=0}^{\infty} b_n a_n.$$

Although the above sum looks like an infinite sum, it is actually always a finite sum because of the defining condition on V . In particular, any given sequence $(a_0, a_1, a_2, \dots) \in V$ is eventually 0, and so the above sum is a finite sum regardless of the sequence $(b_0, b_1, \dots)$. The point of this example is the following. Our vector space V , which consists of all sequences which are eventually 0, has a dual space V^* that can be identified with the set of *all* sequences. Unlike in the finite dimensional case, V^* is not isomorphic to V , and in fact is much larger!

Example 3.6.4. Let $V = \{f : \mathbb{R} \rightarrow \mathbb{R}\}$ be the vector space of all functions $\mathbb{R} \rightarrow \mathbb{R}$. Define $\psi : V \rightarrow \mathbb{R}$ by $\psi(f) := f(0)$. Then $\psi \in V^*$.

Example 3.6.5. Let $V = \{f : [0, 1] \rightarrow \mathbb{R} \mid f \text{ is continuous}\}$. Fix any continuous function $g : [0, 1] \rightarrow \mathbb{R}$ and define $\psi : V \rightarrow \mathbb{R}$ by

$$\psi(f) := \int_0^1 f(x)g(x) dx.$$

Then $\psi \in V^*$.

Next, we formally prove a property of finite dimensional dual spaces suggested by the first example.

Theorem 3.6.6. *Let V be a finite dimensional vector space. Then $V^* \cong V$.*

Proof. Let $n := \dim V$, and let $\mathcal{B} := \{v_1, \dots, v_n\}$ be a basis for V . To prove that $V^* \cong V$ it suffices to prove that $\dim V^* = n$. We will do so by producing a basis of V^* with n elements. For each $1 \leq j \leq n$, define a linear functional $v_j^* : V \rightarrow \mathbb{F}$ on the basis $\mathcal{B}$ by

$$v_j^*(v_i) := \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

and extend it to all of V by linearity. We claim that

$$\mathcal{B}^* := \{v_1^*, \dots, v_n^*\}$$

is a basis for V^* .

First, we prove that $\mathcal{B}^*$ is linearly independent. Suppose that for some scalars $a_1, \dots, a_n \in \mathbb{F}$ we have

$$a_1 v_1^* + \dots + a_n v_n^* = 0.$$

Then $a_1 v_1^*(v) + \dots + a_n v_n^*(v) = 0$ for all $v \in V$. In particular, for each $1 \leq j \leq n$ we have

$$a_1 v_1^*(v_j) + \dots + a_{j-1} v_{j-1}^*(v_j) + a_j v_j^*(v_j) + a_{j+1} v_{j+1}^*(v_j) + \dots + a_n v_n^*(v_j) = 0.$$

By definition of v_j^* , this equality is

$$0 + \dots + 0 + a_j \cdot 1 + 0 + \dots + 0 = 0$$

and so $a_j = 0$. Since this holds for all j , it follows that $\mathcal{B}^*$ is linearly independent.

Next, we prove that $\text{Span}(\mathcal{B}^*) = V^*$. Let $\psi \in V^*$ be an arbitrary linear functional. For each $1 \leq j \leq n$, define $a_j := \psi(v_j)$. We claim that $\psi = a_1 v_1^* + \dots + a_n v_n^*$. To prove this, it suffices to show that

$$\psi(v_j) = a_1 v_1^*(v_j) + \dots + a_n v_n^*(v_j)$$

for all $1 \leq j \leq n$. But this is immediate from the definition of a_j and of v_j^* ; the above equality is

$$a_j = 0 + \dots + 0 + a_j \cdot 1 + 0 + \dots + 0$$

which is evidently true. Therefore, $\psi = a_1 v_1^* + \dots + a_n v_n^*$ and so $\text{Span}(\mathcal{B}^*) = V^*$, as desired. $\square$

Remark 3.6.7. In order to prove that $V^* \cong V$ when V is finite dimensional, we had to choose a basis for V . You could try to find a proof that does not require the use of a basis, but you will struggle! Indeed, the fact that it is *not* true that $V^* \cong V$ if V is infinite dimensional suggests that it is in fact necessary to work with a basis — the existence of which is the defining property of what it means to be finite dimensional — in order to prove the theorem.

The basis $\mathcal{B}^*$ we constructed in the proof of Theorem 3.6.6 is important enough to warrant its own name. In particular, given any basis $\mathcal{B}$ for a vector space V , we can form the basis $\mathcal{B}^*$ as above and we call it the *dual basis*.

Definition 3.6.8. Let V be a finite dimensional vector space with basis $\mathcal{B} = \{v_1, \dots, v_n\}$. For each $1 \leq j \leq n$, define a linear functional $v_j^* : V \rightarrow \mathbb{F}$ on the basis $\mathcal{B}$ by

$$v_j^*(v_i) := \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

and extend it to all of V by linearity. The set

$$\mathcal{B}^* := \{v_1^*, \dots, v_n^*\}$$

is called the **dual basis of V with respect to $\mathcal{B}$** .

3.6.1 The transpose of a linear map

One well-known operation on matrices is the *transpose*. Given $A \in M_{m \times n}(\mathbb{F})$, we can form $A^T \in M_{n \times m}(\mathbb{F})$ by swapping rows and columns. Formally, if $A = (a_{ij})$, then $A^T := (a_{ji})$. Or, in more explicit notation,

$$A = \begin{pmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mn} \end{pmatrix} \quad \text{and} \quad A^T = \begin{pmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1n} & \cdots & a_{mn} \end{pmatrix}.$$

Or in even more down-to-earth form as an explicit example,

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \quad \text{and} \quad A^T = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

Like almost everything in matrix linear algebra, there is a correct generalization of the concept of the transpose to abstract linear transformations between vector spaces. Given an abstract linear map $T : V \rightarrow W$, we will define a new linear map, called the *transpose* of T and denoted T^t , of the form $T^t : W^* \rightarrow V^*$. In words, if a linear map goes from V to W , the transpose will go from the dual space of the codomain to the dual space of the domain.

Definition 3.6.9. Let $T : V \rightarrow W$ be linear. Define the **transpose of T** , denoted T^t , as a map $T^t : W^* \rightarrow V^*$ as follows. For $\psi \in W^*$, define $T^t(\psi) : V \rightarrow \mathbb{R}$ by

$$T^t(\psi)(v) := \psi(T(v)).$$

Remark 3.6.10. A definition like this can take some time to parse, and it can help to talk through some it in words. The map T eats vector in V and spits out vectors in W . The transpose T^t is supposed to eat functions of the form $W \rightarrow \mathbb{F}$ and spit out functions of the form $V \rightarrow \mathbb{F}$. It does so as follows. If $\psi : W \rightarrow \mathbb{F}$ is an element of W^* , then to define the function $T^t(\psi)$, which is supposed to eat vectors in V and spit out scalars in $\mathbb{F}$, we need to define what $T^t(\psi)$ does to a random vector $v \in V$. The natural way to produce a scalar given a vector $v \in V$ is to first feed it to T to get a vector $T(v) \in W$, which we then feed to $\psi : W \rightarrow \mathbb{F}$ to get a scalar $\psi(T(v)) \in \mathbb{F}$.

We will see a down-to-earth example below (Example 3.6.14), but first it is worth investigating some more interesting examples that are out of reach from the perspective of lower-division linear algebra.

Example 3.6.11. Let $C^\infty(\mathbb{R}) = \{f : \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is infinitely differentiable}\}$. Recall that the derivative $\frac{d}{dx} : C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R})$ is a linear map. The transpose of the derivative would be a linear map

$$\left(\frac{d}{dx}\right)^t : (C^\infty(\mathbb{R}))^* \rightarrow (C^\infty(\mathbb{R}))^*.$$

To investigate how this map behaves, choose a linear functional $\psi \in (C^\infty(\mathbb{R}))^*$, that is, a linear map $\psi : C^\infty(\mathbb{R}) \rightarrow \mathbb{R}$. One example of such a map would be the evaluation at 0 map from

above, given by $\psi(f) = f(0)$. If we feed this map ψ to $\left(\frac{d}{dx}\right)^t$, we would get another linear map

$$\left(\frac{d}{dx}\right)^t (\psi) : C^\infty(\mathbb{R}) \rightarrow \mathbb{R}.$$

To study the behavior of this map, we would then feed it an infinitely differentiable function $f : \mathbb{R} \rightarrow \mathbb{R}$. Note that

$$\begin{aligned} \left(\frac{d}{dx}\right)^t (\psi)(f) &= \psi\left(\frac{d}{dx}(f)\right) \\ &= f'(0). \end{aligned}$$

Thus, the transpose $\left(\frac{d}{dx}\right)^t$ eats the evaluation-at-0 linear functional and spits out the evaluate-the-derivative-at-0 linear functional.

Example 3.6.12. Consider again the derivative map $\frac{d}{dx} : C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R})$. Another example of a linear functional on $C^\infty(\mathbb{R})$, i.e., a linear map $\tilde{\psi} : C^\infty(\mathbb{R}) \rightarrow \mathbb{R}$, is given by integration against a fixed function. That is, let $g \in C^\infty(\mathbb{R})$ and define $\tilde{\psi} : C^\infty(\mathbb{R}) \rightarrow \mathbb{R}$ by

$$\tilde{\psi}(f) := \int_0^1 f(x)g(x) dx.$$

Like before, $\left(\frac{d}{dx}\right)^t (\tilde{\psi})$ will be another linear functional on $C^\infty(\mathbb{R})$. Also like before, we can understand what this linear functional does by feeding it a function $f : \mathbb{R} \rightarrow \mathbb{R}$. Note that

$$\begin{aligned} \left(\frac{d}{dx}\right)^t (\tilde{\psi})(f) &= \tilde{\psi}\left(\frac{d}{dx}(f)\right) \\ &= \int_0^1 f'(x)g(x) dx \\ &= [f(1)g(1) - f(0)g(0)] - \int_0^1 f(x)g'(x) dx. \end{aligned}$$

Here we have used integration by parts to get the last expression. In words, this says that if $\tilde{\psi}$ is the integrate-against- g linear functional, then, by integration by parts, $\left(\frac{d}{dx}\right)^t (\tilde{\psi})$ is the integrate-against- $(-g')$ linear functional (ignoring the boundary terms $f(1)g(1) - f(0)g(0)$). This is an indication that integration by parts is really a statement about duality of the derivative operator in the context of linear algebra. This is a fascinating perspective that we will return to more in Chapter 5.

The connection between the transpose of a linear map and the transpose of a matrix is, unsurprisingly, given by coordinate representations with respect to a choice of basis and their dual bases. In words, the following theorem says that the coordinate matrix of T^t is the transpose of the coordinate matrix of T .

Theorem 3.6.13. *Let V, W be finite dimensional vector spaces with basis $\mathcal{B}, \mathcal{C}$, respectively. Let $T : V \rightarrow W$ be linear. Then*

$$[T^t]_{\mathcal{C}^*}^{\mathcal{B}^*} = ([T]_{\mathcal{B}}^{\mathcal{C}})^T.$$

Proof. Let $\mathcal{B} = \{v_1, \dots, v_n\}$ and $\mathcal{C} = \{w_1, \dots, w_m\}$, so $\mathcal{B}^* = \{v_1^*, \dots, v_n^*\}$ and $\mathcal{C}^* = \{w_1^*, \dots, w_m^*\}$. Let $A := [T]_{\mathcal{B}}^{\mathcal{C}}$ and denote the (i, j) -entry of A by A_{ij} . Note that, by definition of a coordinate matrix, this means that the scalars A_{ij} are defined by

$$T(v_j) = A_{1j} w_1 + \dots + A_{mj} w_m.$$

To prove the theorem, it suffices to show that the (j, i) -th entry of the matrix $[T^t]_{\mathcal{C}^*}^{\mathcal{B}^*}$ is A_{ij} . To get the (j, i) -th entry of $[T^t]_{\mathcal{C}^*}^{\mathcal{B}^*}$, we can consider the i th column, which is the coordinate vector

$$[T^t(w_i^*)]_{\mathcal{B}^*}.$$

Because $T^t(w_i^*) \in V^*$, and $\{v_1^*, \dots, v_n^*\}$ is a basis for V^* , we can write

$$T^t(w_i^*) = B_{1i} v_1^* + \dots + B_{ni} v_n^*$$

for some scalars $B_{ji} \in \mathbb{F}$. Feeding $v_j \in V$ to this equation gives, on the left side,

$$\begin{aligned} T^t(w_i^*)(v_j) &= w_i^*(T(v_j)) \\ &= w_i^*(A_{1j} w_1 + \dots + A_{mj} w_m) \\ &= A_{1j} w_i^*(w_1) + \dots + A_{mj} w_i^*(w_m) \\ &= 0 + \dots + 0 + A_{ij} \cdot 1 + 0 + \dots + 0 \\ &= A_{ij}. \end{aligned}$$

On the right side, we have

$$\begin{aligned} B_{1i} v_1^*(v_j) + \dots + B_{ni} v_n^*(v_j) &= 0 + \dots + 0 + B_{ji} \cdot 1 + 0 + \dots + 0 \\ &= B_{ji}. \end{aligned}$$

Thus, $B_{ji} = A_{ij}$. Since B_{ji} is precisely the (j, i) -th entry of $[T^t]_{\mathcal{C}^*}^{\mathcal{B}^*}$, this completes the proof. $\square$

Example 3.6.14. Let's work through the above theorem in a concrete example. Consider the map $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ given by

$$T \begin{pmatrix} x \\ y \\ z \end{pmatrix} := \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} x + 2y + 3z \\ 4x + 5y + 6z \end{pmatrix}.$$

Let $\mathcal{B} = \{e_1, e_2, e_3\}$ and $\mathcal{C} = \{e_1, e_2\}$ denote the standard bases of $\mathbb{R}^3$ and $\mathbb{R}^2$, respectively. Then

$$[T]_{\mathcal{B}}^{\mathcal{C}} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}.$$

Next we investigate the transpose T^t . In particular, we will compute the coordinate matrix

$[T^t]_{\mathcal{C}^*}^{\mathcal{B}^*}$. As usual, this coordinate matrix will be given by

$$[T^t]_{\mathcal{C}^*}^{\mathcal{B}^*} = \begin{pmatrix} | & | \\ [T^t(e_1^*)]_{\mathcal{B}^*} & [T^t(e_2^*)]_{\mathcal{B}^*} \\ | & | \end{pmatrix}.$$

First, consider the abstract linear function $T^t(e_1^*) : \mathbb{R}^3 \rightarrow \mathbb{R}$. Because $\{e_1^*, e_2^*, e_3^*\}$ is a basis for $(\mathbb{R}^3)^*$, we have

$$T^t(e_1^*) = B_{11} e_1^* + B_{21} e_2^* + B_{31} e_3^*$$

for some constants $B_{j1} \in \mathbb{R}$. Feeding e_1 to both sides gives

$$T^t(e_1^*)(e_1) = B_{11} e_1^*(e_1) + B_{21} e_2^*(e_1) + B_{31} e_3^*(e_1)$$

which is equivalent to

$$e_1^*(T(e_1)) = B_{11} \cdot 1 + 0 + 0.$$

Since $T(e_1) = (1, 4)$, we have

$$e_1^*(T(e_1)) = e_1^*(1, 4) = e_1^*(1 \cdot e_1 + 4 \cdot e_2) = 1 \cdot e_1^*(e_1) + 4 \cdot e_1^*(e_2) = 1.$$

Thus, $B_{11} = 1$. Similarly, one can show that $B_{21} = 2$ and $B_{31} = 3$. Thus,

$$[T^t(e_1^*)]_{\mathcal{B}^*} = \begin{pmatrix} B_{11} \\ B_{21} \\ B_{31} \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

An identical computation shows that

$$[T^t(e_2^*)]_{\mathcal{B}^*} = \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$$

and so

$$[T^t]_{\mathcal{C}^*}^{\mathcal{B}^*} = \begin{pmatrix} | & | \\ [T^t(e_1^*)]_{\mathcal{B}^*} & [T^t(e_2^*)]_{\mathcal{B}^*} \\ | & | \end{pmatrix} = \begin{pmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{pmatrix}.$$

Note that, indeed, $[T^t]_{\mathcal{C}^*}^{\mathcal{B}^*} = ([T]_{\mathcal{B}}^{\mathcal{C}})^T$.

Exercises

1. Recall that the **product of V and W** is the set

$$V \times W := \{ (v, w) \mid v \in V, w \in W \}.$$

and that $V \times W$ is a vector space. Let⁸ $f : V \times W \rightarrow \mathbb{F}$ be a function (not necessarily linear). For each $v \in V$, define $f_v : W \rightarrow \mathbb{F}$ by $f_v(w) := f(v, w)$. Similarly, for each $w \in W$, define $f_w : V \rightarrow \mathbb{F}$ by $f_w(v) := f(v, w)$. A function $f : V \times W \rightarrow \mathbb{F}$ is **bilinear** if, for all $v \in V$ and $w \in W$, the functions $f_v : W \rightarrow \mathbb{F}$ and $f_w : V \rightarrow \mathbb{F}$ are linear.

- (a) Let $f : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x, y) = 3xy$. Prove that f is bilinear. Prove also that f is not linear.

Let $\mathcal{B}(V \times W, \mathbb{F})$ denote the set of all bilinear maps $f : V \times W \rightarrow \mathbb{F}$. With the usual notions of function addition and scalar multiplication, $\mathcal{B}(V \times W, \mathbb{F})$ is a vector space over $\mathbb{F}$.

- (b) Let $\Psi : \mathcal{L}(V, W^*) \rightarrow \mathcal{B}(V \times W, \mathbb{F})$ be defined as follows. For each linear map $T : V \rightarrow W^*$, let $\Psi(T) : V \times W \rightarrow \mathbb{F}$ be the map defined by

$$\Psi(T)(v, w) := T(v)(w).$$

Prove that Ψ is an isomorphism of vector spaces to conclude⁹ that $\mathcal{B}(V \times W, \mathbb{F}) \cong \mathcal{L}(V, W^*)$.

[Hint: To prove that Ψ is an isomorphism, you need to prove two things: that Ψ is linear, and that Ψ is invertible. For surjectivity, given a bilinear map $f : V \times W \rightarrow \mathbb{F}$, consider the function $T : V \rightarrow W^*$ defined by $T(v) := f_v$.]

- Define a map $\phi : V \rightarrow V^{**}$ as follows. (Here, $V^{**} := (V^*)^*$.) For $v \in V$, let $\phi(v) : V^* \rightarrow \mathbb{F}$ be the function defined by $\phi(v)(\psi) := \psi(v)$. Prove that ϕ is injective. If V is finite dimensional, conclude that ϕ is in fact an isomorphism.¹⁰
- If A is a matrix, then $(A^T)^T = A$. Is it true that if $T : V \rightarrow W$ is a linear transformation, then $(T^t)^t = T$?

⁸More generally, you can replace $\mathbb{F}$ with another vector space U , but we'll keep things simpler for this problem. See Chapter 6 for a more in depth discussion on bilinearity and more generally multilinearity

⁹If you're wondering what the point of all this nonsense is, here is some motivation, particularly for those of you interested in physics. A **tensor of type** (p, q) is a multilinear map

$$T : \underbrace{V^* \times \cdots \times V^*}_{p \text{ times}} \times \underbrace{V \times \cdots \times V}_{q \text{ times}} \rightarrow \mathbb{F}.$$

Here, *multilinear* is the natural generalization of *bilinear* that will be discussed more in Chapter 6. So for example, a tensor of type $(0, 2)$ is a bilinear map $T : V \times V \rightarrow \mathbb{F}$, a tensor of type $(2, 0)$ is a bilinear map $T : V^* \times V^* \rightarrow \mathbb{F}$, and a tensor of type $(1, 1)$ is a bilinear map $T : V^* \times V \rightarrow \mathbb{F}$. The **rank** of a tensor is $p + q$, so the above examples are all rank 2 tensors. In physics, you usually just pretend that any rank 2 tensor is a matrix. This homework problem justifies this identification! Indeed, suppose that V is finite dimensional of dimension n . Then, for example, the set of tensors of type $(1, 1)$ is $\mathcal{B}(V^* \times V, \mathbb{F})$. By this homework problem, this vector space is isomorphic to $\mathcal{L}(V^*, V^{**})$. By the optional homework problems, $V^* \cong V \cong V^{**}$, so $\mathcal{L}(V^*, V^{**}) \cong \mathcal{L}(V, V) = \mathcal{L}(V)$. But we have previously shown in class that $\mathcal{L}(V) \cong M_n(\mathbb{F})$. Therefore, through this chain of isomorphisms, we've shown that the vector space of type $(1, 1)$ tensors $\mathcal{B}(V^* \times V, \mathbb{F})$ is isomorphic to $M_n(\mathbb{F})$. You could do more or less the same with tensors of type $(2, 0)$ and $(0, 2)$. Therefore, in physics, you're justified in pretending that rank 2 tensors are just $n \times n$ matrices! As a concrete example, the bilinear map $f(x, y) = 3xy$ from above is a tensor, and the associated 1×1 matrix is (3). For more information about this, check out Chapter 7!

¹⁰The slogan for this sentence is a *finite dimensional vector space is canonically isomorphic to its double dual*. Note that this isomorphism does not require us to choose a basis, unlike the isomorphism $V^* \cong V$ in Theorem 3.6.6.

Chapter 4

Eigentheory

In this chapter, we will focus on linear transformations $T : V \rightarrow V$ from a vector space back to itself, i.e., linear operators $T \in \mathcal{L}(V)$. In particular, the goal is to develop the theory of eigenvectors and eigenvalues for abstract linear operators. There are many ways to motivate this theory: It has concrete applications in differential equations, it is crucial in the mathematical formulation of quantum physics, it drives the theory of principal component analysis, and so on. However, the vague underlying reason why eigentheory has widespread uses and applications is that eigenvectors and eigenvalues give us a way to break down the behavior of a linear transformation in simple and fundamental ways. This will be the motivating perspective of this chapter.

That is, given a linear operator $T : V \rightarrow V$, we want to understand the behavior and effect of the transformation T by dissecting and decomposing the vector space V into simpler pieces that interact nicely with T . This will all be made precise in the coming sections. For now, I will give an imperfect and silly analogy that captures the spirit of this motivation. A linear operator $T : V \rightarrow V$ is like an entrée; a combination of various ingredients, seasonings, and cooking techniques, all coming together to produce a single culinary object. Given a delicious entrée, you might want to know *why* it tastes so good, perhaps so that you can recreate it yourself. What are the primary ingredients? How much salt was used? Where does the smokey aftertaste come from, and is that a subtle hint of sweetness? Was this reverse-seared or was it finished in the oven? These are the questions you would ask to understand the fundamental nature of this entrée, and in general this can be described as seeking the *recipe* for the dish. Given a linear operator $T : V \rightarrow V$, we want to find its recipe. What are the fundamental ingredients that, when mixed together, give us the action of T ? How much of each ingredient was used? How are the ingredients arranged?

Eigenvectors and eigenvalues are the ingredients that form the recipe for a linear operator $T : V \rightarrow V$, and this chapter will train you to be a mathematical chef.

4.1 Invariant subspaces

As a reader with prior linear algebra exposure, you likely already know what eigenvectors and eigenvalues of a matrix are and how to compute them. Depending on what your prior linear algebra exposure was like, this computation-centric understanding of eigen-things may be the extent of your familiarity. At this level of linear algebra, we are more interested in the

geometric intuition and structural significance of eigenvectors and eigenvalues. Thus, to skirt around any computation heavy bias you have about the role of eigentheory in linear algebra, we are going to begin by discussing a more general structural feature of a linear operator in this section as a way to set the stage for how to think about eigenvectors and eigenvalues. In particular, this section will cover the notion of *invariant subspaces*, and then in the following section we will specialize to the notion of eigenspaces.

Suppose we are given a linear operator $T : V \rightarrow V$. In order to understand the behavior of T and its effect on V , a natural desire is to break down V into simpler pieces that capture various aspects of the total action of T . For example, one could look for subspaces $W \subset V$ such that $T(W) \subset W$. With such a subspace, it then makes sense to restrict T to W and view it as a linear map $T : W \rightarrow W$. Since W is a subspace of V , this gives us a setting in which we can study the effect of T in a smaller and more fundamental domain. This leads to the definition of an *invariant subspace*.

Definition 4.1.1. Let V be a vector space and $T : V \rightarrow V$ a linear operator. A **T -invariant subspace** is a subspace $W \subset V$ satisfying $T(W) \subset W$. That is, a subspace such that if $w \in W$, then $T(w) \in W$.

Example 4.1.2. Let V be a vector space, and let $T : V \rightarrow V$ be any linear transformation. Then $\{0\}$ and V are always T -invariant subspaces.

Example 4.1.3. Fix $\theta \in [0, 2\pi)$ and let $T_\theta : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be given by

$$T_\theta \begin{pmatrix} x \\ y \\ z \end{pmatrix} := \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix}.$$

Geometrically, T_θ corresponds to a counter-clockwise rotation in the (x, y) -plane by an angle of θ , fixing the z -axis. Then $W := \text{Span}(e_1, e_2)$, the (x, y) -plane, and $U := \text{Span}(e_3)$, the z -axis, are both T_θ -invariant subspaces.

Indeed, let's verify that W is T_θ -invariant. Let $w = a e_1 + b e_2 \in W$. Note that

$$T_\theta(w) = T_\theta \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} a \\ b \\ 0 \end{pmatrix} = \begin{pmatrix} a \cos \theta - b \sin \theta \\ a \sin \theta + b \cos \theta \\ 0 \end{pmatrix} \in \text{Span}(e_1, e_2) = W.$$

Thus, W is T_θ -invariant. Similarly, let $u = c e_3 \in U$. Note that

$$T_\theta(u) = T_\theta \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ c \end{pmatrix} \in \text{Span}(e_3) = U.$$

Thus, U is T_θ -invariant.

Example 4.1.4. Let $D : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be the derivative linear transformation, i.e., the linear map defined by $D(p)(x) := p'(x)$. For each $n \geq 0$, let $W_n \subset P(\mathbb{R})$ be the subspace defined by $W_n := \text{Span}(1, x, \dots, x^n)$. Then W_n is D -invariant.

Example 4.1.5. Let $R : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ be given by $R(x, y) := (x, -y)$. Geometrically, R is described as a reflection across the x -axis. The x -axis, $W := \text{Span}(e_1)$ is R -invariant, since $R(x, 0) = (x, 0)$. Likewise, the y -axis, $U := \text{Span}(e_2)$, is also R -invariant, since $R(0, y) \in (0, -y) \in U$.

Reflection across the x -axis is a simple but suggestive example for why seeking invariant subspaces is a good idea. For instance, in the above example, we can decompose the underlying vector space as the direct sum of two invariant subspaces $\mathbb{R}^2 = W \oplus U$, and so to understand the total effect of R on $\mathbb{R}^2$ it suffices to study its effect on each invariant subspace W and U . The restricted map $R : W \rightarrow W$ is simply the identity map, and $R : U \rightarrow U$ is the map which multiplies every vector by -1 . One can view these invariant subspaces and their restricted R -action as the fundamental building blocks of the map $R : \mathbb{R}^2 \rightarrow \mathbb{R}^2$.

Next, we state and prove a simple proposition that gives one way to find invariant subspaces of any linear map.

Proposition 4.1.6. *Let $T : V \rightarrow V$ be a linear operator. Then $\ker T$ and $\text{im } T$ are both T -invariant subspaces.*

Proof. Let $v \in \ker T$. Note that $T(v) = 0 \in \ker T$, and so $\ker T$ is in fact T -invariant. Next, let $w \in \text{im } T$. Then $T(w) \in \text{im } T$ by definition of $\text{im } T$, and so $\text{im } T$ is T -invariant. $\square$

Note that a consequence of T -invariance is invariance under repeated applications of T . That is, if W is a T -invariant subspace, then it will also be a T^2 -invariant space, a T^3 -invariant subspace, and more generally a T^k -invariant subspace, where $T^k = T \circ \cdots \circ T$. Even more generally, we can consider polynomials of linear operators. This will turn out to be a fruitful concept to explore.

Definition 4.1.7. Let $p \in P(\mathbb{F})$ be given by $p(x) = a_0 + a_1x + \cdots + a_nx^n$. If $T \in \mathcal{L}(V)$, where V is a vector space over $\mathbb{F}$, then we define $p(T) \in \mathcal{L}(V)$ to be the operator

$$p(T) := a_0I_V + a_1T + \cdots + a_nT^n.$$

Remark 4.1.8. Note that this definition implies $T^0 = I_V$.

Proposition 4.1.9. *Let $T \in \mathcal{L}(V)$ and let $W \subset V$ be a T -invariant subspace. Then for any polynomial $p \in P(\mathbb{F})$, W is a $p(T)$ -invariant subspace.*

Proof. ADD $\square$

Finally, we will investigate the role of T -invariant subspaces on coordinate matrix representations of T .

FINISH

Exercises

1. Let $V := W_1 \oplus W_2$, and let $P : V \rightarrow V$ be the projection map defined by $P(w_1 + w_2) := w_1$ for $w_1 \in W_1$ and $w_2 \in W_2$. Prove that W_1 and W_2 are both P -invariant subspaces.
2. Let V be a vector space, and let $v_0 \neq 0 \in V$ be a fixed nonzero vector. Let $T : V \rightarrow V$ be a linear operator. Define

$$W_{T, v_0} := \{ p(T)(v_0) \mid p \in P(\mathbb{F}) \}.$$

- (a) Prove that W_{T,v_0} is a subspace of V .
 - (b) Prove that W_{T,v_0} is T -invariant.
 - (c) Prove that W_{T,v_0} is the smallest T -invariant subspace containing v_0 .
 - (d) If W_{T,v_0} , we call v_0 a **T -cyclic vector**. Does every linear map $T : V \rightarrow V$ necessarily admit a T -cyclic vector?
3. Let $T, S \in \mathcal{L}(V)$ such that $T \circ S = S \circ T$. Prove that $\ker p(S)$ and $\ker p(T)$ are T -invariant for every polynomial $p \in P(\mathbb{F})$.
4. This exercise requires knowledge of dual spaces; see Section 3.5. Let $T : V \rightarrow V$ be a linear operator, and let $W \subset V$ be a T -invariant subspace. Recall that the *annihilator* of W is the subspace $W^0 \subset V^*$ of the dual space defined by

$$W^0 := \{ \psi \in V^* \mid \psi(w) = 0 \text{ for all } w \in W \}.$$

Prove that W^0 is T^t -invariant, where $T^t : V^* \rightarrow V^*$ is the abstract transpose of T .

5. This exercise requires knowledge of quotient vector spaces; see Section 2.6.1. Let $T : V \rightarrow V$ be linear and let $W \subset V$ be a T -invariant subspace. Define $\tilde{T} : V/W \rightarrow V/W$ by

$$\tilde{T}([v]) := [T(v)].$$

Prove that $\tilde{T}$ is a well-defined linear map (i.e., that $\tilde{T}([v])$ does not depend on the choice of representative for the equivalence class $[v]$). Give an example that shows why T -invariance of W is necessary in order for T to descend to a well-defined map $V/W \rightarrow V/W$.

4.2 Diagonalization, eigenvalues and eigenvectors

THIS SECTION NEEDS DECOMPOSING AND REORGANIZING PROBABLY

Motivation

Definition 4.2.1. Let V be a finite dimensional vector space, and let $T \in \mathcal{L}(V)$. The operator T is **diagonalizable** if there is an (ordered) basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}}$ is a diagonal matrix.

Suppose we have a diagonalizable operator $T \in \mathcal{L}(V)$ and a basis $\mathcal{B} = \{v_1, \dots, v_n\}$ such that $[T]_{\mathcal{B}}$ is diagonal. Let a_{ij} denote the (i, j) -th entry of $[T]_{\mathcal{B}}$. Then, as usual, we have

$$T(v_j) = a_{1j}v_1 + \dots + a_{nj}v_n.$$

But since $[T]_{\mathcal{B}}$ is diagonal, $a_{ij} = 0$ if $i \neq j$. So in the above expression, all terms are 0 except for the one with a_{jj} . Thus,

$$T(v_j) = a_{jj}v_j.$$

This suggests that an operator will be diagonalizable if we can find a basis $\{v_1, \dots, v_n\}$ of V such that $T(v_j) = \lambda_j v_j$ for some scalars λ_j , which would ultimately be the diagonal entries of the resulting coordinate matrix. This motivates the following definition.

Definition 4.2.2. Let V be a vector space, and let $T \in \mathcal{L}(V)$. A nonzero vector $v \in V$ is an **eigenvector of T** with associated **eigenvalue λ** if $T(v) = \lambda v$ for some $\lambda \in \mathbb{F}$.

We then have the following theorem.

Theorem 4.2.3. Let V be finite dimensional, and let $T \in \mathcal{L}(V)$. Then T is diagonalizable if and only if there is a basis $\mathcal{B}$ of V consisting of eigenvectors of T .

Proof. First, suppose that T is diagonalizable. Then there is a basis $\mathcal{B} = \{v_1, \dots, v_n\}$ such that $[T]_{\mathcal{B}}$ is diagonal. Then the computation from above shows that $T(v_j) = a_{jj}v_j$, where a_{jj} is the (j, j) -th entry of $[T]_{\mathcal{B}}$. Thus, $\mathcal{B}$ is a basis consisting of eigenvectors of T .

Conversely, suppose that $\mathcal{B} = \{v_1, \dots, v_n\}$ is a basis consisting of eigenvectors of T with associated eigenvalues $\lambda_1, \dots, \lambda_n \in \mathbb{F}$. Then since $T(v_j) = \lambda_j v_j$, it follows that

$$T_{\mathcal{B}} = \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix}$$

and so T is diagonalizable. □

Example 4.2.4. Consider the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T\left(\begin{pmatrix} x \\ y \end{pmatrix}\right) = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Let $v_1 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$. Note that

$$T\left(\begin{pmatrix} 1 \\ 1 \end{pmatrix}\right) = \begin{pmatrix} 3 \\ 3 \end{pmatrix} = 3 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

and

$$T\left(\begin{pmatrix} 1 \\ -1 \end{pmatrix}\right) = \begin{pmatrix} -1 \\ 1 \end{pmatrix} = -1 \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$

Thus, v_1 is an eigenvector of T with eigenvalue 3, and v_2 is an eigenvector of T with eigenvalue -1. Furthermore, it is straightforward to verify that $\mathcal{B} = \{v_1, v_2\}$ is a basis of V . Thus, T is diagonalizable and

$$[T]_{\mathcal{B}} = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix}.$$

Example 4.2.5. Let $C^\infty(\mathbb{R})$ denote the vector space of infinitely differentiable functions $f : \mathbb{R} \rightarrow \mathbb{R}$ (that is, functions whose derivatives of all orders exist). Let $T : C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R})$ be given by $T(f)(x) = f'(x)$. Fix $\lambda \in \mathbb{R}$ and let $f(x) = e^{\lambda x}$. Then

$$T(f)(x) = f'(x) = \lambda e^{\lambda x} = \lambda f(x).$$

Since $T(f) = \lambda f$, it follows that f is an eigenvector of T with eigenvalue λ . Thus, every single real number is an eigenvalue of T ! Note that it does not make sense to ask if T is diagonalizable, because $C^\infty(\mathbb{R})$ is infinite dimensional.

Proposition 4.2.6. *Let V be finite dimensional, and let $T \in \mathcal{L}(V)$. Fix $\lambda \in \mathbb{F}$. The following four statements are equivalent.*

- (i) *The element $\lambda \in \mathbb{F}$ is an eigenvalue of T .*
- (ii) *The operator $T - \lambda I$ is not injective.*
- (iii) *The operator $T - \lambda I$ is not surjective.*
- (iv) *The operator $T - \lambda I$ is not invertible.*

Proof. First, assume (i), so that λ is an eigenvalue of T . Then there is a nonzero vector $v \in V$ such that $T(v) = \lambda v$. This implies that $(T - \lambda I)(v) = 0$. Since $v \neq 0$, it follows that $T - \lambda I$ is not injective. This gives (ii).

Next, assume (ii). Since $T - \lambda I$ is not injective and the dimensions of the domain and codomain are equal, by a previous proposition it follows that $T - \lambda I$ is not surjective. This gives (iii).

Next, assume (iii). Since $T - \lambda I$ is not surjective, it is not invertible. This gives (iv).

Finally, assume (iv). Since $T - \lambda I$ is not invertible, and the dimensions of the domain and codomain are equal, it follows that $T - \lambda I$ is not injective. Thus, there is a nonzero vector $v \in \ker(T - \lambda I)$, and so $T(v) - \lambda v = 0$, which implies $T(v) = \lambda v$. Thus, λ is an eigenvalue of T , and this gives (i). $\square$

Example 4.2.7. Consider the transformation $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by

$$T \begin{pmatrix} x \\ y \end{pmatrix} := \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}.$$

Fix $\lambda \in \mathbb{R}$ and consider the operator $T - \lambda I$. To identify eigenvalues of T , we want to understand when $T - \lambda I$ is not injective. Let $\mathcal{B}$ be the standard basis of $\mathbb{R}^2$. Then

$$[T - \lambda I]_{\mathcal{B}} = [T]_{\mathcal{B}} - \lambda [I]_{\mathcal{B}} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - \lambda & 1 \\ 0 & 1 - \lambda \end{pmatrix}.$$

We can immediately see that if $\lambda = 1$, the operator $T - \lambda I$ will not be invertible. Indeed, note that

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

This implies that $(T - I)(1, 0) = (0, 0)$ and so $T - I$ is not injective. Thus, $\lambda = 1$ is an eigenvalue of T . Next, suppose that $\lambda \neq 1$. Suppose that $(T - \lambda I)(x, y) = (0, 0)$. Then

$$\begin{pmatrix} 0 \\ 0 \end{pmatrix} = \begin{pmatrix} x + y \\ y \end{pmatrix} - \lambda \begin{pmatrix} x \\ y \end{pmatrix} \tag{4.2.1}$$

$$= \begin{pmatrix} (1 - \lambda)x + y \\ (1 - \lambda)y \end{pmatrix}. \tag{4.2.2}$$

Since $1 - \lambda \neq 0$, it follows that $y = 0$ by the second component. This implies that $(1 - \lambda)x = 0$, and so $x = 0$. Thus, $(x, y) = (0, 0)$, and so $T - \lambda I$ is injective. This implies λ is not an eigenvalue of T .

Thus, the only eigenvalue of T is 1.

FINISH

Definition 4.2.8. Let $\lambda \in \mathbb{F}$ be an eigenvalue of $T \in \mathcal{L}(V)$. The **eigenspace of λ** is the subspace of all eigenvectors associated to λ , that is,

$$E_\lambda(T) := \ker(T - \lambda I).$$

The **geometric multiplicity of λ** , when defined, is $\dim E_\lambda(T)$.

Proposition 4.2.9. Let $T \in \mathcal{L}(V)$, and suppose that $v_1, \dots, v_k \in V$ are eigenvectors of T with associated eigenvalues λ_j . If all λ_j 's are distinct (that is, if $\lambda_i \neq \lambda_j$ for all i, j) then $\{v_1, \dots, v_k\}$ is linearly independent.

Proof. We will prove the case $k = 2$, and leave the general case as an exercise. (Hint: use induction!).

Suppose that v_1, v_2 are eigenvectors of T with associated eigenvalues $\lambda_1 \neq \lambda_2$. Suppose that $a_1 v_1 + a_2 v_2 = 0$ for some scalars $a_1, a_2 \in \mathbb{F}$. Apply T to both sides gives

$$T(a_1 v_1 + a_2 v_2) = T(0) \quad \Rightarrow \quad a_1 T(v_1) + a_2 T(v_2) = 0.$$

Since $T(v_j) = \lambda_j v_j$, we have $a_1 \lambda_1 v_1 + a_2 \lambda_2 v_2 = 0$. On the other hand, since $a_1 v_1 + a_2 v_2 = 0$, multiplying both sides by λ_1 gives

$$a_1 \lambda_1 v_1 + a_2 \lambda_1 v_2 = 0.$$

Thus,

$$a_1 \lambda_1 v_1 + a_2 \lambda_1 v_2 = 0 = a_1 \lambda_1 v_1 + a_2 \lambda_2 v_2.$$

This implies $a_2(\lambda_1 - \lambda_2)v_2 = 0$. Since $\lambda_1 - \lambda_2 \neq 0$, it follows that $a_2 = 0$. This implies that $a_1 v_1 = 0$, and since $v_1 \neq 0$ (it is an eigenvector), we have $a_1 = 0$. Thus, since $a_1 = a_2 = 0$, $\{v_1, v_2\}$ is linearly independent. $\square$

Corollary 4.2.10. Let V be finite dimensional, and let $T \in \mathcal{L}(V)$. Then T has at most $\dim V$ distinct eigenvalues.

Proof. Let $\lambda_1, \dots, \lambda_k$ be distinct eigenvalues. Let v_j be an eigenvector associated to λ_j . Then by the proposition, $\{v_1, \dots, v_k\}$ is linearly independent. By a theorem from a long time ago, it then follows that $k \leq \dim V$. $\square$

Corollary 4.2.11. Let V be finite dimensional, and suppose that $T \in \mathcal{L}(V)$ has $\dim V$ distinct eigenvalues. Then T is diagonalizable.

Proof. Let $n := \dim V$ and let $\lambda_1, \dots, \lambda_n$ be the distinct eigenvalues of T . Let v_j be an eigenvector associated to λ_j . Then $\{v_1, \dots, v_n\}$ is a linearly independent set in V . Since $n = \dim V$, it is a basis. Thus, V has a basis of eigenvectors of T , and so T is diagonalizable. $\square$

Proposition 4.2.12. Suppose that $T, S \in \mathcal{L}(V)$ are similar. Then T and S have the same eigenvalues.

Proof. Let $\phi \in \mathcal{L}(V)$ be an isomorphism such that $T = \phi^{-1} \circ S \circ \phi$. Let $\lambda \in \mathbb{F}$ be an eigenvalue of S . Then $S - \lambda I$ is not injective. We claim that $T - \lambda I$ is not injective as well. Note that

$$\begin{aligned} T - \lambda I &= \phi^{-1} \circ S \circ \phi - \lambda I \\ &= \phi^{-1} \circ S \circ \phi - \lambda \phi^{-1} \circ I \circ \phi \\ &= \phi^{-1} \circ (S - \lambda I) \circ \phi. \end{aligned}$$

Let $v \neq 0 \in \ker S - \lambda I$. Let $w := \phi^{-1}(v) \in V$. Note that $w \neq 0$, since ϕ^{-1} is injective. We have

$$\begin{aligned} (T - \lambda I)(w) &= (\phi^{-1} \circ (S - \lambda I) \circ \phi)(w) \\ &= \phi^{-1}((S - \lambda I)(v)) \\ &= \phi^{-1}(0) \\ &= 0 \end{aligned}$$

Since $w \neq 0$, this shows that $T - \lambda I$ is not injective, and so λ is an eigenvalue of T .

By a symmetric argument, if λ is an eigenvalue of T then it is also an eigenvalue of S . Thus, T and S have the same eigenvalues. $\square$

Next, a simple (but important) example to demonstrate that eigenvalues do not necessarily always exist.

Example 4.2.13. Consider the map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (-y, x)$. Geometrically, this map rotates every vector counter clockwise by 90 degree. From this description, it should be clear that there are no eigenvalues, but we will prove this carefully. Fix $\lambda \in \mathbb{R}$ and consider the operator $T - \lambda I$. Note that

$$(T - \lambda I)(x, y) = T(x, y) - \lambda(x, y) = (-y - \lambda x, x - \lambda y).$$

We claim that $T - \lambda I$ is injective. Indeed, suppose that $(T - \lambda I)(x, y) = (0, 0)$. Then $(-y - \lambda x, x - \lambda y) = (0, 0)$, and so

$$\begin{cases} y + \lambda x = 0 \\ x - \lambda y = 0 \end{cases}.$$

The second equation implies $x = \lambda y$, and so $y + \lambda(\lambda y) = 0$. This implies $y(1 + \lambda^2) = 0$. Since $1 + \lambda^2 > 0$, it follows that $y = 0$. Since $x = \lambda y$, $x = 0$. Thus, $(x, y) = (0, 0)$, and so $T - \lambda I$ is injective. Thus, since $T - \lambda I$ is injective for all $\lambda \in \mathbb{R}$, there are no eigenvalues of T .

In contrast, we have the following powerful theorem that suggests one reason why complex numbers are nice.

Theorem 4.2.14. Let V be a finite dimensional vector space over $\mathbb{C}$. Let $T \in \mathcal{L}(V)$. Then T has an eigenvalue.

The proof is a classic argument that uses the fundamental theorem of algebra, which we recall here.

Theorem 4.2.15 (Fundamental theorem of algebra). *Any polynomial $p \in P(\mathbb{C})$ can be completely factored into linear factors.*

Proof of the previous theorem. Let $n = \dim V$. Fix a nonzero vector $v \in V$. Consider the following list of vectors:

$$\{v, T(v), T^2(v), \dots, T^n(v)\}$$

where by T^2 we mean $T \circ T$, and so on. Since this is a list of $n + 1$ vectors, this set is linearly dependent. Thus, there exist scalars $a_0, a_1, \dots, a_n \in \mathbb{C}$, not all 0, such that

$$a_0v + a_1T(v) + a_2T^2(v) + \dots + a_nT^n(v) = 0.$$

Thus,

$$(a_0I + a_1T + a_2T^2 + \dots + T^n)(v) = 0.$$

By the fundamental theorem of algebra, the left-hand side can be factored as

$$c[(T - \lambda_1I) \circ (T - \lambda_2I) \circ \dots \circ (T - \lambda_nI)](v) = 0$$

where $\lambda_1, \dots, \lambda_n$ are the (possibly repeated) roots of the polynomial $a_0 + a_1z + \dots + a_nz^n$. Since $v \neq 0$, this shows that $(T - \lambda_1I) \circ (T - \lambda_2I) \circ \dots \circ (T - \lambda_nI)$ is not injective, and hence not invertible. Thus, at least one of the factors $(T - \lambda_jI)$ is not invertible, otherwise the above composition would be invertible. Thus, λ_j is an eigenvalue of T . $\square$

Exercises

1. Let $T, S \in \mathcal{L}(V)$. Prove that if $v \in V$ is an eigenvector of T with eigenvalue λ , and is also an eigenvector of S with eigenvalue μ , then v is an eigenvector of $S \circ T$ with eigenvalue $\lambda\mu$.
2. Recall that $P(\mathbb{R})$ is the vector space of polynomials (of any degree) with coefficients in $\mathbb{R}$. Let $T : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be the linear operator defined by $T(p(x)) := xp(x)$. Prove that T has no eigenvalues.
3. Recall that $F(\mathbb{R}, \mathbb{R})$ is the vector space of all functions $f : \mathbb{R} \rightarrow \mathbb{R}$. Let $T : F(\mathbb{R}, \mathbb{R}) \rightarrow F(\mathbb{R}, \mathbb{R})$ be the operator defined by $T(f(x)) = xf(x)$. Prove that T does have eigenvalues. In fact, prove that every single real number is an eigenvalue of T .

Why is this problem not a contradiction to the previous problem?

4. Similarly, recall that the derivative operator $\frac{d}{dx} : C^\infty(\mathbb{R}) \rightarrow C^\infty(\mathbb{R})$ has many eigenvalues. Prove that, in contrast, the derivative operator $\frac{d}{dx} : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ does not have any eigenvalues.
5. Prove that a linear operator $T : V \rightarrow V$ on a finite dimensional vector space is invertible if and only if 0 is not an eigenvalue of T .

6. Suppose that $T : V \rightarrow V$ is invertible, and suppose that λ is an eigenvalue of T . Prove that λ^{-1} is an eigenvalue of T^{-1} .
7. Suppose that $T \in \mathcal{L}(V)$ such that $\dim \operatorname{im} T = k$. Prove that T has at most $k + 1$ distinct eigenvalues.
8. Prove that if $T \in \mathcal{L}(V)$ is diagonalizable, then T^{-1} is diagonalizable.
9. Let V be a vector space of dimension 4, and let $T \in \mathcal{L}(V)$. Suppose that $\lambda \in \mathbb{F}$ is an eigenvalue of T such that $\dim E_\lambda(T) = 4$. Let $\mathcal{B}$ be any basis of V . Compute the coordinate matrix $[T]_{\mathcal{B}}$.
10. Suppose that V is a finite dimensional vector space, and that $T \in \mathcal{L}(V)$ is diagonalizable. Prove that

$$V = \ker T \oplus \operatorname{im} T.$$

11. Fix $A \in \mathcal{L}(V)$, and define a linear map $\phi_A : \mathcal{L}(V) \rightarrow \mathcal{L}(V)$ by $\phi_A(T) := A \circ T$. Prove that if $\lambda \in \mathbb{F}$ is an eigenvalue of ϕ_A , then λ is also an eigenvalue of A .
12. Let V be a real vector space, let $T, S \in \mathcal{L}(V)$, and let $\lambda \neq 0 \in \mathbb{R}$.

Suppose that $v \in V$ is an eigenvector of T with eigenvalue λ and an eigenvector of S with eigenvalue λ . Suppose that $w \in V$ is an eigenvector of T with eigenvalue λ and an eigenvector of S with eigenvalue $-\lambda$.

Prove that $\{T, S\} \subset \mathcal{L}(V)$ is linearly independent.

13. Let V be a real vector space of dimension 3, let $T, S \in \mathcal{L}(V)$, and let $v_1, v_2, v_3 \in V$.
Suppose that v_1 is an eigenvector of T with eigenvalue 1, v_2 is an eigenvector of T with eigenvalue 3, and v_3 is an eigenvector of T with eigenvalue 0.
Suppose also that v_1 is an eigenvector of S with eigenvalue 4, v_2 is an eigenvector of S with eigenvalue 7, and v_3 is an eigenvector of S with eigenvalue -3 .
Prove that $T \circ S = S \circ T$.

14. Let V be a real vector space. Let $T, S \in \mathcal{L}(V)$. Suppose that S is invertible, and suppose that 0, 1, 2 are eigenvalues of T .
Prove that $\{T, T^2, S\}$ is linearly independent.

15. Give an example of operators $T, S \in \mathcal{L}(V)$ such that T is diagonalizable, but $S \circ T$ is not diagonalizable.
16. Let $T : F(\mathbb{R}, \mathbb{R}) \rightarrow F(\mathbb{R}, \mathbb{R})$ be the operator defined by $T(f)(x) = x^2 f(x)$. Give an example of an eigenvector of T .
17. For each of the following statements, determine whether it is *true* or *false*. If a statement is true, prove it. If it is false, provide an explicit counter example.

Throughout, assume that V is a nontrivial vector space.

- (a) If $T \in \mathcal{L}(V)$ is invertible, then T has an eigenvector.

- (b) If $T \in \mathcal{L}(V)$ satisfies $T^2 = T$, then T has an eigenvector.
 - (c) If $v, w \in V$ are eigenvectors of $T \in \mathcal{L}(V)$, both with eigenvalue $\lambda \in \mathbb{F}$, then $\{v, w\}$ is linearly dependent.
 - (d) If $T \in \mathcal{L}(V)$ is diagonalizable, then $[T]_{\mathcal{B}}$ is a diagonal matrix for every basis $\mathcal{B}$ of V .
 - (e) Suppose that $T \in \mathcal{L}(V)$ is diagonalizable. Then every element $v \in V$ is an eigenvector of T .
 - (f) If $T \in \mathcal{L}(V)$ and $\mathcal{B}_1, \mathcal{B}_2$ are two (ordered) bases of V such that the coordinate matrix $[T]_{\mathcal{B}_1}^{\mathcal{B}_2}$ is diagonal, then T is diagonalizable.
 - (g) If λ is an eigenvalue of $T + S$, then λ is an eigenvalue of both T and S .
18. Let $T, S \in \mathcal{L}(V)$ such that T and S are diagonalizable via the same basis. That is, there exists a basis $\mathcal{B}$ of V such that $[T]_{\mathcal{B}}$ and $[S]_{\mathcal{B}}$ are diagonal matrices. Prove that $T \circ S = S \circ T$.
19. A subspace U of a vector space V is called **invariant under** $T \in \mathcal{L}(V)$ if $T(U) \subset U$.
- (a) Suppose that E_{λ} is an eigenspace of $T \in \mathcal{L}(V)$. Prove that E_{λ} is invariant under T .
 - (b) Suppose that $T, S \in \mathcal{L}(V)$ such that $T \circ S = S \circ T$. Prove that $\ker S$ and $\text{im } S$ are invariant under T .
20. The **spectrum** of a linear operator is the set of scalars $\lambda \in \mathbb{F}$ such that $T - \lambda I$ is not invertible.
- (a) Prove that if V is finite dimensional then the spectrum of T is precisely the set of eigenvalues of T .
 - (b) Let $T : P(\mathbb{R}) \rightarrow P(\mathbb{R})$ be given by $T(p(x)) = xp(x)$. Recall from last week that T does not have any eigenvalues. On the other hand, prove that 0 is in the spectrum of T .

4.3 Diagonalizability as a direct sum of eigenspaces

Our next goal is to prove the following theorem.

Theorem 4.3.1. *Let V be a finite dimensional vector space. Then $T \in \mathcal{L}(V)$ is diagonalizable if and only if*

$$V = E_{\lambda_1}(T) \oplus \cdots \oplus E_{\lambda_k}(T)$$

where $\lambda_1, \dots, \lambda_k$ are the (distinct) eigenvalues of T .

Before we prove this theorem, we need to discuss the direct sum of multiple subspaces, rather than just two. This is a natural generalization of the normal direct sum.

Definition 4.3.2. Let $W_1, \dots, W_k$ be subspaces of a vector space V . Define

$$W_1 + \cdots + W_k := \{ w_1 + \cdots + w_k \mid w_j \in W_j \}.$$

We say that V is the **direct sum** of $W_1, \dots, W_k$, written

$$V = W_1 \oplus \dots \oplus W_k$$

if $V = W_1 + \dots + W_k$ and if $W_i \cap W_j = \emptyset$ for all $i \neq j$.

Example in $\mathbb{R}^3$

Lemma 4.3.3. Suppose that $\lambda \neq \mu$ are two eigenvalues of $T \in \mathcal{L}(V)$. Then $E_\lambda(T) \cap E_\mu(T) = \{0\}$.

Proof. Suppose that $v \in E_\lambda(T) \cap E_\mu(T)$. Then $T(v) = \lambda v$ and $T(v) = \mu v$. So $(\lambda - \mu)v = 0$. Since $\lambda - \mu \neq 0$, it follows that $v = 0$. This proves that $E_\lambda(T) \cap E_\mu(T) = \{0\}$. $\square$

Proof of the above theorem. First, suppose that T is diagonalizable. Let $\lambda_1, \dots, \lambda_k$ be the distinct eigenvalues of T . Let

$$\{v_1^1, \dots, v_{j_1}^1, \dots, v_1^k, \dots, v_{j_k}^k\}$$

be an eigenbasis of V such that $\{v_1^i, \dots, v_{j_i}^i\}$ are eigenvectors associated to λ_i . Note that $\{v_1^i, \dots, v_{j_i}^i\}$ is a basis of $E_{\lambda_i}(T)$. By the above lemma, $E_{\lambda_i}(T) \cap E_{\lambda_j}(T) = \{0\}$ for all $i \neq j$, so we simply have to show that $V = E_{\lambda_1}(T) + \dots + E_{\lambda_k}(T)$. Fix $v \in V$. Write

$$v = (a_1^1 v_1^1 + \dots + a_{j_1}^1 v_{j_1}^1) + \dots + (a_1^k v_1^k + \dots + a_{j_k}^k v_{j_k}^k).$$

Since $a_1^i v_1^i + \dots + a_{j_i}^i v_{j_i}^i \in E_{\lambda_i}(T)$, this proves that $V = E_{\lambda_1}(T) + \dots + E_{\lambda_k}(T)$. Thus, $V = E_{\lambda_1}(T) \oplus \dots \oplus E_{\lambda_k}(T)$.

Next, suppose that $V = E_{\lambda_1}(T) \oplus \dots \oplus E_{\lambda_k}(T)$. Let $\mathcal{B}_i = \{v_1^i, \dots, v_{j_i}^i\}$ be a basis for the subspace $E_{\lambda_i}(T)$ for each $i = 1, \dots, k$. Note that each element of $\mathcal{B}_i$ is an eigenvector of T with eigenvalue λ_i . We claim that

$$\mathcal{B} := \mathcal{B}_1 \cup \dots \cup \mathcal{B}_k$$

is a basis of V . Fix $v \in V$. Since $V = E_{\lambda_1}(T) + \dots + E_{\lambda_k}(T)$, we can write $v = v_1 + \dots + v_k$ where each $v_i \in E_{\lambda_i}$. We can further write $v_i = a_1^i v_1^i + \dots + a_{j_i}^i v_{j_i}^i$, and so

$$v = (a_1^1 v_1^1 + \dots + a_{j_1}^1 v_{j_1}^1) + \dots + (a_1^k v_1^k + \dots + a_{j_k}^k v_{j_k}^k).$$

Thus, $\text{Span}(\mathcal{B}) = V$. It remains to prove that $\mathcal{B}$ is linearly independent. FINISH $\square$

Corollary 4.3.4. Let V be a finite dimensional vector space. Then $T \in \mathcal{L}(V)$ is diagonalizable if and only if the sum of the geometric multiplicities of all eigenvalues of T is $\dim V$. That is,

$$\sum_{j=1}^k \dim E_{\lambda_j}(T) = \dim V.$$

Proof. Recall from the practice midterm 1 problems that $\dim(W_1 \oplus W_2) = \dim W_1 + \dim W_2$. One can similarly show that

$$\dim(W_1 \oplus \dots \oplus W_k) = \dim W_1 + \dots + \dim W_k.$$

The corollary immediately follows from this observation. $\square$

Example, think about this in the context of the non diagonalizable operator from Friday

Exercises

1. Suppose that $\dim V = n$, and that $T \in \mathcal{L}(V)$ has two eigenvalues $\lambda \neq \mu \in \mathbb{F}$. Prove that if $\dim E_\lambda(T) = n - 1$, then T is diagonalizable.
2. Let V be a finite dimensional vector space such that $V = U \oplus W$ for some nonzero subspaces $U, W \subset V$. Define a map $P : V \rightarrow V$ by $P(u + w) = u$ for $u \in U, w \in W$. Prove that P is diagonalizable.
3. Let V be a finite dimensional real vector space, and suppose that $V = W_1 \oplus W_2$ for some subspaces W_1, W_2 . Define a linear map $T : V \rightarrow V$ as follows. For each $v \in V$, write v uniquely as $v = w_1 + w_2$ for $w_1 \in W_1$ and $w_2 \in W_2$. Then define $T(v) := 3w_1$.
 - (a) Compute $\ker T$.
 - (b) Compute $\operatorname{im} T$.
 - (c) Prove that T is diagonalizable.

4.4 Computational aspects of eigenvalues via the determinant

To close out our discussion of diagonalization and eigenvalue theory (which is much more extensive than what we've seen!), I'll remind you of some facts from lower division linear algebra about the determinant and the characteristic polynomial. Chapter 6 develops the theory of determinants much more thoroughly; as far as these notes are concerned, this brief section is optional.

Proposition 4.4.1. *There exists a map $\det : M_n(\mathbb{F}) \rightarrow \mathbb{F}$ called the **determinant** that satisfies the following properties:*

(i)

$$\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc;$$

(ii) $\det(AB) = (\det A)(\det B)$;

(iii) $\det(A^{-1}) = (\det A)^{-1}$;

(iv) A is invertible if and only if $\det A \neq 0$.

A good question is: given a linear operator, how in general can I go about finding the eigenvalues? Beyond what we've discussed above, another computational option is to use the characteristic polynomial.

Definition 4.4.2. Let V be an n -dimensional vector space. Let $T \in \mathcal{L}(V)$. Let $\mathcal{B}$ be a basis of V . The **characteristic polynomial** of T is

$$c_T(\lambda) := \det([T]_{\mathcal{B}} - \lambda I_n)$$

Proposition 4.4.3. *Let V be a finite dimensional vector space, and let $T \in \mathcal{L}(V)$.*

- (i) *The characteristic polynomial $c_T(\lambda)$ does not depend on the choice of basis in the previous definition.*
- (ii) *The eigenvalues of T are precisely the zeroes of $c_T(\lambda)$, viewed as a polynomial over $\mathbb{F}$.*

Proof.

- (i) Let $\mathcal{B}'$ be another basis of V . Let Q be the change of coordinate matrix from $\mathcal{B}$ to $\mathcal{B}'$. Then

$$\begin{aligned}
 \det([T]_{\mathcal{B}} - \lambda I_n) &= \det(Q^{-1}[T]_{\mathcal{B}'}Q - \lambda Q^{-1}I_nQ) \\
 &= \det(Q^{-1}([T]_{\mathcal{B}'} - \lambda I_n)Q) \\
 &= \det(Q^{-1}) \det([T]_{\mathcal{B}'} - \lambda I_n) (\det Q) \\
 &= (\det Q)^{-1} (\det Q) \det([T]_{\mathcal{B}'} - \lambda I_n) \\
 &= \det([T]_{\mathcal{B}'} - \lambda I_n).
 \end{aligned}$$

- (ii) Note that $T - \lambda I$ is invertible if and only if $[T - \lambda I]_{\mathcal{B}} = [T]_{\mathcal{B}} - \lambda I_n$ is invertible. This matrix is invertible if and only if

$$c_T(\lambda) = \det([T]_{\mathcal{B}} - \lambda I_n) \neq 0.$$

Thus, the eigenvalues of T are precisely the scalars $\lambda \in \mathbb{F}$ such that $c_T(\lambda) = 0$.

□

Exercises

Chapter 5

Inner Product Spaces

Let V be an abstract vector space, as we have studied all throughout this text. Here are some hypothetical questions. Given two vectors $v, w \in V$, can we ask if the vectors are *orthogonal*, or more generally about the *angle* between these two vectors? Can we ask if v is *longer* than w ? Can we ask about the *distance* between v and w ?

The answer is *no*! In an abstract vector space, there is no notion of angle, length, or distance. You might think that these concepts exist in $\mathbb{R}^n$, but that comes from your previous bias about Euclidean space. In other words, I am describing the following principle.

Principle 5.0.1. In an abstract vector space, there is no notion of geometry.

By *geometry* I mean properties like length, angle, distance, and so on. A vector space is a purely algebraic structure, and although we like to visualize vector spaces like $\mathbb{R}^2$ and $\mathbb{R}^3$, this involves cluttering the pure algebra of a vector space with our subconscious understanding of geometry. In reality, no such thing exists in a vector space.

Of course, it would be nice to introduce the notion of geometry into a vector space. In $\mathbb{R}^2$, $\mathbb{R}^3$, and more generally $\mathbb{R}^n$, we already have a familiar notion of Euclidean geometry that is captured almost entirely by the dot product. In particular, recall that we can test orthogonality, compute angles, and compute lengths in the Euclidean sense all in terms of the dot product. We want to introduce geometry into vector spaces, and this is done with something called an *inner product*. An inner product is an extra amount of structure that one can place on an abstract vector space that is meant to generalize the familiar notion of a dot product. Using an inner product, we can breathe geometry into the cold algebraic world of abstract vector spaces. This is the driving principle behind this chapter.

Principle 5.0.2. One can bestow the notion of geometry onto an abstract vector space with a choice of something called an *inner product*.

5.1 Inner products

For now, we will consider vector spaces over either $\mathbb{R}$ or $\mathbb{C}$. We briefly recall some notions about complex numbers. The *conjugate* of $a + bi \in \mathbb{C}$ is

$$\overline{a + bi} := a - bi \in \mathbb{C}$$

and the *modulus* of a complex number $a + bi \in \mathbb{C}$ is

$$|a + bi| := \sqrt{a^2 + b^2}.$$

Note that if $\lambda \in \mathbb{R}$ is a real number, we have $\overline{\lambda} = \lambda$. One can prove that if $z, w \in \mathbb{C}$, then $\overline{zw} = \overline{z} \overline{w}$ and that $z\overline{z} = |z|^2$.

Definition 5.1.1. Let V be a (real or complex) vector space. An **inner product** is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ satisfying the following properties.

- (1) $\langle v + v', w \rangle = \langle v, w \rangle + \langle v', w \rangle$ and $\langle \lambda v, w \rangle = \lambda \langle v, w \rangle$ for all $v, v', w \in V$ and $\lambda \in \mathbb{F}$.
- (2) $\overline{\langle v, w \rangle} = \langle w, v \rangle$ for all $v, w \in V$.
- (3) $\langle v, v \rangle > 0$ if $v \neq 0 \in V$.

Example 5.1.2. Consider the vector space $\mathbb{R}^n$. The **standard inner product**, denoted $\langle \cdot, \cdot \rangle_{\text{std}}$ also known as the **dot product**, is given by

$$\left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\rangle_{\text{std}} := x_1 y_1 + \cdots + x_n y_n.$$

This really is the usual dot product that you are familiar with, and it does indeed satisfy all of the properties of Definition 5.1.1. Indeed, for example, note that

$$\begin{aligned} \left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\rangle_{\text{std}} &= \left\langle \begin{pmatrix} x_1 + x'_1 \\ \vdots \\ x_n + x'_n \end{pmatrix}, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\rangle_{\text{std}} \\ &= (x_1 + x'_1)y_1 + \cdots + (x_n + x'_n)y_n \\ &= x_1 y_1 + x'_1 y_1 + \cdots + x_n y_n + x'_n y_n \\ &= (x_1 y_1 + \cdots + x_n y_n) + (x'_1 y_1 + \cdots + x'_n y_n) \\ &= \left\langle \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\rangle_{\text{std}} + \left\langle \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \end{pmatrix}, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right\rangle_{\text{std}}. \end{aligned}$$

This proves that $\langle v + v', w \rangle_{\text{std}} = \langle v, w \rangle_{\text{std}} + \langle v', w \rangle_{\text{std}}$ for all $v, v', w \in \mathbb{R}^n$, which is half of axiom (1) in Definition 5.1.1. Verification of the rest of the axioms are left as an exercise.

Example 5.1.3. There is also a standard inner product on $\mathbb{C}^n$, but it is necessary to introduce a conjugate in order to satisfy (2) and (3) of Definition 5.1.1. The **standard inner product** on $\mathbb{C}^n$ is given by

$$\left\langle \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}, \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \right\rangle_{\text{std}} := z_1 \bar{w}_1 + \cdots + z_n \bar{w}_n.$$

For example,

$$\left\langle \begin{pmatrix} 1+i \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 1-i \end{pmatrix} \right\rangle = (1+i)\bar{2} + 3\bar{1-i} = (1+i)2 + 3(1+i) = 5 + 5i.$$

One can verify that $\langle \cdot, \cdot \rangle_{\text{std}}$ satisfies all of the properties of Definition 5.1.1. For example, note that

$$\begin{aligned} \overline{\left\langle \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}, \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix} \right\rangle_{\text{std}}} &= \overline{z_1 \bar{w}_1 + \cdots + z_n \bar{w}_n} \\ &= \bar{z}_1 \bar{\bar{w}_1} + \cdots + \bar{z}_n \bar{\bar{w}_n} \\ &= \bar{z}_1 w_1 + \cdots + \bar{z}_n w_n \\ &= \left\langle \begin{pmatrix} w_1 \\ \vdots \\ w_n \end{pmatrix}, \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \right\rangle_{\text{std}} \end{aligned}$$

which verifies (2) of Definition 5.1.1. The rest of the properties are left as an exercise.

Example 5.1.4. Crucially, *there is more than one choice of inner product on any given vector space*. For example, consider the vector space $\mathbb{R}^2$. We could of course choose the standard inner product, i.e., the dot product, as in example 5.1.2:

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle_{\text{std}} = x_1 y_1 + x_2 y_2.$$

However, there are other inner products we could choose. For example, define an inner product $\langle \cdot, \cdot \rangle_2$, where the subscript 2 is just used to differentiate this inner product from the standard one, by

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle_2 := 2x_1 y_1 + x_2 y_1 + x_1 y_2 + 2x_2 y_2.$$

Note that we can alternatively describe $\langle \cdot, \cdot \rangle_2$ by using a 2×2 matrix as follows:

$$\begin{aligned} \left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle_2 &= 2x_1y_1 + x_2y_1 + x_1y_2 + 2x_2y_2 \\ &= y_1(2x_1 + x_2) + y_2(x_1 + 2x_2) \\ &= \begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 \end{pmatrix} \\ &= \begin{pmatrix} y_1 & y_2 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}. \end{aligned}$$

Again, it is possible to verify all of the inner product axioms in Definition 5.1.1 for this non-standard inner product $\langle \cdot, \cdot \rangle$. For the sake of variety, we will prove (3) and leave the rest as an exercise. For any $v = (x_1, x_2) \in \mathbb{R}^2$, note that

$$\langle v, v \rangle_2 = \left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \right\rangle_2 = 2x_1x_1 + x_2x_1 + x_1x_2 + 2x_2x_2 = 2x_1^2 + 2x_1x_2 + 2x_2^2.$$

Now for something sneaky.¹ Note that

$$\frac{1}{2}(2x_1 + x_2)^2 + \frac{3}{2}x_2^2 = \frac{1}{2}(4x_1^2 + 4x_1x_2 + x_2^2) + \frac{3}{2}x_2^2 = 2x_1^2 + 2x_1x_2 + 2x_2^2.$$

Thus,

$$\langle v, v \rangle_2 = \frac{1}{2}(2x_1 + x_2)^2 + \frac{3}{2}x_2^2.$$

In particular, since $x_2^2 \geq 0$ and $(2x_1 + x_2)^2 \geq 0$, it follows that $\langle v, v \rangle \geq 0$ for all $v \in \mathbb{R}^2$. Moreover, if $v \neq 0$, then at least one of x_1 or x_2 will be nonzero, and so in this case $\langle v, v \rangle > 0$. This verifies (3). The rest of the properties are left as an exercise.

Example 5.1.5. Let $V = C[-1, 1] = \{ f : [-1, 1] \rightarrow \mathbb{R} \mid f \text{ continuous} \}$ be the set of continuous real-valued functions on the interval $[-1, 1]$. Define an inner product by

$$\langle f, g \rangle := \int_{-1}^1 f(x)g(x) \, dx.$$

The fact that $\langle \cdot, \cdot \rangle$ is linear in the first argument follows from linearity of integration, and the fact that $\langle f, g \rangle = \langle g, f \rangle$ is² trivial, so this verifies the first two requirements of an inner product. The most nontrivial condition to check is the statement that if $f \neq 0$, then $\langle f, f \rangle > 0$. We leave this as an exercise, since it is more in the realm of analysis. The idea is to observe that

$$\langle f, f \rangle = \int_{-1}^1 [f(x)]^2 \, dx.$$

Since $[f(x)]^2 \geq 0$, and since f is continuous, the only way for the integral to be equal to 0 is for $f = 0$.

¹To see where this sneakiness comes from, check out the exercises!

²Note that this is a real vector space, so (2) in Definition 5.1.1 just becomes $\langle f, g \rangle = \langle g, f \rangle$.

Definition 5.1.6. An **inner product space** is a vector space together with a choice of inner product.

Remark 5.1.7. The above examples should emphasize the fact that a given vector space can be viewed as an inner product space in different ways. For example, $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_{\text{std}})$ and $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$, where the indicated inner products are the ones from the examples above, are *different* inner product spaces. In practice, I will be a bit sloppy and will frequently just refer to V as an inner product space, omitting the explicit choice of inner product.

Proposition 5.1.8. Let V be an inner product space. Then

- (i) $\langle v, w + w' \rangle = \langle v, w \rangle + \langle v, w' \rangle$
- (ii) $\langle v, \lambda w \rangle = \bar{\lambda} \langle v, w \rangle$
- (iii) $\langle v, 0 \rangle = \langle 0, v \rangle = 0$
- (iv) $\langle v, v \rangle = 0$ if and only if $v = 0$.

Proof. We prove (ii), and leave the rest as an exercise. Note that (1) and (2) of the definition of inner product together imply

$$\begin{aligned} \langle v, \lambda w \rangle &= \overline{\langle \lambda w, v \rangle} \\ &= \overline{\lambda \langle w, v \rangle} \\ &= \bar{\lambda} \overline{\langle w, v \rangle} \\ &= \bar{\lambda} \langle v, w \rangle. \end{aligned}$$

□

Remark 5.1.9. We say that an inner product is *conjugate linear* in the second component. When $\mathbb{F} = \mathbb{R}$, (i) and (ii) above, together with (1) in Definition 5.1.1, imply that an inner product is bilinear.

Now we can introduce the first notions of geometry into an inner product space. Again, none of these notions exist a priori — they are *defined* using an inner product.

Definition 5.1.10. Let V be an inner product space. The **length** or **norm** or **magnitude** of a vector $v \in V$ is $\|v\| := \sqrt{\langle v, v \rangle}$. A **unit vector** is a vector such that $\|v\| = 1$.

Remark 5.1.11. Note that this definition makes sense because of the axiom $\langle v, v \rangle > 0$ if $v \neq 0 \in V$ in the definition of an inner product. In particular, even if V is a vector space over the complex numbers, $\langle v, v \rangle \geq 0$ is a non-negative *real* number for all $v \in V$, and hence $\sqrt{\langle v, v \rangle}$ is well-defined.

Here we state some basic properties of the norm associated to an inner product. These should be familiar from your previous understanding of lengths of vectors in multivariable calculus.

Proposition 5.1.12. Let V be an inner product space. Then

- (i) $\|\lambda v\| = |\lambda| \|v\|$

(ii) $v = 0$ if and only if $\|v\| = 0$.

Proof. This is an exercise, which is immediate from the previous proposition about inner products. $\square$

The notion of angle, and in particular of orthogonality, is also central to the geometry of inner product spaces.

Definition 5.1.13. Let V be an inner product space. Two vectors $v, w \in V$ are **orthogonal**, sometimes written $v \perp w$, if $\langle v, w \rangle = 0$.

Example 5.1.14. Consider $\mathbb{C}^2$ with the standard inner product. Let $v = (1, i)$, and let $w = (1, -i)$. Note that

$$\langle v, w \rangle_{\text{std}} = 1 \cdot 1 + i \cdot \overline{-i} = 1 + i \cdot i = 1 + i^2 = 0.$$

Thus, v and w are orthogonal.

Example 5.1.15. Consider $\mathbb{R}^2$ with the non-standard inner product from Example 5.1.4 defined by

$$\left\langle \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \begin{pmatrix} y_1 \\ y_2 \end{pmatrix} \right\rangle_2 := 2x_1y_1 + x_2y_1 + x_1y_2 + 2x_2y_2.$$

Let $v = (1, 0)$ and $w = (1, -2)$. Note that

$$\left\langle \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -2 \end{pmatrix} \right\rangle_2 := 2(1)(1) + (0)(1) + (1)(-2) + 2(0)(-2) = 0.$$

Thus, in the inner product space $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$, the vectors $v = (1, 0)$ and $w = (1, -2)$ are orthogonal! We can also compute the lengths of the vectors v and w in this nonstandard geometric world. Note that, for example,

$$\|v\| = \sqrt{\langle v, v \rangle_2} = \sqrt{2}.$$

Likewise, one can also check that $\|w\| = \sqrt{6}$.

Remark 5.1.16. This example raises a lot of interesting borderline philosophical points about our understanding of geometry and abstract mathematics. It is tempting to look at the previous example and think *this is stupid, the vectors $(1, 0)$ and $(1, -2)$ are clearly not orthogonal because if you draw them they don't form a right angle*; for example, see the left side of Figure 5.1. But in drawing the vectors $(1, 0)$ and $(1, -2)$ in the normal way, you are automatically imposing the *standard* inner product onto your picture. In other words, the way you naturally draw $\mathbb{R}^2$ packages the geometry coming from the standard inner product. If you were to draw v and w in $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$, they would look like the right side of Figure 5.1.

Example 5.1.17. This is an informal example for those readers that are interested in probability theory. I will keep this vague for now, and will refer to Exercise 16 for more information. It turns out that, given a probability space, the set of random variables forms a vector space:

$$V = \{ X \mid X \text{ is a random variable} \}.$$

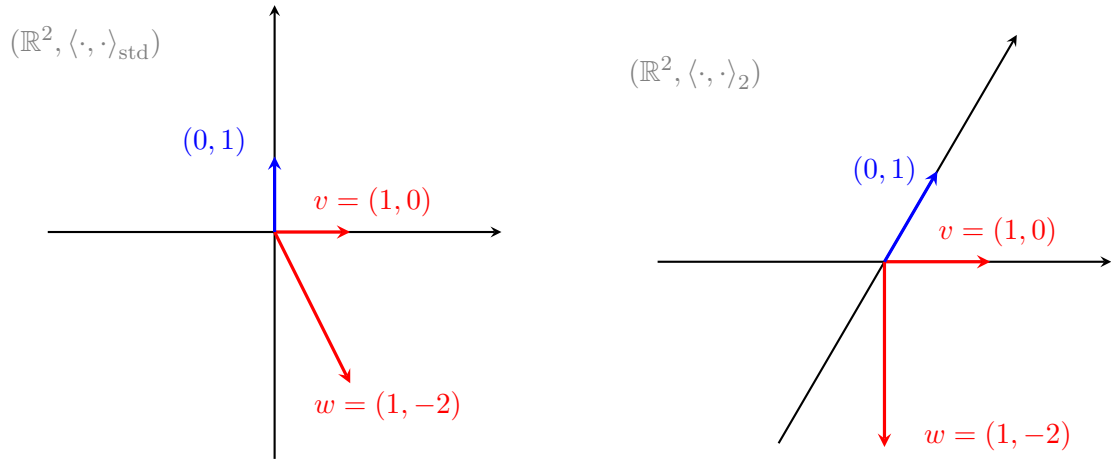


Figure 5.1: On the left is the standard inner product space $\mathbb{R}^2$ with the non-orthogonal vectors $(1, 0)$ and $(1, -2)$ drawn in red. The vector $(0, 1)$ is also drawn in blue for reference. The right picture consists of the same vectors, drawn in the nonstandard inner product space $(\mathbb{R}^2, \langle \cdot, \cdot \rangle_2)$. In both pictures the x -axis, i.e., $\text{Span}(e_1)$, and the y -axis, i.e., $\text{Span}(e_2)$, are drawn in black.

One can define an inner product on V using the covariance:

$$\langle X, Y \rangle := \text{Cov}(X, Y).$$

In this set up, two random variables being *uncorrelated* corresponds to the two random variables being orthogonal in this inner product space!

Exercises

1. Verify the rest of the inner product axioms for the standard inner products on $\mathbb{R}^n$ and $\mathbb{C}^n$ in examples 5.1.2 and 5.1.3.
2. Verify the rest of the inner product axioms for the integral inner product defined in Example 5.1.5.
3. In this exercise we will investigate non-standard inner products like $\langle \cdot, \cdot \rangle_2$ as defined in Example 5.1.4. FINISH
4. Let $C^\infty([0, 2\pi])$ be the vector space of infinitely differentiable functions $f : [0, 2\pi] \rightarrow \mathbb{R}$. Define $\langle \cdot, \cdot \rangle$ by

$$\langle f, g \rangle := \int_0^{2\pi} f'(x)g(x) dx.$$

Prove that $\langle \cdot, \cdot \rangle$ is *not* an inner product on $C^\infty([0, 2\pi])$.

5. Let V be a (real or complex) finite dimensional inner product space. Let $\mathcal{B} = \{v_1, \dots, v_n\}$ be a basis for V . Prove that if $\langle v, v_j \rangle = 0$ for all $j = 1, \dots, n$, then $v = 0$.
6. Let V be a (real or complex) inner product space with inner product $\langle \cdot, \cdot \rangle_1$. For $T \in \mathcal{L}(V)$, define $\langle \cdot, \cdot \rangle_2$ by

$$\langle v, w \rangle_2 := \langle T(v), T(w) \rangle_1.$$

- (a) Prove that if T is injective, then $\langle \cdot, \cdot \rangle_2$ is an inner product.
 (b) Prove that if T is not injective, then $\langle \cdot, \cdot \rangle_2$ is not an inner product.

7. Let $C([0, 2\pi])$ be the real inner product space of continuous functions $f : [0, 2\pi] \rightarrow \mathbb{R}$ with inner product defined by

$$\langle f, g \rangle := \int_0^{2\pi} f(x)g(x) dx.$$

Let $f, g \in C([0, 2\pi])$ be defined by $f(x) = \sin x$ and $g(x) = \cos x$. Prove that f and g are orthogonal.

8. Let V be a (real or complex) inner product space. Let $v, w \in V$. Define the **projection of v along w** , written $\text{proj}_w v \in V$, to be

$$\text{proj}_w v := \frac{\langle v, w \rangle}{\|w\|^2} w.$$

Prove that $v - \text{proj}_w v$ is orthogonal to w . Sketch a picture of what you've proven in $\mathbb{R}^2$.

9. Consider $C([0, 2\pi])$, the real vector space of continuous functions $f : [0, 2\pi] \rightarrow \mathbb{R}$. Prove that

$$\langle f, g \rangle := \int_0^\pi f(x)g(x) dx$$

is *not* an inner product on $C([0, 2\pi])$.

10. Let V be a (real or complex) inner product space and suppose that $v, w \in V$ are orthogonal. Prove that

$$\|v + w\|^2 = \|v\|^2 + \|w\|^2.$$

Draw a picture and identify what theorem you just proved.

11. Prove the *parallelogram law*: For all $v, w \in V$, where V is a (real or complex) inner product space,

$$\|v + w\|^2 + \|v - w\|^2 = 2\|v\|^2 + 2\|w\|^2.$$

Draw a picture and explain why this is called the *parallelogram law*.

12. Let T be an operator on a (real or complex) inner product space V . Prove that if $\|T(v)\| = \|v\|$ for all $v \in V$, then T is injective.

13. Let V be a real inner product space. Prove that

$$\langle v, w \rangle = \frac{1}{4} \|v + w\|^2 - \frac{1}{4} \|v - w\|^2$$

for all $v, w \in V$.

14. Let V and W be inner product spaces with inner products $\langle \cdot, \cdot \rangle_V$ and $\langle \cdot, \cdot \rangle_W$, respectively. Prove that on $V \times W$,

$$\langle (v_1, w_1), (v_2, w_2) \rangle := \langle v_1, v_2 \rangle_V + \langle w_1, w_2 \rangle_W$$

is an inner product.

15. Let V be an inner product space with $\dim V = 3$. Fix vectors $v_0, w_0 \in V$ such that $\langle v_0, w_0 \rangle = 4$. Define $T \in \mathcal{L}(V)$ by

$$T(v) := \langle v, v_0 \rangle w_0.$$

- (a) Compute, with proof, $\dim \operatorname{im} T$ and $\dim \ker T$.
 (b) Prove that T is diagonalizable.

16. ADD

5.2 The Cauchy-Schwarz inequality

The next theorem is one of the most important inequalities in all of math. For us, it will in particular help to justify the definition of angle in an inner product space.

Theorem 5.2.1 (Cauchy-Schwarz inequality). *Let V be an inner product space. Then for all $v, w \in V$,*

$$|\langle v, w \rangle| \leq \|v\| \|w\|.$$

Furthermore, equality holds if and only if v and w are parallel.

Proof. Assume that $\mathbb{F} = \mathbb{R}$. The proof in the complex case is essentially identical, with some annoying technical differences because of complex numbers. Working these details out is left as an exercise.

Fix $v, w \in V$. The theorem is immediate if either v or w is 0, so assume $v, w \neq 0$. Define a function $f : \mathbb{R} \rightarrow \mathbb{R}$ as follows. For $t \in \mathbb{R}$,

$$f(t) := \langle v + tw, v + tw \rangle.$$

Note that $f(t) = \|v + tw\|^2 \geq 0$. On the other hand, note that

$$\begin{aligned} f(t) &= \langle v + tw, v + tw \rangle \\ &= \langle v, v \rangle + \langle tw, v \rangle + \langle v, tw \rangle + \langle tw, tw \rangle \\ &= \|v\|^2 + 2 \langle v, w \rangle t + \|w\|^2 t^2. \end{aligned}$$

Thus, f is a polynomial in t . To clarify and emphasize this point, we are viewing $f(t)$ as a quadratic polynomial expression

$$f(t) = (\|w\|^2) t^2 + (2 \langle v, w \rangle) t + \|v\|^2.$$

Since $f(t) \geq 0$, the discriminant ($b^2 - 4ac$) of this quadratic polynomial must be nonpositive (if it were positive, there would be two distinct real roots of $f(t)$, which would imply that $f(t) < 0$ for some values of t). Thus,

$$(2 \langle v, w \rangle)^2 - 4 (\|w\|^2) \|v\|^2 \leq 0.$$

Rearranging the inequality gives

$$\langle v, w \rangle \leq \|v\| \|w\|$$

as desired. $\square$

The equality if and only if statement is left as an exercise. $\square$

Corollary 5.2.2 (The triangle inequality). *Let V be an inner product space, and let $v, w \in V$. Then*

$$\|v + w\| \leq \|v\| + \|w\|.$$

Proof. Again, we prove the triangle inequality in the real case. The complex version is similar, with some technical differences that don't really matter.

Fix $v, w \in V$. Note that

$$\begin{aligned} \|v + w\|^2 &= \langle v + w, v + w \rangle \\ &= \langle v, v \rangle + \langle v, w \rangle + \langle w, v \rangle + \langle w, w \rangle \\ &= \|v\|^2 + 2\langle v, w \rangle + \|w\|^2. \end{aligned}$$

By the Cauchy-Schwarz inequality, $\langle v, w \rangle \leq \|v\| \|w\|$. Thus,

$$\begin{aligned} \|v + w\|^2 &\leq \|v\|^2 + 2\|v\| \|w\| + \|w\|^2 \\ &= (\|v\| + \|w\|)^2. \end{aligned}$$

Taking square roots of both sides gives

$$\|v + w\| \leq \|v\| + \|w\|$$

as desired. $\square$

Now, we can discuss the notion of *angle* in an abstract inner product space, beyond orthogonality. To continue emphasizing the main principle of this chapter, the notion of angle and orthogonality is *defined* via our choice of an inner product. It wasn't already there.

Definition 5.2.3. Let $v, w \in V$ be vectors in a real inner product space. The **angle** between v and w is

$$\theta_{v,w} := \arccos \left(\frac{\langle v, w \rangle}{\|v\| \|w\|} \right).$$

Remark 5.2.4. Note that when we define angle, we have to specialize to an inner product space over the real numbers. In particular, the domain of $\arccos x$ is $[-1, 1]$, and so this definition is well defined because of Cauchy-Schwarz!

Proposition 5.2.5. For $v, w \in V$ in an inner product space, $\langle v, w \rangle = \|v\| \|w\| \cos \theta_{v,w}$.

Proof. This is immediate from rearranging the definition of $\theta_{v,w}$ above. $\square$

As a remark, we aren't really that interested in angles in general, and I recommend not trying to use the concept of angle, or the formula $\langle v, w \rangle = \|v\| \|w\| \cos \theta_{v,w}$, in exercises and proofs in inner product spaces. It is more out of interest that we can define the notion of angle in complete generality! *Orthogonality* is a more important concept to understand and wield effectively.

Exercises

5.3 Orthonormal bases and orthogonal complements

Suppose that V is a finite dimensional inner product space. Since it is a vector space, we can discuss the notion of a basis, as we have been all quarter. In an inner product space, we can look for a special kind of basis called an *orthonormal basis*.

Definition 5.3.1. Let V be an inner product space. A set of vectors $\{v_1, \dots, v_n\} \subset V$ is **orthonormal** if $\|v_j\| = 1$ for all j , and if $\langle v_i, v_j \rangle = 0$ for $i \neq j$. An **orthonormal basis** is a basis of V which is orthonormal.

Example 5.3.2. Consider $\mathbb{R}^2$ with the standard inner product (the dot product). Then $\{(1, 0), (0, 1)\}$ is an orthonormal basis. Another orthonormal basis is

$$\left\{ \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix}, \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} \right\}.$$

Theorem 5.3.3. Let V be an inner product space.

- (i) There is an orthonormal basis of V .
- (ii) If $\{u_1, \dots, u_n\}$ is an orthonormal basis of V , then for all $v \in V$,

$$v = \sum_{j=1}^n \langle v, u_j \rangle u_j.$$

Proof.

- (i) We only provide the idea of the proof for (i). Recall the Gramm-Schmidt process. Let $\mathcal{B}$ be any basis of V , and then apply Gram-Schmidt.
- (ii) Let $v \in V$. Because $\{u_1, \dots, u_n\}$ is a basis, we can write

$$v = \sum_{j=1}^n a_j u_j$$

for some constants $a_j \in \mathbb{F}$. Note that

$$\begin{aligned} \langle v, u_i \rangle &= \left\langle \sum_{j=1}^n a_j u_j, u_i \right\rangle \\ &= \sum_{j=1}^n \langle a_j u_j, u_i \rangle \\ &= \sum_{j=1}^n a_j \langle u_j, u_i \rangle. \end{aligned}$$

Since $\{u_1, \dots, u_n\}$ is orthonormal, $\langle u_j, u_i \rangle = 0$ if $i \neq j$ and $\langle u_i, u_i \rangle = 1$. Thus, the above equation simplifies to $\langle v, u_i \rangle = a_j$, which proves the theorem.

□

Definition 5.3.4. Let $W \subset V$ be a subspace of an inner product space. The **orthogonal complement** of W is the set

$$W^\perp := \{ v \in V \mid \langle v, w \rangle = 0 \text{ for all } w \in W \}.$$

Examples: R3

Example 5.3.5. Consider the vector space $C[-1, 1] = \{ f : [-1, 1] \rightarrow \mathbb{R} \mid f \text{ is continuous} \}$ with inner product given by

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx.$$

Let $W = \{ f \in C[-1, 1] \mid f(-x) = -f(x) \}$ be the subspace of odd functions. We claim that the orthogonal complement of W is $W^\perp = \{ f \in C[-1, 1] \mid g(-x) = g(x) \}$, the subspace of all even functions. First, we show that

$$\{ g \in C[-1, 1] \mid g(-x) = g(x) \} \subset W^\perp.$$

Let $g \in \{ g \in C[-1, 1] \mid g(-x) = g(x) \}$ and let $f \in W$. Note that $f(-x)g(-x) = -f(x)g(x)$, and so fg is an odd function. Thus,

$$\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx = 0.$$

This implies that $g \in W^\perp$.

Next, we show that

$$W^\perp \subset \{ g \in C[-1, 1] \mid g(-x) = g(x) \}.$$

This part is actually a bit trickier. Let $g \in W^\perp$. Then $\langle g, f \rangle = 0$ for every odd function f . Note that we can write

$$g(x) = \underbrace{\frac{g(x) + g(-x)}{2}}_{=: g_{\text{even}}(x)} + \underbrace{\frac{g(x) - g(-x)}{2}}_{=: g_{\text{odd}}(x)}.$$

Moreover, $g_{\text{even}}(x) = \frac{g(x) + g(-x)}{2}$ is an even function, and $g_{\text{odd}}(x) = \frac{g(x) - g(-x)}{2}$ is an odd function. Then $\langle g, f \rangle = 0$ for every $f \in W$ implies

$$\langle g_{\text{even}} + g_{\text{odd}}, f \rangle = 0 \quad \Rightarrow \quad \langle g_{\text{even}}, f \rangle + \langle g_{\text{odd}}, f \rangle = 0$$

for every $f \in W$. Note that because g_{even} is even and f is odd, the first part of our argument shows that $\langle g_{\text{even}}, f \rangle = 0$. Thus, we have

$$\langle g_{\text{odd}}, f \rangle = 0$$

for every odd $f \in W$. In particular, since $g_{\text{odd}} \in W$, it follows that

$$\|g_{\text{odd}}\| = \sqrt{\langle g_{\text{odd}}, g_{\text{odd}} \rangle} = 0.$$

By properties of norms, it follows that $g_{\text{odd}} = 0$ and so $g = g_{\text{even}}$. In particular, $g \in W^\perp$. This completes the proof that

$$W^\perp = \{ f \in C[-1, 1] \mid g(-x) = g(x) \}.$$

Proposition 5.3.6. *Let $W \subset V$ be a subspace of an inner product space. Then $W^\perp$ is a subspace of V .*

Proof. ADD □

Proposition 5.3.7. *Let $W \subset V$ be a subspace of an inner product space. Then*

$$V = W \oplus W^\perp.$$

Proof. ADD □

Exercises

1. Let V be a finite dimensional inner product space, and let $\{u_1, \dots, u_n\}$ be an orthonormal basis of V . Prove that, for all $v, w \in V$,

$$\langle v, w \rangle = \sum_{j=1}^n \langle v, u_j \rangle \overline{\langle w, u_j \rangle}.$$

2. Let V be a finite dimensional inner product space, and let $W \subset V$ be a subspace. Prove that $(W^\perp)^\perp = W$.
3. Let V be a real inner product space with $\dim V = 3$, let $T \in \mathcal{L}(V)$, and let $\mathcal{B} = \{v_1, v_2, v_3\}$ be a basis of V . Fix $v, w \in V$.

Suppose that $\langle v, v_1 \rangle = 1$, $\langle v, v_2 \rangle = -1$, and $\langle v, v_3 \rangle = -1$. Further suppose that $[T^*(w)]_{\mathcal{B}} = (2, 3, -1)$.

4. Let V and W be finite dimensional inner product spaces with (ordered) orthonormal bases $\mathcal{B} = \{v_1, v_2, v_3\} \subset V$ and $\mathcal{C} = \{w_1, w_2\} \subset W$, respectively. Suppose³ that $T : V \rightarrow W$ satisfies $\|T(v)\|_W \leq \|v\|_V$ for all $v \in V$.

Let a_{ij} denote the (i, j) entry of the coordinate matrix $[T]_{\mathcal{B}}^{\mathcal{C}}$. Prove that $|a_{ij}| \leq 1$ for all i, j .

(a) Compute $\langle T(v), w \rangle$.

(b) Now suppose further that $\mathcal{B}$ is an *orthonormal eigenbasis* of V with respect to T , and suppose that the eigenvalues associated to v_1, v_2, v_3 are $1, 2, -1$, respectively. Compute $[T^4(v)]_{\mathcal{B}}$.

³Here, $\|\cdot\|_V$ and $\|\cdot\|_W$ denote the (necessarily different!) norms on V and W , respectively. Similarly, denote the respective inner products by $\langle \cdot, \cdot \rangle_V$ and $\langle \cdot, \cdot \rangle_W$.

5. Let $V = C([-1, 1])$ be the inner product space of continuous functions $f : [-1, 1] \rightarrow \mathbb{R}$ with inner product given by

$$\langle f, g \rangle := \int_{-1}^1 f(x)g(x) dx.$$

Let

$$W := \{ f \in V \mid f(0) = 0 \}.$$

- (a) Prove that W is a subspace.
 (b) Prove that $W^\perp = \{0\}$. Conclude that $(W^\perp)^\perp \neq W$.
6. Let V be a finite dimensional inner product space, and let $W_1, W_2 \subset V$ be subspaces. Prove that

$$(W_1 + W_2)^\perp = W_1^\perp \cap W_2^\perp.$$

5.4 Adjoints

The concept of *duality* in mathematics is an elusive but pervasive one. It can be hard to pin down exactly what it means, but it is recognizable when it appears. In the context of linear algebra, inner product spaces are a natural arena to discuss duality. In this section we will do so by introducing the notion of an *adjoint* of a linear transformation $T : V \rightarrow V$. This will be another linear transformation $T^* : V \rightarrow V$ that is “dual” to T in some sense.

FTOC MOTIVATION???

Proposition 5.4.1. *Let V be an inner product space. Suppose that $\langle v, w \rangle = \langle v, w' \rangle$ for all $v \in V$. Then $w = w'$.*

Proof. Since $\langle v, w \rangle = \langle v, w' \rangle$ for all $v \in V$, we have $\langle v, w - w' \rangle = 0$ for all $v \in V$. In particular, $\langle w - w', w - w' \rangle = 0$. It follows that $w - w' = 0$, and so $w = w'$. $\square$

The following is a definition/theorem — we need to prove uniqueness and linearity of the defined operator.

Definition 5.4.2. Let V be an inner product space, and let $T \in \mathcal{L}(V)$. The **adjoint** of T , when it exists, is the *unique, linear* map $T^* : V \rightarrow V$ such that

$$\langle T(v), w \rangle = \langle v, T^*(w) \rangle$$

for all $v, w \in V$.

Proof of uniqueness and linearity of T^ .* Assume that T^* exists. First we prove that it is unique. Assume that there is another map $U : V \rightarrow V$ such that $\langle T(v), w \rangle = \langle v, U(w) \rangle$ for all $v, w \in V$. Then $\langle v, U(w) \rangle = \langle v, T^*(w) \rangle$ for all $v, w \in V$, and so by the previous proposition we have $U(w) = T^*(w)$ for all $w \in V$, hence $U = T^*$. Thus, T^* is unique.

Next we prove linearity of T^* . Fix $v \in V$, and let $w, w' \in V$. Then

$$\begin{aligned}\langle v, T^*(w + w') \rangle &= \langle T(v), w + w' \rangle \\ &= \langle T(v), w \rangle + \langle T(v), w' \rangle \\ &= \langle v, T^*(w) \rangle + \langle v, T^*(w') \rangle \\ &= \langle v, T^*(w) + T^*(w') \rangle.\end{aligned}$$

Since this holds for all $v \in V$, we have $T^*(w + w') = T^*(w) + T^*(w')$. The other property is similar, and is left as an exercise. $\square$

Note that adjoints don't always exist! But this phenomenon is restricted to infinite dimensions. In finite dimensions, adjoints do always exist.

Theorem 5.4.3. *Let V be a finite dimensional inner product space, and let $T \in \mathcal{L}(V)$. Then T^* exists.*

Proof. Omitted. $\square$

Whenever we state a proposition or problem involving adjoint, we will just assume implicitly that we are working with a transformation T for which the adjoint exists.

Example: transpose

This example captures the motivation and intuition behind the adjoint. In some sense, it's an abstraction of the transpose operation. In actuality, it's an abstraction of the *conjugate-transpose*. The following proposition makes this precise.

Proposition 5.4.4. *Let V be a finite dimensional inner product space, and let $T \in \mathcal{L}(V)$. Let $\mathcal{B} = \{u_1, \dots, u_n\}$ be an orthonormal basis of V . Then*

$$[T^*]_{\mathcal{B}} = \overline{[T]_{\mathcal{B}}}^T.$$

Proof. Let $A = [T]_{\mathcal{B}}$. We wish to compute $B := [T^*]_{\mathcal{B}}$. Note that, because $\mathcal{B}$ is orthonormal,

$$T^*(u_j) = \langle T^*(u_j), u_1 \rangle u_1 + \dots + \langle T^*(u_j), u_n \rangle u_n.$$

This tells us that $B_{ij} = \langle T^*(u_j), u_i \rangle$. Similarly, $A_{ij} = \langle T(u_j), u_i \rangle$. Note that

$$\overline{A_{ji}} = \overline{\langle T(u_i), u_j \rangle} = \langle u_j, T(u_i) \rangle = \langle T^*(u_j), u_i \rangle = B_{ij}.$$

Since $\overline{A_{ji}} = B_{ij}$ it follows that $B = \overline{A}^T$. $\square$

Examples: integration by parts, shift operator

5.4.1 The adjoint and the transpose

This subsection requires knowledge of dual spaces, and the notion of the (abstract) transpose of a linear transformation; see Section 3.5.

In this section, we have introduced the adjoint of a linear transformation $T : V \rightarrow V$ of an inner product space, which is another linear operator $T^* : V \rightarrow V$ on V . Proposition 5.4.4 suggests that we can think about the adjoint of a linear map as an abstract generalization of the (conjugate) transpose of a matrix, and this is indeed the case. If you have read Section 3.5 about dual spaces, then you would remember that we have already seen a different generalization of the transpose of a matrix. In that setting, given a linear map $T : V \rightarrow V$, the *transpose* is a linear map $T^t : V^* \rightarrow V^*$ between dual spaces. The examples and results that we have studied regarding the transpose and the adjoint suggest that these two notions should be related on some level; for example, compare Theorem 3.6.13 and Proposition 5.4.4. In this subsection we will elucidate this relationship.

First, we recall an important fact about dual spaces in finite dimensions, namely, that if V is a finite dimensional vector space then $V^* \cong V$. In words, the dual space of a vector space is isomorphic to the original vector space. To prove this fact, we had to choose a basis $\mathcal{B}$ of V and then construct the so-called *dual basis* $\mathcal{B}^*$. It is interesting to note that we had no choice but to choose a basis, and the isomorphism $V^* \cong V$ depends on this choice. In other words, there no *natural* way to describe an isomorphism of V^* with V without choosing a basis. While this is not the end of the world, it is always nice to find ways to do things in math without having to make choices. It turns out that if V is an *inner product space*, so that it comes equipped with the additional structure of an inner product, then there is a natural (basis-free) isomorphism of V^* with V . This is given by the following proposition.

Proposition 5.4.5. *Let V be a finite dimensional inner product space. Then the map $\phi : V \rightarrow V^*$ given by*

$$\phi(v) := \langle v, \cdot \rangle$$

is an isomorphism.

Remark 5.4.6. In words, this proposition says that on an inner product space, every linear functional $V \rightarrow \mathbb{F}$ can be described as “inner product against a fixed vector”.

Proof. First, we clarify the notation in the statement of the proposition. For an element $v \in V$, we define $\phi(v) \in V^*$ as the function $\phi(v) : V \rightarrow \mathbb{F}$ given by $\phi(v)(w) := \langle v, w \rangle$. Note that $\phi : V \rightarrow V^*$ is in fact linear. This follows from the fact that an inner product is linear in the first argument.

Thus, it remains to show that ϕ is invertible. Since we have already proven that $\dim V = \dim V^*$, it suffices to prove that ϕ is injective. Suppose that $\phi(v) = 0$. Then $\phi(v)(w) = 0$ for all $w \in V$, and so $\langle v, w \rangle = 0$ for all $w \in V$. Thus, $v = 0$. This proves injectivity. $\square$

With this isomorphism $\phi : V \rightarrow V^*$, we can describe the relation between the adjoint $T^* : V \rightarrow V$ and the transpose $T^t : V^* \rightarrow V^*$.

Proposition 5.4.7. *Let V be a finite dimensional inner product space, and let $T : V \rightarrow V$ be linear. Let $T^* : V \rightarrow V$ be the adjoint of T , and let $T^t : V^* \rightarrow V^*$ be the transpose of T . Let $\phi : V \rightarrow V^*$ be the isomorphism given by $\phi(v) := \langle v, \cdot \rangle$. Then $\phi^{-1} \circ T^t \circ \phi = T^*$.*

Remark 5.4.8. In words, this proposition says that T^t and T^* are indeed the “same”, in the sense that they are *similar*. One is given by conjugating the other with the isomorphism ϕ .

Proof. By composing both sides with ϕ , to prove the desired statement it suffices to prove that $T^t \circ \phi = \phi \circ T^*$. Note that both $T^t \circ \phi$ and $\phi \circ T^*$ are maps of the form $V \rightarrow V^*$. Thus, given $v \in V$ we wish to prove that $(T^t \circ \phi)(v)$ and $(\phi \circ T^*)(v)$ both define the same linear functional on V . To do this, we must prove that

$$(T^t \circ \phi)(v)(w) = (\phi \circ T^*)(v)(w)$$

for all $w \in V$ (and all $v \in V$). So fix $v, w \in V$. Note that

$$\begin{aligned} (T^t \circ \phi)(v)(w) &= T^t(\phi(v))(w) \\ &= \phi(v)(T(w)) \\ &= \langle v, T(w) \rangle. \end{aligned}$$

On the other hand, note that

$$\begin{aligned} (\phi \circ T^*)(v)(w) &= \phi(T^*(v))(w) \\ &= \langle T^*(v), w \rangle \\ &= \langle v, T(w) \rangle. \end{aligned}$$

Thus, $(T^t \circ \phi)(v)(w) = (\phi \circ T^*)(v)(w)$ as desired. This proves that $T^t \circ \phi = \phi \circ T^*$ and hence

$$\phi^{-1} \circ T^t \circ \phi = T^*$$

as desired. □

Exercises

1. Let V be an inner product space, and let $T \in \mathcal{L}(V)$ such that T^* exists. Prove that

$$(\text{im } T^*)^\perp = \ker T.$$

2. Let V be a finite dimensional inner product space. Prove that $\ker(T^* \circ T) = \ker T$.
3. Let V be an inner product space. An operator $T \in \mathcal{L}(V)$ is **normal** if $T^* \circ T = T \circ T^*$. Prove that if T is normal, then

$$\|T(v)\| = \|T^*(v)\|$$

for all $v \in V$.

4. True or false: If $v, w \in V$ are vectors in an inner product space such that $T(v)$ and $T(w)$ are linearly independent, then $T^*(v)$ and $T^*(w)$ are also linearly independent.

5.5 Self-adjoint operators and the spectral theorem

Definition 5.5.1. Let V be an inner product space, and let $T \in \mathcal{L}(V)$. Then T is **self-adjoint** if $T^* = T$.

By unraveling the definition of the adjoint, we see that T is self-adjoint if

$$\langle T(v), w \rangle = \langle v, T(w) \rangle$$

for all $v, w \in V$.

Examples: symmetric matrix, second derivative

Theorem 5.5.2 (Spectral theorem for self-adjoint operators). *Let V be a (real or complex) finite dimensional inner product space. Let $T \in \mathcal{L}(V)$ be a self-adjoint operator. Then all of the eigenvalues of T are real, and T is orthogonally diagonalizable, i.e., there is a basis of orthogonal eigenvectors.*

Can you hear the shape of a drum?

Exercises

5.6 Orthogonalization methods

5.6.1 Householder transformations

In this subsection we will discuss *Householder transformations*, which are abstract generalizations of reflections across a plane. We begin by clarifying exactly what we mean by a “plane” in a general inner product space.

Definition 5.6.1. Let V be a n -dimensional (real or complex) inner product space. A **hyperplane** $P \subset V$ is a subspace of dimension $n - 1$. Equivalently, a hyperplane is any subspace of the form

$$P = \{v_0\}^\perp = \{v \in V \mid \langle v, v_0 \rangle = 0\}$$

for some fixed $v_0 \neq 0 \in V$. In this case, we say that v_0 is a **normal vector** to P .

Remark 5.6.2. Note that normal vectors to a hyperplane P are not unique. If v_0 is a normal vector to P , then λv_0 is also a normal vector for any $\lambda \neq 0 \in \mathbb{F}$.

Given a hyperplane $P \subset V$, we wish to build a linear operator $H_P : V \rightarrow V$ which reflects vectors across the plane. The general definition comes from the following geometric motivation in $\mathbb{R}^2$. Given a line P through the origin and a fixed vector $x \in \mathbb{R}^2$ not on the line, we can reflect x across P by first orthogonally projecting onto P and using the difference vector $x - \text{proj}_P x$ to our advantage:

$$x - 2(x - \text{proj}_P x) = 2 \text{proj}_P x - x.$$

See the left side of Figure 5.2.

However, we wish to describe this operation in terms of the data of a normal vector to P . If $u_0 \in \mathbb{R}^2$ is a such a normal vector, which we may also assume without loss of generality is a unit vector so that $\|u_0\| = 1$, then

$$\text{proj}_P x = x - \text{proj}_{u_0} x = x - \langle x, u_0 \rangle u_0.$$

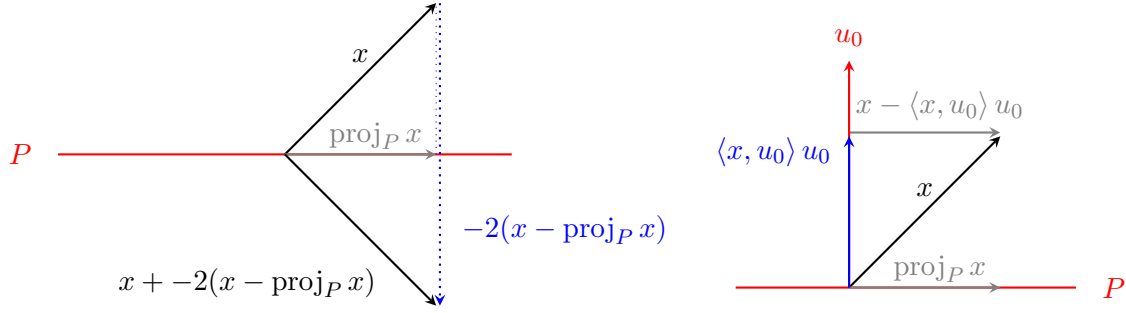


Figure 5.2: The left side depicts a reflection of x across P in terms of the projection onto P , given by the gray vector $\text{proj}_P x$. The right side demonstrates that $\text{proj}_P x = x - \langle x, u_0 \rangle u_0 = x - \text{proj}_{u_0} x$.

See the right side of Figure 5.2. Thus, our desired reflection of x is

$$2 \text{proj}_P x - x = 2(x - \langle x, u_0 \rangle u_0) - x = x - 2 \langle x, u_0 \rangle u_0.$$

This leads to the following abstract definition.

Definition 5.6.3. Let V be a finite dimensional (real or complex) inner product space, and let P be a hyperplane with unit normal vector $u_0 \in V$. The **Householder transformation across P** is the linear operator $H_P : V \rightarrow V$ defined by

$$H_P(x) := x - 2 \langle x, u_0 \rangle u_0.$$

The following proposition summarizes many of the important properties of Householder transformations, including confirmation that they do behave as reflections across P .

Proposition 5.6.4 (Properties of Householder transformations). *Let V be a finite dimensional (real or complex) inner product space, let P be a hyperplane with unit normal vector $u_0 \in V$, and let $H_P : V \rightarrow V$ be the Householder transformation across P .*

- (i) *If $x \in P^\perp$, then $H_P(x) = -x$. If $x \in P$, then $H_P(x) = x$.*
- (ii) *The eigenvalues of H_P are -1 and 1 . The eigenspaces are $E_{-1}(H_P) = P^\perp$ and $E_1(H_P) = P$. In particular, H_P is diagonalizable.*
- (iii) *The operator H_P is self-adjoint, that is, $H_P^* = H_P$.*
- (iv) *The operator H_P is unitary if V is complex, and orthogonal if V is real, that is, $H_P^* = H_P^{-1}$.*

Proof.

- (i) First, assume that $x \in P^\perp$. Then since $P^\perp = \text{Span}(u_0)$, we have $x = \langle x, u_0 \rangle u_0$. Thus,

$$H_P(x) = x - 2 \langle x, u_0 \rangle u_0 = \langle x, u_0 \rangle u_0 - 2 \langle x, u_0 \rangle u_0 = -\langle x, u_0 \rangle u_0 = -x.$$

On the other hand, if $x \in P$, then $\langle x, u_0 \rangle = 0$. Thus,

$$H_P(x) = x - 2 \langle x, u_0 \rangle u_0 = x.$$

- (ii) By part (a), u_0 is an eigenvector of H_P with eigenvalue -1 , and any nonzero vector in P is an eigenvector with eigenvalue 1 . Since $V = P \oplus P^\perp$, it follows that these are the only eigenvalues. This also implies that $E_{-1}(H_P) = P^\perp$ and $E_1(H_P) = P$ and in particular that H_P is diagonalizable.

- (iii) Let $v, w \in V$. To prove that $H_P^* = H_P$, we need to prove that

$$\langle H_P(v), w \rangle = \langle v, H_P(w) \rangle.$$

On the left, we have

$$\begin{aligned} \langle H_P(v), w \rangle &= \langle v - 2 \langle v, u_0 \rangle u_0, w \rangle \\ &= \langle v, w \rangle - 2 \langle v, u_0 \rangle \langle u_0, w \rangle. \end{aligned}$$

On the right, we have

$$\begin{aligned} \langle v, H_P(w) \rangle &= \langle v, w - 2 \langle w, u_0 \rangle u_0 \rangle \\ &= \langle v, w \rangle - 2 \overline{\langle w, u_0 \rangle} \langle v, u_0 \rangle \\ &= \langle v, w \rangle - 2 \langle u_0, w \rangle \langle v, u_0 \rangle. \end{aligned}$$

Thus, $\langle H_P(v), w \rangle = \langle v, H_P(w) \rangle$, as desired.

- (iv) By the previous part, to verify that $H_P^* = H_P^{-1}$, we simply need to check that $H_P^2 = I$. Let $x \in V$. Then

$$\begin{aligned} H_P^2(x) &= H_P(x - 2 \langle x, u_0 \rangle u_0) \\ &= H_P(x) - 2 \langle x, u_0 \rangle H_P(u_0) \\ &= (x - 2 \langle x, u_0 \rangle u_0) - 2 \langle x, u_0 \rangle (-u_0) \\ &= x - 2 \langle x, u_0 \rangle u_0 + 2 \langle x, u_0 \rangle u_0 \\ &= x. \end{aligned}$$

This proves that $H_P^2 = I$, as desired. Here we used the fact that $H_P(u_0) = -u_0$ by part (i). Thus, H_P is orthogonal when V is real, and unitary when V is complex.

□

5.6.2 QR Decomposition

Exercises

1. Suppose that P is a hyperplane with normal vector $v_0 \in V$, where v_0 is not necessarily a unit vector. Prove that

$$H_P(x) = x - 2 \frac{\langle x, v_0 \rangle}{\langle v_0, v_0 \rangle} v_0.$$

2. Consider the standard real inner product space $\mathbb{R}^n$, and let $\mathcal{B}$ be the standard basis. Fix a

unit vector $u_0 \in \mathbb{R}^n$ and let $P = \{u_0\}^\perp$. Prove that

$$[H_P]_{\mathcal{B}} = I - 2u_0u_0^T$$

where u_0 is interpreted as an $n \times 1$ matrix and u_0^T is the transpose.

3. Consider the standard complex inner product space $\mathbb{C}^n$, and let $\mathcal{B}$ be the standard basis. Fix a unit vector $u_0 \in \mathbb{C}^n$ and let $P = \{u_0\}^\perp$. Prove that

$$[H_P]_{\mathcal{B}} = I - 2u_0u_0^*$$

where u_0 is interpreted as an $n \times 1$ matrix and u_0^* is the conjugate transpose.

4. Gramm Schmidt exercise

5.7 An aside on quantum physics

Chapter 6

Determinants

In this chapter, we will let $\mathbb{F} = \mathbb{R}$ or $\mathbb{F} = \mathbb{C}$.

6.1 Multilinear maps

6.2 Definition of the determinant (proof of uniqueness)

Theorem 6.2.1. *Let V be a finite n -dimensional vector space, and fix a basis $\mathcal{B} = \{v_1, \dots, v_n\}$. There is a unique multilinear alternating map $D_{\mathcal{B}} : \underbrace{V \times \dots \times V}_{n \text{ times}} \rightarrow \mathbb{F}$ such that*

$$D_{\mathcal{B}}(v_1, \dots, v_n) = 1.$$

In particular, any other multilinear alternating map $\tilde{D} : \underbrace{V \times \dots \times V}_{n \text{ times}} \rightarrow \mathbb{F}$ satisfies $\tilde{D} = C D_{\mathcal{B}}$ for some $C \in \mathbb{F}$.

Corollary 6.2.2. *There is a unique multilinear alternating function $\det : \underbrace{\mathbb{R}^n \times \dots \times \mathbb{R}^n}_{n \text{ times}} \rightarrow \mathbb{R}$ satisfying*

$$\det(e_1, \dots, e_n) = 1$$

where $\{e_1, \dots, e_n\}$ is the standard basis of $\mathbb{R}^n$.

Definition 6.2.3. The function $\det : \underbrace{\mathbb{R}^n \times \dots \times \mathbb{R}^n}_{n \text{ times}} \rightarrow \mathbb{R}$ in the previous corollary is called the **determinant**.

6.3 Construction of the determinant (proof of existence)

6.4 Computing the determinant

6.5 Applications to eigenvalues

Part II

A Taste of Advanced Topics

Chapter 7

Multilinear Algebra and Tensors

Chapter 8

Symplectic Linear Algebra

I am a symplectic geometer by trade, so I can't help but open a window to a part of the mathematical universe that has occupied so much of my mental real estate in recent years. In any course or text on symplectic geometry, typically at the graduate level, symplectic linear algebra is covered in understandable brevity (see, for example, [dSs01] or [Sac20]). I hope to provide an introduction to the basic ideas of symplectic geometry at the linear level which is accessible to an interested 115A-level undergraduate, of the kind that I haven't seen anywhere else.

Symplectic linear algebra is the study of *symplectic vector spaces*, which, much like inner product spaces,¹ are vector spaces equipped with extra structure. In the case of an inner product space, that extra structure is a choice of *inner product*, which is a fairly natural abstraction and generalization of the familiar dot product. Because an inner product abstracts the dot product, it is the natural way to introduce the notion of geometry into a vector space. As such, inner product spaces are easy to motivate and natural to study. On the other hand, symplectic vector spaces are vector spaces equipped with a *symplectic form* — something like an evil twin of an inner product — and they are a bit harder to motivate and also less natural to a first-time learner.

As you will hopefully see soon, symplectic vector spaces are important and naturally pop up in a variety of places. Historically, the study of symplectic geometry arose as the mathematical foundation of Hamiltonian mechanics. In the 1980's, a mathematician named Mikhail Gromov made some incredible discoveries and launched the field into the modern powerhouse it is today. It is now a sophisticated and intricate area of math that has found use in the study of geometry and topology, the formalization of ideas like mirror symmetry in physics, and in applied math and statistics through Hamiltonian Monte Carlo algorithms, just to name a few of its appearances. Unfortunately, it takes many years of study in a variety of areas of math — topology, abstract algebra, analysis, category theory, etc. — to really learn symplectic geometry and behold its power. Fortunately, even at the level of linear algebra it is possible to develop some of the fundamental ingredients and cook some interesting symplectic mathematical recipes.

I'm going to start out by writing down the definition of a symplectic form, just so you can see it and have it in your mind as you read the motivation sections, but I don't expect you to *get* it, or understand it, or believe that it's interesting.

¹In fact, one might call an inner product space a *Riemannian vector space* to make the relationship between symplectic geometry and Riemannian geometry more apparent at the linear level.

Definition 8.0.1. Let V be a real vector space. A **symplectic form** is a map $\omega : V \times V \rightarrow \mathbb{R}$ that satisfies the following properties.

- (1) It is *bilinear*, i.e., linear in each argument, i.e., for all $\lambda \in \mathbb{R}$ and $v, v', w, w' \in V$, we have

$$\omega(v + v', w) = \omega(v, w) + \omega(v', w) \quad \text{and} \quad \omega(v, w + w') = \omega(v, w) + \omega(v, w')$$

and

$$\omega(\lambda v, w) = \lambda \omega(v, w) \quad \text{and} \quad \omega(v, \lambda w) = \lambda \omega(v, w).$$

- (2) It is *antisymmetric*, i.e., for all $v, w \in V$, we have

$$\omega(v, w) = -\omega(w, v).$$

- (3) It is *nondegenerate*, i.e., for all $v \neq 0 \in V$, there is a $w \in V$ such that $\omega(v, w) \neq 0$.

We will start off with two sections provide some motivation, and then in Section 8.3 we will return to the above definition to study some of its properties and consequences. The two motivation sections 8.1 and 8.2 can be read independently (or not at all, if you already find the above definition sufficiently interesting — but I suggest reading them).

8.1 Motivation via Hamiltonian mechanics

Like much of mathematics, the underlying motivation for symplectic linear algebra comes from physics. Specifically, from Hamiltonian mechanics. Much has been written about this by many others who know more about physics than me (see, for example, REFERENCES) and so I will not attempt to give a thorough exposition here; I just want to indicate how the definition of a symplectic form arises when thinking about certain things in physics.

In mechanics, you are interested in the motion of particles. For the moment, let's consider the motion of a single particle moving around a 1-dimensional world, which we can identify with $\mathbb{R}$, with the position variable q . There are various theories in physics one can use to understand this motion — for example, Newtonian mechanics and Lagrangian mechanics are two of them, and Hamiltonian mechanics is a third. We will focus on the last one.

The first perspective in Hamiltonian mechanics is that we need to keep track of more data than *just* the position of the particle. If I tell you that position of the particle at time $t = 1$ is at $q = 4$, you will not be able to tell me where the particle will go next. We need to also keep track of some additional information, like velocity or momentum. In Hamiltonian mechanics we choose momentum, which I will denote with a variable p . In this way, to track the motion of this one particle in 1-dimensional space we are actually keeping track of 2 dimensions worth of data: the particle's position, q , and the particle's momentum, p . When we view position and momentum together as a copy of $\mathbb{R}^2$, we call it *phase space*.

The punchline of Hamiltonian mechanics is that there is a set of differential equations involving position, momentum, and time, that entirely describes the motion of a particle. To write down this set of differential equations, we consider the *total energy* of the particle as a function $H(q, p)$ of its position and momentum. This function, called the *Hamiltonian*, is given

by

$$H(q, p) = \{\text{the particle's kinetic energy}\} + \{\text{the particle's potential energy}\}.$$

Also note that because our particle is moving around, both position and momentum are functions of time: $q(t), p(t)$.

Hamilton's equations then state:

$$\begin{cases} q'(t) = \frac{\partial H}{\partial p} \\ p'(t) = -\frac{\partial H}{\partial q} \end{cases}.$$

By solving these differential equations, one can understand how the motion of a particle evolves over time (both in phase space, and also in the real world by just ignoring the momentum coordinate).

Let me rephrase Hamilton's equations a little bit using vectors. Currently, we have a function $H(q, p)$ of two variables, q and p . In calculus you could compute the *gradient* of this function:

$$\nabla H = \begin{pmatrix} \frac{\partial H}{\partial q} \\ \frac{\partial H}{\partial p} \end{pmatrix}.$$

On the other hand, we could rewrite Hamilton's equations above in vector form as follows:

$$\begin{pmatrix} q'(t) \\ p'(t) \end{pmatrix} = \begin{pmatrix} \frac{\partial H}{\partial p} \\ -\frac{\partial H}{\partial q} \end{pmatrix}.$$

The vector field on the right hand side looks like some screwed up version of the gradient, where the coordinates have been flipped and the second entry picks up a negative sign. I will call this vector field the *symplectic gradient*.²

$$\nabla_{\omega} H := \begin{pmatrix} \frac{\partial H}{\partial p} \\ -\frac{\partial H}{\partial q} \end{pmatrix}.$$

Note that the normal gradient and the symplectic gradient are related by multiplication by a rotation matrix:

$$\nabla_{\omega} H = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \nabla H \quad \text{i.e.} \quad \begin{pmatrix} \frac{\partial H}{\partial p} \\ -\frac{\partial H}{\partial q} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} \frac{\partial H}{\partial q} \\ \frac{\partial H}{\partial p} \end{pmatrix}.$$

This seems a bit contrived, but the appearance of the matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ in Hamiltonian mechanics uncovers the fundamental nature of symplectic forms. In particular, let me define a function $\omega : \mathbb{R}^2 \times \mathbb{R}^2 \rightarrow \mathbb{R}$ as follows. For $v, w \in \mathbb{R}^2$, let

$$\omega(v, w) := w^T \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} v.$$

²More standard terminology would be the *Hamiltonian vector field* and more standard notation would be X_H ; I'm making up some different notation for storytelling purposes.

Note that we have actually used the *negative* of the previous rotation matrix in the definition. This function ω is an example of what we will eventually call a symplectic form! Writing $v = (q_1, p_1)$ and $w = (q'_1, p'_1)$,³ this gives

$$\begin{aligned}\omega(v, w) &= \begin{pmatrix} q'_1 & p'_1 \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} q_1 \\ p_1 \end{pmatrix} \\ &= \begin{pmatrix} q'_1 & p'_1 \end{pmatrix} \begin{pmatrix} -p_1 \\ q_1 \end{pmatrix} \\ &= q_1 p'_1 - q'_1 p_1.\end{aligned}$$

The point of introducing this symplectic form ω is that we can reformulate many equations, ideas, and principles in Hamiltonian mechanics into statements about symplectic forms. For example, the principle of conservation of energy has a nice interpretation in the context of symplectic forms. If you don't know much about physics (like me!) then you'll have to trust me that reformulating equations and ideas in physics in mathematically sophisticated language is a good idea. Not only does this lead to clarity in the physical theories, but it allows mathematicians to abstract away the physics and only study the mathematical properties that arise! Symplectic geometry is, in some sense, the natural mathematical abstraction and generalization of Hamiltonian mechanics.

Principle 8.1.1. Symplectic geometry and symplectic forms arise as the mathematical abstraction of Hamiltonian mechanics.

For one other example of reformulating known math in symplectic language, suppose you have your Hamiltonian (energy) function $H(q, p)$ and you want to take a directional derivative of this function with respect to some vector $v = (a, b)$. From calculus, we know that the directional derivative is given by the dot product of v with the gradient:

$$D_v H = \nabla H \cdot v = a \frac{\partial H}{\partial q} + b \frac{\partial H}{\partial p}.$$

Using the symplectic form ω from above, we can rephrase this in fact in terms of the symplectic gradient. Note that

$$\begin{aligned}D_v H &= a \frac{\partial H}{\partial q} + b \frac{\partial H}{\partial p} = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} \frac{\partial H}{\partial q} \\ \frac{\partial H}{\partial p} \end{pmatrix} = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{\partial H}{\partial p} \\ -\frac{\partial H}{\partial q} \end{pmatrix} \\ &= \omega(\nabla_\omega H, v).\end{aligned}$$

In words, to take the directional derivative of a function H in the direction v , you feed the *symplectic* gradient $\nabla_\omega H$ and the vector v into the symplectic form. This is one sense in which a symplectic form acts as an evil twin to an inner product. In general, this suggests another very vague principle.

³Here, the primes don't mean any kind of derivative — I'm just using primes to distinguish the quantities q'_1 and p'_1 from the quantities q_1 and p_1 in a way that is suggestive of what happens in the next section.

Principle 8.1.2. Symplectic forms allow for a *dual* description of ideas in calculus.

This is one of those principles that is vague enough that it probably doesn't mean anything to you, unless you already know what I'm describing. It also isn't clear why this a useful thing to do, but it turns out to be a fruitful theme in math!

Finally, I'll just remark that this discussion about phase space can be generalized to any even dimension. In particular, suppose your particle was flying around n -dimensional space. There would be n associated position coordinates and n associated momentum coordinates, and so phase space would be $\mathbb{R}^{2n}$. The symplectic form then describes a dual pairing between corresponding position and momentum coordinates!

8.2 Motivation via complex inner products

In Chapter 5 we studied (real and complex) inner product spaces. For the sake of clarity, let me remind you of the definition of an inner product. Also let me specialize specifically to *complex* inner products, which is what I am most interested in discussing in this section.

Definition 8.2.1. Let V be a complex vector space. An **inner product** is a map $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{F}$ satisfying the following properties.

- (1) It is *linear in the first argument* and *conjugate-linear in the second argument*, i.e., for all $v, v', w, w' \in V$ and $\lambda \in \mathbb{C}$ we have

$$\langle v + v', w \rangle = \langle v, w \rangle + \langle v', w \rangle \quad \text{and} \quad \langle v, w + w' \rangle = \langle v, w \rangle + \langle v, w' \rangle$$

and

$$\langle \lambda v, w \rangle = \lambda \langle v, w \rangle \quad \text{and} \quad \langle v, \lambda w \rangle = \bar{\lambda} \langle v, w \rangle.$$

- (2) It is *conjugate-symmetric*, i.e., for all $v, w \in V$ we have

$$\langle v, w \rangle = \overline{\langle w, v \rangle}.$$

- (3) It is *positive definite*, i.e., for every $v \neq 0 \in V$ we have $\langle v, v \rangle > 0$.

Like I said, this is a definition you are familiar with already — I'm simply reminding you of the definition so that you notice the structural similarities with the definition of a symplectic form at the beginning of the chapter.

To be concrete, let's specialize even further and consider the standard inner product on $\mathbb{C}^n$:

$$\mathbb{C}^n = \left\{ \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix} \mid z_j \in \mathbb{C} \right\}$$

with

$$\left\langle \begin{pmatrix} z_1 \\ \vdots \\ z_n \end{pmatrix}, \begin{pmatrix} z'_1 \\ \vdots \\ z'_n \end{pmatrix} \right\rangle = z_1 \overline{z'_1} + \cdots + z_n \overline{z'_n}.$$

Recall that every complex number z can be written in the form $z = x + iy$ where $x, y \in \mathbb{R}$ are real numbers (the real and imaginary parts of z). Thus, we can equivalently describe $\mathbb{C}^n$ — n -tuples of complex numbers — by n pairs of real numbers:

$$\mathbb{C}^n = \left\{ \begin{pmatrix} x_1 + iy_1 \\ \vdots \\ x_n + iy_n \end{pmatrix} \mid x_j, y_j \in \mathbb{R} \right\}.$$

Let's investigate what the standard inner product looks like with respect to the decomposition $z_j = x_j + iy_j$. We have

$$\begin{aligned} \left\langle \begin{pmatrix} x_1 + iy_1 \\ \vdots \\ x_n + iy_n \end{pmatrix}, \begin{pmatrix} x'_1 + iy'_1 \\ \vdots \\ x'_n + iy'_n \end{pmatrix} \right\rangle &= (x_1 + iy_1) \overline{(x'_1 + iy'_1)} + \cdots + (x_n + iy_n) \overline{(x'_n + iy'_n)} \\ &= (x_1 + iy_1)(x'_1 - iy'_1) + \cdots + (x_n + iy_n)(x'_n - iy'_n) \\ &= [(x_1 x'_1 + y_1 y'_1) + i(x_1 y'_1 - x'_1 y_1)] + \cdots \\ &\quad \cdots + [(x_1 x'_1 + y_1 y'_1) + i(x_n y'_n - x'_n y_n)] \\ &= [x_1 x'_1 + \cdots + x_n x'_n + y_1 y'_1 + \cdots + y_n y'_n] \\ &\quad + i[(x_1 y'_1 - x'_1 y_1) + \cdots + (x_n y'_n - x'_n y_n)]. \end{aligned}$$

This is a lot of notation to unravel, but all I have done is written $z_j = x_j + iy_j$ and simplified the value of the standard complex inner product (which is a complex number) into its real and imaginary parts. In other words, to introduce some notation, the standard complex inner product is given by

$$\operatorname{Re} \left\langle \begin{pmatrix} x_1 + iy_1 \\ \vdots \\ x_n + iy_n \end{pmatrix}, \begin{pmatrix} x'_1 + iy'_1 \\ \vdots \\ x'_n + iy'_n \end{pmatrix} \right\rangle + i \operatorname{Im} \left\langle \begin{pmatrix} x_1 + iy_1 \\ \vdots \\ x_n + iy_n \end{pmatrix}, \begin{pmatrix} x'_1 + iy'_1 \\ \vdots \\ x'_n + iy'_n \end{pmatrix} \right\rangle$$

where

$$\operatorname{Re} \left\langle \begin{pmatrix} x_1 + iy_1 \\ \vdots \\ x_n + iy_n \end{pmatrix}, \begin{pmatrix} x'_1 + iy'_1 \\ \vdots \\ x'_n + iy'_n \end{pmatrix} \right\rangle := x_1 x'_1 + \cdots + x_n x'_n + y_1 y'_1 + \cdots + y_n y'_n$$

and

$$\operatorname{Im} \left\langle \begin{pmatrix} x_1 + iy_1 \\ \vdots \\ x_n + iy_n \end{pmatrix}, \begin{pmatrix} x'_1 + iy'_1 \\ \vdots \\ x'_n + iy'_n \end{pmatrix} \right\rangle := (x_1 y'_1 - x'_1 y_1) + \cdots + (x_n y'_n - x'_n y_n).$$

Let me interpret this mess in one more slightly different way. Because a single complex number $z_j \in \mathbb{C}$ corresponds to two different real numbers $x_j, y_j \in \mathbb{R}$, we can naturally view our description of the complex vector space $\mathbb{C}^n$ as a *real* vector space $\mathbb{R}^{2n}$. Recall that when we discussed vector spaces, we pointed out that $\mathbb{C}^n$ can be given a vector space structure in multiple ways: as a complex vector space, it has dimension n , but as a real vector space it has dimension $2n$. Anyway, all of this is to say that we can view $\mathbb{C}^n$ from above as the real vector space

$$\mathbb{R}^{2n} = \left\{ \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ y_1 \\ \vdots \\ y_n \end{pmatrix} \mid x_j, y_j \in \mathbb{R} \right\}$$

where we are considering $2n$ -tuples, grouping all of the real parts (the x_j 's) in the first n slots and all of the imaginary parts (the y_j 's) in the last n slots. From here on out, for the sake of compressing some notation, let

$$v = \begin{pmatrix} x_1 \\ \vdots \\ x_n \\ y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^{2n} \quad \text{and} \quad w = \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \\ y'_1 \\ \vdots \\ y'_n \end{pmatrix} \in \mathbb{R}^{2n}$$

denote two elements of our vector space. The real and imaginary parts of the standard complex inner product above then give us two functions $g : \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$ and $\omega : \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$ defined by

$$g(v, w) := x_1 x'_1 + \cdots + x_n x'_n + y_1 y'_1 + \cdots + y_n y'_n$$

and

$$\omega(v, w) := (x_1 y'_1 - x'_1 y_1) + \cdots + (x_n y'_n - x'_n y_n).$$

That is, I have introduced even more notation to rewrite the standard complex inner product from above as

$$\left\langle \begin{pmatrix} x_1 + iy_1 \\ \vdots \\ x_n + iy_n \end{pmatrix}, \begin{pmatrix} x'_1 + iy'_1 \\ \vdots \\ x'_n + iy'_n \end{pmatrix} \right\rangle = g(v, w) + i\omega(v, w)$$

where g and ω are maps $\mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$. If you're still with me, I'll finally give some punchlines.

Observation 8.2.2. The map $g : \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$, defined above as the real part of the standard complex inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{C}^n$, is a *real* inner product on $\mathbb{R}^{2n}$. In fact, it is the dot product!

Indeed, just to spell it out one more time, we have

$$g(v, w) = g \left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \\ y_1 \\ \vdots \\ y_n \end{pmatrix}, \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \\ y'_1 \\ \vdots \\ y'_n \end{pmatrix} \right) = x_1 x'_1 + \cdots + x_n x'_n + y_1 y'_1 + \cdots + y_n y'_n$$

which is exactly the dot product you are familiar with from calculus (although in $2n$ dimensions instead of 2 or 3). So what we have discovered is that if you take the standard *complex* inner product on $\mathbb{C}^n$ and decompose it into its real and imaginary parts, the real part gives you the standard *real* inner product (the dot product) on $\mathbb{R}^{2n}$. That's cool, but it also suggests the following question.

Question 8.2.3. If the real part g of the standard complex inner product $\langle \cdot, \cdot \rangle$ is a real inner product, what kind of object is ω , the imaginary part?

Let me remind you again what the function $\omega : \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$ from above looks like. For $v, w \in \mathbb{R}^{2n}$, we have

$$\omega(v, w) = \omega \left(\begin{pmatrix} x_1 \\ \vdots \\ x_n \\ y_1 \\ \vdots \\ y_n \end{pmatrix}, \begin{pmatrix} x'_1 \\ \vdots \\ x'_n \\ y'_1 \\ \vdots \\ y'_n \end{pmatrix} \right) = (x_1 y'_1 - x'_1 y_1) + \cdots + (x_n y'_n - x'_n y_n).$$

This does not look like any kind of object we are familiar with from calculus or earlier in the course. There are some determinant-y looking terms, but beyond that ω is somewhat peculiar. Here is the punchline of this motivation section.

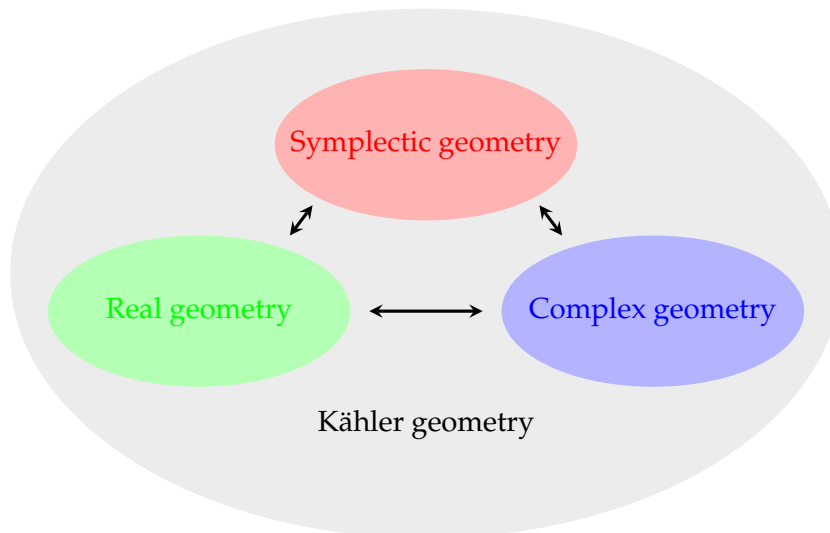
Observation 8.2.4. The map $\omega : \mathbb{R}^{2n} \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$, defined as the imaginary part of the standard complex inner product $\langle \cdot, \cdot \rangle$ on $\mathbb{C}^n$, is a *symplectic form* on $\mathbb{R}^{2n}$!

Because this is a motivation section, I won't verify this observation here — I'll save that for Section 8.3. Instead, I'll just repeat the punchline of all of the work we have done so far: *when you take a complex inner product and decompose it into its real and imaginary parts, the real part becomes a real inner product and the imaginary part becomes a symplectic form.* Complex products are interesting, real inner products are interesting, and symplectic forms are the bizarre glue that relates the two. You can also go the other way: given a real inner product g on some even-dimensional real vector space V , and a symplectic form ω on V , you can combine them form a complex inner product on a complex version of V , roughly given by

$$\langle v, w \rangle := g(v, w) + i \omega(v, w).$$

It takes some care to say what I just said precisely, but it is morally true.

It is an important theme in the study of geometry that these three kinds of geometries — complex⁴ geometry, real⁵ geometry, and symplectic geometry — are intimately related. Mathematicians study each of these three geometries individually, but they also come together in a beautiful part of math called Kähler geometry.



We will explore the relationship between complex geometry, real geometry, and symplectic geometry at the level of linear algebra in the sections below.

8.3 Symplectic vector spaces

With some motivation, we are now ready to officially define the notion of a symplectic form. I will remind you of the definition from the introduction.

Definition 8.3.1. Let V be a real vector space. A **symplectic form** is a map $\omega : V \times V \rightarrow \mathbb{R}$ that satisfies the following properties.

- (1) It is *bilinear*, i.e., linear in each argument, i.e., for all $\lambda \in \mathbb{R}$ and $v, v', w, w' \in V$, we have

$$\omega(v + v', w) = \omega(v, w) + \omega(v', w) \quad \text{and} \quad \omega(v, w + w') = \omega(v, w) + \omega(v, w')$$

and

$$\omega(\lambda v, w) = \lambda \omega(v, w) \quad \text{and} \quad \omega(v, \lambda w) = \lambda \omega(v, w).$$

- (2) It is *antisymmetric*, i.e., for all $v, w \in V$, we have

$$\omega(v, w) = -\omega(w, v).$$

- (3) It is *nondegenerate*, i.e., for all $v \neq 0 \in V$, there is a $w \in V$ such that $\omega(v, w) \neq 0$.

A **symplectic vector space** (V, ω) is a vector space V together with a choice of symplectic form ω .

⁴Or, almost complex geometry, for those in the know.

⁵Riemannian, for those in the know.

Remark 8.3.2. In this chapter, we will assume all vector spaces are finite dimensional, unless otherwise stated.

8.3.1 The standard symplectic vector space

Before we study the properties of symplectic vector spaces, we should play around with the main example of one. This should eventually look familiar, if you read the motivation sections.

Consider the vector space $\mathbb{R}^{2n}$. Denote the usual standard basis by $\{e_1, \dots, e_n, f_1, \dots, f_n\}$. That is, let

$$e_1 = \begin{pmatrix} 1 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \dots, e_n = \begin{pmatrix} 0 \\ \vdots \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, f_1 = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \dots, f_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 0 \\ \vdots \\ 1 \end{pmatrix}.$$

Note that this is a different notation to refer to the standard basis from what we have used in the past. This is intentional, and suggests that $e_1, \dots, e_n$ are *position* directions and $f_1, \dots, f_n$ are the corresponding *momentum* directions. Define the **standard symplectic form** ω_{std} on the standard basis vectors as follows:

$$\begin{cases} \omega_{\text{std}}(e_i, e_j) := 0 & \text{for all } i, j \\ \omega_{\text{std}}(f_i, f_j) := 0 & \text{for all } i, j \\ \omega_{\text{std}}(e_i, f_j) := 0 & \text{for all } i \neq j \\ \omega_{\text{std}}(e_i, f_i) := 1 & \text{for all } i \end{cases}.$$

Extend ω_{std} to all of V by multilinearity (see the next example for an explicit example of what this means). We call $(\mathbb{R}^{2n}, \omega_{\text{std}})$ the **standard symplectic vector space**.

Example 8.3.3. Let's unravel the standard symplectic form a bit more explicitly to get a feel for what the above example is describing. In particular, what I mean by "extend by multilinearity". Let's specialize to $\mathbb{R}^4$. The standard basis above is then

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, f_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, f_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix}.$$

This will be a bit tedious, but let's compute the formula for ω_{std} for any two vector inputs. In particular, let $v = (x_1, x_2, y_1, y_2)$ and let $w = (x'_1, x'_2, y'_1, y'_2)$. We wish to compute $\omega_{\text{std}}(v, w)$. The only way we can possibly do this is to use the multilinearity properties of ω_{std} together with the behavior on the standard basis vectors. Note that

$$\begin{aligned} v &= x_1 e_1 + x_2 e_2 + y_1 f_1 + y_2 f_2 \\ w &= x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2. \end{aligned}$$

Thus,

$$\begin{aligned}\omega_{\text{std}}(v, w) &= \omega_{\text{std}}(x_1 e_1 + x_2 e_2 + y_1 f_1 + y_2 f_2, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) \\ &= \omega_{\text{std}}(x_1 e_1, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) + \omega_{\text{std}}(x_2 e_2, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) \\ &\quad + \omega_{\text{std}}(y_1 f_1, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) + \omega_{\text{std}}(y_2 f_2, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2)\end{aligned}$$

Here the festive coloring is simply to help you keep track of each term, since the notation is fairly overwhelming. Anyway, what I've done so far is used linearity in the first component to break up $\omega_{\text{std}}(v, w)$ into four terms. Let's consider each of these separately. Continuing to use multilinearity, we have

$$\begin{aligned}\omega_{\text{std}}(x_1 e_1, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) &= \omega_{\text{std}}(x_1 e_1, x'_1 e_1) + \omega_{\text{std}}(x_1 e_1, x'_2 e_2) \\ &\quad + \omega_{\text{std}}(x_1 e_1, y'_1 f_1) + \omega_{\text{std}}(x_1 e_1, y'_2 f_2).\end{aligned}$$

Now I'll drop the colors and use linearity in each component to pull out the scalars.

$$\begin{aligned}\omega_{\text{std}}(x_1 e_1, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) &= x_1 x'_1 \omega_{\text{std}}(e_1, e_1) + x_1 x'_2 \omega_{\text{std}}(e_1, e_2) \\ &\quad + x_1 y'_1 \omega_{\text{std}}(e_1, f_1) + x_1 y'_2 \omega_{\text{std}}(e_1, f_2).\end{aligned}$$

Now we can finally use the definition of ω_{std} from above. Note that three of the four terms above are 0:

$$\omega_{\text{std}}(e_1, e_1) = \omega_{\text{std}}(e_1, e_2) = \omega_{\text{std}}(e_1, f_2) = 0.$$

In the remaining term, we know that $\omega_{\text{std}}(e_1, f_1) = 1$. Thus, we have

$$\omega_{\text{std}}(x_1 e_1, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) = x_1 y'_1.$$

I'll leave it to you to verify the values of the other terms in the initial colorful expression in much the same way:

$$\begin{aligned}\omega_{\text{std}}(x_2 e_2, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) &= x_2 y'_2 \\ \omega_{\text{std}}(y_1 f_1, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) &= -x'_1 y_1 \\ \omega_{\text{std}}(y_2 f_2, x'_1 e_1 + x'_2 e_2 + y'_1 f_1 + y'_2 f_2) &= -x'_2 y_2.\end{aligned}$$

Note that in the last two computations, you'll have to use the antisymmetric property of symplectic forms to get

$$\omega_{\text{std}}(f_1, e_1) = -\omega_{\text{std}}(e_1, f_1) = -1 \quad \text{and} \quad \omega_{\text{std}}(f_2, e_2) = -\omega_{\text{std}}(e_2, f_2) = -1.$$

Anyway, putting all of this together to return to our original goal of computing $\omega_{\text{std}}(v, w)$ where $v = (x_1, x_2, y_1, y_2)$ and let $w = (x'_1, x'_2, y'_1, y'_2)$.

We have

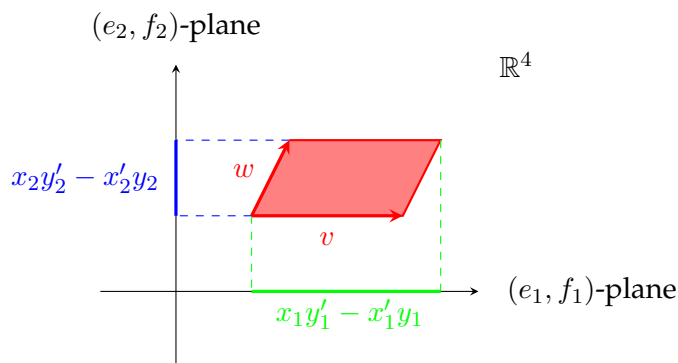
$$\omega_{\text{std}} \left(\begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix}, \begin{pmatrix} x'_1 \\ x'_2 \\ y'_1 \\ y'_2 \end{pmatrix} \right) = (x_1 y'_1 - x'_1 y_1) + (x_2 y'_2 - x'_2 y_2). \quad (8.3.1)$$

Unsurprisingly, this is the exact same formula that arose in Section 8.2 when we considered the imaginary part of a complex inner product!

Here is a geometric interpretation of the above formula. In general, you can think about a symplectic form as a tool that measures a funny kind of “area”, though area can be a misleading word. Note that the expressions $x_1 y'_1 - x'_1 y_1$ and $x_2 y'_2 - x'_2 y_2$ in the above formula are the usual expressions for the determinants of the matrices

$$\begin{pmatrix} x_1 & x'_1 \\ y_1 & y'_1 \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x_2 & x'_2 \\ y_2 & y'_2 \end{pmatrix}.$$

Also recall that a 2×2 determinant measures the (signed) area of the parallelogram determined by the column vectors. Thus, the formula above describes $\omega_{\text{std}}(v, w)$ in the following way. Consider the parallelogram spanned by the vectors v and w in $\mathbb{R}^4$. Project that parallelogram onto the (e_1, f_1) -plane, measure the normal (signed) area, then project the parallelogram onto the (e_2, f_2) -plane and measure the normal (signed) area, and then add up these areas. This gives you $\omega(v, w)$, the *symplectic* area of the parallelogram. Below is a very inaccurate heuristic 2-dimensional representation of this situation in 4-dimensions.



This can be very different from the normal notion of area! For example, consider the parallelogram spanned by $v = (1, 0, 0, 0)$ and $w = (0, 1, 0, 0)$. This parallelogram is like a 2-dimensional unit square sitting in 4 dimensions, and so it has normal area 1. But if you compute using the formula above, you'll find that the *symplectic* area of the square is 0! A shape must have presence in both the position (e_j) direction *and* the paired momentum direction (f_j) in order to accumulate symplectic area. For example, the square spanned by $v = (1, 0, 0, 0)$ and $w = (0, 0, 1, 0)$ has symplectic area 1.

Next, observe that we can write the standard symplectic form on $\mathbb{R}^4$ in terms of a matrix as

follows:

$$\omega_{\text{std}} \left(\begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix}, \begin{pmatrix} x'_1 \\ x'_2 \\ y'_1 \\ y'_2 \end{pmatrix} \right) = \begin{pmatrix} x_1 & x_2 & y_1 & y_2 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x'_1 \\ x'_2 \\ y'_1 \\ y'_2 \end{pmatrix}.$$

This should also look familiar if you read Section 8.1. Note that the matrix A in the above expression is *skew-symmetric*, i.e., $A^T = -A$. This corresponds to the antisymmetry property in Definition 8.3.1. The matrix is also invertible, which corresponds to the nondegeneracy condition. In general, for any symplectic form on $\mathbb{R}^N$, we can always express it in terms of an invertible, skew-symmetric matrix!

Proposition 8.3.4. *Let $(\mathbb{R}^N, \omega)$ be a symplectic vector space. Then $\omega(v, w) = v^T A w$ for an $N \times N$, invertible, skew-symmetric matrix A .*

Proof. The proof extends the idea of the above computation. Define the (i, j) -th entry of $A \in M_{N \times N}(\mathbb{R})$ as follows:

$$A_{ij} := \omega(e_i, e_j)$$

where $\{e_1, \dots, e_N\}$ is the standard basis of $\mathbb{R}^N$.

First, we must show that $\omega(v, w) = w^T A v$ for all $v, w \in \mathbb{R}^N$. Write $v = \sum_{j=1}^N \lambda_j e_j$ and $w = \sum_{i=1}^N \mu_i e_i$ for some scalars $\lambda_j, \mu_i \in \mathbb{R}$. Then, using linearity of ω in each component, we have

$$\begin{aligned} \omega(v, w) &= \omega \left(\sum_{j=1}^N \lambda_j e_j, \sum_{i=1}^N \mu_i e_i \right) \\ &= \sum_{j=1}^N \lambda_j \omega \left(e_j, \sum_{i=1}^N \mu_i e_i \right) \\ &= \sum_{j=1}^N \lambda_j \sum_{i=1}^N \mu_i \omega(e_j, e_i) \\ &= \begin{pmatrix} \lambda_1 & \cdots & \lambda_N \end{pmatrix} \begin{pmatrix} \sum_{i=1}^N \mu_i \omega(e_1, e_i) \\ \vdots \\ \sum_{i=1}^N \mu_i \omega(e_N, e_i) \end{pmatrix} \end{aligned}$$

By the definition of matrix multiplication, we have

$$\begin{pmatrix} \sum_{i=1}^N \mu_i \omega(e_1, e_i) \\ \vdots \\ \sum_{i=1}^N \mu_i \omega(e_N, e_i) \end{pmatrix} = \begin{pmatrix} \omega(e_1, e_1) & \cdots & \omega(e_1, e_N) \\ \vdots & \ddots & \vdots \\ \omega(e_N, e_1) & \cdots & \omega(e_N, e_N) \end{pmatrix} \begin{pmatrix} \mu_1 \\ \vdots \\ \mu_N \end{pmatrix} = A w.$$

Thus, $\omega(v, w) = v^T A w$ as desired.

Next we verify that A is skew-symmetric and invertible. Because

$$A_{ji} = \omega(e_j, e_i) = -\omega(e_i, e_j) = -A_{ij}$$

it follows that $A^T = -A$ and hence A is skew-symmetric. Next, we show that A has a trivial kernel. Suppose that there was some vector $w \in \mathbb{R}^N$ such that $Aw = 0$. Then $v^T Aw = 0$ for all $v \in \mathbb{R}^N$, and hence $\omega(v, w) = 0$ for all $v \in \mathbb{R}^N$. By nondegeneracy of ω , this forces $w = 0$, and hence $\ker A = \{0\}$. Since A is a square matrix, it must be invertible. $\square$

Remark 8.3.5. The converse to this proposition is true, and is left as an exercise. Namely, if A is an invertible, skew-symmetric matrix, then the quantity $v^T Aw$ then describes a symplectic form on $\mathbb{R}^N$.

As a corollary to the above proposition — well, really as a corollary to the proof — we can describe the standard symplectic form on $\mathbb{R}^{2n}$ via a matrix (not just $\mathbb{R}^4$, like we did above).

Corollary 8.3.6. *Let ω_{std} be the standard symplectic form on $\mathbb{R}^{2n}$. Then $\omega_{\text{std}}(v, w) = v^T J_{\text{std}} w$, where J_{std} is the $2n \times 2n$ block matrix given by*

$$J_{\text{std}} := \begin{pmatrix} 0_n & I_n \\ -I_n & 0_n \end{pmatrix}.$$

Here, 0_n denotes the $n \times n$ zero matrix and I_n denotes the $n \times n$ identity.

Proof. By the proof of the previous proposition, we know that the entries of the matrix defining ω_{std} are given by the values $\omega_{\text{std}}(e_i, e_j)$, $\omega_{\text{std}}(f_i, f_j)$, $\omega_{\text{std}}(e_i, f_j)$, and $\omega_{\text{std}}(f_i, e_j)$. We leave it as an exercise to check carefully that the resulting matrix is J_{std} . $\square$

8.3.2 Properties of symplectic vector spaces

If you read the motivation sections, you may have picked up on the following observation. In the setting of Hamiltonian mechanics, symplectic forms lived on *phase space*, which is the vector space that keeps track of both position and momentum. Because there are two types of data in phase space, the overall dimension is even. For example, if a particle is flying around $\mathbb{R}^3$, with 3 position coordinates, then phase space would be $\mathbb{R}^6$ because of the additional 3 momentum coordinates. Likewise, in the section about complex inner products, we naturally identified symplectic forms on $\mathbb{R}^{2n}$, because they came from complex inner products on $\mathbb{C}^n$. Again, the underlying vector space $\mathbb{R}^{2n}$ is even dimensional. This is no accident, and the forced even dimensionality of a symplectic vector space is one of the first fundamental consequence of Definition 8.3.1.

Proposition 8.3.7. *Let (V, ω) be a symplectic vector space. Then $\dim V$ is even.*

Proof. To prove this fact, we will appeal to Proposition 8.3.4. Let $N = \dim V$, and let $T : \mathbb{R}^N \rightarrow V$ be any isomorphism. Define $\omega_0 : \mathbb{R}^N \times \mathbb{R}^N \rightarrow \mathbb{R}$ by precomposition of ω with T . That is, define

$$\omega_0(v, w) := \omega(T(v), T(w)) \quad \text{for all } v, w \in \mathbb{R}^N.$$

Because T is a (linear) isomorphism, one can easily verify that ω_0 is a symplectic form on $\mathbb{R}^N$ (this is an exercise below). Thus, $(\mathbb{R}^N, \omega_0)$ is a symplectic vector space. By Proposition 8.3.4, we may write

$$\omega_0(v, w) = v^T Aw$$

for some $N \times N$ skew-symmetric, invertible matrix.

Note that

$$\det A = \det(A^T) = \det(-A) = (-1)^N \det A.$$

Since A is invertible, $\det A \neq 0$ and so we can divide to get $1 = (-1)^N$. It follows that N must be even. $\square$

This is the first significant difference in the theory of symplectic forms compared to that of inner products. Inner products exist in all dimension (roughly, because an inner product corresponds to an invertible *symmetric* matrix, rather than an invertible *skew-symmetric* matrix), while symplectic structures exist only in even dimensions.

Next, let's continue to investigate the notion of nondegeneracy in Definition 8.3.1. In the previous subsection, we saw that we could interpret the nondegeneracy condition on $\mathbb{R}^N$ as the statement that the corresponding matrix skew-symmetric matrix A is invertible. On the other hand, we can intuitively think of a symplectic form as describing a natural pairing between certain "position" directions and corresponding "momentum" directions, though of course this is just a metaphor in an abstract symplectic vector space. Given a nonzero vector $v \in V$, if you interpret it as a "position" vector then the corresponding vector $w \in V$ in (3) of the above definition roughly represents a corresponding momentum direction.

We can alternatively think about nondegeneracy for a more abstract perspective. Recall the definition of the *dual space* of a vector space V :

$$V^* = \{ f : V \rightarrow \mathbb{R} \mid f \text{ is linear} \}.$$

In words, V^* is the space of all linear maps $V \rightarrow \mathbb{R}$. Also recall that if V is finite dimensional, then V and V^* are isomorphic vector spaces (they have the same dimension). Much like an inner product, nondegeneracy of ω can be viewed as a natural isomorphism between V and V^* .

Proposition 8.3.8. *Let (V, ω) be a symplectic vector space. For any $v \in V$, define $f_v : V \rightarrow \mathbb{R}$ by $f_v(w) := \omega(v, w)$ for all $w \in V$. Then the map*

$$\Phi : V \rightarrow V^*$$

defined by $\Phi(v) := f_v$ is an isomorphism.

Proof. Because $\dim V = \dim V^*$, to prove that Φ is an isomorphism it suffices to prove that Φ is injective. Suppose that $\Phi(v) = 0$. Here, $0 \in V^*$ is the zero map $0 : V \rightarrow \mathbb{R}$. This means that $f_v = 0$, and hence $f_v(w) = 0$ for all $w \in V$. It follows that

$$\omega(v, w) = 0$$

for all $w \in W$. By nondegeneracy of ω , it must be the case that $v = 0$, for otherwise there would be some $w \in W$ such that $\omega(v, w) \neq 0$. Thus, Φ is injective, as desired. $\square$

I want to describe why this proposition is so important in the context of symplectic geometry and Hamiltonian mechanics, but it is a bit difficult to do so without appealing to some

other concepts in math that we have not discussed in these notes. If you have taken multi-variable calculus at a fairly rigorous level (where you understand the derivative as a linear transformation — see, for example, [Shu18], or my notes [Bre20]) then you might find this remark interesting. Otherwise, feel free to skip the following remark.

Remark 8.3.9. To emphasize one perspective of the above proposition, it says that if you pick a linear functional on V (i.e., a linear map $f : V \rightarrow \mathbb{R}$) then there is an associated vector $v \in V$ that corresponds to that functional via the symplectic form. For the moment, let $V = \mathbb{R}^{2n}$. Recall that if $H : V \rightarrow \mathbb{R}$ is any function (not necessarily linear), then the derivative $D_p H : V \rightarrow \mathbb{R}$ is a linear map for all $p \in V$. Thus, one way generate a whole family of linear functionals on V is by picking any differentiable function $H : V \rightarrow \mathbb{R}$ and considering the derivative at each point in V . By the above proposition, for each of the linear functionals $D_p H : V \rightarrow \mathbb{R}$ there is an associated vector $X_H(p) \in V$. This is exactly what I called the *symplectic gradient* in Section 8.1! Collecting all of these vectors for all points $p \in V$ gives us a vector field X_H on V . This vector field is the *symplectic gradient vector field* or the *Hamiltonian vector field* of the function H . If you interpret the function $H : V \rightarrow \mathbb{R}$ as an energy function on the phase space V , then the vector field X_H given by the above proposition is the vector field that determines the dynamics of the resulting system of Hamiltonian mechanics. Describing Hamiltonian mechanics this way in abstract symplectic geometry turns out to be very powerful, and the previous proposition is what makes this formulation possible.

8.3.3 Symplectic forms induce volume measurements

An important property of symplectic vector spaces that I want to describe in this section is related to volume measurement. I discussed above that a symplectic form can be thought of as a tool that measures a funny kind of “area”, a two-dimensional quantity. This area — what I called *symplectic area* — is in general not the same as the usual notion of area. However, a symplectic form does give you a natural way to measure *volume* in the normal sense. To be clear, when I say “volume” I mean something like a $2n$ -dimensional hypervolume, where $\dim V = 2n$. This is the same kind of “volume” that a determinant measures. Indeed, recall that if $v_1, \dots, v_{2n}$ is a set of $2n$ vectors in a $2n$ -dimensional vector space $V = \mathbb{R}^{2n}$, the quantity

$$\det(v_1, \dots, v_{2n})$$

measures the “volume” of the hyperparallelepiped spanned by the vectors. In terms of matrices, this would just be the determinant of the matrix whose columns are the vectors $v_1, \dots, v_{2n}$. I claim that symplectic forms can be used to measure *this* kind of volume.

Describing this in general requires a bit more algebra than we’ve developed in this chapter, so I’ll show you a specific instance of the idea in the case of $\mathbb{R}^4$. If you want to know more about the general algebraic machinery that goes into this, check out Chapter 7. For the moment, consider the vector space $\mathbb{R}^4$ and let ω be any symplectic form (not necessarily the standard one). In words, ω is a function that eats *two* vectors in $\mathbb{R}^4$ and spits out a funny symplectic area measurement of the parallelogram spanned by those two vectors. I’m going to define a new function called $\omega \wedge \omega$, which eats in *four* vectors. You can think about $\omega \wedge \omega$ as being a funny kind of “product”, where we are taking ω “times” itself. But as you’ll see, it will be more complicated than a normal product.

Definition 8.3.10. Let $V = \mathbb{R}^4$, and let $\omega : V \times V \rightarrow \mathbb{R}$ be a symplectic form on V . Define a new function

$$\omega \wedge \omega : V \times V \times V \times V \rightarrow \mathbb{R}$$

by the following formula. For $v_1, v_2, v_3, v_4 \in V$, set

$$(\omega \wedge \omega)(v_1, v_2, v_3, v_4) := \omega(v_1, v_2)\omega(v_3, v_4) - \omega(v_1, v_3)\omega(v_2, v_4) + \omega(v_1, v_4)\omega(v_2, v_3).$$

The punchline of this subsection is that this new function $\omega \wedge \omega$ is actually just the determinant (well, possibly a scalar multiple of the determinant) in disguise! Hence, $\omega \wedge \omega$ is a tool that measures normal (scaled, possibly negative) volumes in a vector space.

Theorem 8.3.11. Let $V = \mathbb{R}^4$, and let $\omega : V \times V \rightarrow \mathbb{R}$ be a symplectic form on V . Then

$$\omega \wedge \omega = C \det$$

for some $C \neq 0$. Moreover, $\omega_{\text{std}} \wedge \omega_{\text{std}} = -\det$.

Proof. We will prove this theorem by appealing to Theorem 6.2.1. In particular, recall that the determinant satisfies the property that it is the *unique* (up to a scalar) multilinear, alternating map $V \times V \times V \times V \rightarrow \mathbb{R}$. Thus, to prove the first part Theorem 8.3.11 it suffices to show that $\omega \wedge \omega$ is multilinear and alternating.

First, we verify multilinearity. We will only explicitly verify additivity in the first component, and will leave the rest of the verification (which is identical) as an exercise. Let $v_1, v'_1, v_2, v_3, v_4 \in V$. Then

$$\begin{aligned} (\omega \wedge \omega)(v_1 + v'_1, v_2, v_3, v_4) &= \omega(v_1 + v'_1, v_2)\omega(v_3, v_4) \\ &\quad - \omega(v_1 + v'_1, v_3)\omega(v_2, v_4) \\ &\quad + \omega(v_1 + v'_1, v_4)\omega(v_2, v_3) \\ &= \omega(v_1, v_2)\omega(v_3, v_4) + \omega(v'_1, v_2)\omega(v_3, v_4) \\ &\quad - \omega(v_1, v_3)\omega(v_2, v_4) - \omega(v'_1, v_3)\omega(v_2, v_4) \\ &\quad + \omega(v_1, v_4)\omega(v_2, v_3) + \omega(v'_1, v_4)\omega(v_2, v_3) \\ &= (\omega \wedge \omega)(v_1, v_2, v_3, v_4) + (\omega \wedge \omega)(v'_1, v_2, v_3, v_4) \end{aligned}$$

as desired. The other multilinearity properties are similar.

Next, we verify that $\omega \wedge \omega$ is alternating. Again, we will only do so explicitly for one swap. That is, we will show that

$$(\omega \wedge \omega)(v_1, v_2, v_3, v_4) = -(\omega \wedge \omega)(v_2, v_1, v_3, v_4)$$

and leave the rest of the similar verification computations as an exercise. Note that

$$\begin{aligned}
(\omega \wedge \omega)(v_2, v_1, v_3, v_4) &= \omega(v_2, v_1)\omega(v_3, v_4) \\
&\quad - \omega(v_2, v_3)\omega(v_1, v_4) \\
&\quad + \omega(v_2, v_4)\omega(v_1, v_3) \\
&= -\omega(v_1, v_2)\omega(v_3, v_4) \\
&\quad - \omega(v_1, v_4)\omega(v_2, v_3) \\
&\quad + \omega(v_1, v_3)\omega(v_2, v_4) \\
&= -[\omega(v_1, v_2)\omega(v_3, v_4) - \omega(v_1, v_3)\omega(v_2, v_4) + \omega(v_1, v_4)\omega(v_2, v_3)] \\
&= -(\omega \wedge \omega)(v_1, v_2, v_3, v_4)
\end{aligned}$$

which gives the desired alternating property. The other swaps are similar.

Thus, $\omega \wedge \omega$ is a multilinear, alternating map. By Theorem 6.2.1 and its immediate corollary, it follows that $\omega \wedge \omega = C \det$ for some $C \in \mathbb{R}$. The fact that $C \neq 0$ follows from nondegeneracy; the details are left as an exercise.

Finally, we verify that $\omega_{\text{std}} \wedge \omega_{\text{std}} = -\det$. To do this, by the corollary of Theorem 6.2.1 it suffices to prove that $(\omega_{\text{std}} \wedge \omega_{\text{std}})(e_1, e_2, f_1, f_2) = -1$, where $\{e_1, e_2, f_1, f_2\}$ is the standard basis. We have

$$\begin{aligned}
(\omega_{\text{std}} \wedge \omega_{\text{std}})(e_1, e_2, f_1, f_2) &= \omega_{\text{std}}(e_1, e_2)\omega_{\text{std}}(f_1, f_2) \\
&\quad - \omega_{\text{std}}(e_1, f_1)\omega_{\text{std}}(e_2, f_2) \\
&\quad + \omega_{\text{std}}(e_1, f_2)\omega_{\text{std}}(e_2, f_1) \\
&= 0 \cdot 0 - 1 \cdot 1 + 0 \cdot 0 \\
&= -1.
\end{aligned}$$

This completes the proof. □

8.3.4 The cotangent symplectic vector space

Finally, in this subsection I want to describe a way to build a symplectic vector space starting with *any* real vector space, and thus gives us a way to build a million examples of symplectic vector spaces.

Theorem 8.3.12 (Cotangent symplectic vector space construction). *Let W be a real vector space of dimension $n \in \mathbb{Z}_{\geq 1}$. Consider the product vector space $W \times W^*$, where $W^* = \{f : W \rightarrow \mathbb{R} \mid f \text{ is linear}\}$ is the dual space of W . Define $\omega_{\text{cot}} : (W \times W^*) \times (W \times W^*) \rightarrow \mathbb{R}$ by*

$$\omega_{\text{cot}}((v, f), (w, g)) := g(v) - f(w).$$

Then ω_{cot} is a symplectic form and hence $(W \times W^, \omega_{\text{cot}})$ is a symplectic vector space of dimension $2n$, called the **cotangent vector space of W** .*

Proof. First, we verify that ω_{cot} is antisymmetric. Note that

$$\omega_{\text{cot}}((w, g), (v, f)) = f(w) - g(v) = -(g(v) - f(w)) = \omega_{\text{cot}}((v, f), (w, g)).$$

Next, we verify multilinearity. We do this explicitly for additivity in the first component and leave the rest of the verification as an exercise. Note that

$$\begin{aligned} \omega_{\text{cot}}((v, f) + (v', f'), (w, g)) &= \omega_{\text{cot}}((v + v', f + f'), (w, g)) \\ &= g(v + v') - (f + f')(w) \\ &= g(v) + g(v') - f(w) - f'(w) \\ &= [g(v) - f(w)] + [g(v') - f'(w)] \\ &= \omega_{\text{cot}}((v, f), (w, g)) + \omega_{\text{cot}}((v', f'), (w, g)). \end{aligned}$$

Finally, we leave nondegeneracy as an exercise. In particular, see Exercise 5. $\square$

Exercises

1. Prove that if ω is a symplectic form, then $\omega(v, v) = 0$ for all $v \in V$.
2. Let A be an $2n \times 2n$ invertible, skew-symmetric matrix. Prove that $\omega(v, w) := v^T A w$ is a symplectic form (this is the converse to Proposition 8.3.4). Determine if $\omega'(v, w) = w^T A v$ is also a symplectic form.
3. Verify that ω_0 as defined in the proof of Proposition 8.3.7 is in fact a symplectic form. That is — to state the problem a bit more generally — let (W, ω) be a symplectic vector space, and let V be a vector space isomorphic to W . Let $T : V \rightarrow W$ be any isomorphism. Prove that

$$\omega_0(v, w) := \omega(T(v), T(w)) \quad \text{for all } v, w \in V$$

defines a symplectic form on V .

4. Let ω be a symplectic form on $\mathbb{R}^4$. Prove that $\omega \wedge \omega$ as in Theorem 8.3.11 is not the zero map.
5. Let W be any real vector space and let $(W \times W^*, \omega_{\text{cot}})$ be the cotangent symplectic vector space. Prove that ω_{cot} is nondegenerate in the following steps below.
 - (a) Let $\{w_1, \dots, w_n\}$ be any basis for W . Define the *dual basis* of W^* in the usual way: For each j , let $w_j^* : W \rightarrow \mathbb{R}$ be the linear map defined by

$$w_j^*(w_i) := \begin{cases} 1 & i = j \\ 0 & i \neq j \end{cases}$$

and extend by linearity. Prove that

$$\{(w_1, 0), \dots, (w_n, 0), (0, w_1^*), \dots, (0, w_n^*)\}$$

is a basis of $W \times W^*$.

- (b) Prove that $\omega_{\text{cot}}((w_j, 0), (0, w_j^*)) = 1$.
- (c) Prove that ω_{cot} is nondegenerate.

8.4 Symplectic linear transformations

In Part I, we introduced vector spaces, and after studying some of their basic properties we proceeded to study their transformations in Chapter 3. This is in line with a general theme in mathematics that I've mentioned, which is that it is fruitful to study the transformations of an object in order to study the object itself. In the case of vector spaces, this amounted to the study of functions $T : V \rightarrow W$ where V and W were vector spaces. However, we didn't want to just study *any* such function — we specifically wanted to study functions that respected the underlying structure of vector spaces. This led to the definition of a *linear transformation*. These are precisely the functions between vector spaces that respect the underlying vector space structure. We want to do the same thing for symplectic vector spaces. That is, given symplectic vector spaces (V_1, ω_1) and (V_2, ω_2) , we want to consider special kinds of functions $T : (V_1, \omega_1) \rightarrow (V_2, \omega_2)$ that respect the structure of the symplectic vector space. By studying such functions, we will learn much more about what it means to be symplectic in general.

What should it mean for a map $T : (V_1, \omega_1) \rightarrow (V_2, \omega_2)$ to “respect the structure of the symplectic vector space”? Well, we are still dealing with vector spaces, so we will still require T to be linear. But we also have symplectic forms to consider now. The correct notion of “respecting the symplectic structure” is given by the following definition.

Definition 8.4.1. Let $T : (V_1, \omega_1) \rightarrow (V_2, \omega_2)$ be a map between symplectic vector spaces. We say that T is a **linear symplectic transformation** if T is linear and if

$$\omega_1(v, w) = \omega_2(T(v), T(w))$$

for all $v, w \in V_1$. If $T : (V_1, \omega_1) \rightarrow (V_2, \omega_2)$ is a linear symplectic transformation which is also an isomorphism, we say that T is a **symplectomorphism** and we say that (V_1, ω_1) and (V_2, ω_2) are **symplectomorphic** vector spaces.

Remark 8.4.2. The word *symplectomorphism* is objectively the coolest word in all of math. In fact, this is the main reason why I became a symplectic geometer!

It is worth investigating this definition a bit more, because the condition

$$\omega_1(v, w) = \omega_2(T(v), T(w))$$

can take a bit of time to absorb. In words, it says that if you precompose the symplectic form ω_2 with T , you actually get ω_1 . In succinct notation, we could⁶ also write $\omega_1 = \omega_2 \circ T$. In picture form, we could consider the following diagram:⁷

⁶I will not use the notation or verbiage here, but it is more standard to call ω_1 the **pullback** of ω_2 by T , and to write $\omega_1 = T^* \omega_2$. If you pursue more math, you will undoubtedly come across this notion.

⁷If you're a more advanced mathematician and you're reading this, I know the diagram is a bit bogus — the domain of ω_j is not V_j , it's $V_j \times V_j$. It's only meant to be a helpful picture, not necessarily a literal commutative diagram.

$$\begin{array}{ccc}
& \mathbb{R} & \\
\omega_1 \uparrow & \nwarrow \omega_2 & \\
V_1 & \xrightarrow{T} & V_2
\end{array}$$

Starting at V_1 , we can feed vectors $v, w \in V_1$ to ω_1 to spit out a real number. Alternatively, we could take the same vectors, first feed them to T — which takes them to V_2 — and then feed the output to ω_2 . This would give us a real number. A symplectic transformation is one such that you get the same number each way! In other words, a symplectic transformation is one that allows you to go from V_1 to $\mathbb{R}$ in the above picture in any which way you want.

To compare this definition with one from the world of inner product spaces, recall that a transformation $T : (V_1, \langle \cdot, \cdot \rangle_1) \rightarrow (V_2, \langle \cdot, \cdot \rangle_2)$ between two (real, for now) inner product spaces is *orthogonal* if

$$\langle v, w \rangle_1 = \langle T(v), T(w) \rangle_2$$

for all $v, w \in V_1$. This is the exact analogue of a symplectic transformation. An orthogonal transformation “preserves the inner product structure” in exactly the same way that a symplectic transformation “preserves the symplectic structure.” In other words, orthogonal transformations preserve notions from our normal understanding of geometry, like length and angle. Likewise, symplectic transformations preserve notions from the bizarre world of symplectic geometry.

Let’s look at some concrete examples and non-examples of symplectic transformations to get a feel for what they are, and then we will prove some general properties about them.

Example 8.4.3. Let (V, ω) be a symplectic vector space. The identity transformation $I_V : V \rightarrow V$ is clearly a symplectic linear transformation, since $\omega(v, w) = \omega(I_V(v), I_V(w))$ for all $v, w \in V$. Since the identity map is an isomorphism, it is in fact a symplectomorphism.

Example 8.4.4. Consider $(\mathbb{R}^4, \omega_{\text{std}})$. Define a linear transformation $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ by $T(v) = Av$, where A is the following 4×4 matrix:

$$A = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}.$$

We claim that T is a symplectomorphism of $(\mathbb{R}^4, \omega_{\text{std}})$ to itself. The matrix A is clearly invertible and so T is an isomorphism. Thus, it suffices to show that T is a symplectic transformation, i.e.,

$$\omega_{\text{std}}(v, w) = \omega_{\text{std}}(T(v), T(w))$$

for all $v, w \in \mathbb{R}^4$. To do this, we will use Equation (8.3.1). Let $v = (x_1, x_2, y_1, y_2)$ and $w = (x'_1, x'_2, y'_1, y'_2)$. Recall by (8.3.1) we have

$$\omega_{\text{std}}(v, w) = (x_1 y'_1 - x'_1 y_1) + (x_2 y'_2 - x'_2 y_2).$$

We have

$$T(v) = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}x_1 \\ \frac{1}{3}x_2 \\ 2y_1 \\ 3y_2 \end{pmatrix}$$

and

$$T(w) = \begin{pmatrix} \frac{1}{2} & 0 & 0 & 0 \\ 0 & \frac{1}{3} & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} x'_1 \\ x'_2 \\ y'_1 \\ y'_2 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}x'_1 \\ \frac{1}{3}x'_2 \\ 2y'_1 \\ 3y'_2 \end{pmatrix}.$$

Thus, again using (8.3.1)

$$\begin{aligned} \omega_{\text{std}}(T(v), T(w)) &= \left(\frac{1}{2}x_1 \cdot 2y'_1 - \frac{1}{2}x'_1 \cdot 2y_1 \right) + \left(\frac{1}{3}x_2 \cdot 3y'_2 - \frac{1}{3}x'_2 \cdot 3y_2 \right) \\ &= (x_1y'_1 - x'_1y_1) + (x_2y'_2 - x'_2y_2) \\ &= \omega_{\text{std}}(v, w). \end{aligned}$$

This verifies that T is a symplectomorphism.

This example is indicative of some general properties of symplectic transformations. Consider the behavior of the above symplectomorphism when restricted to the (e_1, f_1) -plane — that is, the first and third coordinate — and the (e_2, f_2) -plane — the second and fourth coordinate. In the (e_1, f_1) -plane, the e_1 -direction is contracted by a factor of $\frac{1}{2}$, and the f_1 -direction is expanded by a factor of 2. This was necessary to preserve the symplectic structure, and you can see this in the computation above. If you scale the first coordinate by $\frac{1}{2}$, you need to compensate in the third coordinate with a factor of 2. The same thing happens in the (e_2, f_2) -plane, where a contraction by $\frac{1}{3}$ in the e_2 -direction is compensated for by an expansion by 3 in the f_2 -direction.

Another way to describe this phenomenon is using the perspective of position and momentum. As we have seen, the symplectic form exhibits a kind of duality between position and momentum, and the above example suggests that if you squeeze position, you have to expand in the momentum direction, and vice versa. Yet another way to describe this principle is in terms of symplectic area, for which it may be helpful to reference the discussion in 8.3.1. A symplectic transformation must preserve symplectic area!

Example 8.4.5. Consider $(\mathbb{R}^4, \omega_{\text{std}})$. Define a linear transformation $T : \mathbb{R}^4 \rightarrow \mathbb{R}^4$ by $T(v) = Av$, where A is the following 4×4 matrix:

$$A = \begin{pmatrix} 7 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 \\ 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}.$$

Although T is a linear isomorphism, it is not a symplectomorphism.⁸ Heuristically, the trans-

⁸Just to introduce some more fun terminology that we won't discuss — T is not a symplectomorphism, but it is a *conformal symplectomorphism*!

formation scales both position and momentum directions by 7, and thus will not preserve symplectic area. Indeed, one can verify that

$$\omega_{\text{std}}(T(v), T(w)) = 49 \omega_{\text{std}}(v, w)$$

for all $v, w \in V$.

In general, these examples suggest that the property of “respecting the symplectic structure” is a fairly strong one. This is evident by the following proposition, which is the analogue of the fact in inner product spaces that orthogonal transformations are always injective.

Proposition 8.4.6. *Let $T : (V_1, \omega_1) \rightarrow (V_2, \omega_2)$ be a symplectic linear transformation. Then T is injective. In particular, $\dim V_1 \leq \dim V_2$.*

Proof. Let $T : (V_1, \omega_1) \rightarrow (V_2, \omega_2)$ be a symplectic linear transformation, and suppose that $T(v) = 0$. Then for all $w \in V_1$, since $T(v) = 0$ we have

$$\begin{aligned} 0 &= \omega_2(T(v), T(w)) \\ &= \omega_1(v, w). \end{aligned}$$

The second equality follows from the fact that T is a symplectic transformation. By nondegeneracy of ω_1 , the fact that $\omega_1(v, w) = 0$ for all $w \in V$ implies that $v = 0$. Thus, T is injective. $\square$

An important fact in linear algebra is that every finite dimensional vector space (over a given field) is isomorphic. In fact, this is one of the major punchlines of abstract linear algebra. In the world of inner product spaces, it is also a fact that all inner product spaces of a fixed dimension are “isomorphic as inner product spaces”, i.e., given any two inner product spaces $(V_1, \langle \cdot, \cdot \rangle_1)$ and $(V_2, \langle \cdot, \cdot \rangle_2)$ of the same dimension, there is an orthogonal isomorphism $T : V \rightarrow V$. This essentially follows from the fact that every inner product space has an orthonormal basis via Gram-Schmidt. In the symplectic setting, the situation is the same: every symplectic vector space admits a “standard symplectic basis”, which is the analogue of an orthonormal basis, and therefore all symplectic vector spaces of a fixed dimension are symplectomorphic. This is made precise by the following theorem.

Theorem 8.4.7. *Let (V, ω) be a symplectic vector space with $\dim V = 2n$. Then (V, ω) is symplectomorphic to $(\mathbb{R}^{2n}, \omega_{\text{std}})$.*

Remark 8.4.8. The proof of this theorem is fairly involved, but it is a first witness to a number of concepts that we will talk about in the future (symplectic orthogonal complements, symplectic subspaces, standard symplectic bases, symplectic projections, etc).

Proof. We will prove Theorem 8.4.7 by finding a basis $\{v_1, \dots, v_n, w_1, \dots, w_n\}$ of V such that

$$\begin{cases} \omega(v_i, v_j) = 0 & \text{for all } i, j \\ \omega(w_i, w_j) = 0 & \text{for all } i, j \\ \omega(v_i, w_j) = 0 & \text{for all } i \neq j \\ \omega(v_i, w_i) = 1 & \text{for all } i \end{cases} \quad (8.4.1)$$

Indeed, suppose that we have identified such a basis. Then define $T : V \rightarrow \mathbb{R}^{2n}$ by $T(v_i) := e_i$ and $T(w_i) := f_i$, and then extend by linearity, where $\{e_1, \dots, e_n, f_1, \dots, f_n\}$ is the standard basis as described in 8.3.1. Since T maps a basis onto a basis, it is clearly invertible. Moreover, since

$$\omega_{\text{std}}(T(v_i), T(w_j)) = \omega_{\text{std}}(e_i, f_j) = \omega(v_i, w_j)$$

for all i, j by construction of the basis $\{v_1, \dots, v_n, w_1, \dots, w_n\}$, and likewise all other combinations of basis vectors, it follows by multilinearity of the symplectic forms that $\omega(v, w) = \omega_{\text{std}}(T(v), T(w))$ for all $v, w \in V$. Thus, T is a symplectomorphism, as desired.

Therefore, it does indeed suffice to find a basis of V satisfying (8.4.1). We will do this by an analogue of Gram-Schmidt, and we will proceed by induction on n . First, note that the base case $n = 0$ (when $V = \{0\}$) is trivial.

For the inductive step, suppose that for some $k \geq 0$, for any symplectic vector space of dimension $2k$ we can find a basis $\{v_1, \dots, v_k, w_1, \dots, w_k\}$ satisfying (8.4.1). Let (V, ω) be a symplectic vector space of dimension $2(k+1) = 2k+2$. Pick any nonzero vector $v_1 \in V$. By nondegeneracy of ω , there exists a vector $w_1 \in V$ such that $\omega(v_1, w_1) = 1$. Let $W_1 := \text{Span}(v_1, w_1)$.

Next, we will define a different subspace of V as follows. Let

$$W_2 := \{v \in V \mid \omega(v_1, v) = 0 \text{ and } \omega(w_1, v) = 0\}.$$

Note that $W_2 = \ker \Phi(v_1) \cap \ker \Phi(w_1)$, where $\Phi(v_1), \Phi(w_1) : V \rightarrow \mathbb{R}$ are as in Proposition 8.3.8, and thus W_2 is indeed a subspace. We make two further claims about W_2 .

Claim 1: $V = W_1 \oplus W_2$. To prove this, we need to show that $W_1 \cap W_2 = \{0\}$ and that $V = W_1 + W_2$. First, suppose that $v \in W_1 \cap W_2$. Because $v \in W_1$, we have $v = a v_1 + b w_1$ for some $a, b \in \mathbb{R}$. Because $v \in W_2$, we have $\omega(v_1, v) = \omega(w_1, v) = 0$. This implies

$$0 = \omega(v_1, a v_1 + b w_1) = a \omega(v_1, v_1) + b \omega(v_1, w_1) = b.$$

The identical computation with w_1 shows that $a = 0$, and so $v = 0$. This implies $W_1 \cap W_2 = \{0\}$.

Next, let $v \in V$. Define $\tilde{v} := \omega(v, w_1) v_1 + \omega(v_1, v) w_1$, and let $v^\perp = v - \tilde{v}$. Note that $\tilde{v} \in W_1$ and that $v = \tilde{v} + v^\perp$. Thus, to finish the proof of Claim 1 it suffices to show that $v^\perp \in W_2$. Note that

$$\begin{aligned} \omega(v_1, v^\perp) &= \omega(v_1, v - \tilde{v}) \\ &= \omega(v_1, v) - \omega(v_1, \tilde{v}) \\ &= \omega(v_1, v) - \omega(v_1, \omega(v, w_1) v_1 + \omega(v_1, v) w_1) \\ &= \omega(v_1, v) - \omega(v_1, v) \omega(v_1, w_1) \\ &= \omega(v_1, v) - \omega(v_1, v) \\ &= 0. \end{aligned}$$

Similarly,

$$\begin{aligned}
\omega(w_1, v^\perp) &= \omega(w_1, v - \tilde{v}) \\
&= \omega(w_1, v) - \omega(w_1, \tilde{v}) \\
&= \omega(w_1, v) - \omega(w_1, \omega(v, w_1) v_1 + \omega(v_1, v) w_1) \\
&= \omega(w_1, v) - \omega(v, w_1) \omega(w_1, v_1) \\
&= \omega(w_1, v) + \omega(v, w_1) \\
&= 0.
\end{aligned}$$

This verifies that $v^\perp \in W_2$. Thus, $V = W_1 + W_2$, and since $W_1 \cap W_2 = \{0\}$ it follows that $V = W_1 \oplus W_2$.

Claim 2: (W_2, ω) is a symplectic vector space. In other words, this claim says that ω restricts to a symplectic form on the subspace W_2 . The antisymmetry and multilinearity properties immediately follow from those of ω on V , so it simply remains to prove that ω is nondegenerate on W_2 . Let $v \neq 0 \in W_2$. We need to find a $w \in W_2$ such that $\omega(v, w) \neq 0$. By nondegeneracy of ω on V , there is a $w' \in V$ such that $\omega(v, w') \neq 0$. Because $V = W_1 \oplus W_2$ by Claim 1, we can write $w' = \tilde{w} + w$ where $\tilde{w} \in W_1$ and $w \in W_2$. Note then that

$$\omega(v, w) = \omega(v, w) + \omega(v, \tilde{w}) = \omega(v, w') \neq 0.$$

Thus, ω is nondegenerate on W_2 and hence (W_2, ω) is a symplectic vector space.

With these two claims, we can now finish the inductive step of the proof. Since $V = W_1 \oplus W_2$, it follows that $\dim W_2 = \dim V - \dim W_1 = (2k + 2) - 2 = 2k$. Since (W_2, ω) is a $2k$ -dimensional symplectic vector space, by the inductive hypothesis there is a basis

$$\{v_2, \dots, v_k, w_2, \dots, w_k\}$$

satisfying (8.4.1). Adding in v_1 and v_2 from W_1 then gives a basis $\{v_1, v_2, \dots, v_k, w_1, w_2, \dots, w_k\}$ of V satisfying (8.4.1), and we are done. $\square$

8.4.1 Symplectic matrices

In linear algebra, we study abstract linear transformations $T : V \rightarrow W$ between vector spaces. In the finite dimensional case this is more or less equivalent to the study of matrices via the study of coordinates. The nicest types of linear maps are isomorphisms, because these let us classify vector spaces by their dimension. Invariably, we end up being interested in only the invertible linear maps from $\mathbb{R}^n$ to itself, as all finite dimensional vector spaces are isomorphic to some $\mathbb{R}^n$. This leads to the study of the invertible matrices:

$$GL(n) := \{A \in M_n(\mathbb{R}) \mid A \text{ is invertible}\}.$$

An alternative way of describing $GL(n)$, the set of invertible matrices, is as the set of matrices satisfying the condition $\det A \neq 0$.

Similarly, in the world of (say, real) inner product spaces we become interested in the matri-

ces that correspond to invertible orthogonal (inner product preserving) transformations. These matrices are the *orthogonal* matrices:

$$O(n) := \{ A \in M_n(\mathbb{R}) \mid A \text{ is orthogonal} \}.$$

It turns out that preserving the inner product is equivalent to the condition that $A^T A = I = A A^T$. The study of orthogonal matrices is an important one, and leads to interesting theorems like the

We would like to pursue the same program in the symplectic setting. That is, we want to adopt a matrix-centric study of symplectomorphisms. Theorem 8.4.7 shows that all symplectic vector spaces are isomorphic to $(\mathbb{R}^{2n}, \omega_{\text{std}})$, so it suffices to study symplectomorphisms $T : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$; that is, invertible linear maps $T : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ satisfying

$$\omega_{\text{std}}(T(v), T(w)) = \omega_{\text{std}}(v, w)$$

for all $v, w \in \mathbb{R}^{2n}$. Every such symplectic transformation is given by multiplication by some matrix A . That is, $T(v) = Av$ for some $A \in M_{2n}(\mathbb{R})$. The symplectic condition above then reads

$$\omega_{\text{std}}(Av, Aw) = \omega_{\text{std}}(v, w)$$

for all $v, w \in \mathbb{R}^{2n}$. Next, recall by Corollary 8.3.6 that $\omega_{\text{std}}(v, w) = v^T J_{\text{std}} w$, where $J_{\text{std}} \in M_{2n}(\mathbb{R})$ is given by

$$J_{\text{std}} = \begin{pmatrix} 0_n & I_n \\ -I_n & 0_n \end{pmatrix}.$$

Thus, we can further reformulate the symplectic condition as

$$(Av)^T J_{\text{std}} (Aw) = v^T J_{\text{std}} w$$

and thus $v^T (A^T J_{\text{std}} A) w = v^T J_{\text{std}} w$ for all $v, w \in \mathbb{R}^{2n}$. Thus, it must be the case that $A^T J_{\text{std}} A = J_{\text{std}}$. This leads us to the main definition of this subsection.

Definition 8.4.9. A **symplectic matrix** is a $2n \times 2n$ matrix $A \in M_{2n}(\mathbb{R})$ satisfying $A^T J_{\text{std}} A = J_{\text{std}}$. The **symplectic group**, denoted $Sp(2n)$ is the set of all symplectic matrices:

$$Sp(2n) := \{ A \in M_{2n}(\mathbb{R}) \mid A^T J_{\text{std}} A = J_{\text{std}} \}.$$

As is hopefully clear from the preceding discussion, you should think about symplectic matrices as being the matrix analogue of abstract symplectomorphisms. I'll also emphasize that the condition $A^T J_{\text{std}} A = J_{\text{std}}$ is the symplectic analogue of the orthogonal condition $A^T A = A A^T = I$.

Even though abstract linear algebra is elegant and all-encompassing, it is useful to study matrices because they are concrete and they make it easier to discover properties of the corresponding abstract transformations. An example of this is given by the following proposition.

Proposition 8.4.10. Let $A \in Sp(2n)$ be a symplectic matrix. Then $\det A = 1$. In particular, A is invertible. Moreover, $A^{-1} = -J_{\text{std}} A^T J_{\text{std}}$.

Proof. First, we remark that it is actually surprisingly difficult to prove that $\det A = 1$. However, it is easy to prove that $\det A = \pm 1$. Indeed, by properties of the determinant, since $A^T J_{\text{std}} A = J_{\text{std}}$ it follows that

$$\det(A^T J_{\text{std}} A) = \det J_{\text{std}}$$

and hence

$$\det A^T \det J_{\text{std}} \det A = \det J_{\text{std}}.$$

Next, we claim that $\det J_{\text{std}} \neq 0$. Indeed, one can compute that $J_{\text{std}}^2 = -I_{2n}$, and so $(\det J_{\text{std}})^2 = (-1)^{2n} = 1$. It must be the case that $\det J_{\text{std}} \neq 0$. Then dividing the preceding equality by $\det J_{\text{std}}$ gives $(\det A)^2 = 1$, since $\det A^T = \det A$. Thus, $\det A = \pm 1$. Proving that, in fact, $\det A = 1$, is fairly challenging and is left to Exercise 6.

Since $\det A \neq 0$, A is invertible. Note that

$$A(-J_{\text{std}} A^T J_{\text{std}}) = (A(-J_{\text{std}}) A^T) J_{\text{std}} = (A^T J_{\text{std}} A)^T J_{\text{std}} = J_{\text{std}}^T J_{\text{std}} = I_{2n}.$$

This implies that $A^{-1} = -J_{\text{std}} A^T J_{\text{std}}$, as desired. $\square$

Remark 8.4.11. The fact that the determinant of a symplectic matrix is 1 tells us that symplectomorphisms not only preserve symplectic area, but they also preserve the usual notion of (hyper)volume.

There are many more interesting things that can be said about symplectic matrices. If you find this interesting, I suggest consulting [Sac20]. Some of the exercises below also offer a glimpse at more interesting properties of symplectic matrices.

Exercises

1. For each of the following 4×4 matrices A , determine whether the map $T : (\mathbb{R}^4, \omega_{\text{std}}) \rightarrow (\mathbb{R}^4, \omega_{\text{std}})$ defined by $T(v) = Av$ is a symplectomorphism. Try answering the question based heuristically first, and then verify your answer with computations.

(a)

$$A = \begin{pmatrix} 2 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & \frac{1}{7} & 0 \\ 0 & 0 & 0 & 7 \end{pmatrix}.$$

(b)

$$A = \begin{pmatrix} 4 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 \\ 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 4 \end{pmatrix}.$$

(c)

$$A = \begin{pmatrix} 8 & 0 & 0 & 0 \\ 0 & \frac{1}{8} & 0 & 0 \\ 0 & 0 & \frac{1}{8} & 0 \\ 0 & 0 & 0 & 8 \end{pmatrix}.$$

(d)

$$A = \begin{pmatrix} 2 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 5 & 0 & 3 & 0 \\ 0 & 3 & 0 & 4 \end{pmatrix}.$$

2. Define a map $T : (\mathbb{R}^2, \omega_{\text{std}}) \rightarrow (\mathbb{R}^4, \omega_{\text{std}})$ by $T(v) = Av$, where

$$A = \begin{pmatrix} 2 & 1 \\ 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Is T a symplectic transformation? Can you heuristically explain why or why not? Is it a symplectomorphism?

3. Define a map $T : (\mathbb{R}^2, \omega_{\text{std}}) \rightarrow (\mathbb{R}^4, \omega_{\text{std}})$ by $T(v) = Av$, where

$$A = \begin{pmatrix} 2 & 0 \\ 0 & 0 \\ 0 & 2 \\ 0 & 0 \end{pmatrix}.$$

Is T a symplectic transformation? Can you heuristically explain why or why not? Is it a symplectomorphism?

4. Define a map $T : (\mathbb{R}^4, \omega_{\text{std}}) \rightarrow (\mathbb{R}^2, \omega_{\text{std}})$ by $T(v) = Av$, where

$$A = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Is T a symplectic transformation? Can you heuristically explain why or why not? Is it a symplectomorphism?

5. Prove that a 2×2 matrix A is symplectic if and only if $\det A = 1$. Give an example to show that this is not true in higher dimensions.
6. In this exercise I will walk you through a proof of the fact that if A is a symplectic matrix, then $\det A = 1$. This is an interesting elementary proof that is due to [\[Rim18\]](#). Let $A \in \text{Spn}(2n)$ be a symplectic matrix.

(a) Prove that the matrix $A^T A$ has positive, real eigenvalues.

[Hint: Referencing the theory in Chapter 5 would be helpful.]

(b) Prove that the matrix $A^T A + I_{2n}$ has real eigenvalues that are all > 1 . Conclude that $\det(A^T A + I_{2n}) > 1$.

[Hint: Diagonalize $A^T A$ as $A^T A = PDP^{-1}$ and note that $I_{2n} = PP^{-1}$.]

(c) Prove that $A^T A + I_{2n} = A^T(A - J_{\text{std}}AJ_{\text{std}})$.

(d) Let $A_1, A_2, A_3, A_4 \in M_n(\mathbb{R})$ be matrices defined so that

$$A = \begin{pmatrix} A_1 & A_2 \\ A_3 & A_4 \end{pmatrix}.$$

Prove that

$$A - J_{\text{std}} A J_{\text{std}} = \begin{pmatrix} A_1 + A_4 & A_2 - A_3 \\ -A_2 + A_3 & A_1 + A_4 \end{pmatrix}$$

(e) Next, we will cleverly do a block-diagonalization process that involves introducing complex numbers. Let $C = A_1 + A_4$ and $D = A_2 - A_3$. Prove that

$$A - J_{\text{std}} A J_{\text{std}} = \frac{1}{\sqrt{2}} \begin{pmatrix} I_n & I_n \\ iI_n & -iI_n \end{pmatrix} \begin{pmatrix} C + iD & 0 \\ 0 & C - iD \end{pmatrix} \frac{1}{\sqrt{2}} \begin{pmatrix} I_n & iI_n \\ I_n & -iI_n \end{pmatrix}.$$

(f) Use (c) and (e) to prove that

$$\det(A^T A + I_{2n}) = (\det A) |\det(C + iD)|^2.$$

[Hint: Recall that the determinant of a block-diagonal matrix is the product of the determinants of the blocks, and recall that $|z|^2 = z \bar{z}$ for $z \in \mathbb{C}$.]

(g) Use (b) and (f) to conclude that $\det A > 0$ and hence $\det A = 1$.

7. Prove that J_{std} is a symplectic matrix.

8. Prove that if A, B are symplectic matrices, then AB is a symplectic matrix. Likewise, prove that if A is a symplectic matrix, then A^{-1} is also a symplectic matrix.⁹

9. This exercise describes a way to generate all possible symplectic matrices.

(a) Prove that any matrix of the form

$$A = \begin{pmatrix} G & 0_n \\ 0_n & (G^T)^{-1} \end{pmatrix}$$

where $G \in GL(n)$ is any invertible matrix is a symplectic matrix. Roughly, these types of matrices describe all the transformations of a symplectic vector space that transform the position coordinates in some invertible way (multiplication by G), and then compensate for this by transforming the momentum coordinates in the “opposite” way (multiplication by $(G^T)^{-1}$). This kind of an operation preserves the symplectic area.

(b) Prove that any matrix of the form

$$B = \begin{pmatrix} I_n & S \\ 0_n & I_n \end{pmatrix}$$

⁹For those in the know, this roughly proves that $Sp(2n)$ is a group.

where $S \in M_n(\mathbb{R})$ satisfies $S^T = -S$ is a symplectic matrix. These matrices describe a position-momentum *shearing* of sorts. For example, multiplication on $\mathbb{R}^2$ by the 2×2 matrix

$$\begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

is a shear in the horizontal direction. It maps the unit square to the parallelogram spanned by $(1, 0)$ and $(2, 1)$. Note that shearing operations in $\mathbb{R}^2$ are area-preserving. The matrices of the form B are *symplectic shears* that preserve the symplectic area.

- (c) Using the previous exercises, it follows that if you multiply matrices of the form A above, B above, and J_{std} , you get more symplectic matrices. It turns out that *all* symplectic matrices can be formed this way! I think this is difficult to prove, but try it if you want a challenge!

8.5 Special subspaces

One of the interesting features of a symplectic vector space is the presence of interesting kinds of subspaces. That is, subspaces that “behave” differently or “look” differently from the point of view of the symplectic form. This is in contrast to general vector spaces, where all subspaces essentially “look” the same. Even in inner product spaces, there is no inherent property of certain subspaces that distinguish them from other; all subspaces look the same. This is very much not the case in the symplectic world. There are four very special kinds of subspaces in a symplectic vector space: *isotropic* subspaces, *coisotropic* subspaces, *Lagrangian* subspaces, and *symplectic* subspaces.

To define these subspaces, we first need to introduce the notion of the *symplectic orthogonal complement*. This is the analogue of the usual orthogonal complement in the world of inner product spaces. Before I define the symplectic orthogonal complement, let me remind you of the usual orthogonal complement so that you can see the obvious parallels. For a subset $S \subset (V, \langle \cdot, \cdot \rangle)$ of an inner product space, the *orthogonal complement* of S is the set

$$S^\perp = \{ v \in V \mid \langle v, s \rangle = 0 \text{ for all } s \in S \}.$$

In words, $S^\perp$ is the set of vectors which are perpendicular to every element of S , where perpendicularity is detected by the vanishing of the inner product. In the symplectic world, we have the following definition.

Definition 8.5.1. Let (V, ω) be a symplectic vector space, and let $S \subset V$ be any subset. The **symplectic orthogonal complement** of S , denoted S^ω , is the set

$$S^\omega := \{ v \in V \mid \omega(v, s) = 0 \text{ for all } s \in S \}.$$

We have actually already seen and worked with an instance of the symplectic orthogonal complement, though I didn’t use this language at the time. In the long inductive proof of Theorem 8.4.7, we initially defined the subspace $W_1 = \text{Span}(v_1, w_1)$ and then defined

$$W_2 = \{ v \in V \mid \omega(v_1, v) = 0 \text{ and } \omega(w_1, v) = 0 \}.$$

Note that $W_2 = \{v_1, w_1\}^\omega$, from which it follows that $W_2 = W_1^\omega$. In words, the subspace W_2 was defined to be the symplectic orthogonal complement of W_1 !

Though the definition of the symplectic orthogonal complement is in complete analogy with the usual orthogonal complement in inner product spaces, its behavior is actually very different. For example, in an inner product space, it is always the case that $W^\perp \cap W = \{0\}$. Even more, it is always true that $W^\perp \oplus W = V$. This is very much not the case in the symplectic world. Here is an example to demonstrate this.

Example 8.5.2. Consider the standard symplectic vector space $(\mathbb{R}^4, \omega_{\text{std}})$ with standard symplectic basis $\{e_1, e_2, f_1, f_2\}$. Let $W = \text{Span}(e_1, e_2, f_1)$. We claim that $W^\omega = \text{Span}(e_2)$. In particular, $W^\omega \subset W$. That is, the symplectic orthogonal complement of W is *contained in* W .

To verify this, first, note that

$$\omega_{\text{std}}(e_2, e_1) = \omega_{\text{std}}(e_2, e_2) = \omega_{\text{std}}(e_2, f_1) = 0.$$

This implies that $\omega_{\text{std}}(e_2, w) = 0$ for all $w \in W$, and so $\text{Span}(e_2) \subset W^\omega$. On the other hand, suppose that $v \in W^\omega$. Write

$$v = a e_1 + b e_2 + c f_1 + d f_2$$

for some constants $a, b, c, d \in \mathbb{R}$. The fact that $\omega_{\text{std}}(v, e_1) = 0$ then implies that

$$\begin{aligned} 0 &= \omega_{\text{std}}(v, e_1) \\ &= \omega_{\text{std}}(a e_1 + b e_2 + c f_1 + d f_2, e_1) \\ &= a \omega_{\text{std}}(e_1, e_1) + b \omega_{\text{std}}(e_2, e_1) + c \omega_{\text{std}}(f_1, e_1) + d \omega_{\text{std}}(f_2, e_1) \\ &= -c. \end{aligned}$$

Thus, $c = 0$. We leave it as an exercise to similarly show that $d = a = 0$ using the conditions $\omega_{\text{std}}(v, e_2) = \omega_{\text{std}}(v, f_1) = 0$. Thus, $v = b e_2$ and so $v \in \text{Span}(e_2)$. This verifies that $W^\omega = \text{Span}(e_2) \subset W$.

The following definition identifies four important kinds of subspaces, based on their interaction with the operation of taking the symplectic orthogonal complement.

Definition 8.5.3. Let (V, ω) be a symplectic vector space, and let $W \subset V$ be a subspace.

- (1) The subspace W is **isotropic** if $W^\omega \supset W$.
- (2) The subspace W is **coisotropic** if $W^\omega \subset W$.
- (3) The subspace W is **Lagrangian** if $W^\omega = W$.
- (4) The subspace W is **symplectic** if $W^\omega \cap W = \{0\}$.

I'll begin by describing the canonical examples of these types of subspaces in the standard symplectic vector space. Later, we will prove a major theorem (Theorem 8.5.10) that gives us an equivalent way to understand each of them.

Example 8.5.4 (Standard isotropic subspaces). Let $(\mathbb{R}^{2n}, \omega_{\text{std}})$ be the standard symplectic vector space with standard symplectic basis $\{e_1, \dots, e_n, f_1, \dots, f_n\}$. The standard example (not the

only one!) of an isotropic subspace is

$$W = \text{Span}(e_1, \dots, e_k)$$

for some $k \leq n$. Indeed, one can check as an exercise that

$$W^\omega = \text{Span}(e_1, \dots, e_n, f_{k+1}, \dots, f_n)$$

and hence $W \subset W^\omega$. In words, this is an example of a subspace that only contains position directions. Because it only contains position directions, all other position directions — and a handful of non-corresponding momentum directions — kill off everything in the subspace.

Example 8.5.5 (Standard coisotropic subspaces). Let $(\mathbb{R}^{2n}, \omega_{\text{std}})$ be the standard symplectic vector space with standard symplectic basis $\{e_1, \dots, e_n, f_1, \dots, f_n\}$. The standard example (not the only one!) of a coisotropic subspace is

$$W = \text{Span}(e_1, \dots, e_n, f_{k+1}, \dots, f_n)$$

for some $k \leq n$. Indeed, one can check as an exercise that

$$W^\omega = \text{Span}(e_1, \dots, e_k)$$

and so $W^\omega \subset W$. In words, this is an example of a subspace that contains *all* position directions and some of the corresponding momentum directions. Because it only contains some of the momentum directions, there are some position directions $(e_1, \dots, e_k)$ that kill off everything in the subspace. Note that this coisotropic subspace is exactly the symplectic orthogonal complement of the isotropic subspace in the previous example, and vice versa. This is no accident! See Corollary 8.5.9.

Example 8.5.6 (Standard Lagrangian subspaces). Let $(\mathbb{R}^{2n}, \omega_{\text{std}})$ be the standard symplectic vector space with standard symplectic basis $\{e_1, \dots, e_n, f_1, \dots, f_n\}$. The standard (not the only!) example of a Lagrangian subspace is

$$W = \text{Span}(e_1, \dots, e_n).$$

Indeed, one can check that

$$W^\omega = \text{Span}(e_1, \dots, e_n) = W.$$

In words, this is the subspace containing *all* of the position directions, and no momentum directions. Because it contains all of the position directions, no momentum directions can be in the symplectic orthogonal complement (any given momentum direction would pair nontrivially with one of the position directions). Similarly, because there are *no* momentum directions in the subspace, *all* of the position directions are in the symplectic orthogonal complement.

Example 8.5.7 (Standard symplectic subspaces). Let $(\mathbb{R}^{2n}, \omega_{\text{std}})$ be the standard symplectic vector space with standard symplectic basis $\{e_1, \dots, e_n, f_1, \dots, f_n\}$. The standard example (not the only one!) of a symplectic subspace is

$$W = \text{Span}(e_1, \dots, e_k, f_1, \dots, f_k)$$

for some $k \leq n$. Indeed, one can check that

$$W^\omega = \text{Span}(e_{k+1}, \dots, e_n, f_{k+1}, \dots, f_n)$$

and hence $W \cap W^\omega = \{0\}$. In words, this subspace contains a collection of paired position and momentum directions. Since all position and momentum directions in the subspace have their paired direction in the subspace, the only directions that kill everything in W via the symplectic form are those directions which are “orthogonal” in the usual sense.

Before we prove the main classification theorem of these subspaces, it is worth proving some properties of the symplectic orthogonal complement that *do* resemble those of the usual orthogonal complement.

Proposition 8.5.8. *Let (V, ω) be a symplectic vector space, and let $W, W' \subset V$ be subspaces. Then*

- (i) *if $W \subset W'$, then $(W')^\omega \subset W^\omega$,*
- (ii) *$\dim W + \dim W^\omega = \dim V$, and*
- (iii) *$(W^\omega)^\omega = W$.*

Proof.

- (i) Suppose that $W \subset W'$. Let $v \in (W')^\omega$, and let $w \in W$. Since $W \subset W'$, $w \in W'$. Since $v \in (W')^\omega$, it follows that $\omega(v, w) = 0$. Since $w \in W$ was arbitrary, we have $v \in W^\omega$. This implies $(W')^\omega \subset W^\omega$.
- (ii) The proof of (ii) is actually a bit tricky. Even in the case of the normal inner product, the fact that $\dim W + \dim W^\perp = \dim V$ is nontrivial — it may be insightful to reference the relevant sections of Chapter 5 to compare and contrast the argument below.

Consider the map $\Phi : V \rightarrow W^*$, where $\Phi : V \rightarrow V^*$ is the isomorphism from Proposition 8.3.8 given by $\Phi(v) = f_v$, where $f_v(w) = \omega(v, w)$. When we write $\Phi : V \rightarrow W^*$, we are simply viewing the output of Φ as only defining a linear function on $W \subset V$, not all of V . Note that, by definition of W^ω , we have $\ker(\Phi : V \rightarrow W^*) = W^\omega$.

Next, we claim that $\Phi : V \rightarrow W^*$ is surjective. Let $g : W \rightarrow \mathbb{R}$ be linear. Because $W \subset V$ is a subspace of a finite dimensional vector space, we can select a complementary subspace $U \subset V$ such that $W \oplus U = V$. We can then extend $g : W \rightarrow \mathbb{R}$ to a linear function $\tilde{g} : V \rightarrow \mathbb{R}$ by setting $\tilde{g}(w + u) := g(w)$ where $w \in W$ and $u \in U$. Because $\Phi : V \rightarrow V^*$ is an isomorphism, there is some $v \in V$ such that $\Phi(v) = \tilde{g}$. This means that $\tilde{g}(v') = \omega(v, v')$ for all $v' \in V$. Since $V = W \oplus U$, this implies that $g(w) = \omega(v, w)$ for all $w \in W$, and hence $g = \Phi(v)$ where we view $\Phi : V \rightarrow W^*$.

Thus, we have $\ker(\Phi : V \rightarrow W^*) = W^\omega$ and $\text{im}(\Phi : V \rightarrow W^*) = W^*$. By Rank-Nullity, it follows that

$$\dim W^\omega + \dim W^* = \dim V.$$

Since $\dim W^* = \dim W$, we then get the desired equality $\dim W + \dim W^\omega = \dim V$.

(iii) First, let $w \in W$. We wish to show that $w \in (W^\omega)^\omega$. To do this, let $v \in W^\omega$. Note that $\omega(w, v) = 0$. Since $v \in W^\omega$ was arbitrary, it follows that $w \in (W^\omega)^\omega$.

We can get around showing the reverse containment directly by using (ii). Note that, using two separate applications of (ii), we have

$$\dim W + \dim W^\omega = \dim V$$

and

$$\dim W^\omega + \dim (W^\omega)^\omega = \dim V.$$

Equating these two and simplifying gives $\dim W = \dim (W^\omega)^\omega$. Because we have already shown $W \subset (W^\omega)^\omega$, it then must be the case that $W = (W^\omega)^\omega$.

□

Corollary 8.5.9. *Let (V, ω) be a symplectic vector space, and let $W \subset V$ be a subspace. Then W is isotropic if and only if W^ω is coisotropic.*

Proof. By the previous proposition, note that $W^\omega \supset W$ if and only if $(W^\omega)^\omega \subset W^\omega$. Thus, W is isotropic if and only if W^ω is coisotropic. □

Next, we have the main theorem about subspaces of symplectic vector spaces. Observe that statement (ii) below involves the notion of a quotient vector space; see Section 2.6.1 of Chapter 2 as a reference. If you haven't learned about quotient vector spaces, feel free to ignore this aspect of (ii) — we will not use it in the rest of this chapter.

Theorem 8.5.10. *Let (V, ω) be a symplectic vector space with $\dim V = 2n$.*

- (i) *A subspace W is isotropic if and only if $\omega|_W \equiv 0$. That is, for all $v, w \in W$, $\omega(v, w) = 0$. Moreover, if W is isotropic then $\dim W \leq n$.*
- (ii) *A subspace W is coisotropic if and only if ω induces a symplectic form on W/W^ω . Moreover, if W is coisotropic then $\dim W \geq n$.*
- (iii) *A subspace W is Lagrangian if and only if it is both isotropic and coisotropic. In particular, if W is a Lagrangian subspace then $\dim W = n$.*
- (iv) *A subspace W is symplectic if and only if ω restricts to a symplectic form on W .*

Proof.

- (i) First, suppose that W is isotropic. Then $W \subset W^\omega$. Let $v, w \in W$. Since $v \in W$ and $w \in W \subset W^\omega$, it follows that $\omega(v, w) = 0$ and hence $\omega|_W \equiv 0$. Conversely, suppose that $\omega|_W \equiv 0$. Let $v \in W$. We wish to show that $v \in W^\omega$. To do this, fix $w \in W$ and note that $\omega(v, w) = 0$, since $\omega|_W \equiv 0$. It follows that $v \in W^\omega$, and so $W \subset W^\omega$. Thus, W is isotropic.

This establishes the if and only if statement in (i). To establish the moreover statement, note that $\dim W + \dim W^\omega = 2n$, and since $W \subset W^\omega$ we have $\dim W \leq \dim W^\omega$. This forces $\dim W \leq n$, for otherwise $\dim W + \dim W^\omega > 2n$.

- (ii) First, for convenience we briefly recall the definition of W/W^ω and refer to Section 2.6.1 for more details. As a set, we have

$$W/W^\omega = \{ [v] \mid v \in W \text{ and } [v] = [v'] \text{ if } v - v' \in W^\omega \}.$$

The addition and scalar multiplication operations are defined by

$$[v] + [w] := [v + w] \quad \text{and} \quad \lambda \cdot [v] := [\lambda v].$$

One must show that these operations are actually well-defined, in that the choice of representative v of the equivalence class $[v]$ does not matter. To show that ω restricts to a symplectic form on W/W^ω , we have a number of things to do.

Define ω on W/W^ω . For any $[v], [w] \in W/W^\omega$, set

$$\omega([v], [w]) := \omega(v, w).$$

Checking that ω is well-defined on W/W^ω . The first thing we need to check is that ω is actually a well-defined bilinear form on W/W^ω . It suffices to do this in one component. Let $v, v', w \in W$ be vectors such that $v - v' \in W^\omega$. We wish to show that

$$\omega([v], [w]) = \omega([v'], [w])$$

so that the definition above actually makes sense. Because $v - v' \in W^\omega$ and $w \in W$, we know that $\omega(v - v', w) = 0$, and hence $\omega(v, w) = \omega(v', w)$. This gives us the desired equivalence.

Checking that ω is symplectic. The fact that ω is multilinear and antisymmetric on W/W^ω follows immediately from inheriting the same properties from ω on W . We only need to verify that ω is nondegenerate on W/W^ω . If $W^\omega = W$, then this is vacuously true, so for the rest of the proof we may assume $W^\omega \subsetneq W$. Fix $[v] \neq [0] \in W/W^\omega$. Note that, since $[v] \neq [0]$, it is necessarily the case that $v - 0 \notin W^\omega$. We claim that there is some $w \in W$ such that $\omega(v, w) \neq 0$. If this were not the case, then for all $w \in W$ we would have $\omega(v, w) = 0$. This would force $v \in W^\omega$, but we chose v so that $v \notin W^\omega$. Thus, for this choice of $w \in W$ we have

$$\omega([v], [w]) = \omega(v, w) \neq 0$$

and so ω is indeed a symplectic form on W/W^ω .

For the other direction of the if and only if, note that W/W^ω only makes sense as a vector space if $W^\omega \subset W$, and this forces W to be coisotropic.

- (iii) Note that $W = W^\omega$ if and only if $W \subset W^\omega$ and $W^\omega \subset W$. Thus, W is Lagrangian if and only if W is both isotropic and coisotropic. The dimension constraints from (i) and (ii) imply that $\dim W \leq n$ and $\dim W \geq n$, and hence $\dim W = n$.
- (iv) First, suppose that ω restricts to a symplectic form on W . We wish to show that $W \cap W^\omega = \{0\}$. Let $v \in W \cap W^\omega$. Because $v \in W^\omega$, we have $\omega(v, w) = 0$ for all $w \in W$. But ω is nondegenerate on W , and so if $v \neq 0$ then there would be some $w \in W$ such that

$\omega(v, w) = 0$. Thus, it must be the case that $v = 0$. It follows that $W \cap W^\omega = \{0\}$ and so W is a symplectic subspace.

Next, suppose that W is a symplectic subspace, so that $W \cap W^\omega = \{0\}$. We need to prove that ω restricts to a symplectic form on W . Multilinearity and antisymmetry are automatic from the fact that ω is symplectic on V , and so we only need to verify that ω is nondegenerate on W . Fix $v \neq 0 \in W$. Since ω is nondegenerate on V , there is some $\tilde{w} \in V$ such that $\omega(v, \tilde{w}) \neq 0$. Since $\dim W + \dim W^\omega = \dim V$ and $W \cap W^\omega = \{0\}$, we have $W \oplus W^\omega = V$. Thus, we can write $\tilde{w} = w + w'$ for some $w \in W$ and $w' \in W^\omega$. Note that

$$\omega(v, w) = \omega(v, \tilde{w}) - \omega(v, w') = \omega(v, \tilde{w}) \neq 0.$$

Thus, ω is nondegenerate on W , as desired.

□

Exercises

- Let $(\mathbb{R}^{2n}, \omega_{\text{std}})$ be the standard symplectic vector space. Let $\{e_1, \dots, e_n, f_1, \dots, f_n\}$ be the standard symplectic basis.

- Verify that

$$\text{Span}(e_1, \dots, e_k) \quad \text{and} \quad \text{Span}(f_1, \dots, f_k)$$

are isotropic subspaces. What other isotropic subspaces can you find?

- Verify that

$$\text{Span}(e_1, \dots, e_n, f_{k+1}, \dots, f_n) \quad \text{and} \quad \text{Span}(e_1, \dots, e_k, f_1, \dots, f_n)$$

are coisotropic subspaces. What other coisotropic subspaces can you find?

- Verify that

$$\text{Span}(e_1, \dots, e_n) \quad \text{and} \quad \text{Span}(f_1, \dots, f_n)$$

are Lagrangian subspaces. What other Lagrangian subspaces can you find?

- Verify that

$$\text{Span}(e_1, \dots, e_k, f_1, \dots, f_k)$$

is a symplectic subspace. What other symplectic subspaces can you find?

- Let (V^{2n}, ω) be a symplectic vector space and let $W \subset V$ be a subspace with $\dim W = 1$. Prove that W is isotropic.
- Let (V^{2n}, ω) be a symplectic vector space and let $W \subset V$ be a subspace with $\dim W = 2n - 1$. Prove that W is coisotropic.
- Let (V^2, ω) be a 2-dimensional symplectic vector space. Use the previous exercise to prove that every 1-dimensional subspace $L \subset V$ is Lagrangian.
Is it true in general that every half-dimensional subspace $L^n \subset V^{2n}$ is automatically Lagrangian?

5. Let (V, ω) be a symplectic vector space. Prove that $W \subset V$ is a symplectic subspace if and only if W^ω is a symplectic subspace. Moreover, prove that if W is a symplectic subspace, then $V = W \oplus W^\omega$.
6. Let (V, ω) be a symplectic vector space, and let $L \subset V$ be a Lagrangian subspace.
- (a) Let $\{v_1, \dots, v_n\}$ be any basis of L . Show that this basis can be extended to a symplectic basis $\{v_1, \dots, v_n, w_1, \dots, w_n\}$ of V , in the sense that

$$\begin{cases} \omega(v_i, v_j) = 0 & \text{for all } i, j \\ \omega(w_i, w_j) = 0 & \text{for all } i, j \\ \omega(v_i, w_j) = 0 & \text{for all } i \neq j \\ \omega(v_i, w_i) = 1 & \text{for all } i \end{cases}.$$

- (b) Using the previous part, show that (V, ω) is symplectomorphic to $(L \times L^*, \omega_{\text{cot}})$, the cotangent vector space of the Lagrangian L .¹⁰ In words, this exercise shows that *every* symplectic vector space secretly looks like the cotangent vector space of a Lagrangian subspace. This means that the construction in Theorem 8.3.12 is in some sense the only source of examples for symplectic vector spaces.
7. Let (V_1, ω_1) and (V_2, ω_2) be symplectic vector spaces with $\dim V_1 = \dim V_2$. Define $\tilde{\omega}$ on $V_1 \times V_2$ as follows. For $(v_1, v_2), (w_1, w_2) \in V_1 \times V_2$, set

$$\tilde{\omega}((v_1, v_2), (w_1, w_2)) := \omega_1(v_1, w_1) - \omega_2(v_2, w_2).$$

- (a) Prove that $(V_1 \times V_2, \tilde{\omega})$ is a symplectic vector space.
- (b) Prove that $V_1 \times \{0\}$ and $\{0\} \times V_2$ are symplectic vector spaces of $(V_1 \times V_2, \tilde{\omega})$.
- (c) Let $T : (V_1, \omega_1) \rightarrow (V_2, \omega_2)$ be a symplectomorphism. Consider the graph of T inside $V_1 \times V_2$ given by

$$\text{graph}(T) = \{ (v, w) \in V_1 \times V_2 \mid w = T(v) \}.$$

Prove that $\text{graph}(T)$ is a Lagrangian subspace of $(V_1 \times V_2, \tilde{\omega})$.

8. Symplectic gramm-schmidt, symplectic projections?

8.6 Compatible complex structures

In Section 8.2, I motivated the notion of a symplectic form by considering the standard complex inner product on $\mathbb{C}^n$, decomposing it into its real and imaginary parts as a form on $\mathbb{R}^{2n}$, and then taking the imaginary part. In particular, this process gives the standard symplectic form on $\mathbb{R}^{2n}$. I also discussed how this suggested a fairly interwoven relationship between

¹⁰We already know they are symplectomorphic because we proved that all symplectic vector spaces of the same dimension are symplectomorphic, but I want you to write down an explicit symplectomorphism using the basis in the previous part.

symplectic forms, complex numbers, and real inner products. In this section I want to make this discussion a bit more formal. In particular, one thing we will see is how to start with a symplectic vector space (V, ω) and turn it into a “compatible” real and complex inner product space.

8.6.1 Complex structures

The first thing we need to do is to give a new perspective on what we mean by a complex number. In particular, if we start with a symplectic vector space (V, ω) , it is a *real* vector space. It may be even dimensional, but it is not immediately clear how we might begin talking about complex numbers in this setting.

For the moment, let’s reconsider the vector space $\mathbb{C}^n$, viewed as a complex vector space. Because $i \in \mathbb{C}$ is a scalar, we can multiply vectors by it to produce new vectors in $\mathbb{C}^n$. Moreover, this is a linear operation. That is, we can define a linear map $j : \mathbb{C}^n \rightarrow \mathbb{C}^n$ which is given by multiplication by i :

$$j(v) := i v \quad \text{for all } v \in \mathbb{C}^n.$$

Not only is the map $j : \mathbb{C}^n \rightarrow \mathbb{C}^n$ linear, but it has the special property that it squares to $-I$, the negative identity mapping on $\mathbb{C}^n$. Note that, when I say “square” here, I mean with respect to composition. Thus, my claim is that $j^2 = -I$, which is to say $j \circ j = -I$. Let’s check. For any $v \in \mathbb{C}^n$, we have

$$(j \circ j)(v) = j(j(v)) = j(iv) = i \cdot j(v) = i \cdot i \cdot v = i^2 v = -v = -I(v).$$

Thus, $j^2 = -I$. This property of the linear map j is essentially just a rephrasing of the defining property of i as a complex number, namely, that it is defined to be a number such that $i^2 = -1$.

All of this is to say that we can capture the fundamental nature of the complex number i by instead searching for an abstract linear transformation on a real vector space that squares to the negative identity. This leads to the following definition.

Definition 8.6.1. Let V be a real vector space. A **(linear) complex structure** on V is a linear map $J : V \rightarrow V$ such that $J^2 = -I$.

Example 8.6.2. Consider the vector space $\mathbb{R}^{2n}$, and let

$$J_{\text{std}} = \begin{pmatrix} 0_n & I_n \\ -I_n & 0_n \end{pmatrix}$$

as usual. Define $J : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ by $J(v) := J_{\text{std}}v$. Then J defines a complex structure on $\mathbb{R}^{2n}$. Indeed, it suffices to verify that $J_{\text{std}}^2 = -I_{2n}$, where I_{2n} is the $2n \times 2n$ identity matrix. Performing block multiplication gives

$$J_{\text{std}}^2 = \begin{pmatrix} 0_n & I_n \\ -I_n & 0_n \end{pmatrix} \begin{pmatrix} 0_n & I_n \\ -I_n & 0_n \end{pmatrix} = \begin{pmatrix} 0_n^2 - I_n^2 & 0_n I_n + I_n 0_n \\ -I_n 0_n - 0_n I_n & -I_n^2 + 0_n^2 \end{pmatrix} = \begin{pmatrix} -I_n & 0_n \\ 0_n & -I_n \end{pmatrix} = -I_{2n}.$$

Note also that $\tilde{J}(v) := -J_{\text{std}}v$ also defines a linear complex structure on $\mathbb{R}^{2n}$. In general, there may be many different complex structures on a vector space.

Observe that when we defined a linear complex structure, we made no assumption on the dimension of V . A priori, the dimension could even be odd. However, it is an immediate consequence of the definition of a complex structure that the dimension has to be even! This is consistent with the idea that if we view a complex vector space as a real vector space, it has twice (and hence necessarily even) the complex dimension. That is $\mathbb{C}$ is a 1-dimensional complex object, but a 2-dimensional real object.

Proposition 8.6.3. *Let V be a real vector space and let J be a complex structure on V . Then $\dim V$ is even.*

Proof. Suppose for the sake of contradiction that N is odd. Every linear operator on an odd dimensional real vector space has a real eigenvalue, and so there must be some $v \neq 0 \in V$ and a $\lambda \in \mathbb{R}$ such that

$$J(v) = \lambda v.$$

Applying J to both sides gives $-v = \lambda^2 v$. Since $v \neq 0$, it follows that $\lambda^2 = -1$. But λ is a real number, and so this is a contradiction. Thus, $\dim V$ is even. $\square$

The above proof is actually a direct way to compute the eigenvalues of any complex structure, which are $\pm i$ (when viewed over the complex numbers). This gives further mathematical evidence for the validity of the definition of a complex structure.

Next, we build the connection from abstract complex structures to the familiar notion of complex vector spaces more directly.

Proposition 8.6.4. *Let V be a real vector space and $J : V \rightarrow V$ a linear complex structure. Then V can be given the structure of a vector space over $\mathbb{C}$ by defining $\mathbb{C}$ -scalar multiplication as follows:*

$$(a + bi) \cdot v := av + bJ(v) \quad \text{for all } v \in V.$$

Proof. We verify one of the vector space axioms, and leave the rest as an exercise. In particular, we will show that for all $\lambda, \mu \in \mathbb{C}$ and $v \in V$, we have

$$\lambda(\mu v) = (\lambda\mu) v.$$

Fix $v \in V$, and write $\lambda = a + bi \in \mathbb{C}$ and $\mu = c + di \in \mathbb{C}$. Then we have

$$\begin{aligned} \lambda(\mu v) &= (a + bi) [(c + di)v] \\ &= (a + bi)[cv + dJ(v)] \\ &= a[cv + dJ(v)] + bJ([cv + dJ(v)]) \\ &= acv + adJ(v) + bcJ(v) - bdv \\ &= (ac - bd)v + (ad + bc)J(v) \\ &= [(ac - bd) + (ad + bc)i] v \\ &= [(a + bi)(c + di)] v \\ &= (\lambda\mu) v \end{aligned}$$

as desired. The other axioms are left as an exercise. $\square$

8.6.2 Compatible complex structures

Next, we can start discussing the interaction between linear complex structures and symplectic forms.

Definition 8.6.5. Let V be a vector space and ω a symplectic form. A linear complex structure $J : V \rightarrow V$ is **compatible with** ω , also written as **ω -compatible**, if

- (1) $\omega(v, Jv) > 0$ for all $v \neq 0 \in V$, and
- (2) $\omega(Jv, Jw) = \omega(v, w)$ for all $v, w \in V$.

Example 8.6.6. Consider the vector space $\mathbb{R}^{2n}$ and let ω_{std} be the standard symplectic structure. Let J_{std} denote the standard complex structure (here we are abusing notation and writing J_{std} to represent both the abstract transformation $J_{\text{std}} : \mathbb{R}^{2n} \rightarrow \mathbb{R}^{2n}$ and the familiar $2n \times 2n$ matrix J_{std}). We claim that $-J_{\text{std}}$ is an ω_{std} -compatible complex structure.

To verify this, first recall that $\omega_{\text{std}}(v, w) = v^T J_{\text{std}} w$ for all $v, w \in \mathbb{R}^{2n}$. Also, recall that $J_{\text{std}}^2 = -I_{2n}$. We separately consider each condition (1) and (2) above.

- (1) Let $v \neq 0 \in \mathbb{R}^{2n}$. Note that

$$\omega_{\text{std}}(v, -J_{\text{std}}v) = v^T J_{\text{std}}(J_{\text{std}}v) = -v^T J_{\text{std}}^2 v = v^T I_{2n} v = v^T v.$$

Next, recall that the standard inner product on $\mathbb{R}^{2n}$ is given exactly by $\langle v, w \rangle = v^T w$. Thus, this, shows that

$$\omega_{\text{std}}(v, -J_{\text{std}}v) = \|v\|^2 > 0$$

where the norm is given by the standard inner product.

- (2) Next, let $v, w \in \mathbb{R}^{2n}$. We have

$$\begin{aligned} \omega_{\text{std}}(-J_{\text{std}}v, -J_{\text{std}}w) &= (-J_{\text{std}}v)^T J_{\text{std}}(-J_{\text{std}}w) \\ &= v^T (-J_{\text{std}})^T I_{2n} w \\ &= v^T J_{\text{std}} w \\ &= \omega_{\text{std}}(v, w) \end{aligned}$$

as desired.

Thus, $-J_{\text{std}}$ defines a complex structure which is compatible with ω_{std} .

Observe that by the computation in (1) above, for the compatible complex structure $-J_{\text{std}}$, we have

$$\omega_{\text{std}}(v, -J_{\text{std}}w) = \langle v, w \rangle$$

where $\langle \cdot, \cdot \rangle$ denotes the standard inner product on $\mathbb{R}^{2n}$. This is no accident! One of the nice properties of compatible complex structures is that they blend with the symplectic form to give a real inner product.

Proposition 8.6.7. Let (V, ω) be a symplectic vector space, and let $J : V \rightarrow V$ be an ω -compatible complex structure. Then $\langle \cdot, \cdot \rangle_J$ given by $\langle v, w \rangle_J := \omega(v, Jw)$ defines a real inner product on V .

Proof. ADD □

Exercises

1. Let V be a real vector space and $J : V \rightarrow V$ be a linear map satisfying $J^2 = -I$. Give an alternative proof to the fact that $\dim V$ is even using determinants.
2. Let V be a real vector space and a $J : V \rightarrow V$ a complex structure. Verify the rest of the vector space axioms to show that V is a vector space over $\mathbb{C}$ with scalar multipl

8.7 Where to from here? Survey articles on symplectic geometry

In this chapter we have developed thoroughly the theory of symplectic linear algebra. Symplectic linear algebra is itself already interesting, but it is only a brief glimpse into a much deeper part of math called *symplectic geometry*. The main motivation for developing symplectic linear algebra is to use it as a stepping stone to study the more complicated world of symplectic geometry. Unfortunately, symplectic geometry is a subject with a large number of prerequisites, and so discussing it carefully is out of the scope of this text.

Instead, this section will be an informal and imprecise collection of survey articles into the world of symplectic geometry, written for you to see how the theory of symplectic linear algebra evolves as you wade deeper into the mathematical ocean. As such, this very optional section will feel much different than all the others. It is only meant to tell a story in a way that is accessible for an undergraduate linear algebra student.

FINISH

8.7.1 Symplectic manifolds

8.7.2 Contact manifolds

Chapter 9

Fredholm Index Theory

Chapter 10

Lie Algebras

Part III

Appendix

Appendix A

Miscellaneous Review Topics

A.1 Induction

Induction is a well-known and widely used proof technique. It is used to prove statement that come with the descriptor *for all* $n \in \mathbb{N}$, or something similar. That is, suppose you have a collection of mathematical statements $P(n)$, indexed by the natural numbers. As a silly example,

$$\begin{aligned} P(1) &= \text{"The number 1 is not a color."} \\ P(2) &= \text{"The number 2 is not a color."} \\ P(3) &= \text{"The number 3 is not a color."} \\ &\vdots \\ P(n) &= \text{"The number } n \text{ is not a color."} \\ &\vdots \end{aligned}$$

We would like to prove the validity of the statements $P(n)$ for all $n \geq 1$. One possible approach is to prove them one by one. That is, first prove $P(1)$, then $P(2)$, then $P(3)$, and so on, until we have proven all statement. This, unfortunately, is not a very smart idea, because there are infinitely many statements. We would spend the rest of our lives proving the statements and would die before we ever finished. We could conscript all future humans to continue proving statements one by one after we perish, and this would still not be enough. Humanity would go extinct before we can prove infinitely many statements directly. This approach is not feasible.

Another approach is to write down a proof for the validity of $P(n)$ completely abstractly, without making an argument that depends on the specific value of the number n . This *is* a perfectly valid technique. You shouldn't blindly use induction, just because a statement contains the phrase "for all n ." Sometimes a direct and n -abstract argument will work.

However, sometimes a direct n -abstract argument can be difficult, and it can be easier to use induction (which I have yet to explain). The philosophy of induction is to treat the sequence of statements $P(1), P(2), P(3), \dots$ like a chain of standing dominoes. Proving the statement $P(n)$ corresponds to tipping over the n th domino. Instead of tipping over each domino individually (the first approach) or calling upon an earthquake to simultaneously tip over all dominoes at once (the second method), we can take a more refined approach by only tipping over the first

domino, and then arguing that each falling domino will tip over the next. By starting a chain reaction — and proving that a chain reaction must take place — we can delicately finish the job.

To say this in a more mathematical formulation, a proof by induction proceeds as follows. You prove directly the first statement, say, $P(1)$. This corresponds to tipping the first domino. Then, you must prove that if a statement $P(k)$ is true, then the following statement $P(k + 1)$ must also be true. This proves that one falling domino will cause the next to fall. I'll summarize this in the following principle, and then we will work through some examples.

Principle A.1.1 (Principle of induction.). To prove a statement of the form “ $P(n)$ for all $n \geq 1$ ” using induction, you:

- ◊ Establish a *base case* by proving the statement $P(1)$ directly. This tips the first domino.
- ◊ Prove that validity of $P(k)$ implies validity of $P(k + 1)$. That is, assume that $P(k)$ is true — this is the *inductive hypothesis* — and prove that $P(k + 1)$ must also then be true. This shows that if one domino falls, then the next will fall too.

A.1.1 The size of power sets

As a first example, let's consider a concept rooted in the basics of set theory.

Definition A.1.2. Let S be any set (possibly infinite). The **power set of S** , denoted $\mathcal{P}(S)$, is the set consisting of all subsets of S . That is,

$$\mathcal{P}(S) := \{ X \mid X \subset S \}.$$

Example A.1.3. Let $S = \{1, 2\}$. Then

$$\mathcal{P}(S) = \{ \emptyset, \{1\}, \{2\}, \{1, 2\} \}.$$

Note that the empty set is a subset of every set, and hence always an element of the power set of anything! Similarly, S is always a subset of S , and so $S \in \mathcal{P}(S)$.

Example A.1.4. Let $S = \{1, 2, 3\}$. Then

$$\mathcal{P}(S) = \{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{2, 3\}, \{1, 3\}, \{1, 2, 3\} \}.$$

Example A.1.5. Nothing stops us from taking power sets of power sets. Let $S = \{1, 2\}$. Then

$$\begin{aligned} \mathcal{P}(\mathcal{P}(S)) = \{ & \emptyset, \{\emptyset\}, \{\{1\}\}, \{\{2\}\}, \{\{1, 2\}\}, \\ & \{\emptyset, \{1\}\}, \{\emptyset, \{2\}\}, \{\emptyset, \{1, 2\}\}, \{\{1\}, \{2\}\}, \{\{1\}, \{1, 2\}\}, \{\{2\}, \{1, 2\}\}, \\ & \{\emptyset, \{1\}, \{2\}\}, \{\emptyset, \{1\}, \{1, 2\}\}, \{\emptyset, \{2\}, \{1, 2\}\}, \{\{1\}, \{2\}, \{1, 2\}\}, \{\emptyset, \{1\}, \{2\}, \{1, 2\}\} \} \end{aligned}$$

Power sets of power sets and beyond get difficult to itemize explicitly because of the mess of nested parentheses, but I'll point out one thing: the elements $\emptyset$ and $\{\emptyset\}$ are distinct elements of

the double power set!

Now I'll get to the point where we can use some induction. For any (finite) set S , let $|S|$ denote the number of elements in the set. For example, if $S = \{1, 2\}$ then $|S| = 2$, and if $S = \{1, 2, 3\}$ then $|S| = 3$. If you count the elements in the above examples, you'll pick up on the following pattern:

- ◊ When $S = \{1, 2\}$, so that $|S| = 2$, we have $|\mathcal{P}(S)| = 4 = 2^2 = 2^{|S|}$.
- ◊ When $S = \{1, 2, 3\}$, so that $|S| = 3$, we have $|\mathcal{P}(S)| = 8 = 2^3 = 2^{|S|}$.
- ◊ When $S = \{1, 2\}$, so that $|S| = 2$, we have $|\mathcal{P}(S)| = 4$ and $|\mathcal{P}(\mathcal{P}(S))| = 16 = 2^4 = 2^{|\mathcal{P}(S)|}$.

This pattern is no accident. In general, if you have a set of size n , then the size of the power set will be 2^n .

Proposition A.1.6. *Let S be a set with $|S| = n$. Then $|\mathcal{P}(S)| = 2^n$.*

Proof. Before we write a purple-quality proof, I will point out that, although not explicit, this proposition does say implicitly that something is true for all n . In this case, we have to make out the implication that the statement of the proposition implies “for all $n \geq 0$.” Next, because this is our first induction proof, I will intersperse my own commentary (in black print) with the professional-quality writing that would eventually make up my final proof (purple print). Keep in mind that if I were writing a proof for this problem on a test, or on homework, or in a research paper, I would *only* present the purple parts of this proof.

The first step in an induction proof is to directly proof a base case. Above, I hinted that this statement is true for all $n \geq 0$, and so we can take $n = 0$ as our base case.

We proceed by induction on n . We begin with the base case $n = 0$. Let S be a set with $|S| = 0$. The only such set is the empty set, so $S = \emptyset$. Note that

$$\mathcal{P}(S) = \{\emptyset\}$$

and so $|\mathcal{P}(S)| = 1 = 2^0$. This verifies the base case.

A few more comments before proceeding. Note that I began my proof by telling the reader *I am going to use induction*. Comments like this help the readability of your proofs, as they help the reader understanding your argument on a greater scale. Next, I'll once again point out the subtlety in that the empty set $\emptyset$, has *no* elements. It is the set $\emptyset = \{\}$. The set $\{\emptyset\}$ is very different than $\emptyset$.

Next, we perform the inductive step. We assume that we know that for some $k \geq 0$ that the statement is already true. We will prove that this forces the statement to be true for $k + 1$. We are not *proving* the statement is true for k , and we are not even claiming that it is true. We are simply hypothetically assuming that the statement is true for k , and then proving that, if this were the case, it would also have to be true for $k + 1$.

Next, we perform the inductive step. Assume that for some $k \geq 0$, it is true that if S is a set with $|S| = k$, then $|\mathcal{P}(S)| = 2^k$. Let $\tilde{S}$ be a set with $|\tilde{S}| = k + 1$. We must prove that $|\tilde{\mathcal{P}}| = 2^{k+1}$.

This is the set up for our inductive step. Note that we will make significant use of the inductive assumption to prove validity for $k + 1$. This is the point of using induction. To do this, we have to figure out how to use our assumption of sets of size k . Our current set of interest, $\tilde{S}$, contains $k + 1$ elements. It seems like a reasonable idea to me to write $\tilde{S}$ as a union of a subset of size k and one other element. From there, we can understand the power set of $\tilde{S}$ in terms of the power set of the smaller subset and the extra element.

Write $\tilde{S} = S \cup \{x\}$, where $S \subset \tilde{S}$ is a subset with $|S| = k$. By the inductive hypothesis, we know that $|\mathcal{P}(S)| = 2^k$. Thus, we can write

$$\mathcal{P}(S) = \{X_1, X_2, \dots, X_{2^k}\}.$$

Note that

$$\mathcal{P}(\tilde{S}) = \left\{ \underbrace{X_1, X_2, \dots, X_{2^k}}_{2^k \text{ elements}}, \underbrace{X_1 \cup \{x\}, X_2 \cup \{x\}, \dots, X_{2^k} \cup \{x\}}_{2^k \text{ elements}} \right\}.$$

Thus, $|\mathcal{P}(\tilde{S})| = 2^k + 2^k = 2 \cdot 2^k = 2^{k+1}$, as desired. By the principle of induction, this completes the proof.

□

Alternate proof. Before giving you some more examples of proofs by induction, I want to point out that we can prove the above proposition about sizes of power sets directly, without using induction. Arguably, a direct proof is easier.

Let S be a set with $|S| = n$. We can enumerate the elements (in order) as follows:

$$S = \{x_1, \dots, x_n\}.$$

We can identify each element $X \in \mathcal{P}(S)$ with a string of length n of 0s and 1s, where the j th digit, 0 or 1, indicates whether $x_j \notin X$ or $x_j \in X$, respectively. For example, the element $X = \{x_1, x_3, x_n\} \in \mathcal{P}(S)$ would correspond to the string

$$1010 \cdots 01.$$

Note that there are 2^n such strings, as each digit has two possibilities (0 or 1). Thus, $|\mathcal{P}(S)| = 2^n$, as desired.

□

A.1.2 Some numerical inequalities

Proving the size of power sets is a fairly sophisticated example with which to introduce the concept of induction. For some simpler examples, we can consider some numerical inequalities; statements about integers that holds for all n larger than some initial value.

Proposition A.1.7. *For all $n \geq 4$, we have $n! \geq 2^n$.*

Proof. We proceed by induction on n . First we establish the base case $n = 4$. Note that $4! = 4 \cdot 3 \cdot 2 \cdot 1 = 24 \geq 16 = 2^4$, as desired.

Next, we perform the inductive step. Suppose that for some $k \geq 4$ we know that $k! \geq 2^k$. Note that

$$\begin{aligned}(k+1)! &= (k+1) \cdot k! \\ &\geq (k+1) \cdot 2^k \\ &\geq 2 \cdot 2^k \\ &= 2^{k+1}.\end{aligned}$$

In the second inequality we used the inductive hypothesis, and in the third inequality we used the fact that $k+1 \geq 2$, since $k \geq 4$. By the principle of induction this proves that $n! \geq 2^n$ for all $n \geq 4$. $\square$

Proposition A.1.8. *Prove that for all integers $n \geq 0$ and all real numbers $x \geq -1$, we have $(1+x)^n \geq 1 + nx$.*

Proof. We proceed by induction on n . First, we establish the base case $n = 0$. Note that for any $x \geq -1$, we have $(1+x)^0 = 1 \geq 1 = 1 + 0x$ as desired.

Next we perform the inductive step. Suppose that for some $k \geq 1$, we know that $(1+x)^k \geq 1 + kx$ for all $x \geq -1$. Then

$$\begin{aligned}(1+x)^{k+1} &= (1+x)^k(1+x) \\ &\geq (1+kx)(1+x) \\ &= 1 + kx + x + kx^2 \\ &= 1 + (k+1)x + kx^2 \\ &\geq 1 + (k+1)x.\end{aligned}$$

The first inequality comes from the inductive hypothesis and crucially uses the fact that $x \geq -1$, and hence $1+x \geq 0$. Otherwise the inequality would flip directions. The last inequality comes from the fact that $kx^2 \geq 0$. $\square$

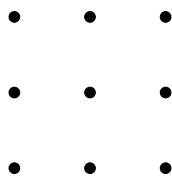
I'll close our discussion of induction with some observations about these two examples. First, note the overall structural similarity between the proofs of these inequalities with the power set example above. I began by telling the reader I would perform induction, and then I verified the base case. After that, I did the inductive step and very explicitly used the inductive hypothesis to make the relevant conclusion. All of these features are common in any induction proof.

A.2 Equivalence relations

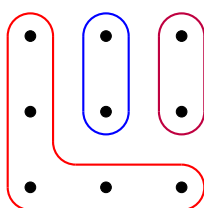
The notion of an *equivalence relation* is ubiquitous in math. In the context of linear algebra at this level, it can help us understand certain finite fields and perform other constructions like quotienting vector spaces by subspaces.

A.2.1 The definition of an equivalence relation

Before giving the precise definition of an equivalence relation, let me describe the intuitive idea. Suppose we have a collection of objects, which I will represent below as a collection of dots.



This collection of dots represents a set of 9 elements. Suppose that for some reason, we wanted to group the objects together so that we could treat groups as one, new object. For example, we could group the objects together like this:

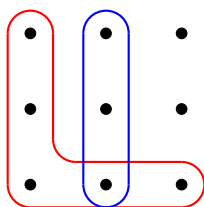


With this grouping, we now think of having 3 total objects, not 9. The red group is a single blob, the blue group is a single blob, and the purple group is a single blob. If you pick up one of the dots in the red blob, you pick up the entire blob. If you tell your friend *hey look at that dot in the middle of the top row*, all your friend will see is the entirety of the blue blob. If you ask your friend, *hey what's the difference between the top right dot and the dot in the middle of the third column*, they will tell you that there is no difference; they both exist as one in the purple blob. This is an equivalence relation! It's just a way to group collections of objects in a way that you treat the groups as one.

Now, there are some features of the above grouping that are important. I will highlight two.

- ◇ Every dot is in a group.
- ◇ No dot is in more than one group.

For example, consider the same collection above but with this attempt at grouping.



This does *not* describe an equivalence relation (which I still have yet to define). This grouping is not great, for a few reasons. First, two of the dots aren't part of a grouping at all; they are left out. We don't want to leave any object out of a grouping. Secondly, the middle dot in the bottom row is part of both the red and blue group. We don't want this to happen, because we want each group to be completely distinct so that we can treat them as separate objects.

With these pictures in mind, here is the official definition of an equivalence relation. After making some sense of the definition, we will consider a few linear-algebra-relevant examples.

Definition A.2.1. Let S be a set. An **equivalence relation** on S is a symbol $\sim$ that satisfies the following axioms.

- (1) (Reflexivity) For all $x \in S$, $x \sim x$.
- (2) (Symmetry) For all $x, y \in S$, we have $x \sim y$ if and only if $y \sim x$.
- (3) (Transitivity) For all $x, y, z \in S$, if $x \sim y$ and $y \sim z$, then $x \sim z$.

For any $x \in S$, let $[x] := \{y \in S \mid y \sim x\}$. The set $[x]$ is called the **equivalence class** of x .

In this definition, you should read the symbol “ $\sim$ ” as “is in the same group as”. So $x \sim x$, the reflexivity condition, means that x must be in the same group as itself. In terms of our above pictures, this is a trivial requirement. The symmetry condition means that if x is in the same group as y , then y must be in the same group as x . This is also a bit tautological in terms of the above groupings. The transitivity condition says that if x and y are in the same group, and if y and z are in the same group, then x and z have to be in the same group. This condition is violated by the figure above with overlapping blue and red groups!

A.2.2 An example involving finite fields

Our first example of an equivalence relation will be on the set of integers $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$. Define an equivalence relation (i.e., a grouping) $\sim$ as follows:

$$n \sim m \quad \text{if and only if } n - m \text{ is divisible by } 3.$$

To be clear, an integer M is **divisible by 3** if $M = 3k$ for some $k \in \mathbb{Z}$. Let’s verify that this is an equivalence relation, and then we’ll discuss how to think about what this gives us.

- (1) (Reflexivity) First we show that $n \sim n$ for all $n \in \mathbb{Z}$. Note that $n - n = 0 = 3 \cdot 0$ and so $n - n$ is divisible by 3. Thus, $n \sim n$.
- (2) (Symmetry) Suppose that $n \sim m$. This means that $n - m = 3k$ for some $k \in \mathbb{Z}$. It follows that $m - n = 3(-k)$, and so $m \sim n$. Thus, the symmetry property is satisfied.
- (3) (Transitivity) Suppose that $n \sim m$ and $m \sim \ell$. We wish to show that $n \sim \ell$. Since $n \sim m$ and $m \sim \ell$, we have $n - m = 3k_1$ and $m - \ell = 3k_2$ for some $k_1, k_2 \in \mathbb{Z}$. Thus,

$$n - \ell = (n - m) + (m - \ell) = 3k_1 + 3k_2 = 3(k_1 + k_2)$$

and so $n \sim \ell$, as desired.

Thus, $\sim$ as defined above is indeed an equivalence relation on $\mathbb{Z}$. But what is it telling us?

To understand this, let’s consider the *set of equivalence classes*. For example, consider the integer $0 \in \mathbb{Z}$. What is the equivalence class $[0]$? By the above definition, $[0]$ is the set of

integers n such that $n \sim 0$. That is, the set of integers n such that $n - 0 = 3k$. This means that $[0]$ is the set of all multiples of 3:

$$[0] = \{\dots, -9, -6, -3, 0, 3, 6, 9, \dots\}.$$

To emphasize the point of an equivalence relation, we also have $[0] = [3] = [6]$ and so on. The notation $[a]$ just means “the group containing the element a .” What about $[1]$, the equivalence class containing 1? Well, this would be the set of integers n such that $n - 1 = 3k$, so that $n = 3k + 1$. This means that

$$[1] = \{\dots, -8, -5, -2, 1, 4, 7, 10, \dots\}.$$

Likewise, one can check that

$$[2] = \{\dots, -7, -4, -1, 2, 5, 8, 11, \dots\}.$$

You’ll notice that with these three equivalence classes (groups), we have now partitioned all of the integers. That is, we have decomposed $\mathbb{Z} = [0] \cup [1] \cup [2]$. This is a grouping of integers according to their remainder after division by 3. This should feel familiar, if you’ve spent some time thinking about finite fields. In particular, the field $\mathbb{F}_3$ can be described precisely using the three equivalence classes of $\mathbb{Z}$ under $\sim$ as above!

A.2.3 An example involving subspaces of vector spaces

Next, here is an equivalence relation in the realm of vector spaces.¹ This equivalence relation is used in defining *quotient vector spaces*, informally, a way to “divide” a vector space by a subspace.

Let V be any vector space, and let $W \subset V$ be a subspace. Define $\sim$ on V as follows:

$$v \sim w \quad \text{if and only if } v - w \in W.$$

Let’s first verify that this is in fact an equivalence relation, and then we will discuss what it means.

- (1) (Reflexivity) For any $v \in V$, note that $v - v = 0 \in W$, since W is a subspace. Thus, $v \sim v$. This proves that $\sim$ is reflexive.
- (2) (Symmetry) Suppose that $v \sim w$. Then $v - w \in W$. Since W is a subspace, and hence is closed under scalar multiplication, we have that $(-1)(v - w) = w - v \in W$, and hence $w \sim v$. This proves that $\sim$ is symmetric.
- (3) (Transitivity) Suppose that $v \sim w$ and $w \sim u$. Then $v - w, w - u \in W$. Note that

$$v - u = (v - w) + (w - u) \in W$$

since W is closed under addition. Thus, $v \sim u$, and so $\sim$ is transitive.

¹To understand this subsection, you should know what a subspace of a vector space is.

Thus, $\sim$ is in fact an equivalence relation on V .

To see what this equivalence class gives us, let's consider the set of equivalence classes. First, consider $[0]$, the equivalence class containing the 0 vector. This is the set of all vectors equivalent to 0, which is the set of all vectors v such that $v - 0 \in W$. But this is precisely all of the vectors in W , and so $[0] = W$. That is, under this equivalence relation we think of the entire subspace W as being one object.

All of the other objects (equivalence classes) are parallel shifts of the subspace W . For example, let $v_0 \in V$ be a vector which is *not* in the subspace W . Then

$$[v_0] = \{ w \in V \mid w - v_0 \in W \} = \{ w \in V \mid w \in \{v_0\} + W \}.$$

In words, the equivalence class containing v_0 is given by adding v_0 to every single vector in W , that is, shifting the subspace W by v_0 . Note that $[v_0]$ is not a subspace of V , because it won't contain the 0 vector — instead, it is a parallel shift of W away from the origin by v_0 . Thus, the equivalence classes of V under $\sim$ foliate the vector space V by parallel copies of W .

A.3 Row reduction

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